Companion volume · Phases 0–10

Catalogue of Proofs

Exact recovery of a differential operator from a single solution

Part I
Phases 0–5, load-bearing
Dim. A1–A31

Part II
Phases 6–10, load-bearing
Dim. A32–A53

Part III
Phases 0–5, the rest
Dim. B1–B72

Part IV
Phases 6–10, the rest
Dim. B73–B97, list cards

Every card carries its ledger identifier, its trust label, and the page of the statement in the companion paper *Exact Recovery of a Differential Operator from a Single Solution, complete edition, Phases 0–10*.

October 2026

How to read this catalogue

This volume contains the proofs of the results stated in the paper *Exact Recovery of a Linear Differential Operator from a Single Solution* (complete edition, Phases 0–10). It is a catalogue: every result has one card, and every card has the same shape.

The shape of a card

Dim. A0 **ID-T0**

Name of the result [proved]

The full statement, with every hypothesis.

> **Idea.** One paragraph that says why the result is true, before any computation.

Proof.

Step 1 First claim

One claim per step, with its computation displayed on its own line.

Step 2 Second claim

...

Why it works. one sentence that names the mechanism of the proof.

■

Uses: the results this proof relies on, by catalogue number.

The four parts

Part I contains the 31 results of Phases 0–5 on which everything else rests (Dim. A1–A31). They were selected from the project's dependency graph: a result is in Part I if many other results depend on it, or if it is a headline theorem of the paper. Their proofs are given in full, step by step.

Part II contains the load-bearing results of Phases 6–10 (Dim. A32–A53): systems and gauge, variable coefficients, convergence and noise, the analytic encoder, the universal dictionary, and the Phase 10 corrections to results of Phases 0–5. They were selected by the same criterion: use in the extended algorithm (Chapter 17 of the paper), dependence in the ledger, and headline status.

Part III contains all the other results of Phases 0–5, grouped by phase and ordered by identifier (Dim. B1–B72). Short and medium proofs are given in full. Proofs that take many pages in the research files (the Burgers asymptotics, the scattering identification, the Hirota computations) are given as a condensed card, with every step and key formula and a pointer to the research file.

Part IV contains the other results of Phases 6–10 (Dim. B73–B155 and list cards). Eighty-three results have a card with a proof (Dim. B73–B155); the others are *list cards*: the statement, its trust label and the research file that holds the proof. All proofs of Part II and Part IV were rewritten from the research packages for this volume and were not re-audited; the trust label refers to the result as recorded in the ledger. This is a deliberate limit of this edition: for Phases 6–10 the research packages are about 150,000 words, and only the proofs that the extended algorithm or a corrected statement depends on were rewritten here.

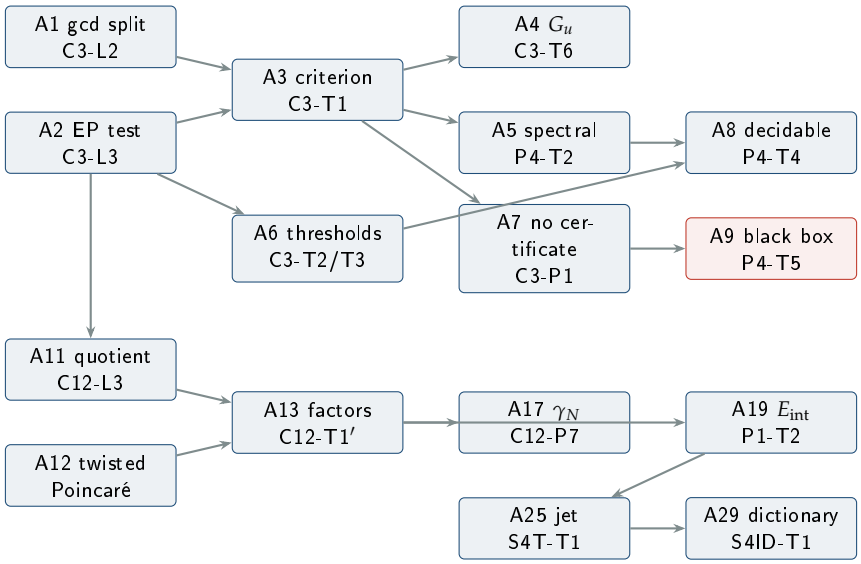
Which edition to read. The companion volumes for Phases 0–5 alone are *paper_fasi05* and *dimostrazioni_fasi05*. In case of a difference, this volume prevails: Phase 10 refuted or corrected several statements of Phases 0–5 (Chapter 10).

Trust labels

Label	Meaning
[proved]	complete proof, passed an independent review
[proved mod. X]	complete except for a cited published theorem X, whose source was checked
[proof written, not audited]	complete proof, not independently reviewed
[from cited source]	a published theorem, used as a black box
[conditional]	depends on a lemma that is not proved
[checked on examples]	checked on examples only
[false]	a counterexample is proved (the card proves the counterexample)
[open]	unsolved

The reviewers were independent AI agents, not human referees. Identifiers such as C3-T1 are those of the project ledger `claims/LEDGER.md`; the research file that holds each original proof is named in the index at the end.

The backbone of Part I (Phases 0–5)



The arrows mean "is used in the proof of". Blue: positive results; red: a negative (impossibility) result. The arrows come from the `depends_on` edges of the project graph.

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Part I

Phases 0–5: the load-bearing proofs

CHAPTER 1

Algebra and identification

In this chapter

Starting from	$\Omega \subset \mathbb{R}^2$ open and connected; the ring $\mathcal{R} = \mathbb{C}[\tau, \xi]$ acting on distributions by $\tau \mapsto \partial_t$, $\xi \mapsto \partial_x$. Classical facts used without proof: unique factorisation in $\mathcal{R}$ (Gauss), the Nullstellensatz, the Chinese remainder theorem, and the fact that distributions agreeing near every point agree.
Why	every identification argument reduces a question about solutions to a question about ideals and exponential-polynomials. This chapter proves those reductions in full, then the identification theorems built on them.
Role	Dim. A1–A9. They prove Chapters 3–5 of the paper: the gcd splitting, the EP test, the identification criterion, the intrinsic operator, the spectral form, the finite-data thresholds, and the decidability results.

Notation for the whole chapter

$V(I) \subset \mathbb{C}^2$ is the common zero set of an ideal I. For a point $z = (\tau_z, \xi_z) \in \mathbb{C}^2$,

$$\mathfrak{m}_z = (\tau - \tau_z,\ \xi - \xi_z)$$

is its maximal ideal, and $e^{\langle z, \cdot \rangle}$ denotes $e^{\tau_z t + \xi_z x}$. $\mathrm{Ann}(w) = \{Q \in \mathcal{R} :\ Q(\partial) w = 0\}$. $\mathrm{EP}(\Omega)$ is the set of restrictions to Ω of finite sums $\sum_k q_k\, e^{\langle z_k, \cdot \rangle}$ with polynomials q_k.

1.1 The algebraic toolkit

Dim. A1 **C3-L2**

Splitting off the gcd [proved]

Let $0 \neq I \subseteq \mathcal{R}$ be an ideal and $G = \gcd(I)$. Then:

(i) $I = G \cdot J$ for an ideal J with $\gcd(J) = 1$;

(ii) $V(J)$ is finite;

(iii) $(\sqrt{J})^N \subseteq J$ for some N, and $J = \mathcal{R}$ if $V(J) = \emptyset$.

Two facts are proved on the way:

- **(V)** two nonzero coprime polynomials have finitely many common zeros;
- **(F)** an irreducible polynomial has infinitely many zeros; hence if $V(\operatorname{Ann} w)$ is finite, some $h \in \operatorname{Ann}(w)$ is not divisible by a given irreducible P_i.

Stated in the paper as no. 3.2, p. 13

Idea. Over the field $\mathbb{C}(\xi)$ two coprime polynomials have a Bézout identity; clearing denominators gives a polynomial in ξ alone that vanishes at every common zero, and likewise in τ. After dividing an ideal by its gcd, a generic combination of its elements is coprime to any fixed element, and (V) applies.

Proof.

Step 1 **Fact (V): coprime polynomials meet in finitely many points**

Let $f, g \neq 0$ be coprime and $K = \mathbb{C}(\xi)$. A common factor of positive τ-degree in $K[\tau]$ would give, by the Gauss lemma, a common factor in $\mathbb{C}[\xi][\tau]$; so f, g are coprime in the principal ideal domain $K[\tau]$. Hence there are $a, b \in K[\tau]$ with

$$af + bg = 1.$$

Multiplying by a common denominator,

$$a'f + b'g = r(\xi), \qquad a', b' \in \mathcal{R}, \quad r \in \mathbb{C}[\xi] \setminus \{0\}.$$

At a common zero (τ_0, ξ_0) the left side vanishes, so $r(\xi_0) = 0$: finitely many ξ_0. Exchanging τ and ξ gives $s(\tau) \neq 0$ with $s(\tau_0) = 0$. Therefore

$$V(f, g) \subseteq \{r = 0\} \times \{s = 0\},$$

a finite set.

Step 2 **Fact (F): an irreducible polynomial has infinitely many zeros**

If $\deg_\tau P_i \geq 1$, then for all ξ_0 outside the finitely many zeros of the leading coefficient, $P_i(\cdot, \xi_0)$ has a root τ_0. This gives infinitely many points. If $\deg_\tau P_i = 0$, then P_i is an irreducible polynomial in ξ, so $P_i = \alpha(\xi - \xi_0)$ and $V(P_i) = \mathbb{C} \times \{\xi_0\}$.

For the consequence: if every $h \in \operatorname{Ann}(w)$ were divisible by P_i, then $V(P_i) \subseteq V(\operatorname{Ann} w)$, an infinite set inside a finite one.

Step 3 (i) Dividing by the gcd

Put $J = \{f/G : f \in I\}$. It is closed under sums and under multiplication by $\mathcal{R}$, so it is an ideal, and $I = G \cdot J$. If a prime π divided every element of J, then $G\pi$ would divide every element of I, against the definition of G. So $\gcd(J) = 1$.

Step 4 (ii) A generic combination is coprime to a fixed element

If $J = \mathcal{R}$ there is nothing to prove. Otherwise pick $0 \neq f \in J$ with distinct prime factors $\pi_1, \dots, \pi_s$. By Step 3, for each a there is $g_a \in J$ with $\pi_a \nmid g_a$. For $c \in \mathbb{C}^s$ put

$$g_c = \sum_{b=1}^{s} c_b \, g_b \in J.$$

For each a the set $H_a = \{c : \pi_a \mid g_c\}$ is a linear subspace, and it is proper because the a-th unit vector is not in it. A finite union of proper subspaces does not cover $\mathbb{C}^s$, so some c lies in no H_a. Then f and g_c are coprime, and by Step 1

$$V(J) \subseteq V(f, g_c) \quad \text{is finite.}$$

Step 5 (iii) Nullstellensatz

The polynomials vanishing on $V(J)$ form $\sqrt{J}$, which is finitely generated, so a power of it lies in J. If $V(J) = \emptyset$, then $\sqrt{J} = \mathcal{R}$ and $1 \in J$.

Why it works. dividing by the gcd leaves an ideal with no common curve, and two curves without common component meet in finitely many points.

■

Uses: Gauss lemma, Nullstellensatz.

Dim. A2 **C3-L3**

Exponential-polynomials are the finite-dimensional functions [proved]

For $w \in \mathcal{D}'(\Omega)$, Ω connected, the following are equivalent:

$$w \in \mathrm{EP}(\Omega) \iff \dim_{\mathbb{C}} \mathcal{R}w < \infty \iff V(\operatorname{Ann} w) \text{ is finite.}$$

Moreover, with distinct z_k:

(a) $Q(\partial)\big(q\, e^{\langle z_k, \cdot \rangle}\big) = \big(Q(z+\partial)q\big)\, e^{\langle z_k, \cdot \rangle}$ for every polynomial q;
(b) if $\sum_k q_k\, e^{\langle z_k, \cdot \rangle} = 0$ on a nonempty open set, every $q_k = 0$;
(c) if $w = \sum_k q_k\, e^{\langle z_k, \cdot \rangle}$ with $q_k \neq 0$, then $V(\operatorname{Ann} w) = \{z_k\}$.

Stated in the paper as no. 3.3, p. 14

Idea. Exponentials are independent, so an exponential-polynomial lives on finitely many points. Conversely, a finite-dimensional derivative-stable space gives two one-

variable ODEs, $p(\partial_t)w = 0$ and $q(\partial_x)w = 0$. On a rectangle they force w to be an exponential-polynomial, and connectedness glues the rectangles together.

Proof.

Step 1 (a) Derivatives of an exponential times a polynomial

For one derivative,

$$\partial_t\big(q\,e^{\langle z,\cdot\rangle}\big) = \big(\partial_t q + \tau_z q\big)\,e^{\langle z,\cdot\rangle},$$

and the same with ∂_x and ξ_z. Iterating and using linearity gives (a).

Step 2 (b) Exponentials are linearly independent

The sum is real-analytic on $\mathbb{R}^2$ and vanishes on an open set, so it vanishes everywhere. Suppose some $q_k \neq 0$, and let K' be the set of such k. Choose $v \in \mathbb{R}^2$ with $\langle z_k - z_l, v\rangle \neq 0$ for $k \neq l$, and y_0 with $\prod_{K'} q_k(y_0) \neq 0$. Restricting to the line $y_0 + sv$ gives

$$f(s) = \sum_{k\in K'} r_k(s)\,e^{\mu_k s} \equiv 0, \qquad \mu_k = \langle z_k, v\rangle \text{ distinct}, \quad r_k(0) \neq 0.$$

We remove one term at a time. Multiply by $e^{-\mu_1 s}$ and differentiate $\deg r_1 + 1$ times. The first term disappears. Each other term becomes

$$\Big(\big(D + \mu_k - \mu_1\big)^{\deg r_1 + 1} r_k\Big)\,e^{(\mu_k - \mu_1)s},$$

and since $\mu_k \neq \mu_1$ the operator $(D + c)^N$ with $c \neq 0$ preserves the degree, so the new coefficient is nonzero. Induction ends with $r(s)e^{\mu s} \equiv 0$, $r \neq 0$: impossible.

Step 3 (c) The zero set of the annihilator is the set of exponents

By (a), the derivatives of $q\,e^{\langle z,\cdot\rangle}$ stay in $\mathbb{C}[t,x]_{\leq \deg q}\cdot e^{\langle z,\cdot\rangle}$, which is finite-dimensional. If $|\beta| > \deg q$, the product $(\tau - \tau_z)^{\beta_1}(\xi - \xi_z)^{\beta_2}$ acts on q as ∂^β and kills it. So

$$\prod_k \mathfrak{m}_{z_k}^{\deg q_k + 1} \subseteq \operatorname{Ann}(w), \qquad V(\operatorname{Ann} w) \subseteq \{z_k\}.$$

Conversely, if $Q \in \operatorname{Ann}(w)$ then $Q(z_k + \partial)q_k = 0$ by (a) and (b). The top-degree part of $Q(z_k + \partial)q_k$ is $Q(z_k)$ times the top-degree part of q_k, so $Q(z_k) = 0$.

Step 4 Finite zero set implies finite dimension

If $V(\operatorname{Ann} w) = \varnothing$ then $\operatorname{Ann} w = \mathcal{R}$ and $w = 0$. Otherwise, by A1(iii),

$$\operatorname{Ann}(w) \supseteq \mathfrak{I}(V)^N,$$

where $\mathfrak{I}(V)$ is the ideal of the finite set $V = V(\operatorname{Ann} w)$. By the Chinese remainder theorem

$$\mathcal{R}/\mathfrak{I}(V)^N \cong \bigoplus_{z\in V} \mathcal{R}/\mathfrak{m}_z^N,$$

which is finite-dimensional. Hence $\mathcal{R}w \cong \mathcal{R}/\operatorname{Ann}(w)$ is finite-dimensional too.

Step 5 Finite dimension gives two ODEs

Let $W = \mathcal{R}w$, stable under ∂_t and ∂_x. Let p and q be the minimal polynomials of $\partial_t|_W$ and $\partial_x|_W$. Then

$$p(\partial_t)\, w = 0, \qquad q(\partial_x)\, w = 0.$$

Step 6 On a rectangle the ODE in t separates the variables

Let $Q_0 = I' \times J' \subset \Omega$ be an open rectangle. We claim that every $v \in \mathcal{D}'(Q_0)$ with $p(\partial_t)v = 0$ has the form

$$v = \sum_{j,l} c_{jl} \otimes t^l e^{\lambda_j t}, \qquad c_{jl} \in \mathcal{D}'(J'),$$

with λ_j the roots of p. Induction on $\deg p$. Write $p = (\tau - \lambda)\tilde{p}$. By induction $(\partial_t - \lambda)v = \sum \tilde{c}_{jl} \otimes t^l e^{\lambda_j t}$. Each summand has a particular solution $\tilde{c} \otimes g(t)$:

$$g = r(t)\, e^{\lambda_j t}\ (\deg r = l)\ \text{ if } \lambda_j \neq \lambda, \qquad g = \frac{t^{l+1} e^{\lambda t}}{l+1}\ \text{ if } \lambda_j = \lambda.$$

With v_p their sum, $\partial_t\big(e^{-\lambda t}(v - v_p)\big) = 0$.

A distribution V on $J' \times I'$ with $\partial_t V = 0$ is $c \otimes 1$. Indeed, fix $\chi \in C_c^\infty(I')$ with $\int \chi = 1$. For a test function φ put $\Phi(x) = \int \varphi(x,t)\,\mathrm{d}t$ and

$$\psi(x,t) = \int_{-\infty}^{t} \big(\varphi(x,s) - \chi(s)\Phi(x)\big)\,\mathrm{d}s.$$

Since I' is an interval, ψ has compact support, and $\varphi = \chi \otimes \Phi + \partial_t \psi$. Hence $V(\varphi) = V(\chi \otimes \Phi)$, which depends only on Φ. So $v = v_p + c \otimes e^{\lambda t}$, completing the induction.

Step 7 The ODE in x makes the coefficients exponential-polynomials

Applying $q(\partial_x)$:

$$\sum_{j,l} q(\partial_x) c_{jl} \otimes t^l e^{\lambda_j t} = 0.$$

The functions $g_a = t^l e^{\lambda_j t}$ are independent on I', so there are test functions ψ_b with $\int g_a \psi_b = \delta_{ab}$. Testing against $\varphi(x)\psi_b(t)$ gives $q(\partial_x)c_{jl} = 0$. The one-variable case of Step 6 gives $c_{jl} \in \mathrm{EP}(J')$. So w restricted to Q_0 is the restriction of an exponential-polynomial E_{Q_0} on $\mathbb{R}^2$.

Step 8 Connectedness glues the rectangles

Fix Q_0 and $E = E_{Q_0}$. Let $A = \{y \in \Omega :\ w = E \text{ near } y\}$. It is open and nonempty. It is closed in Ω: if $y \in \bar{A}$, take a rectangle $Q_y \ni y$. On the open set $Q_y \cap A$ we have $E_{Q_y} = w = E$, so by Step 2 the exponential-polynomial $E_{Q_y} - E$ vanishes identically, and $w = E$ near y. By connectedness $A = \Omega$, so $w = E|_\Omega \in \mathrm{EP}(\Omega)$.

Why it works. finite dimension turns differentiation into two commuting matrices; their eigenvalues are the exponents, and rectangles plus connectedness globalise the local form.

■

Uses: A1.

Why rectangles in Step 6

On a U-shaped region, $v = \phi(x)\mathbf{1}_{t>2}$ (with $\phi = 0$ for $x < 1$, $\phi = 1$ for $x > 2$) is smooth and satisfies $\partial_t v = 0$, but it is not of the form $c(x) \otimes 1$. So the local form is proved on rectangles and glued in Step 8. This v is not a counterexample to A2: its derivative space is infinite-dimensional.

1.2 The identification theorems

Dim. A3 **C3-T1**

Identification criterion

[proved]

Let Ω be connected, $u \in \mathcal{D}'(\Omega)$, $P(\partial)u = 0$ with $P = c\prod_i P_i^{e_i}$ nonconstant, and

$$w_i = \Big(\frac{P}{P_i}\Big)(\partial)u, \qquad w_i' = \Big(\frac{P}{P_i^{e_i}}\Big)(\partial)u.$$

(a) $\operatorname{Ann}(u) = (P) \iff w_i \notin \mathrm{EP}(\Omega)$ for every i.

(b) $V(\operatorname{Ann} u) = V(P) \iff w_i' \notin \mathrm{EP}(\Omega)$ for every i.

(c) If P is squarefree, the two conditions coincide.

(d) If P is monic in τ of degree m: $\operatorname{Ann}(u) = (P)$ iff the only $(b_0, \dots, b_{m-1}) \in \mathbb{C}[\xi]^m$ with $\sum_j b_j(\partial_x)\partial_t^j u = 0$ is zero.

(e) In general $\operatorname{Ann}(u) = G \cdot J$ with $G \mid P$, $V(J)$ finite, and $G \sim P \iff \operatorname{Ann}(u) = (P)$.

Stated in the paper as no. 4.2, p. 19

Idea. Write $\operatorname{Ann}(u) = G \cdot J$. If G misses a copy of some P_i, then w_i is killed by a power of the ideal of the finite set $V(J)$, so it is an exponential-polynomial. Conversely, an exponential-polynomial w_i is killed by some h coprime to P_i, and $h \cdot P/P_i$ is an annihilator of u that P does not divide.

Proof.

Let $I = \operatorname{Ann}(u)$. It contains P. By A1, $I = G \cdot J$ with $G = \gcd I$ dividing P, so

$$G = c' \prod_i P_i^{f_i}, \qquad 0 \le f_i \le e_i.$$

Step 1 **(e) $G \sim P$ exactly when $I = (P)$**

If $G \sim P$, then $I = PJ \subseteq (P) \subseteq I$. If $I = (P)$, then P divides every element of I and is itself in I, so it is the gcd.

Step 2 (a), first direction: if $I \neq (P)$, some w_i is an exponential-polynomial

If $I \neq (P)$, then $G \not\sim P$, so some $f_i < e_i$, and $P/P_i = G\,H$ for some $H \in \mathcal{R}$. By A1(iii), $J \supseteq \mathfrak{n}^N$ with $\mathfrak{n}$ the ideal of $V(J)$. For $h \in \mathfrak{n}^N$:

$$h \cdot \frac{P}{P_i} = G\,(Hh) \in G\,J = I, \qquad \text{so} \qquad h(\partial)\,w_i = 0.$$

Hence $\mathrm{Ann}(w_i) \supseteq \mathfrak{n}^N$, its zero set lies in the finite $V(J)$, and $w_i \in \mathrm{EP}$ by A2.

Step 3 (a), second direction: an exponential-polynomial w_i gives a new annihilator

Suppose $w_i \in \mathrm{EP}$. Then $V(\mathrm{Ann}\,w_i)$ is finite, so by A1(F) some $h \in \mathrm{Ann}(w_i)$ has $P_i \nmid h$. Put

$$Q = h \cdot \frac{P}{P_i}.$$

Then $Q(\partial)u = h(\partial)w_i = 0$, so $Q \in I$. If $Q = PA$, cancelling P/P_i in the domain $\mathcal{R}$ gives $h = P_i A$, a contradiction. So $Q \in I \setminus (P)$.

Step 4 (b) The variety sees only whether each $f_i \geq 1$

We have $V(I) = V(G) \cup V(J)$, with $V(J)$ finite. If $f_i \geq 1$, then $V(P_i) \subseteq V(G) \subseteq V(I)$. If $f_i = 0$, then $V(P_i) \cap V(I)$ lies in the finite set

$$\bigcup_{f_j \geq 1} \big(V(P_i) \cap V(P_j)\big) \;\cup\; V(J),$$

while $V(P_i)$ is infinite, so $V(P_i) \not\subseteq V(I)$. Hence $V(I) = V(P)$ iff all $f_i \geq 1$. Finally, $f_i = 0$ iff $w_i' \in \mathrm{EP}$: one direction is Step 2 with $P/P_i^{e_i}$ in place of P/P_i; for the other, an $h \in \mathrm{Ann}(w_i')$ with $P_i \nmid h$ gives $G \mid h\,P/P_i^{e_i}$, and P_i divides neither factor, so $f_i = 0$.

Step 5 (c) and (d)

For $e_i = 1$, $w_i = w_i'$, which is (c). For (d), divide by the monic P in $\mathbb{C}[\xi][\tau]$:

$$Q = AP + \sum_{j<m} b_j(\xi)\,\tau^j \quad \text{uniquely.}$$

Since $A(\partial)P(\partial)u = 0$, $Q \in I$ iff $\sum_j b_j(\partial_x)\partial_t^j u = 0$, and $Q \in (P)$ iff all $b_j = 0$.

Why it works. the gcd of the annihilator records the infinite part of u; any factor it misses leaves behind a function killed by a zero-dimensional ideal, that is, an exponential-polynomial.

■

Uses: A1, A2.

> **Example 1.1 The proof on a wave**
>
> Let $P = \tau^2 - \xi^2$ and $u = e^{x-t} + e^{-(x+t)^2}$. For the factor $P_i = \tau + \xi$:
>
> $$w_i = (\partial_t - \partial_x)u = -2e^{x-t} \in \mathrm{EP}.$$

Step 3 takes $h = \xi - 1$, which kills e^{x-t} and is not a multiple of $\tau + \xi$, and produces

$$Q = (\xi - 1)(\tau - \xi) \in \mathrm{Ann}(u) \setminus (P).$$

Dim. A4 **C3-T6**

The intrinsic operator [proved]

Let $0 \neq P \in \mathrm{Ann}(u)$, $P = c\prod_i P_i^{e_i}$, and

$$S_i = \Big\{k \in \{0, \dots, e_i\} : \Big(\frac{P}{P_i^k}\Big)(\partial)u \in \mathrm{EP}\Big\}.$$

Then $G_u = \gcd \mathrm{Ann}(u) \sim \prod_i P_i^{f_i}$ with $f_i = e_i - \max S_i$. Moreover $\mathrm{Ann}(u)$ is principal iff $G_u(\partial)u = 0$, and then $\mathrm{Ann}(u) = (G_u)$.

Stated in the paper as no. 4.3, p. 22

Idea. The multiplicity f_i of P_i in G_u is detected by stripping copies of P_i from P one at a time: as long as fewer than $e_i - f_i$ copies are removed the leftover is an exponential-polynomial, and as soon as more are removed it is not.

Proof.

Step 1 The key equivalence

By A1, $\mathrm{Ann}(u) = G_u J_u$ with $G_u \mid P$. We show, for $1 \le k \le e_i$,

$$f_i \le e_i - k \iff \Big(\frac{P}{P_i^k}\Big)(\partial)u \in \mathrm{EP}.$$

($\Rightarrow$) Then $G_u \mid P/P_i^k$; write $P/P_i^k = G_u H$. As in A3, Step 2, $h \cdot P/P_i^k \in \mathrm{Ann}(u)$ for every h in a power of the ideal of $V(J_u)$, so the function is an exponential-polynomial.

($\Leftarrow$) If it is an exponential-polynomial, pick h in its annihilator with $P_i \nmid h$ (A1(F)). Then G_u divides $h \cdot P/P_i^k$. Comparing multiplicities of P_i gives $f_i \le e_i - k$.

Step 2 Reading off f_i

The equivalence holds trivially for $k = 0$. So $S_i = \{0, 1, \dots, e_i - f_i\}$, and $\max S_i = e_i - f_i$. (The set is an interval because EP is stable under derivatives:

$$\Big(\frac{P}{P_i^{k-1}}\Big)(\partial)u = P_i(\partial)\Big[\Big(\frac{P}{P_i^k}\Big)(\partial)u\Big].)$$

Step 3 Principality

If $G_u(\partial)u = 0$, then $(G_u) \subseteq \mathrm{Ann}(u) = G_u J_u \subseteq (G_u)$. Conversely, if $\mathrm{Ann}(u) = (Q)$, then Q is the gcd: $Q \sim G_u$, and $G_u(\partial)u = 0$.

Why it works. each copy of P_i that u does not need leaves an exponential-polynomial behind; counting them gives the exponent.

■

Uses: A1, A2, A3.

Dim. A5 **P4-T2**

Spectral theorem for the cyclic module [proved]

Let $W = \mathcal{R}u \cong \mathcal{R}/I$ with $I = \mathrm{Ann}(u) = G \cdot J$ as in A1. A *joint eigenvector* for $z \in \mathbb{C}^2$ is $w \in W$, $w \neq 0$, with $\mathfrak{m}_z(\partial)w = 0$; $\sigma_p(u)$ is the set of z having one.

(i) Every joint eigenvector for z is $c\,e^{\langle z,\cdot\rangle}$ with $c \neq 0$, and $\sigma_p(u) = V(J)$.

(ii) $W \cap \mathrm{EP}(\Omega) = \mathcal{R}\,G(\partial)u \cong \mathcal{R}/J$, of finite dimension $\dim \mathcal{R}/J$.

(iii) $\mathrm{Ann}(u)$ is principal $\iff J = \mathcal{R} \iff \sigma_p(u) = \emptyset \iff W \cap \mathrm{EP} = 0$.

(iv) Hence, if $P(\partial)u = 0$: $\mathrm{Ann}(u) = (P) \iff \sigma_p(u) = \emptyset$ and $G \sim P$.

Stated in the paper as no. 4.3, p. 21

Idea. Translate statements about the quotient ring $\mathcal{R}/I$ (maximal ideals, socle) into statements about W. The only analytic input is the EP test A2.

Proof.

Two small facts are used.

- **(LG)** If $G \mid (\tau - \tau_z)Q$ and $G \mid (\xi - \xi_z)Q$, then $G \mid Q$: in a unique factorisation domain the gcd of the two products is Q.
- **(LS)** If $M \neq 0$ is a module with $\mathfrak{m}_z^N M = 0$, some $v \neq 0$ in M has $\mathfrak{m}_z v = 0$: take $y \neq 0$ and k maximal with $\mathfrak{m}_z^k y \neq 0$.

Step 1 Joint eigenvectors are single exponentials

If $w = Q(\partial)u \neq 0$ and $\mathfrak{m}_z^N(\partial)w = 0$, then $V(\mathrm{Ann}\, w) \subseteq \{z\}$, so $w \in \mathrm{EP}$ and by A2 $w = q\,e^{\langle z,\cdot\rangle}$. The eigenvector condition gives $\partial_t q = \partial_x q = 0$: q is a constant.

Step 2 $\sigma_p(u) \subseteq V(J)$

Let $Q(\partial)u$ be a joint eigenvector at z. Then $(\tau - \tau_z)Q$ and $(\xi - \xi_z)Q$ lie in $I \subseteq (G)$, so $G \mid Q$ by (LG). Write $Q = GQ'$. Cancelling G:

$$(\tau - \tau_z)Q' \in J, \qquad (\xi - \xi_z)Q' \in J, \qquad Q' \notin J.$$

So $\mathfrak{m}_z \subseteq (J : Q') \neq \mathcal{R}$. As $\mathfrak{m}_z$ is maximal, $(J : Q') = \mathfrak{m}_z \supseteq J$, and $z \in V(J)$.

Step 3 $V(J) \subseteq \sigma_p(u)$

By A1(iii), $J \supseteq \prod_{z' \in V(J)} \mathfrak{m}_{z'}^N$, and by the Chinese remainder theorem

$$\mathcal{R}/J \cong \prod_{z' \in V(J)} \mathcal{R}/\left(J + \mathfrak{m}_{z'}^N\right).$$

The factor at z is nonzero and killed by $\mathfrak{m}_z^N$. By (LS) it contains $v \neq 0$ with $\mathfrak{m}_z v = 0$. Through the isomorphism of Step 4 this gives a joint eigenvector $(vG)(\partial)u \neq 0$.

Step 4 The exponential-polynomial part is $\mathcal{R}G(\partial)u$

Since $\mathcal{R}$ is a domain,

$$\operatorname{Ann}\big(G(\partial)u\big) = (I : G) = \{Q : \ QG \in GJ\} = J.$$

$V(J)$ is finite, so $G(\partial)u \in \mathrm{EP}$ and $\mathcal{R}G(\partial)u \subseteq \mathrm{EP}$. Conversely, let $w = Q(\partial)u \in \mathrm{EP}$. For each prime factor P_i of G, with multiplicity f_i, A1(F) gives $h \in \operatorname{Ann}(w)$ with $P_i \nmid h$. Then $hQ \in I \subseteq (G)$, so $P_i^{f_i} \mid hQ$ and $P_i^{f_i} \mid Q$. Hence $G \mid Q$ and $w \in \mathcal{R}G(\partial)u$. The map $Q \mapsto (QG)(\partial)u$ is onto with kernel J, so the space is $\cong \mathcal{R}/J$.

Step 5 (iii) and (iv)

$J = \mathcal{R} \iff V(J) = \emptyset$ (A1(iii)) $\iff \sigma_p(u) = \emptyset$ (Steps 2–3). Also $J = \mathcal{R} \iff I = (G)$ is principal, and $W \cap \mathrm{EP} \cong \mathcal{R}/J = 0 \iff J = \mathcal{R}$. Finally $\operatorname{Ann}(u) = (P)$ iff I is principal and $G \sim P$.

Why it works. the finite part J of the annihilator is exactly the module of exponentials inside $\mathcal{R}u$; its points are the eigenvalues.

■

Uses: A1, A2.

An empty spectrum alone is not enough

Take $u = e^{x^2}$ and $P = \tau^2$. For $Q = Q_0(\xi) + \tau Q_1$ one has $Q(\partial)u = Q_0(\partial_x)e^{x^2}$, which vanishes only if $Q_0 = 0$. So $\operatorname{Ann}(u) = (\tau)$ is principal and $\sigma_p(u) = \emptyset$, but $\operatorname{Ann}(u) \neq (\tau^2)$. The spectrum sees J_u, not G_u; condition (iv) needs both.

1.3 Finite data

Dim. A6 **C3-T2**

Bézout threshold and jet threshold [proved]

(C3-T2) Let $u = \sum_{k=1}^{M} c_k\, e^{\langle z_k, \cdot \rangle}$ with distinct $z_k \in V(P)$ and $c_k \neq 0$, P squarefree, $n \geq \deg P$. If

$$\big|\{z_k\} \cap V(P_i)\big| > n \deg P_i \qquad \text{for every } i,$$

then $\mathrm{Ann}(u) \cap \mathcal{R}_{\leq n} = P \cdot \mathcal{R}_{\leq n - \deg P}$. The threshold cannot be lowered uniformly.

(C3-T3) Let $0 \neq u \in \mathrm{EP}(\Omega)$, $d = \dim \mathcal{R}u$, $p_0 \in \Omega$.

(i) $\mathrm{Ann}(u) \cap \mathcal{R}_{\leq n} = \{Q \in \mathcal{R}_{\leq n} : \partial^\beta(Q(\partial)u)(p_0) = 0 \ \forall |\beta| \leq d-1\}$.

(ii) $\mathrm{Ann}(u)$ is generated in degree $\leq d$, so the jet of order $2d-1$ determines it.

(iii) The order $2d-1$ is optimal in the worst case.

Stated in the paper as no. 5.2, p. 26

Idea. For modes: an annihilator vanishes at every exponent, and Bézout bounds how many exponents a curve that does not contain $V(P_i)$ can pass through. For jets: inside the d-dimensional space $\mathcal{R}u$, the subspaces of elements vanishing to order k at p_0 shrink strictly, so after $d-1$ steps they are zero.

Proof.

Step 1 Annihilators of a sum of modes

By A2(a)(b), $Q(\partial)u = \sum_k c_k Q(z_k)\, e^{\langle z_k, \cdot \rangle}$, which is zero iff $Q(z_k) = 0$ for all k.

Step 2 Bézout forces every P_i to divide Q

Let $Q \in \mathrm{Ann}(u)$ with $\deg Q \leq n$. If $Q = 0$ there is nothing to prove; a nonzero constant cannot kill $u \neq 0$. Suppose $P_i \nmid Q$. Then Q and P_i are coprime, and Bézout's theorem (the homogenisations have no common component, since the line at infinity divides neither) gives

$$\big|V(Q) \cap V(P_i)\big| \leq \deg Q \cdot \deg P_i \leq n \deg P_i.$$

But by Step 1, $\{z_k\} \cap V(P_i) \subseteq V(Q) \cap V(P_i)$ has more points. So $P_i \mid Q$ for every i, and since the P_i are distinct primes, $P \mid Q$.

Step 3 The threshold is sharp

Take $Z_n = \{(j^2, \pm j) : j = 1, \dots, n\}$, $2n$ points on $\tau = \xi^2$, and

$$Q_n = \prod_{j=1}^{n} (\tau - j^2).$$

It has degree n, vanishes on Z_n, and is not a multiple of $\tau - \xi^2$ because it does not involve ξ.

Step 4 (i) Vanishing subspaces shrink strictly

Let $W = \mathcal{R}u$ and

$$K_k = \{w \in W : \partial^\beta w(p_0) = 0 \text{ for all } |\beta| \le k\}.$$

If $K_k = K_{k+1}$, then K_k is stable under derivatives (if $w \in K_{k+1}$ then $\partial w \in K_k$), so every $w \in K_k$ has all derivatives zero at p_0; being analytic on a connected set, $w = 0$. So the dimensions drop strictly until they reach 0. Since $\dim K_0 = d - 1$, we get $K_{d-1} = 0$. For $Q \in \mathcal{R}_{\le n}$, $Q(\partial)u \in W$ vanishes iff it lies in K_{d-1}. Any upper bound $d' \ge d$ works as well.

Step 5 (ii) Generators of low degree

$\mathcal{R}/\operatorname{Ann}(u) \cong W$ has dimension d. In a graded monomial order, the standard monomials form a set of d elements closed under division. A standard monomial $\tau^a \xi^b$ has $(a+1)(b+1) \ge a+b+1$ divisors, all standard, so $a+b+1 \le d$. The minimal generators of the initial ideal therefore have degree at most d, and so do the elements of the reduced Gröbner basis, which generate $\operatorname{Ann}(u)$. Apply (i) with $n = d$: the jet of order $2d-1$ suffices.

Step 6 (iii) Optimality

Let $u = \sum_{j=1}^{d} e^{jx}$, so $d = \dim \mathcal{R}u$, and $n = d$. Then $\operatorname{Ann}(u) \cap \mathbb{C}[\xi]_{\le d} = \mathbb{C} \cdot Q_0$ with $Q_0 = \prod_j (\xi - j)$. Put $m_k = \partial_x^k u(p_0)$. With only $|\beta| \le d-2$ the test of (i) is the Hankel system

$$\sum_{i=0}^{d} q_i \, m_{k+i} = 0, \qquad k = 0, \dots, d-2,$$

which has $d-1$ equations in $d+1$ unknowns, so a kernel N of dimension at least 2. Take $Q_1 \in N \setminus \mathbb{C}Q_0$ and $Q' = Q_0 + sQ_1$ with s small: it has degree d and d simple roots μ_j. Choose w_j with $\sum_j w_j \mu_j^k = m_k$ for $k < d$ (Vandermonde) and set

$$u' = \sum_j w_j \, e^{\mu_j (x - x_0)}.$$

The moment sequences of u and u' satisfy the same recursion given by Q' and agree for $k \le d-1$, hence for $k \le 2d-2$. So u and u' have the same jet of order $2d-2$, but $Q' \in \operatorname{Ann}(u') \setminus \operatorname{Ann}(u)$.

Why it works. finitely many exponentials are finitely many points; Bézout counts how many points a curve must miss, and the strict drop of vanishing subspaces counts how many derivatives are needed.

■

Uses: A1, A2.

Dim. A7 **C3-P1**

No finite certificate

[proved] (jets) / [from cited source] (functionals)

Let Ω be connected, $p_0 \in \Omega$, P nonconstant, u smooth with $P(\partial)u = 0$, $K \geq 0$, and $E_P = \ker P \cap \mathrm{EP}(\Omega)$.

(i) $\bigcap_{v \in E_P} \mathrm{Ann}(v) = (P)$.

(ii) There is $v \in E_P$ with the same jet of order K at p_0 as u; in particular $\mathrm{Ann}(v) \supsetneq (P)$.

On convex Ω the same holds for any finitely many continuous linear functionals (Malgrange); on a non-convex region it can fail.

Stated in the paper as no. 5.3, p. 27

Idea. By duality, the possible jets of solutions are cut out by the multiples of P. Exponential-polynomial solutions already cut out the same space, because for every Q that is not a multiple of P one builds a solution $s^{e-1}e^{\langle z,\cdot\rangle}$ that Q does not kill.

Proof.

Pair an operator Q of degree at most K with a jet by

$$\langle Q, J_K w\rangle = \big(Q(\partial)w\big)(p_0).$$

Step 1 **Vanishing on all of E_P means annihilating all of E_P**

If $(Q(\partial)v)(p_0) = 0$ for every $v \in E_P$, then, since E_P is stable under derivatives, all derivatives of the exponential-polynomial $Q(\partial)v$ vanish at p_0. It is entire with zero Taylor series, so it is 0. Hence $Q \in \bigcap_{v \in E_P} \mathrm{Ann}(v)$.

Step 2 **(i) A non-multiple of P misses some exponential solution**

Let $Q \neq 0$ with $P \nmid Q$. Some i has $f = v_{P_i}(Q) < e_i$; write $Q = P_i^f Q'$ with $P_i \nmid Q'$. The sets $V(P_i) \cap V(\nabla P_i)$ and $V(P_i) \cap V(Q')$ are finite (A1(V)), while $V(P_i)$ is infinite. Pick $z \in V(P_i)$ with

$$\nabla P_i(z) \neq 0, \qquad Q'(z) \neq 0,$$

and a coordinate $s \in \{t, x\}$ with $\partial_s P_i(z) \neq 0$. On polynomials in s, $P_i(z + \partial)$ lowers the degree by exactly one and $Q'(z + \partial)$ preserves it. Let $v = s^{e_i-1}e^{\langle z,\cdot\rangle}$. Then $P(z + \partial)s^{e_i-1} = 0$, so $v \in E_P$, while $Q(z + \partial)s^{e_i-1}$ has degree $e_i - 1 - f \geq 0$ and is nonzero. So $Q \notin \mathrm{Ann}(v)$.

Step 3 **(ii) Every jet of a solution is a jet of an exponential solution**

By Steps 1–2, every Q of degree $\leq K$ orthogonal to the jets of E_P is a multiple of P, hence orthogonal to the jet of u. In finite dimension this means

$$J_K(u) \in J_K(E_P),$$

so a suitable $v \in E_P$ exists. Finally $\mathrm{Ann}(v) \supsetneq (P)$: if $v = 0$ it is $\mathcal{R}$; otherwise $V(\mathrm{Ann}\, v)$ is finite while $V(P)$ is infinite.

Step 4 General functionals and convexity

For general continuous functionals the analogue on convex Ω is Malgrange's approximation theorem (Ann. Inst. Fourier 6, Chap. I §2 Thm 2) [from cited source]. Without convexity it fails: on a U-shaped region take $P = \tau$, $u = \phi(x)\mathbf{1}_{t>2}$, and

$$\ell(w) = w(5/2, 5/2) - w(1/2, 5/2).$$

Then $\ell(u) = 1$ while $\ell(v) = 0$ for every $v \in \ker \partial_t$ with $\mathrm{Ann}(v) \neq (\tau)$.

Why it works. finite data see a finite-dimensional shadow of $\ker P$, and exponential solutions already fill every such shadow.

■

Uses: A1, A2.

Dim. A8 **P4-T4**

Decidability with a bound [proved]

In the model M2$^+$ (exact jets at p_0 in a computable field K, plus a zero-test oracle):

(i) for every finite-dimensional $\mathcal{L} \subseteq \mathcal{R}$ with a basis over K, $\mathrm{Ann}(u) \cap \mathcal{L}$ is computable;

(ii) "$\mathrm{Ann}(u) \neq (P)$" is semi-decidable;

(iii) if an integer $B \geq \dim(\mathcal{R}u \cap \mathrm{EP})$ is given, then with $N = B + \deg P - 1$,

$$\mathrm{Ann}(u) = (P) \iff \mathrm{Ann}(u) \cap \mathcal{R}_{\leq N} = (P) \cap \mathcal{R}_{\leq N},$$

which is decidable. The test "$Q(\partial)u \in \mathrm{EP}$?" is then decidable, and G_u is computable.

Stated in the paper as no. 5.4, p. 28

Idea. Linear algebra on jets gives a decreasing chain of candidate spaces that stabilises at the true annihilators; the zero oracle confirms when it has stabilised. If identification fails, a witness of small degree exists, and the bound B says how small.

Proof.

Step 1 (i) A decreasing chain of linear spaces

Let

$$A_k = \{Q \in \mathcal{L} : \ \partial^\beta \big(Q(\partial)u\big)(p_0) = 0 \text{ for all } |\beta| \leq k\}.$$

It is the kernel of a matrix with entries $\partial^{\beta+\alpha}u(p_0) \in K$, so it has a computable basis. The chain $A_0 \supseteq A_1 \supseteq \cdots \supseteq \mathrm{Ann}(u) \cap \mathcal{L}$ stabilises, and its limit is $\mathrm{Ann}(u) \cap \mathcal{L}$, because $Q(\partial)u$ is analytic with zero Taylor series at p_0. Algorithm: for $k = 0, 1, \ldots$ compute a basis of A_k and ask the oracle about each element; stop when all answers are "zero". It stops only when $A_k \subseteq \mathrm{Ann}(u)$, so it is correct, and it stops at the stabilisation index.

Step 2 (ii) Semi-decision

For $n = \deg P, \deg P + 1, \ldots$ compute $\mathrm{Ann}(u) \cap \mathcal{R}_{\le n}$ by (i) and compare its dimension with that of $P \cdot \mathcal{R}_{\le n-\deg P}$. If $\mathrm{Ann}(u) \neq (P)$, a witness Q exists and the loop stops at $n = \max(\deg Q, \deg P)$.

Step 3 (iii) A witness of degree at most N

Suppose $\mathrm{Ann}(u) \neq (P)$. By A3, some $w_i = (P/P_i)(\partial)u$ is an exponential-polynomial in $\mathcal{R}u$, so

$$d_i = \dim \mathcal{R}w_i \le B.$$

By A6(ii), $\mathrm{Ann}(w_i)$ is generated in degree $\le d_i$, and not all generators are divisible by P_i (A1(F)). Pick such a generator h, so $\deg h \le B$. Then

$$Q = h \cdot \frac{P}{P_i} \in \mathrm{Ann}(u) \setminus (P), \qquad \deg Q \le B + \deg P - 1 = N.$$

The comparison in degree N is computable by (i).

Step 4 The EP test from the bound

Let $w = Q(\partial)u$; its oracles follow from those of u. If $w \in \mathrm{EP}$, then $\dim \mathcal{R}w \le B$, so the minimal polynomials of ∂_t and ∂_x on $\mathcal{R}w$ have degree $\le B$: there are nonzero $a(\tau), b(\xi)$ of degree $\le B$ in $\mathrm{Ann}(w)$. Conversely, such a, b give a finite zero set and $w \in \mathrm{EP}$ (A2). Both searches are computable by (i). With this test, A4 computes G_u.

Why it works. a failure of identification is always witnessed in bounded degree, and the bound B makes that degree explicit.

■

Uses: A1, A2, A3, A4, A6.

Dim. A9 **P4-T5**

Black-box impossibility [proved]

Let $P \in K[\tau, \xi]$ be nonconstant (K the algebraic numbers), Ω connected, $p_0 \in \Omega$ with coordinates in $K \cap \mathbb{R}$, and u real-analytic with $P(\partial)u = 0$, jets in K, and $\mathrm{Ann}(u) = (P)$ (such u exists for every P, (P4-L7)). For every order k_0 and every finite set $\mathcal{Q} \subset K[\tau, \xi]$ there is an exponential-polynomial $v \in \ker P$ with

- the same jet of order k_0 at p_0 as u;
- for every $Q \in \mathcal{Q}$: $Q(\partial)v = 0 \iff Q(\partial)u = 0$;
- $\mathrm{Ann}(v) \supsetneq (P)$.

Hence no algorithm in the model M2$^+$ decides "$\mathrm{Ann}(\cdot) = (P)$" correctly on all inputs.

Stated in the paper as no. 5.3, p. 27

Idea. Run a correct algorithm on u: it halts after finitely many questions. An adversary builds v among the exponential-polynomial solutions so that it gives the same answer to every one of those questions. First match the jet, then, for each Q with $Q(\partial)u \neq 0$, add a solution with zero jet that Q does not kill, with generic coefficients.

Proof.

Step 1 Witness points with algebraic coordinates

Each $V(P_i)$ has infinitely many points with coordinates in K: roots of $P_i(\cdot, \xi_0)$ for $\xi_0 \in K$, or $K \times \{\xi_0\}$ if P_i does not involve τ. For such z with $\nabla P_i(z) \neq 0$, pick a coordinate s_z with $\partial_s P_i(z) \neq 0$ and put

$$g_{i,z,j} = s_z^j \, e^{\langle z, \cdot - p_0 \rangle}, \qquad j < e_i.$$

As in A7, Step 2, $P(z+\partial)s_z^j = 0$, so $g_{i,z,j} \in \ker P$.

Step 2 Matching the jet

Let V_{k_0} be the span of the jets of order k_0 of all $g_{i,z,j}$. With the pairing of A7,

$$V_{k_0}^{\perp} = (P) \cap \mathcal{R}_{\leq k_0};$$

the inclusion "$\subseteq$" is A7, Step 2, with points chosen in K^2. Since $\langle PA, J_{k_0} u \rangle = 0$, $J_{k_0}(u) \in V_{k_0}$. A linear system with coefficients in K that is solvable over $\mathbb{C}$ is solvable over K, so some combination v_1 of the g's has the jet of u.

Step 3 A zero-jet solution for each test

For $Q \in \mathcal{Q} \setminus (P)$ (so $Q(\partial)u \neq 0$), pick i with $f = v_{P_i}(Q) < e_i$, $Q = P_i^f Q'$, and more than $\dim \mathcal{R}_{\leq k_0}$ points $z \in V(P_i) \cap K^2$ with $\nabla P_i(z) \neq 0$ and $Q'(z) \neq 0$. The map

$$c \;\mapsto\; J_{k_0}\Big(\sum_z c_z \, s_z^{e_i - 1} e^{\langle z, \cdot - p_0 \rangle} \Big)$$

goes from a larger space to a smaller one, so it has a nonzero kernel vector c. Let e_Q be the corresponding sum: its jet is 0, while $Q(\partial)e_Q \neq 0$ by A2(b).

Step 4 A generic combination answers like u

Let $v_\lambda = v_1 + \sum_Q \lambda_Q \, e_Q$. It has the jet of u for every λ. For each Q with $Q(\partial)u \neq 0$, the set $\{\lambda : \; Q(\partial)v_\lambda = 0\}$ is a proper affine subspace. Finitely many proper subspaces do not cover the space, so some λ avoids all of them. For $Q \in (P)$ both answers are "zero". So $v = v_\lambda$ answers every question as u does, and $v \in \mathrm{EP} \cap \ker P$ gives $\mathrm{Ann}(v) \supsetneq (P)$.

Step 5 Conclusion

The run of the algorithm on u depends only on the answers it receives. They coincide for v, so it outputs "Ann $= (P)$" on v as well, which is wrong.

Why it works. a finite conversation with a black box can always be imitated by a finite sum of exponentials, and such sums never identify.

■

Uses: A2, A7.

CHAPTER 2

Decomposition and regularity

In this chapter

Starting from	$P = c\prod_{j=1}^{s} Q_j$ with $Q_j = P_j^{m_j}$ pairwise coprime; $\Omega \subset \mathbb{R}^2$ open; $\mathcal{A}$ is $\mathcal{D}'(\Omega)$ or $C^\infty(\Omega)$; the toolkit of Chapter 1.
Why	to prove that a solution splits into one piece per factor, with an ambiguity of exactly known size, that holes in Ω can break this, and exactly when the pieces keep the Sobolev regularity of u.
Role	Dim. A10–A18. They prove Chapters 6 and 7 of the paper.

Notation for the whole chapter

For a module M over $\mathcal{R}$, $\mathrm{Sol}_{\mathcal{A}}(M) = \mathrm{Hom}_{\mathcal{R}}(M, \mathcal{A})$ is its solution space; for example $\mathrm{Sol}(\mathcal{R}/(f)) = \ker_{\mathcal{A}} f$. For $z = (p, q) \in \mathbb{C}^2$ we write $e_z = e^{pt+qx}$ and $D_t = \partial_t - p$, $D_x = \partial_x - q$. Note that $D_t(e_z \tilde{w}) = e_z\, \partial_t \tilde{w}$.

2.1 Terms and factors

Dim. A10 R2-T3

Terms do not give components [proved]

Let $L = L_1 + L_2 \in \mathcal{R}$.

(a) In every $\mathcal{R}$-module $\mathcal{A}$:

$$\ker L \cap \big(\ker L_1 + \ker L_2\big) \subseteq \mathrm{Sol}\big(\mathcal{R}/(L,\ L_1L_2)\big) = \mathrm{Sol}\big(\mathcal{R}/(L_1 + L_2,\ L_1^2)\big).$$

(b) If $\mathcal{A} = C^\infty$ or $\mathcal{D}'$ of a convex Ω and L_1, L_2 are coprime, equality holds.

(c) $V(L_1 + L_2,\ L_1^2) = V(L_1) \cap V(L_2)$. For coprime L_1, L_2 the space is finite-dimensional, of dimension $2\sum_p I_p(L_1, L_2)$, and consists of exponential-polynomials with exponents at the common zeros.

Stated in the paper as no. 6.1.1, p. 31

Idea. A solution that splits by terms satisfies one extra equation, $L_1L_2u = 0$, because the operators commute. Modulo L that extra equation is L_1^2, and the common zeros of L and L_1^2 are where both terms vanish.

Proof.

Step 1 (a) The extra equation

Let $u = u_1 + u_2$ with $L_1 u_1 = 0$ and $L_2 u_2 = 0$. Since the operators commute,

$$L_1 L_2\, u = L_2(L_1 u_1) + L_1(L_2 u_2) = 0.$$

Together with $Lu = 0$ this gives $u \in \mathrm{Sol}(\mathcal{R}/(L, L_1 L_2))$. Modulo L, $L_2 \equiv -L_1$, so $L_1 L_2 \equiv -L_1^2$ and the two ideals coincide.

Step 2 (b) Equality from the decomposition theorem

For coprime L_1, L_2 the finite module $\mathcal{R}/(L_1, L_2)$ has no obstruction (A12), so the sequence of A13 is exact: every element of $\mathrm{Sol}(\mathcal{R}/(L, L_1 L_2))$ is killed by $L_1 L_2$, hence by A13 splits as a sum of solutions of L_1 and of L_2.

Step 3 (c) The variety

On $V(L_1)$ we have $L_1 + L_2 = 0 \iff L_2 = 0$. Off $V(L_1)$, $L_1^2 \neq 0$. So

$$V(L_1 + L_2,\ L_1^2) = V(L_1) \cap V(L_2).$$

For coprime terms this set is finite (A1(V)). The dimension follows from the additivity of intersection numbers, $I_p(F, GH) = I_p(F, G) + I_p(F, H)$, and $I_p(L_1 + L_2, L_1) = I_p(L_2, L_1)$.

Why it works. commuting terms turn a split by terms into an extra equation, which cuts the solution space down to the finitely many points where both terms vanish.

■

Uses: A1, A12, A13.

Example 2.1 The heat equation as a sum of two terms

Take $L_1 = \partial_x^2$ and $L_2 = -\partial_t$. The ideal is $(\xi^2 - \tau,\ \xi^4) = (\tau^2,\ \xi^2 - \tau)$, of dimension $4 = 2 \cdot I_0(\xi^2, \tau)$. The solutions that split by terms are

$$1, \qquad x, \qquad x^2 + 2t, \qquad x^3 + 6tx.$$

The generic heat solution $e^{s^2 t + sx}$ with $s \neq 0$ has $L_1 L_2 u = -s^4 u \neq 0$, so it does not split.

2.2 The factor decomposition

Dim. A11 **C12-L3**

The finite quotient of two coprime factors [proved]

Let $f, g, h \in \mathcal{R} \setminus \{0\}$.

(a) If f, g are coprime, (f, g) contains a nonzero $r(\xi)$ and a nonzero $s(\tau)$; hence $\dim \mathcal{R}/(f, g) < \infty$.

(b) If f is coprime to g and to h, then

$$\dim \mathcal{R}/(f, gh) = \dim \mathcal{R}/(f, g) + \dim \mathcal{R}/(f, h).$$

(c) $(f, g) = \mathcal{R} \iff V(f) \cap V(g) = \emptyset$.

Stated in the paper as no. 6.2, p. 33

Idea. (a) is the Bézout identity of A1, Step 1. (b) is an exact sequence of finite-dimensional spaces. (c) uses that commuting operators on a nonzero finite-dimensional space have a common eigenvector.

Proof.

Step 1 (a) A finite spanning set

A1, Step 1 gives $r(\xi) \in (f, g)$ and symmetrically $s(\tau) \in (f, g)$, both nonzero. Then $\mathcal{R}/(f, g)$ is spanned by the monomials

$$\tau^a \xi^b, \qquad a < \deg s, \quad b < \deg r.$$

Step 2 (b) An exact sequence

The sequence

$$0 \to \mathcal{R}/(f, h) \xrightarrow{\cdot g} \mathcal{R}/(f, gh) \to \mathcal{R}/(f, g) \to 0$$

is well defined because $g\,(f, h) \subseteq (f, gh)$, and its middle is exact because the kernel of the projection is $(f, g)/(f, gh)$, the image of multiplication by g. Injectivity: if $ga = \alpha f + \beta gh$, then $g(a - \beta h) = \alpha f$; since $\gcd(f, g) = 1$, $f \mid a - \beta h$, so $a \in (f, h)$. Dimensions add.

Step 3 (c) A common eigenvector gives a common zero

($\Leftarrow$) If f, g shared a nonconstant factor π, then $\emptyset \neq V(\pi) \subseteq V(f) \cap V(g)$ (A1(F)). So disjoint zero sets force coprimality, and (a) applies. If $(f, g) \neq \mathcal{R}$, then $N = \mathcal{R}/(f, g)$ is nonzero and finite-dimensional. The commuting operators τ, ξ on N have a common eigenvector v, with eigenvalues (p, q). Then $fv = gv = 0$ gives $f(p, q) = g(p, q) = 0$, a common zero.

($\Rightarrow$) If (p, q) is a common zero, $(f, g) \subseteq \mathfrak{m}_{(p,q)} \neq \mathcal{R}$.

Why it works. coprime curves meet in finitely many points, and the quotient ring counts them with multiplicity.

■

Uses: A1.

Dim. A12 **C12-L1'**

Twisted Poincaré lemma; no obstruction for finite modules [proved]

Let Ω be open, connected, nonempty, with $H^1_{dR}(\Omega) = 0$.

(i) **(C12-L1′)** If $\varphi, \psi \in \mathcal{A}$ satisfy $D_x\varphi = D_t\psi$, there is $w \in \mathcal{A}$ with

$$D_t w = \varphi, \qquad D_x w = \psi.$$

Moreover $\{w \in \mathcal{A} : D_t w = D_x w = 0\} = \mathbb{C}\, e_z$.

(ii) **(C12-L2′)** For every $\mathcal{R}$-module M with $n = \dim M < \infty$:

$$\operatorname{Ext}^1_{\mathcal{R}}(M, \mathcal{A}) = 0, \qquad \dim \operatorname{Sol}_{\mathcal{A}}(M) = n,$$

and every solution is the restriction of a unique exponential-polynomial solution on $\mathbb{R}^2$.

Stated in the paper as no. 6.3, p. 35

Idea. Multiplying by e_z^{-1} turns the twisted derivatives into ordinary ones: (i) becomes "a closed 1-form is exact". For distributions, solve locally on squares and glue; the gluing error is a smooth closed form, exact by hypothesis. (ii) follows from (i) for the simplest modules and by induction along a composition series.

Proof.

Step 1 Removing the twist

Put $\tilde\varphi = e_z^{-1}\varphi$ and $\tilde\psi = e_z^{-1}\psi$. The condition becomes

$$\partial_x \tilde\varphi = \partial_t \tilde\psi,$$

and we look for $\tilde w$ with $\partial_t \tilde w = \tilde\varphi$, $\partial_x \tilde w = \tilde\psi$; then $w = e_z \tilde w$.

Step 2 Smooth case

The 1-form $\eta = \tilde\varphi\, \mathrm{d}t + \tilde\psi\, \mathrm{d}x$ is closed, hence exact because $H^1_{dR}(\Omega) = 0$.

Step 3 Distributions: local primitives glued by a partition of unity

Cover Ω by open squares U_j with closure in Ω, and take a locally finite partition of unity $\chi_j \in C^\infty_c(U_j)$. On each square a primitive w_j exists (integration in one variable, as in A2, Step 6). On an overlap $w_j - w_k$ has zero gradient, so it is a constant c_{jk}. Set $w = \sum_j \chi_j w_j$. On U_k,

$$w - w_k = \sum_j \chi_j\, c_{jk} =: h_k \quad \text{is smooth.}$$

So $a = \tilde\varphi - \partial_t w$ and $b = \tilde\psi - \partial_x w$ equal $-\partial_t h_k$, $-\partial_x h_k$ locally. They form a smooth closed 1-form, which is $d\theta$ for a smooth θ by hypothesis. Then $w + \theta$ is the required primitive.

Step 4 The kernel

If $\partial_t \tilde{w} = \partial_x \tilde{w} = 0$, then $\tilde{w}$ is locally constant, hence constant on the connected Ω.

Step 5 (ii) The simplest module

For $M = \mathcal{R}/\mathfrak{m}_z$ use the Koszul resolution

$$0 \to \mathcal{R} \xrightarrow{\beta} \mathcal{R}^2 \xrightarrow{\alpha} \mathcal{R} \to \mathcal{R}/\mathfrak{m}_z \to 0.$$

Applying $\mathrm{Hom}(\cdot, \mathcal{A})$, the solutions are $\ker(D_t, D_x) = \mathbb{C} e_z$ (Step 4) and $\mathrm{Ext}^1 = \ker \beta^* / \operatorname{im} \alpha^*$, which is exactly the compatibility problem of (i): it vanishes.

Step 6 (ii) Induction on the dimension

A finite-dimensional M has a composition series with factors $\mathcal{R}/\mathfrak{m}_z$. The long exact sequence of Ext and induction on n give $\mathrm{Ext}^1(M, \mathcal{A}) = 0$ and $\dim \mathrm{Sol}(M) = n$. On $\mathbb{R}^2$ the solutions are exponential-polynomials; restriction to Ω is injective (analytic functions vanishing on an open set vanish), and the dimensions agree.

Why it works. on a region without holes, closed forms are exact, so finite systems of first-order equations are always solvable.

■

Uses: A2.

Dim. A13 **C12-T1′**

Factor decomposition [proved]

Let Ω be open, connected, nonempty, with $H^1_{dR}(\Omega) = 0$, $\mathcal{A} \in \{\mathcal{D}'(\Omega), C^\infty(\Omega)\}$, $P = c \prod_{j=1}^{s} Q_j$ with pairwise coprime Q_j.

(i) $\ker_{\mathcal{A}} P = \ker_{\mathcal{A}} Q_1 + \cdots + \ker_{\mathcal{A}} Q_s$.

(ii) The kernel $\mathcal{N}$ of the sum map $\Sigma : \bigoplus_j \ker Q_j \to \ker P$ has dimension

$$\delta(P) = \sum_{i<j} \dim \mathcal{R}/(Q_i, Q_j) < \infty,$$

and its elements are tuples of exponential-polynomials. For $s = 2$, $\mathcal{N} = \{(w, -w) : w \in \ker Q_1 \cap \ker Q_2\}$.

(iii) The decomposition is unique $\iff$ $\delta(P) = 0$ $\iff$ $V(Q_i) \cap V(Q_j) = \emptyset$ for all $i \neq j$.

(iv) If u is smooth, every decomposition of u in $\mathcal{D}'$ consists of smooth functions.

Stated in the paper as no. 6.3, p. 34

Idea. For two coprime factors there is a short exact sequence of modules. Applying $\mathrm{Hom}(\cdot, \mathcal{A})$ gives a sequence of solution spaces that ends with Ext^1 of the finite module $\mathcal{R}/(f, g)$. That group vanishes on a region without holes (A12), so the sum map is onto, and its kernel is the solution space of the finite quotient.

Proof.

Step 1 Two factors: the module sequence

Let $f = Q_1$, $g = Q_2$. In a unique factorisation domain $(f) \cap (g) = (fg)$, and the sequence

$$0 \to \mathcal{R}/(fg) \xrightarrow{\iota} \mathcal{R}/(f) \oplus \mathcal{R}/(g) \xrightarrow{\pi} \mathcal{R}/(f,g) \to 0, \qquad \iota(1) = (1,1), \quad \pi(a,b) = a - b,$$

is exact.

Step 2 Two factors: the solution sequence

Since $\ker_{\mathcal{A}} f = \mathrm{Hom}(\mathcal{R}/(f), \mathcal{A})$, applying $\mathrm{Hom}(\cdot, \mathcal{A})$ gives the exact sequence

$$0 \to \mathrm{Sol}\big(\mathcal{R}/(f,g)\big) \xrightarrow{\pi^*} \ker f \oplus \ker g \xrightarrow{\iota^*} \ker fg \to \mathrm{Ext}^1_{\mathcal{R}}\big(\mathcal{R}/(f,g), \mathcal{A}\big),$$

with $\iota^*(u_1, u_2) = u_1 + u_2$ and $\pi^*(w) = (w, -w)$.

Step 3 Two factors: the obstruction vanishes

By A11(a), $\mathcal{R}/(f,g)$ is finite-dimensional. By A12(ii), $\mathrm{Ext}^1 = 0$ and $\dim \mathrm{Sol} = \dim \mathcal{R}/(f,g)$. So ι^* is onto, which is (i), and its kernel has dimension $\dim \mathcal{R}/(f,g)$, which is (ii) for $s = 2$.

Step 4 Any number of factors: induction

$P' = Q_2 \cdots Q_s$ is coprime to Q_1, so $\ker P = \ker Q_1 + \ker P'$, and by induction $\ker P' = \sum_{j \geq 2} \ker Q_j$. Write $\Sigma = \Sigma_2 \circ (\mathrm{id} \oplus \Sigma')$ with Σ' onto. For B onto, $\dim \ker(AB) = \dim \ker B + \dim \ker A$, and A11(b) gives $\dim \mathcal{R}/(Q_1, P') = \sum_{j \geq 2} \dim \mathcal{R}/(Q_1, Q_j)$. Hence

$$\delta(P) = \delta(P') + \sum_{j \geq 2} \dim \mathcal{R}/(Q_1, Q_j) = \sum_{i<j} \dim \mathcal{R}/(Q_i, Q_j).$$

Step 5 The ambiguity consists of exponential-polynomials

If $(w_j) \in \mathcal{N}$, then

$$w_j = -\sum_{i \neq j} w_i \ \in \ \ker Q_j \cap \ker \prod_{i \neq j} Q_i = \mathrm{Sol}\Big(\mathcal{R}/\big(Q_j, \textstyle\prod_{i \neq j} Q_i\big)\Big),$$

a finite-dimensional solution space, made of exponential-polynomials by A12(ii).

Step 6 (iii) and (iv)

$\delta(P) = 0 \iff$ every $\mathcal{R}/(Q_i, Q_j) = 0 \iff (Q_i, Q_j) = \mathcal{R} \iff V(Q_i) \cap V(Q_j) = \emptyset$ (A11(c)). For (iv), take a smooth decomposition from (i) in C^∞; any other decomposition in $\mathcal{D}'$ differs from it by an element of $\mathcal{N}$, which is smooth.

Why it works. the Chinese remainder sequence is exact as modules; its only possible failure on functions is an Ext^1 of a finite module, and that vanishes when Ω has no holes.

■

Uses: A11, A12.

Where the exact number $\delta(P)$ comes from

In terms of local intersection numbers,

$$\delta(P) = \sum_{i<j} m_i m_j \sum_z I_z(P_i, P_j).$$

For $s \geq 3$ the pairwise formula is exact, with no inclusion–exclusion correction; it was checked on three concurrent lines, three tangent parabolas, a triple point with multiplicity, and four factors.

Dim. A14 **C12-P10**

Holes break the decomposition [proved]

Let Ω contain a closed curve γ and a point $\alpha = (t_0, x_0) \notin \Omega$ with winding number $n = \mathrm{Ind}_\gamma(\alpha) \neq 0$. For

$$P = \tau^2 + \xi^2 = (\tau - i\xi)(\tau + i\xi) = Q_1 Q_2,$$

the function $u = \log\left((t-t_0)^2 + (x-x_0)^2\right)$ lies in $\ker_{C^\infty(\Omega)} P$ but not in $\ker_{\mathcal{D}'} Q_1 + \ker_{\mathcal{D}'} Q_2$.

Stated in the paper as no. 6.4, p. 35

Idea. A splitting $u = h + k$ would make dh equal to the form dw/w with $w = \overline{z - \alpha}$. An exact form integrates to zero around γ, but dw/w integrates to $-2\pi i\, n \neq 0$.

Proof.

Step 1 **The candidate component**

u is harmonic off α. Put $w = (t - t_0) - i(x - x_0)$. Then

$$F = Q_2 u = \frac{2}{w}$$

is smooth on Ω.

Step 2 **A splitting would make a form exact**

Suppose $u = h + k$ with $Q_1 h = 0$ and $Q_2 k = 0$. Then $Q_2 h = F$ and $Q_1 h = 0$, so

$$\partial_t h = \frac{F}{2}, \qquad \partial_x h = \frac{F}{2i}.$$

So the smooth form $\eta = \frac{F}{2}\,\mathrm{d}t + \frac{F}{2i}\,\mathrm{d}x = \dfrac{dw}{w}$ equals dh.

Step 3 The integral around γ

Mollify: $\eta_\epsilon = \eta * \rho_\epsilon = d(h * \rho_\epsilon)$ is exact near γ, so $\oint_\gamma \eta_\epsilon = 0$. Since η is smooth, $\eta_\epsilon \to \eta$ uniformly on γ, hence $\oint_\gamma dw/w = 0$. But

$$\oint_\gamma \frac{dw}{w} = \overline{\oint_\gamma \frac{dz}{z-\alpha}} = -2\pi i\, n \neq 0,$$

a contradiction.

Why it works. the logarithm has monodromy around the hole, and no single-valued component can carry it.

■

Uses: none.

For a fixed P the condition is subtler

On the punctured disc, $P = \tau\xi$ decomposes in C^∞ but not in $\mathcal{D}'$ (take $u = H(x)H(t)$, with H the Heaviside function). The hypothesis $H^1_{dR} = 0$ is what makes the theorem hold for *every* P; the exact condition for a fixed P is the cohomological criterion (OPF-T4) (Part II).

2.3 Regularity

Dim. A15 **C12-C1**

A derivative can be lost [false] (counterexample proved)

Let $\Omega = (0,1)^2$, $f = \tau - \xi$, $g = \tau - \xi - 1$. For each $m \in \mathbb{N}$ there is

$$u \in H^m(\Omega) \cap C^\infty(\Omega), \qquad f(\partial)g(\partial)u = 0,$$

whose unique decomposition $u = u_1 + u_2$ ($u_1 \in \ker f$, $u_2 \in \ker g$) has $u_1, u_2 \notin H^m(\Omega)$, while $u_2 = f(\partial)u \in H^{m-1}(\Omega)$. The loss is exactly one derivative.

Stated in the paper as no. 7.1, p. 38

Idea. Near the corner $(0,0)$ the characteristic chords of direction $(1,1)$ are short. On a short chord the shapes 1 and e^t of the two components are nearly equal, so a function that is small on the chord can have large components. A suitable profile along the chords makes u smooth while its components blow up by one derivative.

Proof.

Step 1 Characteristic coordinates

Use (t,s) with $s = x + t$ (determinant 1, so Sobolev norms are equivalent). Then

$$f = \partial_t\big|_s, \qquad g = \partial_t\big|_s - 1.$$

Near the corner the region is $\{0 < t < s < 1\}$: the chord at height s has length s.

Step 2 The function

Let $G_0(s) = 1/s$ and $G_m(s) = s^{m-1}\log s$ for $m \geq 1$, and

$$u = G(s)\,(e^t - 1).$$

Then $fu = G(s)e^t$ and $g(Ge^t) = 0$, so $fgu = 0$. Since $f - g = 1$, $(f, g) = \mathcal{R}$ and the decomposition is unique:

$$u_1 = -g(\partial)u = -G(s), \qquad u_2 = f(\partial)u = e^t\,G(s).$$

Step 3 u is in H^m

The derivatives are $\partial_t^k \partial_s^j u = G^{(j)}(s)\,\partial_t^k(e^t - 1)$. Integrating over $t \in (0, s)$ and using $|e^t - 1| \leq et$:

$$\int_0^s |\partial_t^k \partial_s^j u|^2 \,\mathrm{d}t \leq \begin{cases} e^2 s\,|G^{(j)}(s)|^2, & k \geq 1, \\ \frac{e^2}{3} s^3\,|G^{(j)}(s)|^2, & k = 0. \end{cases}$$

By induction $G^{(j)} = s^{m-1-j}(\alpha_j \log s + \beta_j)$ for $j \leq m-1$ and $G^{(m)} = (m-1)!/s$. So the integrals $\int_0^1 |G^{(j)}|^2 s\,\mathrm{d}s$ $(j \leq m-1)$ and $\int_0^1 |G^{(j)}|^2 s^3\,\mathrm{d}s$ $(j \leq m)$ are all finite.

Step 4 u_2 is not in H^m

$\partial_s^m u_2 = e^t G^{(m)}$ and

$$\int \left|e^t G^{(m)}\right|^2 \;\geq\; c \int_0^1 s^{-2}\, s\,\mathrm{d}s = \infty.$$

Hence also $u_1 = u - u_2 \notin H^m$, while $u_2 = f(\partial)u \in H^{m-1}$.

Why it works. the component shapes become collinear on short chords, and the profile G is chosen to sit exactly at the edge of integrability.

■

Uses: none.

Dim. A16 **C12-T3**

Chord criterion

[proved]

Let Ω be bounded, open and convex, $c \in \mathbb{R}$, $d_1 \neq d_2$, $f = \partial_t + c\partial_x - d_1$, $g = \partial_t + c\partial_x - d_2$. In coordinates (t, s) with $s = x - ct$, let $\ell(s)$ be the length of the chord at height s, and $\mathcal{H}_i = \ker_{L^2} f_i$. Equivalent:

(a) $\inf_s \ell(s) > 0$;

(b) every $u \in L^2$ with $fgu = 0$ has components in L^2 with $\|u_1\|^2 + \|u_2\|^2 \leq C\|u\|^2$;

(c) $\mathcal{H}_1 + \mathcal{H}_2$ is closed;

(d) the Friedrichs cosine $c_F(\mathcal{H}_1, \mathcal{H}_2) < 1$.

Stated in the paper as no. 7.2, p. 39

Idea. On each chord u is $F(s)e^{d_1 t} + G(s)e^{d_2 t}$. The squared L^2 norm on the chord is a Gram quadratic form in (F, G), uniformly positive exactly when the chords do not shrink. The other equivalences are Hilbert-space facts.

Proof.

Step 1 The components on each chord

Since $f - g = \kappa := d_2 - d_1 \neq 0$, the decomposition is unique:

$$u_2 = \kappa^{-1} f(\partial) u, \qquad u_1 = -\kappa^{-1} g(\partial) u,$$

and $u_1 = e^{d_1 t} F(s)$, $u_2 = e^{d_2 t} G(s)$ with $F, G \in L^2_{loc}$.

Step 2 (a)⇒(b): a uniform Gram bound

Let $\ell_0 = \inf \ell > 0$. The Gram matrix

$$\Gamma(\alpha, \beta) = \Big(\int_\alpha^\beta e^{d_a t}\, \overline{e^{d_b t}}\, \mathrm{d}t \Big)_{a,b=1,2}$$

is positive definite and continuous; on the compact set $\{-M \le \alpha,\ \alpha + \ell_0 \le \beta \le M\}$ its smallest eigenvalue is at least some $\gamma > 0$. By Fubini,

$$\int_{\alpha(s)}^{\beta(s)} |u|^2 \,\mathrm{d}t \ \ge\ \gamma \left(|F(s)|^2 + |G(s)|^2 \right),$$

while $\|e^{d_i t} F\|^2 \le B \int |F|^2$. Integrating in s gives (b).

Step 3 (b)⇒(c)⇒(d)

$\ker_{L^2} fg$ is closed, and by (b) it equals $\mathcal{H}_1 + \mathcal{H}_2$. If the sum is closed, the sum map $\mathcal{H}_1 \oplus \mathcal{H}_2 \to \mathcal{H}_1 + \mathcal{H}_2$ is a continuous bijection of Banach spaces (the intersection is $\{0\}$), so by the open mapping theorem $\|x\|^2 + \|y\|^2 \le C\|x + y\|^2$. If $c_F = 1$, unit vectors x_k, y_k with $\mathrm{Re}\langle x_k, y_k \rangle \to -1$ would give $\|x_k + y_k\| \to 0$, a contradiction.

Step 4 (d)⇒(a), by contraposition

Suppose $\inf \ell = 0$. Since ℓ is concave, $\ell \to 0$ at an end s_-. Choose disjoint intervals $E_k \to s_-$ with $\sup_{E_k} \ell^2 \le 4^{-k}$, constants $c_k = (\int_{E_k} \ell)^{-1/2}$, and $G = \sum_k c_k \mathbf{1}_{E_k}$. With

$$h_s(t) = e^{d_2 t} - e^{\lambda \alpha(s)} e^{d_1 t}, \qquad |h_s| \le C_1 \ell(s) \text{ on the chord},$$

the function $u = G(s) h_s(t) \in \ker fg$ has

$$\|u\|^2 \le C_1^2 \sum_k c_k^2 \int_{E_k} \ell^3 \le C_1^2 \sum_k 4^{-k} < \infty, \qquad \|e^{d_2 t} G\|^2 \ge e^{-2|d_2| M} \sum_k c_k^2 \int_{E_k} \ell = \infty.$$

So u is a limit of its truncations, which lie in $\mathcal{H}_1 + \mathcal{H}_2$, but u itself does not: the sum is not closed, and $c_F = 1$.

Why it works. the conditioning of the split on a chord is a 2×2 Gram matrix whose smallest eigenvalue is controlled by the chord length.

■

Uses: A13.

Dim. A17 **C12-P7**

Regularity criterion: closed and dense [proved]

Let $P = c\prod_{j=1}^{N} Q_j$ with pairwise coprime Q_j, $\mathcal{H}_j = \ker_{H^m(\Omega)} Q_j$, $W = \bigoplus_j \mathcal{H}_j$, $S(\omega) = \sum_j \omega_j$, and

$$\gamma_N = \inf \sigma\big(S^*S\big|_{(\ker S)^\perp}\big).$$

(a) Every $u \in \ker_{H^m} P$ has components in $H^m \iff \sum_j \mathcal{H}_j$ is closed and dense in $\ker_{H^m} P$.

(b) $\sum_j \mathcal{H}_j$ is closed $\iff \gamma_N > 0$; then the minimal-energy split satisfies, optimally,

$$\sum_j \|u_j\|_{H^m}^2 \le \gamma_N^{-1}\|u\|_{H^m}^2.$$

(c) For $N = 2$: $\gamma_2 = 1 - c_F$, the Friedrichs cosine after removing $\mathcal{I} = \mathcal{H}_1 \cap \mathcal{H}_2$.

(d) For $N \ge 3$ the pairwise cosines do not decide.

Stated in the paper as no. 7.3, p. 40

Idea. The components exist in H^m iff the sum map is onto $\ker_{H^m} P$. A bounded map has closed range iff it is bounded below on the orthogonal complement of its kernel, and γ_N is the square of that lower bound. For (d), glue infinitely many 3×3 examples whose conditioning tends to 0 while all pairwise angles stay large.

Proof.

Step 1 (a) Closed and dense means equal

$\ker_{H^m} P$ is closed in H^m (because $P(\partial) : H^m \to \mathcal{D}'$ is continuous) and contains the sum. A closed dense subspace is the whole space; the converse is clear.

Step 2 (b) Closed range iff bounded below

S^*S is self-adjoint, positive, and leaves $(\ker S)^\perp$ invariant. On it

$$\|S\omega\|^2 = \langle S^*S\omega, \omega\rangle \ge \gamma_N\|\omega\|^2.$$

If $\gamma_N > 0$, S is bounded below on $(\ker S)^\perp$, so its range is closed. If the range is closed, $S : (\ker S)^\perp \to \operatorname{im} S$ is a continuous bijection between Banach spaces, so its inverse is bounded, $\|S\omega\| \ge c\|\omega\|$, and $\gamma_N \ge c^2 > 0$. The minimal-energy preimage $S^\dagger u$ satisfies $\|S^\dagger u\|^2 \le \gamma_N^{-1}\|u\|^2$, with near-equality along spectral vectors near γ_N.

Step 3 (c) Two spaces

$(\ker S)^\perp = (A \oplus B) \oplus \{(z,z) : z \in \mathcal{I}\}$ with $A = \mathcal{H}_1 \ominus \mathcal{I}$, $B = \mathcal{H}_2 \ominus \mathcal{I}$. On the second summand $S^*S = 2$. On $A \oplus B$,

$$S^*S = I + K, \qquad K(a,b) = (\Pi_A b,\ \Pi_B a),$$

with $\|K\| = c_F$ and spectrum symmetric about 0. So the bottom of the spectrum is $1 - c_F$.

Step 4 (d) Three lines in $\mathbb{C}^3$

For $\varepsilon > 0$ let

$$v_1 = \frac{(1, 0, \varepsilon)}{\sqrt{1+\varepsilon^2}}, \qquad v_2 = \Big(-\tfrac{1}{2}, \tfrac{\sqrt{3}}{2}, 0\Big), \qquad v_3 = \Big(-\tfrac{1}{2}, -\tfrac{\sqrt{3}}{2}, 0\Big),$$

and $\ell_i = \mathbb{C}v_i$. Here S^*S is the Gram matrix of (v_1, v_2, v_3), with

$$\det = \frac{3\varepsilon^2}{4(1+\varepsilon^2)} > 0, \qquad \gamma_3(\varepsilon) \sim \frac{\varepsilon^2}{3} \to 0,$$

while the pairwise cosines are at most $\frac{1}{2}$.

Step 5 (d) Gluing the blocks

Take the Hilbert direct sum over blocks with $\varepsilon_k \to 0$, and $\mathcal{L}_i = \bigoplus_k \ell_i^{(\varepsilon_k)}$. Pairwise cosines are suprema over blocks, so they stay $\le \frac{1}{2}$ and all pairwise sums are closed. But $S^*S = \bigoplus_k G_{\varepsilon_k}$ has bottom of spectrum 0 and trivial kernel, so the triple sum is not closed.

Why it works. regularity of the components is the bounded inverse theorem for the sum map, and γ_N is its smallest singular value squared.

■

Uses: A13.

Dim. A18 **OPF5B-T1**

Density on bounded convex regions

[proved mod. Hörmander 10.2.1, Stein VI.3]

Let $\Omega \subset \mathbb{R}^2$ be bounded, open, convex, $P = c\prod_{j=1}^{N} Q_j \neq 0$ as in A13, $m \ge 0$, and $\mathcal{H}_j = \ker_{H^m(\Omega)} Q_j$. Then $\sum_j \mathcal{H}_j$ is dense in $\ker_{H^m(\Omega)} P$. Consequently the components of every $u \in \ker_{H^m} P$ lie in H^m iff $\gamma_N > 0$.

Stated in the paper as no. 7.3, p. 41

Idea. Approximate u by solutions of P defined on a neighbourhood of $\bar{\Omega}$, then apply the decomposition A13 on a slightly larger convex set, where the components are smooth up to $\partial\Omega$. The approximants are built by shifting u slightly inward near each boundary point, gluing with a partition of unity, and correcting the gluing error with a fundamental solution. The correction gains exactly as much as the error loses, for every P.

Tools for this proof

Write $p(\zeta) = P(i\zeta)$, $p^{(\alpha)} = \partial_\zeta^\alpha p$, and

$$\tilde{P}(\zeta) = \Big(\sum_\alpha |p^{(\alpha)}(\zeta)|^2\Big)^{1/2}, \qquad k_m(\zeta) = (1+|\zeta|^2)^{m/2}.$$

$B_{2,k}$ is the space of f with $k\hat{f} \in L^2$; $B_{2,k_m} = H^m$. A weight k is *temperate* if $k(\zeta+\eta) \le C(1+|\eta|)^N k(\zeta)$.

- **[F1]** (Hörmander, ALPDO II, Thm. 10.2.1) [from cited source]: P has a fundamental solution E with $\varphi E \in B_{\infty,\tilde{P}}$ for every $\varphi \in C_c^\infty$.
- **[F2]** (Stein, Ch. VI §3, Thm. 5) [from cited source]: a bounded convex set has a bounded extension operator $H^m(\Omega) \to H^m(\mathbb{R}^2)$.
- **(L1)** $\tilde{P}$ and $1/\tilde{P}$ are temperate; $|p^{(\alpha)}| \le \tilde{P}$, so $P^{(\alpha)}(\partial)$ maps B_{2,k_m} to $B_{2,k}$ with $k = k_m/\tilde{P}$; multiplication by $\varphi \in C_c^\infty$ is bounded on $B_{2,k}$ for temperate k; and $\|(\varphi E) * g\|_{2,k\tilde{P}} \le \sup|\tilde{P}\,\widehat{\varphi E}| \cdot \|g\|_{2,k}$.

Proof.

Let $u \in \ker_{H^m(\Omega)} P$ and $\varepsilon \in (0,1]$.

Step 1 Fixed choices

Fix $o \in \Omega$ and $\rho > 0$ with $B(o,\rho) \subset \Omega$. Cover $\partial\Omega$ by balls $B(p_i,\rho/8)$, $i = 1,\dots,n$. Take $\Omega_0 \Subset \Omega$ containing the rest of $\bar{\Omega}$, and a partition of unity $\theta_0 \in C_c^\infty(\Omega_0)$, $\theta_i \in C_c^\infty(B(p_i,\rho/4))$, with $\sum_i \theta_i = 1$ on a neighbourhood V of $\bar{\Omega}$. Put

$$\tau_0 = 0, \qquad \tau_i = \varepsilon\,(o - p_i).$$

Step 2 Geometry: inward shifts stay inside

For $y_0 \in \bar{\Omega} \cap \bar{B}(p_i,\rho/2)$,

$$B\big(y_0 + \tau_i,\ \varepsilon\rho/2\big) \subset \Omega.$$

Indeed $y_0 + \tau_i + q = (1-\varepsilon)y_0 + \varepsilon(z + q/\varepsilon)$ with $z = o + y_0 - p_i \in \bar{B}(o,\rho/2)$, and $z + q/\varepsilon \in B(o,\rho) \subset \Omega$ for $|q| < \varepsilon\rho/2$; convexity does the rest. Consequently, on a neighbourhood V_ε of $\bar{\Omega}$, every point of $\operatorname{supp}\theta_i$ shifted by τ_i lies in Ω.

Step 3 Shifted copies solve P where they are used

Let $\tilde{u} = \operatorname{Ext} u \in H^m(\mathbb{R}^2)$ [F2], $\tilde{u}_i = \tilde{u}(\cdot + \tau_i)$, $w_i = \tilde{u}_i - \tilde{u}$. Translations commute with $P(\partial)$, so $P(\partial)\tilde{u}_i = 0$ wherever the shifted point lies in Ω, in particular on $V_\varepsilon \cap \operatorname{supp}\theta_i$.

Step 4 The shifts are small

By continuity of translations in H^m (in Fourier variables, dominated convergence with $|\tau_i| \le \varepsilon \operatorname{diam}\Omega$),

$$\|w_i\|_{H^m(\mathbb{R}^2)} \to 0 \qquad (\varepsilon \to 0).$$

Step 5 The gluing error

Let

$$g = \sum_{i=0}^{n} [P(\partial), \theta_i]\, w_i = \sum_i \sum_{\alpha \ne 0} \frac{\partial^\alpha \theta_i}{\alpha!}\, (\partial_z^\alpha P)(\partial)\, w_i.$$

It is supported in a fixed compact set, and by (L1)

$$\|g\|_{2,k} \le C \sum_i \|w_i\|_{H^m}, \qquad k = \frac{k_m}{\tilde{P}},$$

with C independent of ε.

Step 6 The correction gains what the error loses

Let $z = E * g$, so $P(\partial)z = g$. With $\varphi = 1$ near the relevant bounded set, $z = (\varphi E) * g$ near $\bar{\Omega}$, and by (L1) and [F1]

$$\|z\|_{H^m(\Omega)} \le C' \sup |\tilde{P}\,\widehat{\varphi E}| \cdot \|g\|_{2,k} \le C'' \sum_i \|w_i\|_{H^m}.$$

Step 7 A solution on a neighbourhood of $\bar{\Omega}$, close to u

Put $v = \sum_i \theta_i \tilde{u}_i - z$. On V_ε,

$$P(\partial)\Big(\sum_i \theta_i \tilde{u}_i\Big) = \sum_i [P, \theta_i]\tilde{u}_i = \sum_i [P, \theta_i] w_i + \Big[P, \sum_i \theta_i\Big]\tilde{u} = g,$$

because $\sum_i \theta_i \equiv 1$ there. So $P(\partial)v = 0$ on $V_\varepsilon \supset \bar{\Omega}$. On Ω, $v - u = \sum_i \theta_i w_i - z$, hence

$$\|v - u\|_{H^m(\Omega)} \le C''' \sum_i \|w_i\|_{H^m} \to 0.$$

Step 8 Decompose on a slightly larger convex set

For fixed ε, a metric neighbourhood W' of $\bar{\Omega}$ inside V_ε is convex. Mollify, $v_\eta = v * \rho_\eta$: it is smooth on W', solves P there, and $v_\eta \to v$ in $H^m(\Omega)$. By A13 on W', $v_\eta = \sum_j v_{\eta,j}$ with smooth $v_{\eta,j} \in \ker Q_j$. Since $\bar{\Omega} \subset W'$ is compact, the $v_{\eta,j}$ have bounded derivatives on Ω, so $v_{\eta,j}|_\Omega \in \mathcal{H}_j$. Choosing first ε, then η, makes $\|v_\eta - u\|_{H^m(\Omega)}$ as small as we like.

Why it works. the commutator error contains only derivatives of P, which are dominated by $\tilde{P}$, and Hörmander's fundamental solution gains exactly $\tilde{P}$; no assumption on the factors is needed.

■

Uses: A13; [F1] Hörmander, [F2] Stein.

CHAPTER 3

The interaction term

In this chapter

Starting from	an evolution factor $Q = \tau - a(\xi)$, $a = \sum_{n=1}^{N} a_n$ with $a_n \in \mathbb{C}[\xi]$ the terms; a bounded rectangle $\Omega = I \times J$; $u \in \ker Q$, base time t_0, $u_0 = u(t_0, \cdot)$, $s = t - t_0$.
Why	the interaction term compares the full solution with the sum of the single-term evolutions. We prove its exact form and cost in a class of exponential type, that it is not continuous in Sobolev norms and cannot be regularised, that the Dyson series converges, and that non-commuting terms with an infinite adjoint orbit admit no finite product formula.
Role	Dim. A19–A24. They prove Chapter 8 of the paper.

The class $\mathfrak{B}$ and two lemmas used throughout

For $\rho > 0$,

$$\|f\|_{B_\rho} = \sup_{j \geq 0} \rho^{-j} \|f^{(j)}\|_{L^2(J)}, \qquad B_\rho(J) = \{f : \|f\|_{B_\rho} < \infty\},$$

and $\mathfrak{B}_\rho(\Omega)$ likewise with $\sup_{i,j} \rho^{-i-j} \|\partial_t^i \partial_x^j u\|_{L^2}$; $\mathfrak{B} = \bigcup_\rho \mathfrak{B}_\rho$. For $A(\xi) = \sum_l \alpha_l \xi^l$ write $|A|(\rho) = \sum_l |\alpha_l| \rho^l$.

(P1-L1) B_ρ and $\mathfrak{B}_\rho$ are Banach spaces with $\|\partial_x\| \leq \rho$; $\mathfrak{B}_\rho \hookrightarrow H^m$ for every m; every exponential-polynomial lies in $\mathfrak{B}$; for $u \in \mathfrak{B}_\rho(\Omega)$ the trace $t \mapsto u(t, \cdot)$ is continuous into $B_{\rho'}(J)$ for every $\rho' > \rho$ [proved].

(P1-L2) On B_ρ, $\|A(\partial_x)\| \leq |A|(\rho)$, and $e^{sA(\partial_x)} = \sum_k \frac{s^k}{k!} A(\partial_x)^k$ converges in operator norm. These exponentials commute, satisfy $e^{s(A+A')(\partial_x)} = e^{sA(\partial_x)} e^{sA'(\partial_x)}$, and

$$e^{sA(\partial_x)} \big(x^l e^{\xi x}\big) = \partial_\xi^l \big(e^{sA(\xi)} e^{\xi x}\big) \qquad \text{[proved]}.$$

Both are proved in Part II (Dim. B for P1-L1, P1-L2).

3.1 Exact form and cost

Dim. A19 **P1-T2**

Closed form of the interaction [proved]

Let $u \in \mathfrak{B}_\rho(\Omega)$ with $Q(\partial)u = 0$, $\rho' > \rho$, $u_n(t) = e^{s\,a_n(\partial_x)}u_0$.

(i) u_n is the unique solution in $\mathfrak{B}$ of $(\partial_t - a_n(\partial_x))v = 0$ with $v(t_0) = u_0$, and $u(t) = e^{s\,a(\partial_x)}u_0$.

(ii) In $B_{\rho'}(J)$,

$$E_{\mathrm{int}}^{\mathrm{add}}(t) = \sum_{|S|\geq 2} \prod_{n\in S} \left(e^{s\,a_n(\partial_x)} - \mathrm{Id}\right) u_0, \qquad E_{\mathrm{int}}^{\mathrm{mult}} \equiv 0.$$

(iii) If $u_0 = \int e^{\zeta x}\,\mathrm{d}\mu_0(\zeta)$ with μ_0 compactly supported, then with $z_n = e^{s\,a_n(\zeta)} - 1$,

$$E_{\mathrm{int}}^{\mathrm{add}}(t) = \int e_{\geq 2}(z)\, e^{\zeta x}\,\mathrm{d}\mu_0(\zeta), \qquad e_{\geq 2}(z) = \prod_n (1+z_n) - 1 - \sum_n z_n.$$

Stated in the paper as no. 8.2, p. 44

Idea. Everything happens in a commutative Banach algebra of bounded operators. The single-term flows commute and their product is the full flow. Subtracting the "first-order" part $\mathrm{Id} + \sum_n (e^{s a_n} - \mathrm{Id})$ from the expanded product leaves exactly the subsets with at least two elements.

Proof.

Step 1 (i) Flows are unique in $\mathfrak{B}$

By P1-L1, u is a continuous curve in $B_{\rho'}$. A solution in $\mathfrak{B}$ of $\partial_t v = A(\partial_x)v$ with datum $v(t_0)$ is $e^{(t-t_0)A}v(t_0)$; uniqueness follows from the group property on B_{ρ_2} for large ρ_2. Apply this with $A = a$ and with $A = a_n$.

Step 2 (ii) The full flow is the product of the single flows

By P1-L2,

$$e^{s\,a(\partial_x)} = \prod_{n=1}^{N} e^{s\,a_n(\partial_x)},$$

so $E_{\mathrm{int}}^{\mathrm{mult}} = 0$.

Step 3 (ii) Expanding the product

Write $e^{s\,a_n} = \mathrm{Id} + \Delta_n$ with $\Delta_n = e^{s\,a_n} - \mathrm{Id}$. The Δ_n commute, so

$$\prod_{n=1}^{N} (\mathrm{Id} + \Delta_n) = \sum_{S\subseteq\{1,\dots,N\}} \prod_{n\in S} \Delta_n.$$

The empty set gives u_0; the singletons give $u_n(t) - u_0$. Subtracting them from $u(t)$ leaves the sum over $|S| \geq 2$.

Step 4 (iii) Mode by mode

$\zeta \mapsto e^{\zeta x} \in B_{\rho'}$ is continuous on the support of μ_0, so the integral is a Bochner integral and bounded operators pass under it. On one mode,

$$\prod_{n\in S}\left(e^{s\,a_n(\partial_x)} - \mathrm{Id}\right)e^{\zeta x} = \prod_{n\in S} z_n\; e^{\zeta x}.$$

Summing over $|S| \geq 2$ gives $e_{\geq 2}(z)$.

Why it works. commuting bounded operators obey the algebra of numbers, and the interaction is the part of a product of N factors that involves at least two of them.

■

Uses: P1-L1, P1-L2.

Example 3.1 A stationary state is pure interaction

Take $a_1 = -c\xi$, $a_2 = D\xi^2$ and the stationary mode $u = e^{cx/D}$, so $a(c/D) = 0$. With $\theta = c^2 s/D$:

$$z_1 = e^{-\theta} - 1, \qquad z_2 = e^{\theta} - 1, \qquad E_{\mathrm{int}}^{\mathrm{add}} = z_1 z_2\, u = -4\sinh^2\!\left(\frac{\theta}{2}\right) e^{cx/D}.$$

Dim. A20 **P1-T3**

Cost: $O(N)$ versus $\Theta(2^N)$ [proved]

Model: $u_0 = \sum_{k=1}^{M} p_k(x) e^{\zeta_k x}$ with $\deg p_k \leq L$; terms a_n of degree d_n; unit-cost arithmetic and exp.

(a) One value of $E_{\mathrm{int}}^{\mathrm{add}}(t)$ costs

$$O\Big(M\big[N(L+1)^2 + (L+1)\textstyle\sum_n (d_n+1)\big]\Big).$$

(b) All order aggregates $E^{(k)} = \sum_{|S|=k}\prod_{n\in S}\Delta_n u_0$, $2 \leq k \leq N$, cost $O\big(M[N^2(L+1)^2 + (L+1)\sum_n(d_n+1)]\big)$.

(c) The full table of cross terms has $M(2^N - N - 1)$ entries and costs $\Theta\big(M\,2^N(L+1)^2\big)$.

Stated in the paper as no. 8.3, p. 46

Idea. By A19(iii), each mode contributes $e_{\geq 2}(z)$ times the mode, and $e_{\geq 2}$ is a product minus a sum: it can be evaluated without ever expanding the product. Polynomial prefactors are handled by truncated power series in ζ.

Proof.

Step 1 The quantity to compute for one summand

By A19(iii) and P1-L2, the part of E_{int} coming from $x^l e^{\zeta_k x}$ is

$$\partial_\zeta^l \left[e_{\geq 2}\big(z(\zeta)\big)\, e^{\zeta x} \right]_{\zeta=\zeta_k}.$$

So we need the Taylor coefficients of $e_{\geq 2}(z(\zeta))$ at ζ_k up to order L. Put $\eta = \zeta - \zeta_k$ and work modulo η^{L+1}.

Step 2 (a) Four operations per summand

(1) $a_n(\zeta_k + \eta) \bmod \eta^{L+1}$ by repeated synthetic division: $O((L+1)(d_n+1))$.
(2) $\exp\big(s\, a_n\big) \bmod \eta^{L+1}$ by the recurrence $f' = g'f$: $O((L+1)^2)$.
(3) $\prod_n (1+z_n)$: $N-1$ truncated products, $O(N(L+1)^2)$.
(4) Subtract $1 + \sum_n z_n$ and read off the coefficients.

Summing over n and over the M summands gives (a).

Step 3 (b) All orders by dynamic programming

The aggregate of order k is the elementary symmetric function $e_k(z)$, the coefficient of λ^k in $\prod_n (1+\lambda z_n)$. Multiplying in one factor at a time, step n updates at most n coefficients, each a truncated product: $O(N^2(L+1)^2)$ per summand.

Step 4 (c) The full table

Each cross term c_S is a separate output, so the output alone has $M(2^N - N - 1)$ entries of size $L+1$. They are obtained by $c_S = c_{S\setminus\{\max S\}} \cdot z_{\max S}$, one truncated product per subset, which matches the lower bound up to the factor $L+1$.

Why it works. a product can be evaluated in linear time; only listing its expansion is exponential.

■

Uses: A19, P1-L2.

3.2 Instability

Dim. A21 **P1-P1(e)**

The interaction is not continuous in H^m [false] (continuity is false; proof of the counterexample)

Let $Q = \tau + c\xi - D\xi^2$ with $c, D > 0$, terms $a_1 = -c\xi$ and $a_2 = D\xi^2$, $I = (0, T)$, and any base time $t_0 \in \bar{I}$. For every $m \in \mathbb{N}$,

$$\sup \left\{ \frac{\|E_{\mathrm{int}}^{\mathrm{add}}\|_{L^2(\Omega)}}{\|u\|_{H^m(\Omega)}} : u \in \mathrm{EP},\ Qu = 0,\ u \neq 0 \right\} = \infty.$$

Hence $u \mapsto E_{\mathrm{int}}^{\mathrm{add}}$ has no continuous extension from $\mathfrak{B} \cap \ker Q$ to $\ker_{H^m} Q$.

Stated in the paper as no. 8.5, p. 47

Idea. On a bounded interval, real exponentials $e^{\zeta x}$ with large ζ are legitimate data. The diffusion flow alone amplifies them by $e^{D\zeta^2 s}$, while the true solution grows only like $e^{(D\zeta^2 - c\zeta)s}$. The interaction term must cancel the difference, and the ratio grows like $e^{c\zeta T}$, faster than any power of ζ.

Proof.

Step 1 The test solutions

For real $\zeta > c/D$ let

$$u_\zeta = e^{\alpha t + \zeta x}, \qquad \alpha = D\zeta^2 - c\zeta > 0.$$

Then $Qu_\zeta = 0$ and $\|u_\zeta\|_{H^m(\Omega)} \le C_m (1+\zeta)^m\, e^{\alpha T}\, \|e^{\zeta x}\|_{L^2(J)}$.

Step 2 The interaction of one mode

By A19(iii),

$$E_{\mathrm{int}}^{\mathrm{add}} = e^{\alpha t_0}\, z_1 z_2\, e^{\zeta x}, \qquad z_1 = e^{-c\zeta s} - 1, \quad z_2 = e^{D\zeta^2 s} - 1.$$

Step 3 Case $t_0 < T$

Put $T' = T - t_0 > 0$. For $s \in [T'/2, T']$ and ζ large, $|z_1| \ge \frac{1}{2}$ and $|z_2| \ge \frac{1}{2} e^{D\zeta^2 s}$. Integrating over $t \in [T - \zeta^{-2}, T]$:

$$\|E_{\mathrm{int}}^{\mathrm{add}}\|^2 \ge c\, \zeta^{-2}\, e^{2\alpha t_0 + 2D\zeta^2 T'}\, \|e^{\zeta x}\|^2.$$

Dividing by $\|u_\zeta\|_{H^m}^2$, and using $2D\zeta^2 T' - 2\alpha(T - t_0) = 2c\zeta T'$, the ratio of squares is at least

$$c_m\, \zeta^{-4m-2}\, e^{2c\zeta(T - t_0)} \longrightarrow \infty.$$

Step 4 Case $t_0 = T$

Now $s \in [-T, 0]$, $z_1 = e^{c\zeta|s|} - 1$, and $|z_2| = 1 - e^{-D\zeta^2|s|} \ge \frac{1}{2}$ for large ζ. For $t \in [0, T/2]$, $|z_1| \ge \frac{1}{2} e^{c\zeta T/2}$, and the ratio is at least $c_m'(1+\zeta)^{-4m} e^{c\zeta T} \to \infty$.

Step 5 No continuous extension

$u \mapsto E_{\text{int}}^{\text{add}}$ is linear and unbounded on a dense subspace, so it has no continuous extension.

Why it works. a bounded interval admits real exponential modes, and on them one term alone grows much faster than the full equation.

■

Uses: A19.

Where this does not happen

On $\mathbb{R}$ with L^2 data the modes are $e^{i\kappa x}$, every $|z_n| \le 2$ for dissipative or dispersive terms, and the interaction is bounded: $\|E_{\text{int}}\| \le 2\|u_0\|$ for advection–diffusion (P2-P5). The failure is a feature of *bounded* domains.

Dim. A22 **P2-T2**

No regularisation removes the interaction [proved]

(P2-T2) Let $\Omega = I \times J$, $a = \sum_{n=1}^{N} a_n$ with distinct a_n and $a - a_n \neq 0$ for all n, $D_n = \ker_{H^m}(\partial_t - a_n(\partial_x))$, $M = \overline{\sum_n D_n}$, and $u \in \ker_{H^m}(\partial_t - a(\partial_x))$. Put

$$g = \prod_{n=1}^{N}(a - a_n) \in \mathbb{C}[\xi], \qquad K = \deg g, \qquad Q = \prod_{n}(\tau - a_n), \qquad k = \deg Q.$$

(a) $Q(\partial)u = g(\partial_x)u$, and $\|E^{\perp}(u)\|_{H^m} \ge \|g(\partial_x)u\|_{H^{m-k}}/C_Q$, where $E^{\perp}(u)$ is the component of u orthogonal to M.

(b) $E^{\perp}(u) = 0 \iff g(\partial_x)u = 0$; this space has dimension K and consists of exponential-polynomials.

(c) Every split by terms, with any regulariser, leaves a residual of norm at least $\|E^{\perp}(u)\|$.

(P2-T4) If maps R_α with Lipschitz constants L_α converge pointwise to Φ on a set V, then $\mathrm{Lip}(\Phi) \le \liminf L_\alpha$. Every exact selection of components has Lipschitz constant at least $\gamma_N^{-1/2}$.

Stated in the paper as no. 8.5, p. 47

Idea. Every per-term piece is killed by $Q = \prod_n(\tau - a_n)$, so Q kills the closed span M. On a solution of the full equation, Q acts as the polynomial $g(\partial_x)$. So $\|Q(\partial)u\|$ measures the distance of u from M, and only the finitely many solutions killed by g have distance zero. P2-T4 is the fact that a pointwise limit of uniformly Lipschitz maps is Lipschitz with the same constant.

Proof.

Step 1 (a) Q acts as g on solutions

Divide $Q \in \mathbb{C}[\xi][\tau]$ by the monic $\tau - a(\xi)$:

$$Q = (\tau - a)H + r(\xi), \qquad r(\xi) = Q(a(\xi), \xi) = \prod_n (a - a_n)(\xi) = g(\xi).$$

Since constant-coefficient operators commute in $\mathcal{D}'$, $Q(\partial)u - g(\partial_x)u = H(\partial)(\partial_t - a(\partial_x))u = 0$.

Step 2 (a) The distance from M

Each $v \in D_n$ is killed by its own factor $\tau - a_n$ of Q, hence by Q; by continuity Q kills M. Write $u = P_M u + E^\perp(u)$. The map $Q(\partial) : H^m \to H^{m-k}$ is bounded with norm C_Q ((P2-L1)), so

$$\|g(\partial_x)u\|_{H^{m-k}} = \|Q(\partial)E^\perp(u)\|_{H^{m-k}} \le C_Q \, \|E^\perp(u)\|_{H^m}.$$

Step 3 (b) The exceptional space

If $E^\perp(u) = 0$, Step 2 gives $g(\partial_x)u = 0$, so u is killed by the ideal $\mathfrak{J} = (\tau - a,\ g)$. The map $\mathcal{R} \to \mathbb{C}[\xi]$, $\tau \mapsto a(\xi)$, is onto with kernel $(\tau - a)$, so

$$\mathcal{R}/\mathfrak{J} \cong \mathbb{C}[\xi]/(g), \qquad \dim = K.$$

By A12, the solutions form a K-dimensional space of exponential-polynomials. Conversely, such u split by factors of Q (A13) with components in $\mathfrak{B} \subset H^m$ ((P1-P2(c))), so $u \in M$.

Step 4 (c) Any regulariser

For any representation $S\omega$, $u - S\omega \in u + M$, so $\|u - S\omega\| \ge \operatorname{dist}(u, M) = \|E^\perp(u)\|$.

Step 5 (P2-T4) Lipschitz constants pass to the limit

For $v, w \in V$: $\|R_\alpha v - R_\alpha w\| \le L_\alpha \|v - w\|$, and the left side tends to $\|\Phi v - \Phi w\|$. Taking the supremum over $v \ne w$ gives $\operatorname{Lip}(\Phi) \le \liminf_\alpha L_\alpha$. For an exact selection σ ($S\sigma(u) = u$), $\sigma(u) - \sigma(0)$ is a preimage of u, so its norm is at least that of the minimal preimage, $\|S^\dagger u\|$; hence $\operatorname{Lip}(\sigma) \ge \|S^\dagger\| = \gamma_N^{-1/2}$ (A17).

Why it works. the per-term pieces all satisfy one extra equation, and on solutions of the full equation that extra equation is a nonzero polynomial in ∂_x.

■

Uses: A12, A13, A17, P2-L1, P1-P2.

Example 3.2 Advection–diffusion

For $a_1 = -c\xi$ and $a_2 = D\xi^2$: $g = (D\xi^2)(-c\xi) = -cD\xi^3$, $K = 3$, and $Q(\partial)u = -cD\, u_{xxx}$. The exceptional space is

$$\{E^\perp = 0\} = \operatorname{span}\{1,\ x - ct,\ (x - ct)^2 + 2Dt\},$$

with, for example,

$$(x-ct)^2+2Dt=\underbrace{\Big[(x-ct)^2-\tfrac{2D}{c}(x-ct)\Big]}_{\in D_1}+\underbrace{\tfrac{2D}{c}\,x}_{\in D_2}.$$

3.3 Perturbation and non-commuting terms

Dim. A23 **P3-T1**

Dyson–Phillips converges for every time [proved]

(a) On $X=L^2(\mathbb{R})$ let $A_0=a(\partial_x)$, $a(\xi)=\sum_{l\le q}\alpha_l\xi^l$, $\alpha_q\neq 0$, and $V=\sum_{k\le r}b_k(x)\partial_x^k$ with bounded b_k and $r<q$.

(a1) The condition (Par): $\operatorname{Re}a(i\kappa)\le\omega-\delta|\kappa|^q$ for some $\delta>0$, holds iff q is even and $\operatorname{Re}(\alpha_q i^q)<0$.

(a2) Under (Par), the Dyson–Phillips series of $e^{t(A_0+\varepsilon V)}$ converges in operator norm, uniformly on every $[0,T]$, is entire in ε, and its sum is the semigroup generated by $A_0+\varepsilon V$.

(b) The same holds on $L^2(0,L)$ for $A_0=D\partial_x^2$ with Dirichlet conditions and $V=b_1\partial_x+b_0$.

Stated in the paper as no. 8.6, p. 49

Idea. The principal term smooths: $\|\partial^k S_0(s)\|$ blows up only like $s^{-k/q}$. Since the perturbation has order $r<q$, the singularity $s^{-r/q}$ is integrable. Iterating gives Beta integrals, and the series is bounded by a Mittag-Leffler function, which is entire.

Proof.

Step 1 The abstract estimate (P3-L2)

Suppose $\|S_0(t)\|\le Me^{\omega t}$ and, for $s\in(0,T]$,

$$\|VS_0(s)\|\le K\,s^{-\beta}e^{\omega s},\qquad \beta\in[0,1),\quad \gamma=1-\beta.$$

Define $S^{(0)}=S_0$ and $S^{(n)}(t)x=\int_0^t S^{(n-1)}(t-s)\,VS_0(s)x\,\mathrm{d}s$. By induction,

$$\|S^{(n)}(t)\|\le Me^{\omega t}\,\frac{\big(K\Gamma(\gamma)\big)^n\,t^{n\gamma}}{\Gamma(1+n\gamma)}.$$

The induction step is the Beta integral

$$\int_0^t(t-s)^{(n-1)\gamma}s^{\gamma-1}\,\mathrm{d}s=t^{n\gamma}\,\frac{\Gamma\big((n-1)\gamma+1\big)\Gamma(\gamma)}{\Gamma(n\gamma+1)}.$$

Step 2 Convergence and analyticity

The bound sums to $Me^{\omega t}E_\gamma\big(K\Gamma(\gamma)t^\gamma\big)$, where $E_\gamma(w)=\sum_n w^n/\Gamma(1+n\gamma)$ is the Mittag-Leffler function, entire by Stirling. Replacing V by εV gives a power series in ε with infinite radius.

Step 3 The sum is the right semigroup

By Miyadera–Voigt (Engel–Nagel, Thm. III.3.14, Cor. III.3.15–16) [from cited source], $A_0+\varepsilon V$ generates a semigroup S satisfying the variation-of-constants formula. The series satisfies the same formula; the difference Δ satisfies $\Delta(t)=\int_0^t \Delta(t-s)\,\varepsilon V S_0(s)\,\mathrm{d}s$, and iterating with the Beta integral gives $\|\Delta\|\le C\,(|\varepsilon|K\Gamma(\gamma))^n t^{n\gamma}/\Gamma(1+n\gamma)\to 0$.

Step 4 (a1) The symbol criterion

Let $p(\kappa)=\operatorname{Re}a(i\kappa)=\operatorname{Re}(\alpha_q i^q)\kappa^q+p_{<q}(\kappa)$. If q is even and $\operatorname{Re}(\alpha_q i^q)=-2\delta<0$, then $p(\kappa)\le -2\delta\kappa^q+C(1+|\kappa|^{q-1})\le\omega-\delta|\kappa|^q$. Conversely, (Par) forces $p\to-\infty$ like $-\delta|\kappa|^q$ on both sides, so $\deg p=q$, the leading coefficient is negative, and q cannot be odd.

Step 5 (a2) The hypothesis of Step 1 holds with $\beta=r/q$

By Plancherel, for $s>0$,

$$\|\partial^k S_0(s)\|=\sup_\kappa|\kappa|^k e^{s\,p(\kappa)}\le e^{\omega s}\sup_{y\ge0}y^k e^{-s\delta y^q}=e^{\omega s}\Big(\frac{k}{q\delta e\,s}\Big)^{k/q}.$$

With $\|b_k\cdot\|\le\|b_k\|_\infty$ and $s^{-k/q}\le T^{(r-k)/q}s^{-r/q}$ on $(0,T]$, this gives the bound of Step 1 with $\beta=r/q<1$.

Step 6 (b) Dirichlet

In the sine basis $e_n=\sqrt{2/L}\sin(n\pi x/L)$,

$$\|\partial S_0(s)f\|^2=\sum_n|f_n|^2\mu_n^2e^{-2D\mu_n^2 s}\le\frac{\|f\|^2}{2eDs},$$

so Step 1 applies with $\beta=\frac12$.

Why it works. a smoothing principal term pays for a lower-order perturbation, and the payment is an integrable singularity in time.

■

Uses: Engel–Nagel (Miyadera–Voigt).

The hypotheses cannot be dropped (P3-C1)

With a perturbation of the same order ($A_0=\partial^2$, $V=-2\partial^2$) the terms have norm $(2n)^ne^{-n}/n!\sim 2^n/\sqrt{2\pi n}$ and the series diverges. With a dispersive principal term ($A_0=i\partial^2$, $V=i\partial$) the sum does not even generate a semigroup on L^2.

Dim. A24 **P1-T5**

No finite product of flows for an infinite adjoint orbit [proved]

Let $\mathcal{A}$ be a unital associative $\mathbb{C}$-algebra and $X, Y \in \mathcal{A}$ such that

$$X, \quad Y_0 = Y, \quad Y_1 = \operatorname{ad}_X Y, \quad Y_2 = \operatorname{ad}_X^2 Y, \ \dots$$

are linearly independent. Then there is no finite word $(Z_1, \dots, Z_d) \in \{X, Y\}^d$, no common polydisc $U = \{|t| < r_0\} \times \{|\varepsilon| < \varepsilon_0\}$, and no functions g_i holomorphic on U with $g_i(0, \varepsilon) = 0$ and $g_i(t, 0) \equiv 0$ whenever $Z_i = Y$, such that

$$e^{t(X+\varepsilon Y)} = \prod_{i=1}^{d} e^{g_i(t,\varepsilon) Z_i} \qquad \text{in } \mathcal{A}[[t]] \text{ for all } |\varepsilon| < \varepsilon_0.$$

Stated in the paper as no. 8.7, p. 50

Idea. Differentiate the identity in ε at $\varepsilon = 0$. The left side produces the moments $t^{k+1}/(k+1)$ of the independent elements Y_k. The right side, being a finite word, produces a sum of at most p exponential-type terms, whose Hankel matrices have rank at most p. But the Hankel matrices of the moments are Hilbert matrices, of full rank.

Proof.

Step 1 At zero coupling

For $\varepsilon = 0$ the Y-factors are the identity, so $e^{tX} = e^{G(t)X}$ with $G = \sum_{Z_i = X} g_i(t, 0)$. If $G \neq t$, let m be the order of $G - t$; the coefficient of t^m in $e^{(G-t)X} = 1$ gives $(G - t)_m X = 0$, impossible since $X \neq 0$. So $G = t$.

Step 2 Derivative in ε: the left side

Coefficients of t^m on both sides live in the finite-dimensional span of words of length $\leq m$, and depend holomorphically on ε, so we may differentiate. By the formal Duhamel formula,

$$\partial_\varepsilon e^{t(X+\varepsilon Y)}\Big|_{\varepsilon=0} = e^{tX} \int_0^t e^{-\sigma X} Y e^{\sigma X} \, \mathrm{d}\sigma = e^{tX} \sum_{k \geq 0} (-1)^k \frac{t^{k+1}}{(k+1)!} Y_k.$$

Step 3 Derivative in ε: the right side

At $\varepsilon = 0$ all factors are exponentials of X. The X-factors contribute $r(t) X e^{tX}$ with $r = \sum_{Z_i = X} \partial_\varepsilon g_i(t, 0)$. The Y-factors contribute $q_i(t)\, e^{G_i X} Y e^{(t - G_i)X}$, with $q_i = \partial_\varepsilon g_i(t, 0)$ and G_i the sum of the earlier X-times. Multiply on the left by e^{-tX} and use

$$e^{-\tau X}\, Y\, e^{\tau X} = \sum_k (-1)^k \frac{\tau^k}{k!} Y_k, \qquad \tau_i = t - G_i.$$

Step 4 Comparing coefficients

The result is

$$rX + \sum_k (-1)^k Y_k \sum_{i:Z_i=Y} q_i \frac{\tau_i^k}{k!} = \sum_k (-1)^k \frac{t^{k+1}}{(k+1)!} Y_k.$$

By independence, $r = 0$ and, for every $k \geq 0$,

$$\sum_{i:Z_i=Y} q_i(t)\, \tau_i(t)^k = \frac{t^{k+1}}{k+1}.$$

Step 5 A rank contradiction

Fix $t \in (0, r_0)$ and let $p = \#\{i : Z_i = Y\}$. The sequence $m_k = t^{k+1}/(k+1) = \sum_{i=1}^{p} q_i \tau_i^k$ has Hankel matrices

$$H_n = [m_{i+j}]_{i,j<n} = W^{\mathsf{T}} \operatorname{diag}(q)\, W, \qquad W_{ij} = \tau_i^j,$$

of rank at most p. On the other hand

$$H_n = t \operatorname{diag}(t^i) \operatorname{Hilb}_n \operatorname{diag}(t^j), \qquad \operatorname{Hilb}_n = \Big[\frac{1}{i+j+1}\Big],$$

and the Hilbert matrix is positive definite (the Gram matrix of monomials in $L^2(0,1)$), so H_n has rank n. For $n = p+1$ this is a contradiction.

Why it works. a finite product of flows can only produce finitely many "exponential moments", while the true first-order interaction needs infinitely many independent ones.

■

Uses: none.

> **Example 3.3 Where the hypothesis holds, and where it does not**
>
> In the Weyl algebra, $X = x^a \partial$ and $Y = \partial$ with $a \geq 3$ give $Y_k = c_k\, x^{k(a-1)} \partial$ with $c_k \neq 0$: distinct degrees, so the hypothesis holds. For ∂^2 and x^k with $k \geq 3$ it fails (each is locally nilpotent on the other), and the question is open (P1-Q3).

CHAPTER 4

Nonlinear equations

In this chapter

Starting from	scalar evolutions $u_t = \sum_n T_n(u)$ with T_n differential polynomials (Burgers: $T_1 = -uu_x$, $T_2 = \nu u_{xx}$); smooth periodic data u_0; for shocks, the Hopf–Lax and Cole–Hopf representations.
Why	to prove the nonlinear results of the paper: the t^2 jet of the interaction, its closed form before the shock, its sizes $\Theta(\nu)$ and $\nu\lvert\log\nu\rvert$ after the shock, the exact sequence of dictionary identification, and the two identification thresholds (Burgers N-shocks, KdV N-solitons).
Role	Dim. A25–A31. They prove Chapter 9 of the paper. Three proofs (A27, A28, A31) are long in the research files; here they are given as a complete chain of steps with every key formula, and marked *condensed*.

Notation for the whole chapter

For a differential polynomial P in the jet variables $y_0 = u$, $y_1 = u_x$, …, the *evolutionary derivation* is

$$D_P = \sum_k \partial_x^k(P)\,\frac{\partial}{\partial y_k},$$

so that $D_P(Q) = Q'[P]$, the derivative of Q in the direction P. For Burgers with datum u_0 on the circle $\mathbb{T}$ we use the inviscid flow $v = S_B(t)u_0$ (of $-uu_x$ alone), the heat flow $h = S_A(t)u_0$ (of νu_{xx} alone), and the Lagrangian map

$$X(t,a) = a + t\,u_0(a), \qquad J(t,a) = 1 + t\,u_0'(a), \qquad t^* = -\frac{1}{\min u_0'}.$$

[F1] (Lax–Oleinik = Kružkov entropy solution) and **[F2]** (Hopf's formula for viscous Burgers) are classical [from cited source]; [F2] is proved in the research files, [F1] is Kružkov 1970, Thm. 1 §3.

4.1 The interaction term

Dim. A25 **S4T-T1**

Jet of the nonlinear interaction [proved]

Let $\hat{S}_n(s) = \sum_k \frac{s^k}{k!} D_{T_n}^k$ be the formal flow of T_n, $\hat{u}(s)$ the Taylor series of u at t_0, and

$$\hat{E}^{\mathrm{add}}(s) = \hat{u}(s) - u_0 - \sum_n \big(\hat{S}_n(s)u_0 - u_0\big).$$

(i) The coefficient of $s^k/k!$ in $\hat{E}^{\mathrm{add}}$ is $\sum_{\mathbf{w}} D_{\mathbf{w}} u_0$, over the words $\mathbf{w} = (n_1, \dots, n_k)$ with at least two distinct letters, $D_{\mathbf{w}} = D_{T_{n_1}} \cdots D_{T_{n_k}}$. In particular

$$[s^2]\,\hat{E}^{\mathrm{add}} = \frac{1}{2} \sum_{m \neq n} T_n'[u_0]\big(T_m(u_0)\big).$$

(ii) For Burgers ($A = \nu u_{xx}$, $B = -uu_x$):

$$[s^2]\,\hat{E}^{\mathrm{add}} = -\nu\,\big(2u_0'u_0'' + u_0u_0'''\big),$$

and for $u_0 = \varepsilon \sin x$ this is $\frac{3}{2}\nu\varepsilon^2 \sin 2x$.

Stated in the paper as no. 9.1, p. 51

Idea. The full flow is the exponential of the *sum* of the derivations. A power of a sum expands over all words in the letters, while each single-term flow is the exponential of one letter. They agree on the constant words, so the difference is exactly the mixed words.

Proof.

Step 1 The Taylor series of the true solution

Since $u_t = F(u) = \sum_n T_n(u)$, by induction

$$\partial_t^k u(t_0) = D_F^k\, u_0, \qquad D_F = \sum_n D_{T_n},$$

using that $P \mapsto D_P$ is linear.

Step 2 Expanding the power of a sum

In the associative algebra of operators on differential polynomials,

$$\Big(\sum_n D_{T_n}\Big)^k = \sum_{\mathbf{w} \in \{1,\dots,N\}^k} D_{\mathbf{w}}.$$

The constant words $(n, \dots, n)$ contribute $D_{T_n}^k u_0$, which are exactly the Taylor coefficients of $\hat{S}_n(s)u_0$. What remains is the sum over the mixed words: this is (i).

Step 3 Order two

For $k = 2$ the mixed words are the ordered pairs (m, n) with $m \neq n$, and

$$D_{T_m} D_{T_n} u_0 = D_{T_m}(T_n) = T_n'[u_0]\big(T_m(u_0)\big).$$

Dividing by 2! gives the formula.

Step 4 Burgers

With $A'[w] = \nu w_{xx}$ and $B'[w] = -(u_0 w)_x$:

$$A'[B] + B'[A] = -\nu\,(u_0 u_0')'' - \nu\,(u_0 u_0'')' = -\nu\,\big(4u_0' u_0'' + 2u_0 u_0'''\big).$$

Half of it is $-\nu(2u_0' u_0'' + u_0 u_0''')$. For $u_0 = \varepsilon \sin x$,

$$2u_0' u_0'' + u_0 u_0''' = -3\varepsilon^2 \sin x \cos x = -\tfrac{3}{2}\varepsilon^2 \sin 2x.$$

Why it works. the single-term flows reproduce exactly the words with one letter, so the interaction is the sum of the mixed words, which start at order two.

■

Uses: none.

> **Why every coefficient is $O(\nu\varepsilon^2)$**
>
> A word with a letters A and b letters B, applied to $u_0 = \varepsilon w_0$, has degree $1 + b$ in ε and a factor ν^a. A mixed word has $a, b \geq 1$, so every coefficient is $O(\nu\varepsilon^2)$: there is no term of order ε.

Dim. A26 S4T-T2

Closed form before the shock [proved]

Let u_0 be smooth and real, $0 < T < t^*$, $X = X(t,a)$, $J = J(t,a)$. For $t \in [0,T]$,

$$E_{\text{int}}^{\text{add}}(t,X) = \nu t\Big[\frac{u_0''(a)}{J^2} - u_0''(X)\Big] + R^{\text{add}},$$

with $\|R^{\text{add}}(t)\|_\infty \leq \nu^2 C(t)$. In the small-data regime $u_0 = \varepsilon w_0$, $\varepsilon T \max(-w_0') \leq \theta < 1$, the remainder is $\leq C(\theta, w_0, T)\,\nu^2\varepsilon^2 t^3$.

Stated in the paper as no. 9.2.1, p. 53

> **Idea.** Expand the viscous solution around the inviscid one, $u = v + \nu\delta + O(\nu^2)$. The first-order coefficient δ solves a linear transport equation along the characteristics, and in Lagrangian coordinates it integrates exactly. Every remainder is controlled by a maximum principle along characteristics, with constants that do not depend on ν before the shock.

Proof.

Two tools are proved in Part II: the comparison principle (S4T-L0) (for $g_t + bg_x + cg - \nu g_{xx}$ with $|c| \leq K$, $\|g(t)\|_\infty \leq e^{Kt}(\|g(0)\|_\infty + \int_0^t \|Lg\|_\infty)$) and the pre-shock derivative bounds (S4T-L1) ($\|\partial_x^j w\|_\infty \leq K_j$ on $[0,T]$, independent of ν, for both the viscous and the inviscid solution).

Step 1 The difference $e = u - v$ is $O(\nu)$

Since $uu_x - vv_x = ue_x + v_x e$,

$$e_t + u\,e_x + v_x\,e = \nu\,u_{xx}, \qquad e(0) = 0.$$

The comparison principle with $c = v_x$ gives $\|e(t)\|_\infty \le \nu\,\varepsilon_0(t)$, $\varepsilon_0 = K_2 t e^{K_1 t}$. Differentiating once and twice gives in the same way $\|e_x\| \le \nu\varepsilon_1$ and $\|e_{xx}\| \le \nu\varepsilon_2$.

Step 2 The first-order coefficient in Lagrangian form

Let δ solve the linear transport equation

$$\delta_t + (v\,\delta)_x = v_{xx}, \qquad \delta(0) = 0.$$

Along $X(t,a)$ we have $J_t = v_x(t,X)\,J$ and $v_{xx}(t,X) = u_0''(a)/J^3$, so

$$\frac{d}{dt}(\delta J) = J\,v_{xx} = \frac{u_0''(a)}{J^2}, \qquad \delta(t,X) = \frac{t\,u_0''(a)}{J^2}.$$

Step 3 The second-order remainder

$w = e - \nu\delta$ solves $w_t + (vw)_x = \nu e_{xx} - e e_x$, $w(0) = 0$. The comparison principle gives $\|w(t)\|_\infty \le \nu^2 \rho(t)$ with $\rho = t e^{K_1 t}(\varepsilon_2 + \varepsilon_0\varepsilon_1)$.

Step 4 The heat flow

By Taylor's formula with integral remainder for $h = S_A(t)u_0$,

$$\big\|h(t) - u_0 - \nu t\,u_0''\big\|_\infty \le \frac{\nu^2 t^2}{2}\,\|u_0''''\|_\infty.$$

Step 5 Assembling

$E_{\text{int}}^{\text{add}} = (u - u_0) - (v - u_0) - (h - u_0) = e - (h - u_0)$, so

$$E_{\text{int}}^{\text{add}} = \nu\big(\delta - t\,u_0''\big) + w - \big(h - u_0 - \nu t u_0''\big),$$

and at (t,X) the first term is $\nu t\big[u_0''(a)/J^2 - u_0''(X)\big]$.

Step 6 The small-data refinement

The crude remainder contains $\frac{t^2}{2}\|u_0''''\|$, which is linear in ε. The remainder $\hat{w} = w - (h - u_0 - \nu t u_0'')$ solves

$$\hat{w}_t + (v\hat{w})_x - \nu\hat{w}_{xx} = \nu^2\partial_x^2\big(\delta - t u_0''\big) - (v\,r_h)_x - e\,e_x, \qquad \hat{w}(0) = 0,$$

and each source term is $O(\nu^2 t^2)$ and quadratic in ε: the first by a mixed Taylor expansion in (t,λ) for the datum λu_0 (it vanishes to second order in t and in λ), the other two directly. The comparison principle gives the bound $C\nu^2\varepsilon^2 t^3$.

Why it works. before the shock the characteristics do not cross, so the viscous correction is a linear transport along them that integrates in closed form.

■

Uses: S4T-L0, S4T-L1 (Part II), [F2].

4.2 After the shock *(condensed proofs)*

The Hopf–Lax picture

Put $U_0(y) = \int_0^y u_0$ and, for fixed t,

$$\Phi_x(y) = U_0(y) + \frac{(x-y)^2}{2t}, \qquad m(x) = \min_y \Phi_x(y).$$

[F1]: the inviscid solution is $v(t,x) = (x-a)/t$ with a a minimiser of Φ_x. **[F2]**: the viscous solution is

$$u(t,x) = \frac{x - \langle y \rangle_x}{t}, \qquad \langle f \rangle_x = \frac{\int f(y)\, e^{-\Phi_x(y)/2\nu}\, \mathrm{d}y}{\int e^{-\Phi_x(y)/2\nu}\, \mathrm{d}y}.$$

So $u - v = (a - \langle y \rangle_x)/t$: the whole difference is the distance between the minimiser and the Gibbs mean at temperature 2ν. Hypothesis (H_t): every minimiser has $J > 0$, and only finitely many $x_1, \dots, x_K$ (the shocks) have two minimisers $a_i^- < a_i^+$. Put

$$\rho_i = \tfrac{1}{2} \log \frac{J(a_i^-)}{J(a_i^+)}, \qquad \kappa(y) = \frac{t\, u_0''(y)}{J(y)^2}.$$

Dim. A27 — **OPA-T1**

The interaction after the shock is $\Theta(\nu)$

[proved mod. [F1] Kružkov]

Under (H_t), for fixed $t > 0$ and $\nu \to 0$,

$$\| E_{\mathrm{int}}^{\mathrm{add}}(t) \|_{L^1(\mathbb{T})} = \nu \Lambda_{\mathrm{add}} + o(\nu),$$

$$\Lambda_{\mathrm{add}} = \int_0^{2\pi} t \left| \frac{u_0''(a(x))}{J(a(x))^2} - u_0''(x) \right| \mathrm{d}x + 4 \sum_{i=1}^{K} \log\left(2 \cosh \frac{\rho_i}{2}\right).$$

If $t > t^*$ then $K \geq 1$ and $\Lambda_{\mathrm{add}} \geq 4 \log 2 > 0$: the order is exactly ν. The estimate is not uniform in t ($\Lambda \to \infty$ as $t \downarrow t^*$).

Stated in the paper as no. 9.2.2, p. 53

Idea. Away from the shocks the Gibbs measure concentrates at a single minimiser, and Laplace's method gives $u - v = \nu\kappa + O(\nu^{3/2})$. Near a shock it concentrates on two minimisers, with weights given by a logistic function of $(x - x_i)/\nu$. The difference between that smooth step and the sharp jump integrates to $4\nu \log(2\cosh(\rho_i/2))$.

Proof.

Step 1 Uniform structure (compactness)

Under (H_t) there are $\delta, j_0, \eta, r_0 > 0$ such that: $J \geq j_0$ near every minimiser; away from the shock intervals $(x_i - r_0, x_i + r_0)$ the minimiser $a(x)$ is unique and $\Phi_x - m(x) \geq \eta$ at distance $\geq \delta$ from it; near x_i there are two C^3 branches $a_i^{\pm}(x)$, and

$$\Delta_i(x) = \Phi_x(a_i^+(x)) - \Phi_x(a_i^-(x)), \qquad \Delta_i(x_i) = 0, \qquad \Delta_i'(x) = -\frac{a_i^+ - a_i^-}{t} \leq -\frac{4\delta}{t}.$$

Step 2 Laplace with explicit remainders

If $\psi(0) = \psi'(0) = 0$ and $\lambda \leq \psi'' \leq \Lambda$ near 0, then with $\langle\cdot\rangle$ the mean for $e^{-\psi/\varepsilon}$,

$$\langle z \rangle = -\varepsilon \frac{\psi'''(0)}{2\psi''(0)^2} + O(\varepsilon^{3/2}), \qquad \langle |z|^k \rangle \leq m_k \varepsilon^{k/2}.$$

(Integrate by parts twice against $e^{-\psi/\varepsilon}$.) The tails where $\Phi_x - m \geq \eta$ contribute $O(e^{-\eta/4\nu})$.

Step 3 Smooth region

With $\varepsilon = 2\nu$, $\psi(z) = \Phi_x(a+z) - m$, $\psi'' = J/t$, $\psi''' = u_0'''$:

$$u - v = \frac{a - \langle y \rangle_x}{t} = \nu\, \kappa(a(x)) + O(\nu^{3/2})$$

uniformly away from the shocks.

Step 4 Shock layer: logistic weights

Near x_i, the Gibbs mass splits between the two wells with weight $\sigma = L(\ell(x))$ on the right one, $L(z) = 1/(1+e^{-z})$, and

$$\ell(x) = -\frac{\Delta_i(x)}{2\nu} + \rho_i(x) + O(\nu^{1/2}),$$

the term ρ_i coming from the ratio of the two Gaussian widths $\sqrt{J(a^{\mp})}$. With $H = \mathbf{1}_{x > x_i}$,

$$u - v = \frac{a^+ - a^-}{t}\,(H - \sigma) + \nu\kappa(a(x)) + R, \qquad |R| \leq C\nu |H - \sigma| + C\nu^{3/2}.$$

Since $|H - \sigma| \leq Ce^{-2\delta|x - x_i|/t\nu}$, the two pieces overlap only on a set of size $O(\nu|\log\nu|)$ and can be integrated separately.

Step 5 The layer integral

Put $x = x_i + \nu w$. Then $\ell \to D_i w/2 + \rho_i$ with $D_i = (a_i^+ - a_i^-)/t$, and by dominated convergence

$$\int \frac{a^+ - a^-}{t} |H - \sigma|\, \mathrm{d}x = 2\nu \Big[\int_{-\infty}^{0} L(z + \rho_i)\, \mathrm{d}z + \int_0^{\infty} \big(1 - L(z + \rho_i)\big)\, \mathrm{d}z \Big] + o(\nu).$$

Using $\int_{-\infty}^{c} L = \log(1 + e^c)$ and $\int_c^{\infty}(1 - L) = \log(1 + e^{-c})$, this is

$$2\nu \log\big((1 + e^{\rho_i})(1 + e^{-\rho_i})\big) = 4\nu \log\Big(2\cosh\frac{\rho_i}{2}\Big).$$

Step 6 From $u - v$ to E_{int}

Summing Steps 3–5 gives $\|u - v\|_{L^1} = \nu\Lambda_{uv} + o(\nu)$. Since $E_{\mathrm{int}}^{\mathrm{add}} = (u - v) - (h - u_0)$ and $h - u_0 = \nu t u_0'' + O(\nu^2)$ uniformly (A26, Step 4), the same computation with $\kappa(a(x)) - tu_0''(x)$ in place of $\kappa(a(x))$ gives Λ_{add}.

Why it works. at small viscosity the solution is a Gibbs average over the minimisers of the Hopf–Lax functional; a shock is a tie between two minimisers, smoothed into a logistic step of width ν.

■

Uses: [F1], [F2]; research file `work/OP-ANA`, §2 (Lemmas 1–3).

Dim. A28 **OPF5A-T2**

Uniform rate $\nu|\log\nu|$ across the shock time [proved mod. [F1] Kružkov]

Assume the shock is born at a single generic cusp (condition (G*)): u_0' has a unique global minimum at a^*, nondegenerate, so that $\kappa_3 = u_0'''(a^*) > 0$. Then there are τ_0, C, ν_0 such that, uniformly for $t \in (0, t^* + \tau_0]$ and $\nu \le \nu_0$,

$$\left| \|S(t)u_0 - S_B(t)u_0\|_{L^1} - 2\nu \log \frac{1}{\max(|t - t^*|, \sqrt{\nu})} \right| \le C\nu.$$

The same holds for $\|E_{\mathrm{int}}^{\mathrm{add}}\|_{L^1}$. In particular $\sup_t \|E_{\mathrm{int}}^{\mathrm{add}}\|_{L^1} = \nu|\log\nu| + O(\nu)$, with coefficient exactly 1. At $t = t^*$ the next term is $\Gamma\nu$, with $\Gamma = \Gamma_{\mathrm{out}} + \log\frac{\kappa_3}{4} + P$ and the universal constant $P = 1.188654\ldots$ (OPF5A-T1).

Stated in the paper as no. 9.2.2, p. 53

Idea. Near the cusp the Hopf–Lax functional is quartic instead of quadratic. The Laplace expansion of A27 still holds outside a shrinking window, and there it integrates to a logarithm of J, which vanishes like α^2 at the cusp. Inside the window a rescaling turns the problem into a universal tilted double well, whose contribution is $O(\nu)$.

Proof.

Step 1 Exact expansion of the Gibbs mean

Integration by parts on $\mathbb{R}$ gives $\langle\psi'\rangle = 0$ and $\langle z\psi'\rangle = 2\nu$. With $\psi'(z) = pz + \frac{q}{2}z^2 + \rho_3(z)$, $p = J_t(a)/t$, $q = u_0''(a)$, $|\rho_3| \le C|z|^3$:

$$\langle z\rangle = -\frac{\nu q}{p^2} + E_1, \qquad E_1 = \frac{q^2}{4p^2}\langle z^3\rangle + \frac{q}{2p^2}\langle z\rho_3\rangle - \frac{\langle\rho_3\rangle}{p}.$$

Step 2 Geometry of the cusp

Write $a = a^* + \alpha$ and $\eta = 1 - t/t^*$. For $t \le t^*$,

$$\eta + c\alpha^2 \le J_t \le \eta + C\alpha^2, \qquad c(z^4 + A^2z^2) \le \psi(z) \le C(z^4 + A^2z^2),$$

with $A^2 = \eta + \alpha^2$, so the Gibbs width is $w = \min(\nu^{1/4}, \sqrt{\nu}/A)$ and $\langle|z|^k\rangle \le C_k w^k$.

Step 3 Outer region $A \geq \nu^{1/4}$

There $p \geq cA^2$, $|q| \leq CA$, $|E_1| \leq C\nu^{3/2}/A^5$, and with $\mathrm{d}x = J_t\,\mathrm{d}a$,

$$\int_{\mathcal{O}} |E_1|\,\mathrm{d}x \leq C\nu^{3/2} \int_{A\geq\nu^{1/4}} A^{-3}\,\mathrm{d}\alpha \leq C\nu.$$

The main part is $\nu \int t|u_0''|/J_t\,\mathrm{d}a$. Near a^*, $tu_0'' = \partial_a J_t$, so the integral is a logarithm of J_t between the inner edge ($J_t \asymp \max(\sqrt{\nu}, \eta)$) and $J_t \asymp 1$:

$$\nu \int_{\mathcal{O}} |\kappa_t|\,\mathrm{d}x = 2\nu \log \frac{1}{\max(\sqrt{\nu}, \eta)} + O(\nu).$$

Step 4 Inner region $A < \nu^{1/4}$

Its measure in x is $\leq C\nu^{3/4}$ and there $|u - v| \leq \langle |z| \rangle / t \leq C\nu^{1/4}$: contribution $O(\nu)$. Together with Step 3 this proves the formula for $t \leq t^*$ (the result (OPF5A-T1)).

Step 5 After the cusp: the normal form

For $t = t^* + \tau$ the outer region $|\alpha| \geq K'\sqrt{\tau}$ works as before and gives $2\nu \log(1/\tau) + O(\nu)$, which is also the lower bound. In the inner region rescale

$$a = a^* + \sqrt{\tau}A, \qquad x = X_t(a^*) - t\tau^{3/2}\xi, \qquad \Psi_\tau(\xi, A) = \tau^{-2}\big[\Phi_x(a^* + \sqrt{\tau}A) - \Phi_x(a^*)\big].$$

Then $\Psi_\tau \to \Psi_0 = \xi A + \frac{\kappa_3}{24}A^4 - \frac{A^2}{2t^{*2}}$ in C^4 on compacts: a tilted double well whose global minimisers are nondegenerate. The effective viscosity is $\hat{\nu} = \nu/\tau^2$, and the A27 estimate applied uniformly to this compact family ((OPF5A-L3′)) gives

$$\int_{\text{inner}} |u - v|\,\mathrm{d}x = \tau^2 \int |A_m - \langle A \rangle_\xi|\,\mathrm{d}\xi \leq \tau^2 C\hat{\nu} = C\nu.$$

Step 6 From $u - v$ to E_{int}

As in A27, Step 6, with $\|h - u_0\|_{L^1} \leq 2\pi\nu t\|u_0''\|_\infty + O(\nu^2)$.

Why it works. at the cusp the Lagrangian Jacobian vanishes quadratically, so integrating $|\partial_a J|/J$ produces exactly one logarithm of $1/\nu$, with coefficient one.

■

Uses: A27, [F1], [F2]; research file `work/OP-F5A` (L2, L3′, Lemma U).

4.3 Identification in a dictionary

Dim. A29 **S4ID-T1**

Dictionary criterion [proved]

Let u be smooth on an open Ω and solve $F(u) = 0$ with $F = y_t - N(y, y_x, \dots)$. Let $[F]$ be the differential ideal generated by F, V a linear space of differential polynomials, $I(u) = \{G : G(u) = 0\}$, ψ the normal-form map that eliminates y_t and its derivatives using F (so $\ker\psi = [F]$ and $G - \psi(G) \in [F]$), and $\mathcal{J}_u$ the relations satisfied by u among the remaining jet variables.

(a) $I(u) \cap V = (\psi|_V)^{-1}(\mathcal{J}_u)$.

(b) The sequence

$$0 \to [F] \cap V \to I(u) \cap V \xrightarrow{\psi} \mathcal{J}_u \cap \psi(V) \to 0$$

is exact; for finite-dimensional V, $\dim(I(u) \cap V) = \dim([F] \cap V) + \dim(\mathcal{J}_u \cap \psi(V))$.

(c) $I(u) \cap V = [F] \cap V \iff \mathcal{J}_u \cap \psi(V) = 0$.

Stated in the paper as no. 9.3, p. 55

Idea. On a solution, every differential polynomial equals its normal form, because their difference lies in $[F]$. So a relation of u is a polynomial whose normal form is a relation among the free jet variables. The part that the equation explains ($[F]$) is the kernel; the part that only u explains is the image.

Proof.

Step 1 (a) Reduction to normal forms

For $G \in V$, $G - \psi(G) \in [F]$ vanishes on the solution u, so

$$G(u) = \psi(G)(u).$$

Hence $G \in I(u)$ iff $\psi(G)$ vanishes on u, that is, $\psi(G) \in \mathcal{J}_u$.

Step 2 (b) Exactness

The kernel of ψ on $I(u) \cap V$ is $\ker\psi \cap V = [F] \cap V$. By (a) the image lies in $\mathcal{J}_u \cap \psi(V)$. Conversely, every element of $\mathcal{J}_u \cap \psi(V)$ is $\psi(G)$ with $G \in V$, and $G \in I(u)$ by (a). The dimension formula follows from exactness.

Step 3 (c)

Read off from (b): the middle term equals the left one iff the right one is zero.

Why it works. the normal form splits every relation of u into a part forced by the equation and a part forced only by the particular solution.

■

Uses: none.

The literal statement is false

In a complete dictionary $\mathcal{D}_{d,k}$, $[F] \cap V$ already contains $\partial_x F$ and other consequences, so "$I(u) \cap V = \mathbb{C}F$" fails for every u, except in the special case $(d,k) = (2,n)$ (S4ID-P1). The correct statement is (c).

Dim. A30 **S4ID-T2**

Burgers with N shocks [proved]

Let $F_B = y_t + y_0 y_1 - \nu y_2$, distinct nonzero $k_1, \dots, k_N$, $E_i = \exp(k_i x + \nu k_i^2 t + c_i)$, $\varphi = 1 + \sum_i E_i$ (nonvanishing on Ω), non-resonance (NR): $\sum_i m_i(\nu k_i^2, k_i) \neq 0$ for $m \in \mathbb{Z}^N \setminus \{0\}$, and $u = -2\nu\varphi_x/\varphi$. Let e_m be defined by $z\prod_i(z-k_i) = \sum_{m=1}^{N+1} e_m z^m$, Y_m the complete Bell polynomials, and

$$R_N = \sum_{m=1}^{N+1} e_m\, Y_m\Big(-\frac{y_0}{2\nu}, \dots, -\frac{y_{m-1}}{2\nu}\Big).$$

(a) $\mathcal{J}_u \cap \mathbb{C}[y_0, \dots, y_{N-1}] = 0$.

(b) $\mathcal{J}_u \cap \mathbb{C}[y_0, \dots, y_N] = R_N \cdot \mathbb{C}[y_0, \dots, y_N]$, with $\deg R_N = N+1$.

(c) In the evolutionary dictionary of order $K = N$ and degree d, u identifies Burgers iff $d \leq N$; with the conjecture S4ID-K1, now proved as (OP-T7), the same holds for every $K \geq N$.

Stated in the paper as no. 9.3, p. 55

Idea. By Cole–Hopf, the jet of $w = u/(-2\nu)$ is the sequence of cumulants of a measure with N atoms at the k_i (plus the atom 0). N atoms leave N free parameters, so the first N jet variables are independent, and the moments of $N+1$ atoms satisfy exactly one linear relation, which in cumulants is R_N.

Proof.

Step 1 Reduction to formal variables

By (NR) the monomials E^m have distinct exponents, so they are linearly independent (A2). Hence $X_i \mapsto E_i$ embeds $\mathbb{C}(X)$ into the meromorphic functions on Ω, intertwining $D(X_i) = k_i X_i$ with ∂_x. So $p \in \mathcal{J}_u$ iff p vanishes on the formal jet of $u = -2\nu\, D\varphi/\varphi$, $\varphi = 1 + \sum_i X_i$.

Step 2 Moments

Put $p_i = X_i/\varphi$ and $r_j = D^j\varphi/\varphi = \sum_i k_i^j p_i$. The p_i generate $\mathbb{C}(X)$, so they are algebraically independent. From $Dr_j = r_{j+1} - w\, r_j$ with $w = D\varphi/\varphi$, induction gives

$$r_j = Y_j(w, Dw, \dots), \qquad Y_j = w_{j-1} + (\text{polynomial in } w_0, \dots, w_{j-2}).$$

Step 3 (a) Independence

By the triangular form, $\mathbb{C}[w,\dots,D^{N-1}w] = \mathbb{C}[r_1,\dots,r_N]$. And $(r_1,\dots,r_N)^{\mathsf{T}} = V\,(p_1,\dots,p_N)^{\mathsf{T}}$ with the Vandermonde matrix $V = (k_i^j)$, invertible. So $r_1,\dots,r_N$, hence $w,\dots,D^{N-1}w$, are algebraically independent.

Step 4 (b) The relation

The moments satisfy

$$\sum_{m=1}^{N+1} e_m\, r_m = \sum_i p_i \sum_m e_m k_i^m = \sum_i p_i\, k_i \prod_l (k_i - k_l) = 0.$$

So $R_N \in \mathcal{J}_u$. Only Y_{N+1} contains w_N, with coefficient 1, and $e_{N+1} = 1$: $R_N = -y_N/(2\nu) + \beta(y_0,\dots,y_{N-1})$, of degree $N+1$. For $p \in \mathcal{J}_u \cap \mathbb{C}[y_0,\dots,y_N]$, divide by R_N as a polynomial of degree one in y_N: the remainder lies in $\mathbb{C}[y_0,\dots,y_{N-1}] \cap \mathcal{J}_u = 0$ by (a).

Step 5 (c) The criterion

By A29(c), identification in $V = \mathbb{C}y_t \oplus A_{\le d,\le N}$ holds iff $\mathcal{J}_u \cap A_{\le d,\le N} = 0$. By (b) this is the set of multiples of R_N of degree $\le d$, which is zero iff $d < N+1$.

Why it works. Cole–Hopf turns the N-shock solution into a mixture of N exponentials, and N exponentials satisfy exactly one moment relation of order N.

■

Uses: A2, A29; (OP-T7) for $K > N$ (Part II).

Dim. A31 **OPF5C-T2**

KdV N-solitons: degree $(N+2)2^{N-1}$ [proved mod. Novikov 1974]

For the N-soliton of KdV with generic distinct wave numbers ($k \in U_0$), the relations among the jet variables $y_0,\dots,y_N$ form a principal ideal (Π_N). Normalising Π_N monic in y_N:

$$\deg_{y_N} \Pi_N = 2^N, \qquad \deg \Pi_N = (N+2)\,2^{N-1}.$$

($N = 1,2,3$: degrees 3, 8, 20.) Consequently the evolutionary dictionary of order N and degree d identifies KdV iff $d < (N+2)2^{N-1}$.

Stated in the paper as no. 9.3, p. 55

Idea. The jets of N-solitons lie on a variety cut out by Novikov's stationary equations, written in Mumford coordinates. A weighted Bézout argument shows that this variety is a free module of rank 2^N over the first N jet variables: y_N takes 2^N values over each point, one for each choice of signs $k_i \mapsto \pm k_i$. The minimal polynomial of y_N is therefore of degree 2^N in y_N, and its total degree is read from the weights.

Proof.

Step 1 Principal ideal

As in A30, Step 1, reduce to formal variables. The first N jet variables are independent (Vandermonde in $2k_i^2k_i^j$), and $y_0,\dots,y_N$ are $N+1$ elements of a field of transcendence degree N. So the ideal of relations in $\mathbb{C}[y_0,\dots,y_N]$ is a nonzero prime of height one, hence principal, (Π_N) (S4ID-T3(b)).

Step 2 Novikov's variety (Nov′)

In Mumford coordinates (U,V), with $f_k(\lambda)=\lambda\prod_i(\lambda+\kappa_i)^2$, the jets of the N-soliton satisfy

$$U \mid V^2 + f_k$$

(Novikov 1974) [from cited source]. The change of variables from jets $(y_0,\dots,y_{2N-1})$ to Mumford coordinates is a weighted triangular automorphism with weights $\operatorname{wt} y_j = j+2$ (OPF5C-L1).

Step 3 A free module of rank 2^N

The N equations of the variety are weighted homogeneous of weights $4N+2-2m$. Their only common zero with $k=0$ and $y_0=\cdots=y_{N-1}=0$ is the origin (OPF5C-L3). By the weighted Bézout theorem (Hironaka's criterion) the coordinate ring is free over $\mathbb{C}[k,y_0,\dots,y_{N-1}]$ of rank

$$\frac{\prod_{m=0}^{N-1}(4N+2-2m)}{\prod_{j=N}^{2N-1}(j+2)} = \frac{\prod_{i=N+2}^{2N+1}2i}{\prod_{i=N+2}^{2N+1}i} = 2^N.$$

Step 4 The 2^N branches are distinct

Replacing k_i by $\pm k_i$ does not change f_k, so the 2^N sign choices give 2^N local sections of the variety near the zero potential. Their values of y_N have distinct linear parts (the multisets $\{\pm k_i\}$ differ), so the characteristic polynomial χ_k of multiplication by y_N is squarefree (OPF5C-L5). Hence $\Pi_N=\chi_k$, of degree 2^N in y_N.

Step 5 Total degree from the weights

Multiplication by y_N has weight $N+2$, so the coefficient of z^{2^N-i} in the universal χ has weight $i(N+2)$, and every monomial has weight at most $W=(N+2)2^N$. Since $\operatorname{wt} y_j\ge 2$, the degree is at most $W/2$. The monomial $y_0^{W/2}$ appears with a nonzero constant coefficient, because no point of the rational fibre has jet $(c,0,\dots,0)$ with $c\neq 0$ (OPF5C-L4). So $\deg\Pi_N = W/2 = (N+2)2^{N-1}$.

Step 6 The criterion

As in A30, Step 5, using A29(c) and Step 1.

Why it works. each soliton can be read with either sign of its wave number, which gives 2^N branches over the jet of order $N-1$, and the weighted grading turns that count into the total degree.

■

Uses: A29, A30 (Step 1), Novikov (Nov′), Blum-Smith Cor. 2.4.19 (Hironaka criterion); research file `work/OP-F5C`, §1.

Part II

Phases 6–10: the load-bearing proofs

CHAPTER 5

Systems and gauge

In this chapter

Starting from	$R = \mathbb{C}[\zeta_1, \dots, \zeta_n]$ with $\zeta_j \leftrightarrow \partial_j$; $P \in R^{l\times k}$ acts on the left on $u \in \mathcal{A}^k$, with $\mathcal{A} = C^\infty(\Omega)$ or $\mathcal{D}'(\Omega)$ and $\Omega \subset \mathbb{R}^n$ open, connected, non-empty.
Why	to prove when a vector solution u determines a matrix operator P, when a sum of kernels is the kernel of a least common left multiple, and how gauge changes what a single field can reveal.
Role	Dim. A32–A35. They prove Chapter 11 of the paper.

Notation for the whole chapter

$U_P = R^{1\times l}P \subseteq R^{1\times k}$ is the row module and $M_P = R^{1\times k}/U_P$ the module of the system, so $\mathrm{Sol}_{\mathcal{A}}(P) \cong \mathrm{Hom}_R(M_P, \mathcal{A})$. For $u \in \mathcal{A}^k$, $\mathrm{Ann}(u) = \{r \in R^{1\times k} : r(\partial)u = 0\}$ and $M_u = R^{1\times k}/\mathrm{Ann}(u) \cong Ru \subseteq \mathcal{A}$. We say u *identifies* P if $\mathrm{Ann}(u) = U_P$, i.e. if $M_P \to \mathcal{A}$, $[r] \mapsto r(\partial)u$, is injective. The *socle* of M at a maximal ideal $\mathfrak{m}$ is $\mathrm{soc}_{\mathfrak{m}}(M) = \mathrm{Hom}_R(R/\mathfrak{m}, M)$. $\mathrm{Ass}(M)$ is the set of associated primes.

5.1 Existence of an identifying solution

Dim. A32 — **SYS-EX**

Existence of an identifying solution: socles of dimension at most one [proved]

Let $\Omega \subseteq \mathbb{R}^n$ be a non-empty open connected set and M a finitely generated R-module. The following are equivalent:

(i) there is an injective homomorphism $M \hookrightarrow \mathcal{A}$;

(ii) $\dim_{\mathbb{C}} \mathrm{soc}_{\mathfrak{m}}(M) \leq 1$ for every maximal ideal $\mathfrak{m}$ of R.

In particular, for $P \in R^{l\times k}$: *there exists* $u \in \mathrm{Sol}_{\mathcal{A}}(P)$ *that identifies* P *if and only if* $\dim \mathrm{Hom}_R(R/\mathfrak{m}, M_P) \leq 1$ *for every maximal* $\mathfrak{m}$. The solution can be taken in $C^\infty(\Omega)$.

Status of novelty. For $\Omega = \mathbb{R}^n$ this is known (Oberst; Pommaret; Macaulay). The only new point is that Ω is arbitrary, also not P-convex (SYS-EX, Ω arbitrary; see A35).

Stated in the paper as no. 11.2, p. 65

Idea. A socle of dimension ≥ 2 at $\mathfrak{m}_\lambda$ would have to embed in the one-dimensional space $\mathbb{C}e^{\lambda\cdot x}$: that is the obstruction. Conversely, one builds explicit entire solutions, one

> family of integrals of exponentials over small real patches of each prime variety, and extends from an essential submodule using that $C^\infty(\mathbb{R}^n)$ is injective.

Proof.

Step 1 (i)⇒(ii): the socle obstruction (SYS-N1)

A homomorphism φ maps $\operatorname{soc}_{\mathfrak{m}_\lambda}(M)$ into $\{f \in \mathcal{A} : (\partial_j - \lambda_j)f = 0\ \forall j\}$. On a connected Ω this space is $\mathbb{C}e^{\lambda\cdot x}$: writing $f = e^{\lambda\cdot x}g$ gives $\nabla g = 0$ in $\mathcal{D}'(\Omega)$, so g is constant. A linear map from a space of dimension ≥ 2 to a line is not injective.

Step 2 Two standard lemmas

SYS-L0. Let $\varphi : M \to F$ be a homomorphism and $E \subseteq M$ essential (every nonzero submodule meets E nontrivially). Then φ is injective iff $\varphi|_E$ is: $\ker\varphi \cap E = \ker\varphi|_E = 0$ forces $\ker\varphi = 0$.

SYS-L1. A finitely generated M has a finite family $x_{\mathfrak{p},j}$, $\mathfrak{p} \in \operatorname{Ass}(M)$, $j \leq \mu_\mathfrak{p}$, with $Rx_{\mathfrak{p},j} \cong R/\mathfrak{p}$, whose sum $E = \bigoplus Rx_{\mathfrak{p},j} \cong \bigoplus_\mathfrak{p}(R/\mathfrak{p})^{\mu_\mathfrak{p}}$ is direct and essential. For $\mathfrak{p} = \mathfrak{m}$ maximal, $\mu_\mathfrak{m} \leq \dim\operatorname{soc}_\mathfrak{m}(M)$.

Step 3 (ii)⇒(i): entire solutions on each prime variety (SYS-EX-1)

Two facts are needed. (EX-L1) If ν is a complex measure of compact support in $\mathbb{C}^n$ and $\Lambda\nu(x) = \int e^{\zeta\cdot x}\,\mathrm{d}\nu(\zeta)$ vanishes on a non-empty open Ω, then $\int h\,\mathrm{d}\nu = 0$ for every polynomial h: $\Lambda\nu$ is entire, hence zero, and its Taylor coefficients at 0 are the moments $\int \zeta^\alpha\,\mathrm{d}\nu$. (EX-L2) For a prime $\mathfrak{q}$ with $d = \dim V(\mathfrak{q}) \geq 1$ there are linear forms ℓ, a connected open $U \subseteq \mathbb{C}^d$ meeting $\mathbb{R}^d$ and a holomorphic section $s : U \to V(\mathfrak{q})$ with $\ell \circ s = \mathrm{id}$, such that a polynomial vanishing on $s(B)$ for a real ball $B \subset U \cap \mathbb{R}^d$ lies in $\mathfrak{q}$. (By Noether normalisation, $\ell|_{V(\mathfrak{q})}$ is finite and surjective; it is an unramified covering away from a proper Zariski-closed Δ, which contains no real open set.)

For every non-maximal $\mathfrak{q} \in \operatorname{Ass}(M)$ choose disjoint closed real balls $B_{\mathfrak{q},1}, \dots, B_{\mathfrak{q},\mu_\mathfrak{q}} \subset U \cap \mathbb{R}^d$ and set

$$f_{\mathfrak{q},j}(x) = \int_{B_{\mathfrak{q},j}} e^{s(y)\cdot x}\,\mathrm{d}y.$$

For $r \in \mathfrak{q}$, $r(\partial)f_{\mathfrak{q},j} = \int r(s(y))e^{s(y)\cdot x}\,\mathrm{d}y = 0$. For maximal $\mathfrak{m}_\lambda$ put $f_\lambda = e^{\lambda\cdot x}$ (here $\mu = 1$ by (ii)). For the heat equation, $\mathfrak{q} = (\tau - \xi^2)$, $s(y) = (y^2, y)$ and $f(t,x) = \int_0^1 e^{y^2 t + yx}\,\mathrm{d}y$ solves $f_t = f_{xx}$.

Step 4 The map $\psi : E \to C^\infty(\Omega)$ is injective

Suppose $\ker\psi \neq 0$. It has an associated prime $\mathfrak{q} \in \operatorname{Ass}(E) = \operatorname{Ass}(M)$ and contains $z = \sum c_{\mathfrak{p},j}x_{\mathfrak{p},j}$ with $\operatorname{Ann}(z) = \mathfrak{q}$. Since each $R/\mathfrak{p}$ is a domain, $\operatorname{Ann}(z) = \bigcap_{c_{\mathfrak{p},j}\neq 0}\mathfrak{p}$; a finite intersection of primes equal to a prime $\mathfrak{q}$ contains their product, hence contains one of them, so $\mathfrak{q}$ itself occurs and all other primes present contain $\mathfrak{q}$. This gives a relation on Ω

$$\sum_j r_{\mathfrak{q},j}(\partial)f_{\mathfrak{q},j} + \sum_{\mathfrak{p}\supsetneq\mathfrak{q}}\sum_j r_{\mathfrak{p},j}(\partial)f_{\mathfrak{p},j} = 0, \qquad \text{some } r_{\mathfrak{q},j} \notin \mathfrak{q}.$$

If $\mathfrak{q} = \mathfrak{m}_\lambda$ is maximal, only one term remains: $r(\lambda)e^{\lambda\cdot x} = 0$, so $r \in \mathfrak{m}_\lambda$, a contradiction. If $\mathfrak{q}$ is not maximal, the left side is $\Lambda\nu$ for $\nu = \nu_\mathfrak{q} + \nu'$ with ν' supported on $\bigcup_{\mathfrak{p}\supsetneq\mathfrak{q}} V(\mathfrak{p})$. By EX-L1,

$\int h\,d\nu = 0$ for all polynomials h. Pick $h_0 \in \prod_{\mathfrak{p}\supsetneq\mathfrak{q}} \mathfrak{p}$ with $h_0 \notin \mathfrak{q}$ (possible since $\mathfrak{q}$ is prime); then $h_0\nu' = 0$. Taking $h = h_0 \cdot g(\ell)$ with g an arbitrary polynomial gives

$$\sum_j \int_{B_{\mathfrak{q},j}} g(y)F_j(y)\,dy = 0, \qquad F_j = (h_0 \circ s)(r_{\mathfrak{q},j} \circ s).$$

By Stone–Weierstrass the polynomials are dense in $C(\bigcup_j B_{\mathfrak{q},j})$, so every $F_j \equiv 0$. By EX-L2, $h_0 \circ s$ is real-analytic and not identically zero on $B_{\mathfrak{q},j}$, so its zero set is null; hence $r_{\mathfrak{q},j} \circ s = 0$ on a dense set, hence everywhere, and EX-L2 gives $r_{\mathfrak{q},j} \in \mathfrak{q}$: a contradiction.

Step 5 Extension from E to M

$C^\infty(\mathbb{R}^n)$ is an injective R-module (Ehrenpreis–Malgrange–Palamodov; cited, [from cited source]). So ψ extends to $\Psi : M \to C^\infty(\mathbb{R}^n)$, and $\rho_\Omega \circ \Psi$ is injective by SYS-L0, because its kernel meets E in $\ker\psi = 0$. Taking $M = M_P$ and $u = (\varphi(e_1), \dots, \varphi(e_k))$ gives $r(\partial)u = \varphi([r])$, hence $\mathrm{Ann}(u) = U_P$.

Why it works. the only thing a single function can do to a socle is send it to a line; and entire solutions built from different real patches of a prime variety are as independent as the polynomials, which is exactly what injectivity needs.

■

Uses: SYS-L0, SYS-L1, EX-L1, EX-L2; Malgrange (injectivity of $C^\infty(\mathbb{R}^n)$, [from cited source]).

Example 5.1 An ODE system

For $n = 1$ and $P = \mathrm{diag}(s-1, s-1)$ the socle at $(s-1)$ has dimension 2: no solution ever identifies P. For $P = \begin{pmatrix} s-1 & 1 \\ 0 & s-2 \end{pmatrix}$ the socles have dimension 1, and $u = (ae^x - be^{2x},\ be^{2x})$ with $ab \neq 0$ identifies it. In general (SYS-T0, known) an ODE system of rank r with invariant factors $e_1 \mid \cdots \mid e_r$ has an identifying solution iff $e_1 = \cdots = e_{r-1} = 1$.

Dim. A33 **SYS-C1**

Square non-singular systems in $n \geq 2$ variables

[proved]

Let $n \geq 2$ and $P \in R^{k\times k}$ with $\det P \neq 0$. Then $\mathrm{soc}_{\mathfrak{m}}(M_P) = 0$ for every maximal $\mathfrak{m}$, so on every open connected Ω there is $u \in \mathrm{Sol}(P)$ that identifies P. Moreover $\mathrm{Ass}(M_P)$ is the set of prime factors of $\det P$.

Stated in the paper as no. 11.2, p. 65

Idea. Because $\det P \neq 0$, the module M_P has projective dimension 1 and $\mathfrak{m}$ has grade $n \geq 2$, so there is no room for a socle.

Proof.

Step 1 An exact sequence

Right multiplication by P gives the exact sequence $0 \to R^{1\times k} \to R^{1\times k} \to M_P \to 0$ (P is injective since $\det P \neq 0$). Apply $\mathrm{Hom}_R(R/\mathfrak{m}, -)$:

$$\mathrm{Hom}(R/\mathfrak{m}, R^k) \to \mathrm{Hom}(R/\mathfrak{m}, M_P) \to \mathrm{Ext}^1_R(R/\mathfrak{m}, R)^k.$$

The left term is 0 (R is a domain). The right term is 0 because $\mathrm{grade}(\mathfrak{m}, R) = n \geq 2$ ($\mathfrak{m}$ is generated by the regular sequence $\zeta_j - \lambda_j$), so $\mathrm{Ext}^i(R/\mathfrak{m}, R) = 0$ for $i < n$. Hence $\mathrm{soc}_{\mathfrak{m}}(M_P) = 0$ and A32 applies.

Step 2 The associated primes

($\subseteq$) $\mathrm{pd}\, M_P \leq 1$, so by Auslander–Buchsbaum $\mathrm{depth}\, M_{\mathfrak{p}} \geq \mathrm{ht}\,\mathfrak{p} - 1$. If $\mathfrak{p} \in \mathrm{Ass}(M_P)$, the depth is 0, so $\mathrm{ht}\,\mathfrak{p} \leq 1$; $\mathfrak{p} \neq 0$ since M_P is torsion, and $\mathfrak{p} \supseteq \mathrm{Ann}(M_P) \ni \det P$. So $\mathfrak{p}$ is a prime factor of $\det P$.

($\supseteq$) $\mathrm{Supp}\, M_P = V(\mathrm{Fitt}_0) = V(\det P)$. A prime factor (f) of $\det P$ is minimal in the support (the only smaller prime is $0 \notin \mathrm{Supp}$), and minimal primes of the support are associated.

Why it works. a non-singular square system is a torsion module of grade one; the socle sits where the depth is zero, and in dimension ≥ 2 no maximal ideal is such a point.

■

Uses: A32.

> **Example 5.2 Dirac 3+1, Klein–Gordon, Lamé**
>
> These systems have $\det P \neq 0$ and $n = 4$ (or 3 for the Lamé system in space), so each has an identifying field, even when a prime of $\mathrm{Ass}(M_P)$ has multiplicity 2. Multiplicity on a non-maximal prime never prevents existence; it only constrains which u works, through an independence test over $R/\mathfrak{p}$.

5.2 Sums of kernels

Dim. A34 **SYS-S1**

Sum of kernels and the least common left multiple [proved]

Let $Q_1 \in R^{a\times k}$, $Q_2 \in R^{b\times k}$ with $U_P \subseteq U_{Q_i}$, put $U_i = U_{Q_i}$ and $N = R^{1\times k}/(U_1 + U_2)$.

(a) On every non-empty open Ω: $\mathrm{Sol}(Q_1) + \mathrm{Sol}(Q_2) = \mathrm{Sol}(P) \;\Rightarrow\; U_1 \cap U_2 = U_P$.

(b) If $U_1 \cap U_2 = U_P$ and $\mathrm{Ext}^1_R(N, \mathcal{A}) = 0$, then $\mathrm{Sol}(Q_1) + \mathrm{Sol}(Q_2) = \mathrm{Sol}(P)$. This holds when Ω is convex, and when N has finite length and $H^1_{dR}(\Omega) = 0$ (for example Ω simply connected).

(c) $\mathrm{Sol}(Q_1) \cap \mathrm{Sol}(Q_2) = \mathrm{Sol}(N)$; for connected Ω it is finite-dimensional iff $\ell(N) < \infty$, and then it consists of exponential-polynomials and has dimension $\delta_{\mathrm{sys}} = \ell(N)$.

For $k = 1$, $n = 2$ and coprime Q_1, Q_2 one recovers C12-T1′ (A13), with $\delta_{\mathrm{sys}} = \delta(P)$.

Stated in the paper as no. 11.3, p. 65

Idea. The two statements (a) and (b) are the two halves of the exactness of $0 \to \mathrm{Sol}(N) \to \mathrm{Sol}(Q_1) \oplus \mathrm{Sol}(Q_2) \to \mathrm{Sol}(P) \to \mathrm{Ext}^1(N, \mathcal{A})$. The obstruction to surjectivity is the connecting map into Ext^1, which de Rham cohomology kills when N has finite length.

Proof.

Step 1 (a) A common multiple not in U_P gives a separating solution

Let $r \in (U_1 \cap U_2) \setminus U_P$. Then $r(\partial)$ kills $\mathrm{Sol}(Q_1) + \mathrm{Sol}(Q_2)$. Since $0 \neq [r] \in M_P$ and $R[r] \cong R/\operatorname{Ann}([r])$, choose a maximal $\mathfrak{m}_\lambda \supseteq \operatorname{Ann}([r])$ and the map $R[r] \to R/\mathfrak{m}_\lambda \to C^\infty(\mathbb{R}^n)$, $[r] \mapsto e^{\lambda\cdot x}$. Extend it to $M_P \to C^\infty(\mathbb{R}^n)$ (injectivity, as in A32 Step 5) and restrict to Ω: we get $v \in \mathrm{Sol}_\Omega(P)$ with $r(\partial)v = e^{\lambda\cdot x} \neq 0$, so $v \notin \mathrm{Sol}(Q_1) + \mathrm{Sol}(Q_2)$.

Step 2 (b) The exact sequences

If $U_1 \cap U_2 = U_P$, the sequence $0 \to M_P \to M_{Q_1} \oplus M_{Q_2} \to N \to 0$, $[r] \mapsto ([r], [r])$ and $([a], [b]) \mapsto [a - b]$, is exact: in the middle, $a - b = c_1 + c_2$ gives $([a], [b]) = ([r], [r])$ for $r = a - c_1$; on the left the kernel is $(U_1 \cap U_2)/U_P = 0$. Applying $\mathrm{Hom}_R(-, \mathcal{A})$,

$$0 \to \mathrm{Sol}(N) \to \mathrm{Sol}(Q_1) \oplus \mathrm{Sol}(Q_2) \to \mathrm{Sol}(P) \to \mathrm{Ext}^1_R(N, \mathcal{A}).$$

So if $\mathrm{Ext}^1(N, \mathcal{A}) = 0$ the sum map is onto. On a convex Ω, $\mathcal{A}$ is injective and Ext vanishes.

Step 3 (b) Finite length and de Rham cohomology

The Koszul resolution of $R/\mathfrak{m}_\lambda$ gives $\mathrm{Ext}^p_R(R/\mathfrak{m}_\lambda, \mathcal{A}) = H^p$ of the Koszul complex of $(\partial_j - \lambda_j)$ on $\mathcal{A}$. Conjugating by $e^{-\lambda\cdot x}$ turns it into the Koszul complex of $(\partial_1, \dots, \partial_n)$, i.e. the de Rham complex of Ω (with currents when $\mathcal{A} = \mathcal{D}'$; de Rham's theorem identifies the cohomologies). So $\mathrm{Ext}^1(R/\mathfrak{m}_\lambda, \mathcal{A}) \cong H^1_{dR}(\Omega; \mathbb{C}) = 0$. N has a composition series with factors $R/\mathfrak{m}$, and for $0 \to N' \to N \to R/\mathfrak{m} \to 0$ the long exact sequence contains $\mathrm{Ext}^1(R/\mathfrak{m}, \mathcal{A}) \to \mathrm{Ext}^1(N, \mathcal{A}) \to \mathrm{Ext}^1(N', \mathcal{A})$; by induction on the length, $\mathrm{Ext}^1(N, \mathcal{A}) = 0$.

Step 4 (c) Intersection and ambiguity

u is killed by U_1 and U_2 iff it is killed by $U_1 + U_2$, so $\mathrm{Sol}(Q_1) \cap \mathrm{Sol}(Q_2) = \mathrm{Sol}(N)$. If $\ell(N) < \infty$, induction with $\dim \mathrm{Hom}(R/\mathfrak{m}, \mathcal{A}) = 1$ (A32 Step 1) gives $\dim \mathrm{Sol}(N) \le \ell(N)$; and since $\mathrm{Ann}(N) \supseteq \prod \mathfrak{m}_i^{e_i}$ has finite colength, $\mathrm{Sol}(N) \subset \mathrm{EP}$ (A2). Equality holds because the $\mathfrak{m}_\lambda$-torsion submodule of $\mathcal{A}$ is $e^{\lambda\cdot x}\mathbb{C}[x] \cong E_R(R/\mathfrak{m}_\lambda)$ (Macaulay inverse systems) and Matlis duality gives $\dim \mathrm{Hom}(N_\lambda, E(R/\mathfrak{m}_\lambda)) = \ell(N_\lambda)$. Conversely, if $\ell(N) = \infty$, $\mathrm{Ass}(N)$ contains a non-maximal $\mathfrak{p}$ and $R/\mathfrak{p} \hookrightarrow N$; the entire functions $f_g = \int_B e^{s(y)\cdot x} g(y)\,\mathrm{d}y$ (g a polynomial) are linearly independent solutions (EX-L1 and Stone–Weierstrass), and they extend to N by injectivity of $C^\infty(\mathbb{R}^n)$. So $\dim \mathrm{Sol}(N) = \infty$.

Why it works. kernels add up exactly when the connecting map into Ext^1 vanishes; for finite-length quotients that Ext is a copy of H^1_{dR}, which is why simple connectivity, and nothing more, is needed there.

■

Uses: A2, A13, A32.

Counterexample 5.3 Simple connectivity is not enough when N has infinite length

[false] **SYS-S1-C** (counterexample). There exist a simply connected open $\Omega \subset \mathbb{R}^2$ and $Q_1, Q_2 \in R^{1\times 2}$ with $U_1 \cap U_2 = U_P$ but $\mathrm{Sol}(Q_1) + \mathrm{Sol}(Q_2) \subsetneq \mathrm{Sol}(P)$, in $C^\infty(\Omega)$ and in $\mathcal{D}'(\Omega)$.

Construction in C^∞.

Coordinates (t, x). Let $\Omega = (0,3)\times(0,3) \setminus ([1,3)\times[1,2])$, a "U" domain (simply connected: it retracts onto the base $(0,1)\times(0,3)$). Take $k = 2$, $Q_1 = (1, 0)$, $Q_2 = (1, \partial_x)$. Then $U_1 \cap U_2 = 0 = U_P$ and $\mathrm{Sol}(P) = \mathcal{A}^2$, while $U_1 + U_2 = R(1,0) + R(0,\xi)$, so $N \cong R/(\xi)$ has infinite length. Moreover

$$\mathrm{Sol}(Q_1) + \mathrm{Sol}(Q_2) = \{(v_1, v_2) : \ v_1 \in \partial_x C^\infty(\Omega)\}.$$

∂_x is not onto $C^\infty(\Omega)$: fix $\rho \in C_c^\infty((1,2))$ with $\int \rho = 1$ and $\beta \in C^\infty(\mathbb{R})$, $\beta = 0$ on $(-\infty, \frac{1}{2}]$, $\beta = 1$ on $[\frac{3}{4}, \infty)$, and put $f(t,x) = \beta(t)\rho(x)/(1-t)$ for $t < 1$, $f = 0$ elsewhere; $f \in C^\infty(\Omega)$. If $\partial_x h = f$, then for $t \in (\frac{3}{4}, 1)$

$$h(t, \tfrac{5}{2}) - h(t, \tfrac{1}{2}) = \int_{1/2}^{5/2} f(t,x)\,\mathrm{d}x = \frac{1}{1-t} \to \infty \quad (t \to 1^-),$$

but h is continuous at $(1, \frac{1}{2})$ and $(1, \frac{5}{2}) \in \Omega$: contradiction. Hence $(-f, 0) \in \mathrm{Sol}(P) \setminus (\mathrm{Sol}(Q_1) + \mathrm{Sol}(Q_2))$.

Variant with a genuine equation: $k = 3$, $Q_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & \tau \end{pmatrix}$, $Q_2 = \begin{pmatrix} 1 & \xi & 0 \\ 0 & 0 & \tau \end{pmatrix}$, $P = (0, 0, \tau)$.

The case $\mathcal{D}'(\Omega)$. With the f above, $\partial_x h = f$ is solvable in $\mathcal{D}'$. Use instead $f = \beta(t)\rho(x)e^{1/(1-t)}$ for $t < 1$. On $A = (0,1)\times(0,3)$, $B = (0,3)\times(0,1)$, $C = (0,3)\times(2,3)$ the distribution solutions of $\partial_x(\cdot) = 0$ are $k(t) \otimes 1$, so $h = F + k_A$ on A (with $F = \int_{1/2}^x f$), $h = k_B$ on B, $h = k_C$ on C; on $A \cap B$, $F = 0$, and on $A \cap C$, $F = \beta e^{1/(1-t)}$. Hence $k_C - k_B = \beta e^{1/(1-t)}$ on $(0,1)$ with $k_C - k_B \in \mathcal{D}'((0,3))$. But $e^{1/(1-t)}$ extends to no distribution near $t = 1$: testing against

$\varphi_\varepsilon \geq 0$ supported in $[1-2\varepsilon, 1-\varepsilon]$ gives $\geq c\varepsilon e^{1/(2\varepsilon)}$, whereas an extension of order N would give $O(\varepsilon^{-N})$.

■

Why it works. the obstruction is not topological: it is the failure of Ω to be P-convex for ∂_x. Simple connectivity suffices only when the quotient N is a finite-length module.

5.3 Hamiltonian Dirac systems

Dim. A35 **H-DIRAC**

Dirac coefficients are read exactly from an identifying field [proved]

Let $\mathcal{C} = \{L = I_4\partial_t + \sum_{k=1}^{3} \alpha^k \partial_k + B : \alpha^k, B \in \mathrm{Mat}_4(\mathbb{C})\}$ (no Clifford relations assumed), $\Omega \subset \mathbb{R}^{1+3}$ open connected non-empty.

(a) If $\psi \in \mathrm{Sol}(L_0)$ identifies $L_0 \in \mathcal{C}$ and $L \in \mathcal{C}$ satisfies $L\psi = 0$, then $L = L_0$.

(b) For every $L_0 \in \mathcal{C}$ there is an identifying $\psi \in C^\infty(\Omega)^4$.

(c) For massive Dirac in $3+1$, in chiral form $D = \begin{pmatrix} -m & X \\ Y & -m \end{pmatrix}$, $M_D \cong (R/\mathfrak{q})^{1\times 2}$ with $\mathfrak{q} = (KG - m^2)$ prime and $KG = p_0^2 - |p|^2$; ψ identifies D iff (ψ_3, ψ_4) are $R/\mathfrak{q}$-independent. In $1+1$, $M_{D_2} \cong R/\mathfrak{q}$ and ψ identifies D_2 iff $\psi_2 \notin \mathrm{EP}$.

Stated in the paper as no. 11.4, p. 66

Idea. The coefficient of ∂_t is normalised to I, so any other first-order operator killing ψ differs from L_0 by a left multiple WL_0 with W of lower degree; comparing the top-degree part forces $W = 0$ because $\tau I + \alpha \cdot \xi$ is not a zero divisor.

Proof.

Step 1 (a) $L = VL_0$ and $W = V - I$

The rows of L lie in $\mathrm{Ann}(\psi) = R^{1\times 4}L_0$, so $L = VL_0$ with $V \in R^{4\times 4}$. Put $W = V - I$, so that $L - L_0 = WL_0$. The left side contains no τ and has degree ≤ 1.

Step 2 The leading part cannot cancel

Suppose $W \neq 0$ and let $W_d \neq 0$ be its top homogeneous component, of degree d. The degree-$(d+1)$ part of WL_0 is $W_d H$ with $H = \tau I + \sum \alpha_0^k \xi_k$. Since $\det H = \tau^4 + \cdots \neq 0$, $W_d H = 0$ would give $W_d \det H = W_d H \,\mathrm{adj} H = 0$, hence $W_d = 0$. If $d \geq 1$ the degree $d+1 \geq 2$ contradicts "degree ≤ 1"; if $d = 0$, W is constant and WL_0 has τ-coefficient $W \neq 0$, contradicting the absence of τ in $L - L_0$. So $W = 0$ and $L = L_0$.

Step 3 (b) Existence

$\det L_0$ is monic of degree 4 in τ, so non-zero, and A33 ($n = 4$) gives an identifying ψ.

Step 4 (c) The chiral form

In M_D the rows of D vanish. The first two give $e_{\text{top}} = \frac{1}{m} X e_{\text{bot}}$; substituting in the last two gives $\frac{1}{m}(YX - m^2)e_{\text{bot}} = \frac{KG-m^2}{m} e_{\text{bot}} = 0$. So $M_D \cong R^{1\times 2}/R^{1\times 2}(KG - m^2)I_2 = (R/\mathfrak{q})^2$, free over $R/\mathfrak{q}$ with basis e_3, e_4; SYS-L0 on $E = M_D$ gives the criterion. In $1+1$, $M_{D_2} \cong R/\mathfrak{q}$ is cyclic, and in $\mathbb{C}[\tau, \xi]$ every ideal strictly containing the height-one prime $\mathfrak{q}$ has finite colength, so $\mathrm{Ann}(\psi_2) \supsetneq \mathfrak{q}$ iff $\psi_2 \in \mathrm{EP}$ (A2).

Why it works. normalising the time derivative removes the left-GL ambiguity $D \sim VD$ of the covariant form $\gamma^\mu \partial_\mu$; what is left is rigid.

■

Uses: A2, A32, A33.

Scope

The criterion is an independence statement, not a test: for a given ψ it cannot be certified from finitely many data (A7). The statement needs Ω connected: on $B_1 \sqcup B_2$ take $\psi_2 = e^{\lambda \cdot x}$ on B_1 and 0 on B_2.

CHAPTER 6

Variable coefficients

In this chapter	
Starting from	a rectangle $\Omega = I \times J \subset \mathbb{R}^2$; the free Schrödinger operator $L_0 = i\partial_t + \partial_x^2$; potentials $V(x)$ or $V(t,x)$; solutions u in the class (SF): $u \in C^{1,2}(\Omega)$ and $L_0 u = Vu$ almost everywhere. $\mathcal{K}$ is an algebra of functions closed under ∂_t, ∂_x.
Why	to prove what a single solution reveals about a variable coefficient: that for a huge coefficient class nothing is revealed, that for the class $V(x)$ every non-zero solution reveals V, that for $V(t,x)$ the data can fall silent, and how stable the reconstruction is.
Role	Dim. A36–A40. They prove Chapter 12 of the paper.

6.1 Structured identification

Dim. A36 — **VAR-N1**

Rank-one annihilator for large classes (VAR-N1); identification is injectivity of Φ_u (VAR-T1(a)) [proved]

(VAR-N1) Let $\Omega \subset \mathbb{R}^2$ be open, $\mathcal{K}$ a unital commutative algebra of complex functions on Ω closed under ∂_t, ∂_x, and $\mathcal{K}[\partial] = \mathcal{K}\langle \partial_t, \partial_x \rangle$. Let $u \in C^\infty(\Omega)$ have no zeros, with $a = u_t/u$ and $b = u_x/u$ in $\mathcal{K}$. Then

$$\mathrm{Ann}_{\mathcal{K}[\partial]}(u) = \mathcal{K}[\partial](\partial_t - a) + \mathcal{K}[\partial](\partial_x - b), \qquad \mathcal{K}[\partial]/J \cong \mathcal{K} \cdot u,$$

the isomorphism being the residue map $\rho_u(Q) = (Qu)/u$.

(VAR-T1(a)) Let $L_\theta = L_0 + \sum_k \theta_k T_k$ be an affine family and $\Phi_u(\theta) = \sum_k \theta_k (T_k u)$. The set $\{\theta : L_\theta u = 0\}$ is empty or equal to $\theta_0 + \ker \Phi_u$; so θ is identified by u iff $\ker \Phi_u = 0$.

Stated in the paper as no. 12.1, p. 68

Idea. Every derivative of u is u times an element of $\mathcal{K}$, and the elements $\partial_t - a, \partial_x - b$ already reduce each derivative to its coefficient. So nothing but u itself is contained in the module $\mathcal{K}[\partial]u$.

Proof.

Step 1 **The coefficients $r_\alpha = \partial^\alpha u/u$ lie in $\mathcal{K}$**

By induction on $|\alpha|$: $r_0 = 1$ and $r_{\alpha+e_x} = \partial_x(r_\alpha u)/u = (r_\alpha)_x + r_\alpha b$, similarly in t with a; $\mathcal{K}$ is closed under derivation and product.

Step 2 $\partial^\alpha - r_\alpha \in J$

True for $\alpha = 0$. If $\partial^\beta - r_\beta \in J$ and $\alpha = \beta + e_x$, then

$$\partial^\alpha = \partial_x(\partial^\beta - r_\beta) + \partial_x r_\beta, \qquad \partial_x r_\beta = r_\beta(\partial_x - b) + r_\beta b + (r_\beta)_x \equiv r_\alpha \pmod{J}.$$

So $Q = \sum q_\alpha \partial^\alpha \equiv \sum q_\alpha r_\alpha = \rho_u(Q) \pmod{J}$. If $Qu = 0$ then $\rho_u(Q) = 0$, so $Q \in J$; conversely $J \subseteq \operatorname{Ann}(u)$ since $(\partial_t - a)u = (\partial_x - b)u = 0$.

Step 3 **The isomorphism and part (a)**

$Q \mapsto Qu = \rho_u(Q)u$ is onto $\mathcal{K}u$ (because $\rho_u(f) = f$) with kernel J. For (a): $L_\theta u - L_{\theta_0} u = \Phi_u(\theta - \theta_0)$ and Φ_u is linear; on $\{u \neq 0\}$, $\Phi_u(\theta) = u\sum_k \theta_k \rho_u(T_k)$.

Why it works. if the class of coefficients is closed under differentiation, a zero-free u can absorb any coefficient into a and b: the pair $(\mathcal{D}u, u)$ is isomorphic to $(\mathcal{K}, 1)$.

■

Uses: none beyond the definitions.

The hypothesis "no zeros" cannot be dropped

With zeros the statement fails: $u = t - x^2$ near 0 satisfies no law $\partial_t + \alpha\partial_x + \beta$ with smooth α, β (at 0 it would give $1 = 0$).

6.2 The potential $V(x)$ is read from any non-zero solution

Dim. A37 **VAR-T1b**

Every non-zero solution identifies $V(x)$ [proved]

Let $\Omega = I \times J$, $V \in L^\infty_{\mathrm{loc}}(J)$ complex, u in (SF) with $L_0 u = V(x)u$ a.e., $u \not\equiv 0$, and $X_0 = \{x \in J : u(t,x) = 0 \ \forall t \in I\}$. Then:

(i) X_0 is closed and discrete in J; in particular $\operatorname{int} X_0 = \varnothing$ and $|X_0| = 0$;

(ii) if $W \in L^\infty_{\mathrm{loc}}(J)$ and $L_0 u = W(x)u$ a.e., then $W = V$ a.e. (everywhere if both are continuous);

(iii) $V(x) = (iu_t + u_{xx})/u$ at every (t,x) with $u(t,x) \neq 0$ (a.e., or everywhere if $u \in C^{1,2}$ and V is continuous).

Stated in the paper as no. 12.2, p. 69

Idea. If u vanished on infinitely many lines $x = x_n \to x_0$, cutting u at x_0 would give a new solution vanishing on a half-strip; then unique continuation across a graph $x = g(t)$, proved with a Carleman weight, forces $u \equiv 0$.

Proof.

Step 1 A Carleman estimate (VAR-L0)

Fix $g \in C^\infty(\mathbb{R})$, $t_0 \in \mathbb{R}$, $a, b > 0$; put $s = x - g(t)$, $B = \{|t - t_0| \le a, |s| \le b\}$, $\mu = 1/(4b)$ and the weight

$$\varphi(t,x) = -s + \mu s^2 - (t - t_0)^2.$$

There is λ_0 such that for $\lambda \ge \lambda_0$ and every w with compact support in $\operatorname{int} B$ (with $w, w_t, w_{xx} \in L^2$)

$$\mu\lambda^3 \int e^{2\lambda\varphi}|w|^2 \le \int e^{2\lambda\varphi}|L_0 w|^2.$$

For $z \in C_c^\infty$ one conjugates, $e^{\lambda\varphi}L_0 e^{-\lambda\varphi} = S + A$ with $S = \partial_x^2 + \lambda^2\varphi_x^2 + i\partial_t$ symmetric and $A = -2\lambda\varphi_x\partial_x - \lambda\varphi_{xx} - i\lambda\varphi_t$ antisymmetric, and

$$\|(S+A)z\|^2 = \|Sz\|^2 + \|Az\|^2 + ([S,A]z,z), \qquad ([S,A]z,z) = \int \left[4\lambda\varphi_{xx}|z_x|^2 + 4\lambda\varphi_{xt}\operatorname{Im}(z_x\bar{z}) + (4\lambda^3\varphi_x^2\varphi_{xx} - \lambda$$

On B, $\varphi_x \in [-\frac{3}{2}, -\frac{1}{2}]$, $\varphi_{xx} = 2\mu$, $\varphi_{xxxx} = 0$; absorbing the cross term in $4\lambda\varphi_{xx}|z_x|^2 = 8\lambda\mu|z_x|^2$ gives $([S,A]z,z) \ge (2\mu\lambda^3 - \lambda C_1)\|z\|^2 \ge \mu\lambda^3\|z\|^2$ for $\lambda \ge (C_1/\mu)^{1/2}$. Mollifying extends the estimate to w as stated.

Step 2 Unique continuation across a graph (VAR-L1)

Let u vanish on $\{x < g(t)\}$ near $p_0 = (t_0, g(t_0))$. Choose ε small and $\chi \in C^\infty$ with $\chi = 1$ on $[-\varepsilon, \infty)$, $\chi = 0$ on $(-\infty, -2\varepsilon]$; set $w = \chi(\varphi)u$. Then $\operatorname{supp} w \subset \{\varphi \ge -2\varepsilon,\ s \ge 0\} \subset \operatorname{int} B$ and $L_0 w = \chi(\varphi)Vu + R$ with R supported in $\{-2\varepsilon \le \varphi \le -\varepsilon\}$. The Carleman estimate gives, for λ large,

$$\tfrac{1}{2}(\mu\lambda^3)^{1/2}\|e^{\lambda\varphi}w\| \le \|e^{\lambda\varphi}R\| \le e^{-\lambda\varepsilon}\|R\|_{L^2}.$$

On $\Lambda = \{\varphi \ge -\varepsilon/2\}$, $w = u$ and $\|e^{\lambda\varphi}w\| \ge e^{-\lambda\varepsilon/2}\|u\|_{L^2(\Lambda)}$, so $\|u\|_{L^2(\Lambda)} \le 2(\mu\lambda^3)^{-1/2}e^{-\lambda\varepsilon/2}\|R\| \to 0$: $u = 0$ near p_0.

Step 3 Globalisation (VAR-L2)

If $u = 0$ on $I \times (\alpha, \beta)$ then $u \equiv 0$ on Ω: the set W of points near which $u = 0$ is open; Step 2 with $g \equiv \beta$ puts $I \times (\alpha, \beta]$ in W, and a continuation argument along the graphs $\varphi_s(t) = \beta + (s - \beta)\psi(t)$ (with ψ a bump on a time interval) pushes the zero region to the right until it fills Ω; the left side is symmetric under $x \mapsto -x$.

Step 4 (i) X_0 is discrete

Let $x_n \to x_0$ in X_0, $x_n \ne x_0$. X_0 is closed, so $u(\cdot, x_0) \equiv 0$, and $u_x(\cdot, x_0) = \lim(u(\cdot, x_n) - u(\cdot, x_0))/(x_n - x_0) = 0$. Put $\tilde{u} = 1_{x \ge x_0}u$: it is continuous, $\tilde{u}_x = 1_{x \ge x_0}u_x$ is continuous (no jump), so $\tilde{u}_{xx} = 1_{x \ge x_0}u_{xx}$ and $\tilde{u}$ is in (SF) with potential $V1_{x \ge x_0} \in L^\infty_{\mathrm{loc}}$. $\tilde{u}$ vanishes on the strip $I \times (\alpha, x_0)$, hence by Step 3 $\tilde{u} \equiv 0$, i.e. $u = 0$ on $I \times [x_0, \sup J)$; this contains a non-empty strip,

so Step 3 applied to u itself gives $u \equiv 0$, a contradiction. (The accumulation of the zero lines is used only to get $u_x(\cdot, x_0) = 0$, which makes $\tilde{u}$ a strong solution.)

Step 5 (ii) and (iii)

By A36(a) with $T_1 = -1$: $(V - W)u = 0$ a.e. For $x \notin X_0$ there is t with $u(t,x) \neq 0$, hence $u \neq 0$ on a rectangle around (t,x), and by Fubini $V = W$ a.e. there. By (i), $J \setminus X_0$ has full measure; if V, W are continuous they agree on a dense set, so everywhere. (iii) is the equation itself where $u \neq 0$.

Why it works. a Schrödinger solution cannot hide: it has infinite propagation speed, and unique continuation from a strip is strong enough to forbid isolated vertical zero lines from accumulating.

■

Uses: A36.

> **Example 6.1 The harmonic oscillator and Airy**
>
> For $V = x^2$ and the Hermite functions $h_0, \dots, h_3$ (and $h_0 + h_1$), X_0 is empty or $\{0\}$ and $V = (iu_t + u_{xx})/u$ recovers x^2 exactly. The same formula recovers the Pöschl–Teller and Airy potentials (SymPy checks).

6.3 Time-dependent potentials: silent strips

Dim. A38 **VAR-T1c**

Time-dependent potentials: identification outside silent strips [proved]

Let $\Omega = I \times J$, u in (SF), $Z(u) = \{u = 0\}$.

(i) In the class $V \in C(\Omega)$ the potentials compatible with u are $V + \{W : W = 0 \text{ on } \Omega \setminus Z(u)\}$; so V is identified iff $\operatorname{int} Z(u) = \emptyset$ (a.e. version: iff $|Z(u)| = 0$).

(ii) If $L_0 u = Vu$ with $V \in L^\infty_{\mathrm{loc}}$ and $\operatorname{int} Z(u) \neq \emptyset$, there is an open interval $T' \subset I$ with $u = 0$ on $T' \times J$ (a *silent strip*).

(iii) Silent strips exist for $u \not\equiv 0$: Tychonoff's series gives a solution vanishing for $t \leq 0$ and non-zero for $t > 0$.

The claim "every $u \not\equiv 0$ identifies $V(t,x)$" is **false**.

Stated in the paper as no. 12.3, p. 69

Idea. Apply A36(a) to the difference of two compatible potentials; for the strips use A37 Step 3; for the counterexample use the classical Tychonoff solution of the heat equation, adapted to Schrödinger.

Proof.

Step 1 (i) The difference has to vanish where $u \neq 0$

Two compatible potentials differ by W with $Wu = 0$, i.e. $W = 0$ on $\Omega \setminus Z(u)$. If $\operatorname{int} Z(u) \neq \emptyset$ take $W \in C_c(\operatorname{int} Z(u)) \neq 0$; if $\operatorname{int} Z(u) = \emptyset$, a continuous W vanishing on the dense set $\Omega \setminus Z(u)$ is zero.

Step 2 (ii) Silent strips

A ball $B((t_0, x_0), r) \subset Z(u)$ gives $u = 0$ on $T' \times (x_0 - \frac{r}{2}, x_0 + \frac{r}{2})$ with $T' = (t_0 - \frac{r}{2}, t_0 + \frac{r}{2})$; A37 Step 3 on $T' \times J$ gives $u = 0$ on $T' \times J$.

Step 3 (iii) Tychonoff's series

Let $g(t) = e^{-t^{-2}}$ for $t > 0$, $g = 0$ for $t \leq 0$, and

$$u(t,x) = \sum_{k \geq 0} (-i)^k \frac{g^{(k)}(t)}{(2k)!} x^{2k} \quad \text{on } (-1,1)^2.$$

Termwise, $u_{xx} = \sum_k (-i)^{k+1} g^{(k+1)} x^{2k}/(2k)! = -iu_t$, so $L_0 u = 0$. The Cauchy estimate $|g^{(k)}(t)| \leq k!(\theta t)^{-k} e^{-1/(2t^2)}$ ($\theta = 1/10$) bounds the kth term by $(x^2/(\theta t))^k e^{-1/(2t^2)}/k!$ (since $k!/(2k)! \leq 1/k!$), a locally uniformly convergent majorant with sum tending to 0 as $t \to 0^+$. So $u \in C^\infty$, $u \equiv 0$ for $t \leq 0$ and $u(t,0) = g(t) \neq 0$ for $t > 0$. Every $V(t,x) = W(t,x) 1_{t<0}$ is then compatible with u.

Why it works. unique continuation holds in x but not in t: a bounded domain without boundary conditions can let data switch on from the boundary, and while the data is silent the potential is not seen.

■

Uses: A36, A37.

Open

For $V \in L^\infty$ it is not known whether $|Z(u)| > 0$ with $\operatorname{int} Z(u) = \emptyset$ can occur (VAR-Q10); the case $V \in L^\infty$, $|Z| > 0$, $\operatorname{int} Z = \emptyset$ was settled in Phase 10 with an explicit example (Chapter 9).

6.4 Stability

Dim. A39 **VAR-P3**

Stability of the reconstruction of V [proved]

Let $K = [t_1, t_2] \times [x_1, x_2]$, $u_i \in C^{1,2}(K)$ with $L_0 u_i = V_i u_i$, $|u_1| \geq \delta > 0$ on K, $e = u_1 - u_2$.

(a) (Lipschitz) $\|V_1 - V_2\|_{L^\infty(K)} \leq \delta^{-1}(1 + \|V_2\|_{L^\infty})\|e\|_{C^{1,2}(K)}$.

(b) (No continuity from C^1) with $u_1 = e^{-iEt}$, $u_2 = e^{-iEt}\varphi_2$, $\varphi_2 = 1 + \varepsilon k^{-1}\sin kx$, $\varepsilon = k^{-1/2}$: $\|e\|_{C^1} \leq 2k^{-1/2} \to 0$ but $\|V_1 - V_2\|_\infty \geq k^{1/2}/2 \to \infty$.

(c) (Hölder under an a priori bound) if $M \geq \|e_{tt}\|_\infty, \|e_{xxx}\|_\infty$, then

$$\|V_1 - V_2\|_\infty \leq C\delta^{-1}(1 + \|V_2\|_\infty)\big(\|e\|^{1/3}M^{2/3} + \|e\|^{1/2}M^{1/2} + \|e\|\big),$$

and the exponent $1/3$ is sharp.

It is a *modulus of continuity*, not a certificate: it gives no algorithm that certifies an error from noisy data without a priori bounds.

Stated in the paper as no. 12.4, p. 70

Proof.

Step 1 (a) A direct identity

From $L_0 u_i = V_i u_i$: $L_0 e = (V_1 - V_2)u_1 + V_2 e$, so $V_1 - V_2 = (L_0 e - V_2 e)/u_1$ on K; take sup norms.

Step 2 (b) An explicit oscillation

$V_2 = E + \varphi_2''/\varphi_2$ with $\varphi_2'' = -\varepsilon k \sin kx$, so $\|V_1 - V_2\|_\infty = \sup|\varepsilon k \sin kx/(1 + \varepsilon k^{-1}\sin kx)| \geq k^{1/2}/2$ for $\varepsilon = k^{-1/2}$.

Step 3 (c) Interpolation

Two Taylor inequalities on an interval of length ℓ: for $f \in C^2$, $\|f'\| \leq 2(\|f\|\|f''\|)^{1/2} + 4\|f\|/\ell$ (one-sided Taylor with $h = \min(\ell/2, 2(\|f\|/\|f''\|)^{1/2})$); for $f \in C^3$, $\|f''\| \lesssim \|f\|^{1/3}\|f'''\|^{2/3} + \cdots$ (second difference with $h \leq \ell/4$). Applied to e_t and e_{xx} in the formula of (a) they give the stated Hölder bound.

Why it works. the reconstruction $V = (iu_t + u_{xx})/u$ divides by u and uses two derivatives: the loss is of order two and the division amplifies near zeros, hence a modulus rather than a stability constant.

■

Uses: A36.

Dim. A40 **VAR-P4**

When the phase is unobserved only V modulo constants is read [proved]

Under the hypotheses of A37, observe $[u] = \{e^{i\theta(t)}u : \theta \in C^1(I;\mathbb{R})\}$ (model DM-φ).

(a) The potentials $W \in L^\infty_{\mathrm{loc}}(J)$ such that some element of $[u]$ solves $L_0 v = W(x)v$ are exactly $\{V + c : c \in \mathbb{R}\}$.

(b) With the phase observed, V is unique (A37).

(c) For $u = e^{-iEt}\phi(x)$, ϕ alone determines $V - E = \phi''/\phi$ and nothing else: $(V, E) \mapsto (V + c, E + c)$.

(d) For $P(\partial_t, L_q)u = 0$ with $L_q = -\partial_x^2 + q$ and P unknown, (P, q) is defined only up to $q \to q + c$, $P(\tau, \ell) \to P(\tau, \ell - c)$: a redundancy of the parametrisation (present with DM3 too), removed by a normalisation such as $q(x_0) = 0$.

Stated in the paper as no. 12.4, p. 70

Proof.

Step 1 (a) The phase changes the potential by $\theta'(t)$

If $v = e^{i\theta}u$, $L_0 v = e^{i\theta}(-\theta' u + L_0 u) = (V(x) - \theta'(t))v$. If also $L_0 v = W(x)v$, then $V(x) - W(x) = \theta'(t)$ a.e. on $\{u \neq 0\}$ *only*: u may be silent on strips (A38), where θ' is free.

Step 2 θ' is constant on each component of $\{t : u(t,\cdot) \not\equiv 0\}$

Let $S_+ = \{t \in I : u(t,\cdot) \not\equiv 0\}$, open and non-empty. Around each point with $u \neq 0$ there is a rectangle on which $V(x) - W(x) = \theta'(t)$; by Fubini θ' is locally constant on S_+, so equal to c_i on each component I_i.

Step 3 The constants agree

On $I_i \times J$, u is a non-zero (SF) solution, so by A37(i) the set $X_i = \{x : u(t,x) \neq 0 \text{ for some } t \in I_i\}$ is open with discrete complement, and $V - W = c_i$ a.e. on X_i. Two such sets are co-discrete, so they intersect in positive measure and $c_i = c_j$. Hence $W = V - c$ a.e.; conversely $\theta(t) = -ct$ gives $W = V + c$.

Step 4 (c) and (d)

(c) is the stationary case of (a). (d): $L_{q+c} = L_q + c$ and $P(\partial_t, L_{q+c} - c) = P(\partial_t, L_q)$ as operators.

Why it works. a global phase costs a function of t only, and a potential depending on x alone can absorb only constants; the silent strips cannot interfere because the phase is free there but the data say nothing.

■

Uses: A37, A38.

CHAPTER 7

Convergence, noise and limits of learning

In this chapter

Starting from	$\Omega \subset \mathbb{R}^n$ open connected, $p_0 \in \Omega$, u real-analytic on Ω (hypothesis (H-an)); the datum at step K is the jet $j_K u = (\partial^\gamma u(p_0))_{\lvert\gamma\rvert \le K}$ (model DM2), or u known up to a noise of norm δ (model DM-δ). $R_{\le m}$ is the space of operators of degree $\le m$.
Why	to prove what can and cannot be learned from a growing amount of data: identification in the limit holds with an explicit rule, but it never comes with a certificate, no learner can decide whether a law exists, and from noisy data one gets stability only relative to a finite level of the dictionary.
Role	Dim. A41–A43. They prove Chapter 13 of the paper.

Notation for the whole chapter

For $m \le K$ put

$$N_{m,K} = \{Q \in R_{\le m} : \partial^\beta (Q(\partial)u)(p_0) = 0 \ \text{ for all } |\beta| \le K - m\}.$$

It is computable from $j_K u$: in matrix form, the kernel of the matrix $M_{m,K}(j)$ with entries $j_{\alpha+\beta}$ (rows β, columns α). $\nu_m(u) = \max\{\mathrm{ord}_{p_0}(Qu) : Q \in R_{\le m} \setminus \mathrm{Ann}(u)\}$.

7.1 Identification in the limit

Dim. A41 **CONV-T1**

Stabilisation by level (CONV-L1), and identification in the limit (CONV-T1)

[proved]

Assume (H-an).

(a) (CONV-L1) $N_{m,K+1} \subseteq N_{m,K}$; $\bigcap_{K \geq m} N_{m,K} = \mathrm{Ann}(u) \cap R_{\leq m}$; and $\nu_m(u) < \infty$ with $N_{m,K} = \mathrm{Ann}(u) \cap R_{\leq m}$ iff $K \geq K^*(m) = m + \nu_m(u)$.

(b) (CONV-T1) The *minimal-degree rule* $m^*(K) = \min\{m \leq K : N_{m,K} \neq 0\}$, $O_K = \gcd(N_{m^*(K),K})$ satisfies: if $\mathrm{Ann}(u) \neq 0$ with minimal degree d, then $m^*(K) = d$ and $O_K = g_d = \gcd(\mathrm{Ann}(u) \cap R_{\leq d})$ for $K \geq K^*(d)$; if $\mathrm{Ann}(u) = (P)$ then $O_K = P$; if $\mathrm{Ann}(u) = 0$ then $m^*(K) \to \infty$.

(c) With an a priori bound B such that $\mathrm{Ann}(u)$ is generated in degree $\leq B$, $\gcd(N_{B,K}) = G_u$ for $K \geq K^*(B)$.

(d) The output sequence (O_K) takes at most $(d+1)(d+2)/2$ distinct values, for every u.

(e) The time of convergence is not bounded by any function of d (for $n = 2$: $u = 1 + x^{N+2}e^{x^2}$ has $\mathrm{Ann}(u) = (\tau)$ and $K^* = N + 2$).

(f) (No correct certificate) For $n \geq 2$ there is no K and no $\{0,1\}$-valued function c of $j_K u$ with: $c(j_K u) = 1 \Rightarrow \mathrm{Ann}(u) = (O_K)$ for every analytic u, and $c = 1$ for some u with non-trivial principal Ann.

Stated in the paper as no. 13.1, p. 71

Idea. The conditions defining $N_{m,K}$ shrink as K grows, and their limit is the annihilator because an analytic function with zero Taylor series vanishes. Finite dimensionality of $R_{\leq m}$ makes the stabilisation happen at a finite step; but nothing says when, and any finite jet is shared with a polynomial, which kills certificates.

Proof.

Step 1 (a) A decreasing chain that stabilises

The conditions of $N_{m,K+1}$ contain those of $N_{m,K}$. Q lies in all of them iff the Taylor series of Qu at p_0 vanishes iff $Qu \equiv 0$ (analyticity and connectedness). For finiteness put $W_m = R_{\leq m}/(\mathrm{Ann}(u) \cap R_{\leq m})$ and $F_k = \{[Q] : \mathrm{ord}_{p_0}(Qu) \geq k\}$: a decreasing chain of subspaces of a finite-dimensional space with intersection 0, so $F_{k_0} = 0$ for some k_0 and $\nu_m \leq k_0 - 1$. Finally, for $Q \notin \mathrm{Ann}(u)$, $Q \in N_{m,K} \iff \mathrm{ord}(Qu) \geq K - m + 1$.

Step 2 (b) The rule

For $m < d$, $\mathrm{Ann}(u) \cap R_{\leq m} = 0$ so $N_{m,K} = 0$ for $K \geq K^*(m)$; for $m = d$ and $K \geq K^*(d)$, $N_{d,K} = \mathrm{Ann}(u) \cap R_{\leq d} \neq 0$. So $m^*(K) = d$ and $O_K = g_d$. If $\mathrm{Ann}(u) = (P)$, every non-zero element has degree $\geq \deg P$, so $\mathrm{Ann} \cap R_{\leq d} = \mathbb{C}P$. If $\mathrm{Ann}(u) = 0$, for fixed m all $N_{m',K}$ with $m' \leq m$ vanish for large K, so $m^*(K) > m$.

Step 3 (c) and (d)

(c) is (a) with $m = B$: $N_{B,K} = \mathrm{Ann}(u) \cap R_{\leq B}$ generates $\mathrm{Ann}(u)$, and the gcd of a generating set of an ideal of a UFD is the gcd of the ideal. (d) For $K \geq 1$, $N_{K,K} \neq 0$ ($\dim R_{\leq K} \geq 3$, one linear equation), so $m^*(K) \leq K$; $N_{m,K} = 0 \Rightarrow N_{m,K+1} = 0$, so m^* is non-decreasing; at fixed m^*, $N_{m,K+1} \subseteq N_{m,K}$ makes $\deg O_K$ non-decreasing, and equal degrees give associated O_K. So O_K is a function of the pair $(m^*(K), \deg O_K)$, monotone in lexicographic order, with at most $(d+1)(d+2)/2$ values.

Step 4 (e) Non-uniform time

Let $f(x) = 1 + x^{N+2}e^{x^2}$. τ kills u. If $Q = \tau Q_1 + q(\xi) \in \mathrm{Ann}(u)$ with $q \neq 0$, then $q(\partial_x)f = 0$ on an interval, so f would be an exponential-polynomial (analyticity), contradicting the growth e^{x^2}; so $\mathrm{Ann}(u) = (\tau)$. Computing orders: $\nu_0 = 0$, and for $m = 1$, $Qu = af + cf'$ with $f' = (N+2)x^{N+1}e^{x^2} + \cdots$ of order $N+1$, so $K^*(1) = N+2$.

Step 5 (f) No certificate

Suppose c and u with $\mathrm{Ann}(u) = (P)$, P non-constant, $c(j_K u) = 1$. Let $u' = T_K u$ be the Taylor polynomial of order K: $j_K u' = j_K u$ and $\mathrm{Ann}(u')$ contains all monomials of degree $K+1$, so $R_n/\mathrm{Ann}(u')$ is finite-dimensional, whereas $R_n/(P)$ is infinite-dimensional for $n \geq 2$. So $\mathrm{Ann}(u') \neq (P)$, while the rule and c see the same data: $c(j_K u') = 1$ would assert $\mathrm{Ann}(u') = (O_K(u')) = (P)$, false.

Why it works. the data at each finite stage are shared by a polynomial, whose annihilator is a different ideal; hence no stage can be certified, even though from some stage on the rule is right.

■

Uses: A2, A7.

> **Example 7.1 Convergence at an unpredictable time**
>
> For $u = 1 + x^{N+2}e^{x^2}$ the rule outputs $O_K = 1$ ("no law of degree 1 found") for $1 \leq K \leq N+1$ and $O_K = \tau$ from $K = N+2$ on, for every N. Nothing in the first $N+1$ derivatives announces τ.

Dim. A42 **CONV-T2**

No learner decides whether a law exists: analytic Gold theorem (CONV-T2, T2′) [proved]

Let $n \geq 1$ and $(\varphi_K)_{K \geq 0}$ be *any* sequence of maps $\varphi_K : \mathbb{R}^{J_K} \to \{\text{law}, \text{no law}\}$ on jets of order $\leq K$ (no computability, continuity or measurability assumed). Then there is a real entire function u on $\mathbb{R}^n$ such that $\varphi_K(j_K u)$ does not converge to the correct answer ("law" if $\mathrm{Ann}(u) \neq 0$, "no law" if $\mathrm{Ann}(u) = 0$). (T2′) For $n \geq 2$ the same holds with u restricted to the class $\{\mathrm{Ann}(u) \text{ principal non-zero}\} \cup \{\mathrm{Ann}(u) = 0\}$; hence no learner identifies Ann on that class.

Stated in the paper as no. 13.2, p. 72

Idea. Build u as a limit of polynomials: at each stage, add Taylor coefficients, chosen outside a proper algebraic set, that make every operator of degree $\le k$ fail to annihilate u; but wait until the learner has said "law" on the current polynomial, which it must eventually do. The learner then says "law" infinitely often on a function with Ann $= 0$.

Proof.

Step 1 Step zero

If some polynomial p has $\varphi_K(j_K p) \not\to$ "law", take $u = p$ ($\partial_t^{\deg p+1} \in \mathrm{Ann}(p)$). Assume $\varphi_K(j_K p) \to$ "law" for every polynomial p.

Step 2 An avoidance lemma

Let $g_0(z) = e^{|z|^2}$. (i) $Q(\partial)(h\, g_0) \neq 0$ for every non-zero polynomial h and $Q \neq 0$: its top-degree part is $Q_{\mathrm{top}}(2z)h_{\mathrm{top}}\, e^{|z|^2} \neq 0$. (ii) For m, L put $N_{m,K}^{>L}(g_0) = \{Q \in R_{\le m} : \partial^\beta(Qg_0)(0) = 0 \text{ for } L < |\beta| \le K - m\}$; the intersection over K consists of the Q with Qg_0 a polynomial of degree $\le L$, so $\tau^{L+1}Q \in \mathrm{Ann}(g_0) = 0$ and $Q = 0$. By finite dimension there is $\bar K(m, L)$ with $N_{m,\bar K}^{>L}(g_0) = 0$: the submatrix of rows $|\beta| > L$ has full column rank, so some minor $\Delta \neq 0$ at the jet of g_0. (iii) Δ is a real polynomial in the coefficients $y = (j_\gamma)_{L<|\gamma|\le \bar K}$, not identically zero, so its zero set has empty interior and there is y^* with $|y_\gamma^*| \le \gamma!/|\gamma|!$ and $\Delta(y^*) \neq 0$. Any analytic u' whose jet has coefficients y^* in orders $(L, \bar K]$ has $\mathrm{Ann}(u') \cap R_{\le m} = 0$.

Step 3 Construction and limit

Build polynomials p_k and integers $H_0 < H_1 < \cdots$ with $\varphi_{H_k}(j_{H_k} p_k) =$ "law". At step k apply the lemma with $m = k$, $L = H_k$, add the block y_k^* in orders $(H_k, L_k']$, and by Step 1 pick $H_{k+1} > L_k'$ with "law" on the new polynomial. Put $u = \lim p_k$. Its Taylor coefficients satisfy $|c_\gamma| \le 1/|\gamma|!$, so u is entire and real. For each k, $j_{H_{k+1}} u = j_{H_{k+1}} p_{k+1}$, hence $\varphi_{H_{k+1}}(j_{H_{k+1}} u) =$ "law" infinitely often, while the block y_k^* gives $\mathrm{Ann}(u) \cap R_{\le k} = 0$ for every k: $\mathrm{Ann}(u) = 0$. So φ is wrong on u.

Step 4 (T2′) The principal class

Test functions $v = q + \varepsilon t^{N-1} e^{|x|^2}$ with q a real polynomial of degree $\le M$, $N = M + 2$, have $\mathrm{Ann}(v) = (\tau^N)$ (divide Q by τ^N; the remainder gives a polynomial plus $e^{|x|^2}$ times a polynomial, impossible unless each coefficient vanishes) and $j_M v = j_M q$. Redo the construction with $v_k = q_k + \varepsilon_k t^{N_k - 1} e^{|x|^2}$, $\varepsilon_k = 1/(N_k - 1)!$, $N_k = L_k' + 2$, which lies in the class and on which the learner must say "law"; the tails occupy orders $> H_{k+1}$, so the limit u is entire (the tails are dominated by $e^{|t|} e^{\sum |x_i|^2}$), the learner says "law" infinitely often, and $\mathrm{Ann}(u) = 0$.

Why it works. any rule that is eventually right on all polynomials can be fooled by a function that agrees with polynomials longer and longer yet is differentially transcendental.

■

Uses: A7, A41.

Reading

"Ann$(u) \neq 0$" is *verifiable in the limit* (the rule "say law at step K if $m^*(K) = m^*(K-1)$" is eventually right when Ann $\neq 0$ and wrong infinitely often when Ann $= 0$), but not *decidable in the limit*.

7.2 Noisy data

Dim. A43 **CONV-T4**

Discontinuity (T3), conditional stability (T4), and a discrepancy rule (T5)

[proved]

(T3) Let X be a topological vector space of functions on Ω continuously embedded in $\mathcal{D}'(\Omega)$, in which polynomials are dense and $e^{|z|^2} \in X$ (for example $H^m(\Omega)$ on a bounded Lipschitz domain). Then $\{\mathrm{Ann} = 0\}$ and $\{\mathrm{Ann} \neq 0\}$ are both dense, and $u \mapsto \mathrm{Ann}(u)$ is discontinuous everywhere.

(T4) Let Ω be bounded Lipschitz, $\Phi(u) : \mathbb{C}^M \to L^2$, $\lambda \mapsto \sum_{|\alpha| \le n_0} \lambda_\alpha \partial^\alpha u$, with $\Phi(u)\lambda^* = 0$, $|\lambda^*| = 1$ and second singular value $\sigma_2(\Phi(u)) \ge \rho > 0$.

(a) If $\|\tilde{u} - u\|_{H^{n_0}} \le \delta$ and $\tilde{\lambda}$ minimises $\|\Phi(\tilde{u})\lambda\|$ on $|\lambda| = 1$, then $\sin \angle(\tilde{\lambda}, \lambda^*) \le 2\delta/\rho$; the order δ/ρ is optimal.

(b) If $\|\tilde{u}_0 - u\|_{L^2} \le \delta$ and $\|u\|_{H^{n_0+s}} \le \mathfrak{M}$, the same holds with δ replaced by $C(2\delta)^{s/(n_0+s)}(2\mathfrak{M})^{n_0/(n_0+s)}$.

(c) (Misfit certificate) If $\|\tilde{u} - u\|_{H^{n_0}} \le \delta$ and $\sigma_1(\Phi(\tilde{u})) > \delta$, then $\mathrm{Ann}(u) \cap \mathcal{D}_{n_0} = 0$.

(T5) With $d \ge 1$ the minimal degree of $\mathrm{Ann}(u) \neq 0$ and $s_m = \sigma_1(\Phi_m(u)) > 0$ for $m < d$, the discrepancy rule $m(\delta) = \min\{m : \sigma_1(\Phi_m(\tilde{u})) \le \delta_m\}$ gives $m(\delta) = d$ and $\sin \angle(\tilde{\lambda}(\delta), P) \to 0$ at rate δ (or $\delta^{s/(d+s)}$) when $\delta < \delta_0 = \frac{1}{2}\min_{m<d} s_m$; if $\mathrm{Ann}(u) = 0$, $m(\delta) \to \infty$. The threshold δ_0 cannot be bounded below from the data.

Stated in the paper as no. 13.3, p. 72

Idea. Singular-value perturbation (Weyl) gives (a) and (c); interpolation between L^2 and H^{n_0+s} via Stein's extension gives (b); the discrepancy rule picks the first level at which the smallest singular value is below the noise level; the threshold fails to be computable because a tiny smooth perturbation creates a law of low degree with tiny margin.

Proof.

Step 1 (T3) Both sets are dense

Polynomials are dense and have Ann $\neq 0$. For $\{\mathrm{Ann} = 0\}$: the map $Q \mapsto Q(\partial)e^{|z|^2}|_\Omega$ from $R_{\le m}$ to $\mathcal{D}'(\Omega)$ is injective, so choose test functions $\psi_i \in C_c^\infty$ with $\det(\langle Q_\alpha(\partial)g_0, \psi_i\rangle) \neq 0$. The map $v \mapsto \langle Q_\alpha(\partial)v, \psi_i\rangle$ is continuous on X, and $D(\varepsilon) = \det\langle Q_\alpha(\partial)(u + \varepsilon g_0), \psi_i\rangle$ is a polynomial in ε not identically zero, so non-zero for all but finitely many ε. For each m only finitely many ε are bad; the union over m is countable, so there are arbitrarily small ε that are good for every

m, and then $\mathrm{Ann}(u+\varepsilon g_0)=0$ (*condensed*: the package §2.1 writes the determinant argument in full).

Step 2 (T4a) A singular-vector estimate

Put $E=\Phi(\tilde u)-\Phi(u)$; by Cauchy–Schwarz $\|E\|\le\delta$. Write $\tilde\lambda=c\lambda^*+w$, $w\perp\lambda^*$, so $|w|=\sin\angle$. Then

$$\rho|w|\le\|\Phi(u)w\|=\|\Phi(u)\tilde\lambda\|\le\|\Phi(\tilde u)\tilde\lambda\|+\delta\le\|\Phi(\tilde u)\lambda^*\|+\delta\le\|\Phi(u)\lambda^*\|+2\delta=2\delta.$$

Optimality: for $A=\mathrm{diag}(0,\rho)$, $E=\begin{pmatrix}0&0\\ \delta&0\end{pmatrix}$ the minimal singular vector of $A+E$ has sine $\delta/\sqrt{\rho^2+\delta^2}$.

Step 3 (T4b) and (T4c)

(b) $e=\tilde u-u$ has $\|e\|_{L^2}\le 2\delta$, $\|e\|_{H^{n_0+s}}\le 2\mathfrak{M}$. With Stein's universal extension operator and Hölder's inequality in Fourier variables with exponents $(n_0+s)/s$ and $(n_0+s)/n_0$,

$$\int(1+|\xi|^2)^{n_0}|\hat f|^2\le\Big(\int|\hat f|^2\Big)^{\frac{s}{n_0+s}}\Big(\int(1+|\xi|^2)^{n_0+s}|\hat f|^2\Big)^{\frac{n_0}{n_0+s}},$$

so $\|e\|_{H^{n_0}}\le C\|e\|_{L^2}^{s/(n_0+s)}\|e\|_{H^{n_0+s}}^{n_0/(n_0+s)}$; then (a). (c) Weyl: $|\sigma_1(\Phi(\tilde u))-\sigma_1(\Phi(u))|\le\|E\|\le\delta$, so $\sigma_1(\Phi(u))>0$.

Step 4 (T5) The discrepancy rule

In the branch $\|\tilde u-u\|_{H^d}\le\delta$: for $m<d$, $\sigma_1(\Phi_m(\tilde u))\ge s_m-\delta>\delta$ when $\delta<s_m/2$, so m is not chosen; for $m=d$, $\sigma_1(\Phi_d(\tilde u))\le\|\Phi_d(\tilde u)q\|\le\delta$ for a unit $q\in\mathrm{Ann}\cap R_{\le d}$. So $m(\delta)=d$, and the angle estimate of (T4a) with the kernel of dimension r gives $\mathrm{dist}(\tilde\lambda,\mathrm{Ann}\cap R_{\le d})\le 2\delta/\rho_d$. If $\mathrm{Ann}(u)=0$, every $s_m>0$, so for each M and small δ no level $m\le M$ is chosen.

Step 5 The threshold is not computable from the data

Let $p=t^d x_1^d\cdots x_{n-1}^d$; one checks $\mathrm{Ann}(p)\cap R_{\le d}=0$ (the lowest-degree part Q_{low} of Q maps p to a nonzero polynomial). For almost every ε the same determinant argument as in Step 1 gives $\mathrm{Ann}(u+\varepsilon p)\cap R_{\le d}=0$: a small smooth multiple of p destroys the law of degree d and leaves a margin that tends to 0 with ε. The package builds from this, for every N and $\eta>0$, a function u' with $u'-u\in C^\infty(\bar\Omega)$, $\|u'-u\|_{H^N}\le\eta$, $\mathrm{Ann}(u')\ne 0$ and $\delta_0(u')\le\eta$ (*condensed*: the explicit choice of u' is in the package, §2.3 item 3). Hence no function $\Psi(\tilde u,\delta)>0$ of the data alone is a lower bound for $\delta_0(v)$ over all admissible v.

Why it works. noise can only be handled relative to a finite level of the dictionary and with a margin ρ; both the level and the margin are properties of the unknown u, not of the data.

■

Uses: A7, A41, A42.

Example 7.2 The heat equation from noisy data

For the heat kernel $u = (t+1)^{-1/2} e^{-x^2/(4(t+1))}$ (minimal degree $d = 2$, $\mathrm{Ann}(u) = (\tau - \xi^2)$), the discrepancy rule returned $m(\delta) = 2$ for $\delta = 10^{-3}$ and 10^{-4} in a SymPy/numerical check, and the observed ratio $\sin\angle / (2\delta/\sigma_2)$ never exceeded 0.27 in 4000 random trials.

CHAPTER 8

The analytic encoder

In this chapter	
Starting from	a symmetric interval $J = [x_0 - a, x_0 + a]$ (translated so that $x_0 = 0$), $q \in C^1(J;\mathbb{C})$, the operators $L_q = -\partial_x^2 + q$ and $L_0 = -\partial_x^2$; an *encoder* is a linear map W acting in x only, with $\dim\ker W < \infty$ on its domain, intertwining L_q with $L_0 + c$ for a constant c, and explicit.
Why	to prove that a variable potential q can be removed by an explicit change of variables (a Volterra transformation operator), which reduces every equation $P(\partial_t, L_q)u = 0$ to the constant-coefficient one; that this encoder is unique and stable; and that no time-independent encoder exists for a genuinely time-dependent potential.
Role	Dim. A44–A47. They prove Chapter 14 of the paper.

Notation for the whole chapter

$\Delta = \{(x,s) : |s| \le |x| \le a\} = \Delta_+ \cup \Delta_-$ is the union of two triangles meeting at the origin. The Goursat problem is

$$K_{xx} - K_{ss} = q(x)K \text{ on } \Delta, \qquad K(x,x) = \tfrac{1}{2}\int_0^x q, \qquad K(x,-x) = 0. \tag{G}$$

The operator $Tf(x) = f(x) + \int_{-x}^{x} K(x,s)f(s)\,\mathrm{d}s$ is the *Volterra transformation operator*, and $W = T^{-1}$.

8.1 The Volterra encoder

Dim. A44 **ENC-1**

The symmetric Volterra transformation operator (ENC-1, ENC-U1) [proved]

Let $q \in C^1(J;\mathbb{C})$ (q complex is allowed).

(1) There is a unique $K \in C(\Delta)$, of class C^2 on each $\Delta_\pm$, solving (G).

(2) For every $f \in C^2(J)$, without boundary conditions, $Tf \in C^2(J)$ and

$$L_q Tf = TL_0 f, \qquad (Tf)(0) = f(0), \qquad (Tf)'(0) = f'(0).$$

(3) T is a bijection of $C(J)$ and of $C^2(J)$; its inverse is $W = I + \int_{-x}^{x} H(x,s)\cdot \mathrm{d}s$ with $H_{xx} - H_{ss} = -q(s)H$, $H(x,x) = -\frac{1}{2}\int_0^x q$, $H(x,-x) = 0$; and $WL_q = L_0 W$ on $C^2(J)$.

(4) If $q \in C^\infty(J)$, T and W are bijections of $C^\infty(J)$.

(5) With $Q = \|q\|_\infty$: $\|K\|_\infty \le \frac{aQ}{2}e^{a\sqrt{Q}}$; $\|K_q - K_{q'}\|_\infty \le C(a,Q,Q')\|q-q'\|_\infty$; $\|T\|, \|W\| \le 1 + 2a\|K\|_\infty$ on $C(J)$ and on $L^2(J)$.

(6) T depends on the base point x_0 already for constant q.

(ENC-U1) Among operators $T = I + \int_{-x}^{x} K(x,s)\cdot \mathrm{d}s$ with $K \in C(\Delta)\cap C^2(\Delta_\pm)$ and $L_q T = TL_0$, the set is the line $\{T_{k_0} : k_0 \in \mathbb{C}\}$ with $K(x,-x) \equiv k_0$, $K(x,x) = k_0 + \frac{1}{2}\int_0^x q$, and $(T_{k_0}f)'(0) = f'(0) + 2k_0 f(0)$. So $T = T_0$ is the unique one preserving Cauchy data at x_0.

Stated in the paper as no. 14.2, p. 74

Idea. Pass to characteristic coordinates, where the Goursat problem becomes a Volterra integral equation solved by successive approximations; the intertwining is a Leibniz computation in which every boundary and interior term cancels; the inverse is the same construction with the opposite sign of q.

Proof.

Step 1 (1) Goursat problem

Put $\xi = x+s$, $\eta = x-s$, $K(x,s) = \Phi(\xi,\eta)$; then $K_{xx} - K_{ss} = 4\Phi_{\xi\eta}$ and (G) is equivalent to

$$\Phi(\xi,\eta) = \tfrac{1}{2}\int_0^{\xi/2} q + \tfrac{1}{4}\int_0^{\xi}\int_0^{\eta} q\big(\tfrac{\alpha+\beta}{2}\big)\Phi(\alpha,\beta)\,\mathrm{d}\beta\,\mathrm{d}\alpha. \tag{G′}$$

Successive approximations $\Phi_0 = \frac{1}{2}\int_0^{\xi/2} q$, $\delta_n = \Phi_n - \Phi_{n-1}$ satisfy $|\delta_n| \le \frac{aQ}{2}\frac{(Q|\xi\eta|/4)^n}{(n!)^2}$ (using $\int_0^{|\xi|}\int_0^{|\eta|}(\alpha\beta)^n = \frac{|\xi\eta|^{n+1}}{(n+1)^2}$), and $|\xi\eta| \le a^2$ on Δ; the series converges uniformly with the bound of (5), and the same bound on a difference gives uniqueness. Differentiating (G′) gives $\Phi_\xi, \Phi_\eta, \Phi_{\xi\eta} = \frac{1}{4}q(\frac{\xi+\eta}{2})\Phi$ and $\Phi_{\xi\xi}, \Phi_{\eta\eta}$ continuous because $q \in C^1$.

Step 2 (2) Intertwining

Fix $x \neq 0$, $g = Tf$. By Leibniz with $K(x,-x) \equiv 0$,

$$g' = f' + K(x,x)f(x) + \int_{-x}^{x} K_x f\,\mathrm{d}s,$$

$$g'' = f'' + \tfrac{\mathrm{d}}{\mathrm{d}x}K(x,x)f(x) + K(x,x)f'(x) + K_x(x,x)f(x) + K_x(x,-x)f(-x) + \int K_{xx}f.$$

Two integrations by parts give $\int_{-x}^{x} Kf'' = K(x,x)f'(x) - K_s(x,x)f(x) + K_s(x,-x)f(-x) + \int K_{ss}f$. In $L_q g - TL_0 f$ the coefficient of $f(x)$ is $q - \frac{\mathrm{d}}{\mathrm{d}x}K(x,x) - K_x(x,x) - K_s(x,x) = q - 2\frac{\mathrm{d}}{\mathrm{d}x}K(x,x) = 0$; that of $f(-x)$ is $-K_x(x,-x) + K_s(x,-x) = -\frac{\mathrm{d}}{\mathrm{d}x}K(x,-x) = 0$; those of $f'(x), f'(-x)$ cancel; and under the integral $-K_{xx} + K_{ss} + qK = 0$. At $x = 0$: g' is continuous with value $f'(0)$ from both sides, $g'' = qg + T(f'')$ has equal limits, hence $g \in C^2(J)$; and $g(0) = f(0)$, $g'(0) = f'(0) + 2K(0,0)f(0) = f'(0)$.

Step 3 (3) The inverse

Let $T' = I + \int H$ with H the kernel of the same type and $-q$; the computation of Step 2 with roles exchanged gives $L_0 T' = T' L_q$ and T' preserves Cauchy data. Put $S = T'T$: it commutes with L_0, preserves Cauchy data and has the form $I + \int_{-x}^{x} R(x,s)\,\mathrm{d}s$ with R continuous. For each $k \in \mathbb{C}$, $Se^{k\cdot}$ solves $y'' = k^2 y$ with data $(1,k)$, so $Se^{k\cdot} = e^{k\cdot}$, i.e. $\int_{-x}^{x} R(x,s)e^{ks}\,\mathrm{d}s = 0$ for all k; by uniqueness of the Fourier–Laplace transform of a compactly supported continuous function, $R \equiv 0$ and $S = I$. The operator $V : f \mapsto \int_{-x}^{x} K(x,s)f(s)\,\mathrm{d}s$ is quasi-nilpotent ($|V^n f(x)| \le \|K\|^n (2|x|)^n \|f\|/n!$), so T is invertible on $C(J)$ by a Neumann series; hence $T' = T^{-1}$ and T is bijective on $C^2(J)$.

Step 4 (4) and (5)

(4) The identity $g'' = qg + T(f'')$ and induction give $f \in C^{2m}, q \in C^{2m-2} \Rightarrow Tf \in C^{2m}$.
(5) The difference $\delta\Phi$ of two solutions satisfies $\delta\Phi = \delta\Phi_0 + \frac{1}{4}\iint [q\,\delta\Phi + (q-q')\Phi_{q'}]$ with $|\delta\Phi_0| \le a\|q - q'\|/2$; the same iteration gives $\|\delta\Phi\| \le (\frac{a}{2} + \frac{a^2\|\Phi_{q'}\|}{4})e^{a\sqrt{Q}}\|q - q'\|$. On L^2 the integral part is Hilbert–Schmidt with norm $\le \sqrt{2}\,a\|K\|_\infty$.

Step 5 (6) Dependence on the base point

For $q \equiv c$, $T^{(x_0)}e^{ik\cdot}(x) = e^{ikx_0}\big[\cos\mu(x - x_0) + ik\frac{\sin\mu(x-x_0)}{\mu}\big]$ with $\mu^2 = k^2 - c$, which differs from $e^{ikx} = T^{(x_1)}e^{ik\cdot}(x)$ at $x = x_1$ when $\mu \neq k$.

Step 6 (ENC-U1) Classification

For any K of this regularity, $L_q Tf - TL_0 f = A(x)f(x) + B(x)f(-x) + \int_{-x}^{x} C(x,s)f(s)\,\mathrm{d}s$ with $A = q - 2\frac{\mathrm{d}}{\mathrm{d}x}K(x,x)$, $B = -2\frac{\mathrm{d}}{\mathrm{d}x}K(x,-x)$, $C = K_{ss} - K_{xx} + qK$. If this vanishes for all $f \in C^2$, taking f supported inside $(-|x|,|x|)$ gives $C \equiv 0$; then $f(x) = 1$, $f(-x) = 0$ gives $A = 0$, and similarly $B = 0$. So $K(x,-x) = k_0$ is constant and $K(x,x) = k_0 + \frac{1}{2}\int_0^x q$; Goursat has a unique solution with $\Phi_0 = k_0 + \frac{1}{2}\int_0^{\xi/2} q$. Conversely each T_{k_0} works.

Why it works. a Volterra operator with a triangular kernel has no spectrum, so it is invertible; and the Goursat conditions are exactly those that make the interior and boundary terms of the Leibniz expansion cancel.

■

Uses: none beyond Fourier–Laplace uniqueness. The status is [proof written, not audited]: the statement is classical (Marchenko, Povzner, Levitan) but the exact theorem numbers and constants were not checked against the sources.

Example 8.1 *q constant*

For $q \equiv c \neq 0$ no differential intertwiner of any order satisfies $WL_c = L_0 W$ (ENC-K0: the recursion on the coefficients w_j forces w_0 to be a polynomial of degree n with non-zero leading coefficient, contradicting $w_0'' + cw_0 = 0$). With the relaxed requirement $WL_c = (L_0 + c)W$ the identity $W = I$ works. The Volterra operator with $T_0 e^{ik\cdot} = e^{ikx_0}[\cos \mu(x - x_0) + ik \sin \mu(x - x_0)/\mu]$ is the genuinely non-trivial one.

Dim. A45 **ENC-C1**

One encoder for every equation in (∂_t, L_q) [proved]

Let $q \in C^\infty(J^\circ)$, $\Omega = I \times J^\circ$ and $u \in C^\infty(\Omega)$. For every polynomial P,

$$P(\partial_t, L_q)u = 0 \iff P(\partial_t, L_0)(Wu) = 0.$$

The same W serves Schrödinger, heat, wave, Klein–Gordon and every polynomial in (∂_t, L_q).

Stated in the paper as no. 14.2, p. 75

Proof.

Step 1 **The kernel does not depend on t**

W acts in x and commutes with ∂_t; differentiation under the integral is legitimate because u is smooth on compacts. Exhaust J° by intervals $J_{a'}$, $a' < a$; the kernels agree by uniqueness of (G).

Step 2 **Powers of L_q**

From $WL_q = L_0 W$ on C^∞, induction gives $WL_q^j = L_0^j W$, hence $W\,P(\partial_t, L_q) = P(\partial_t, L_0)\,W$. Since W is bijective on $C^\infty(J^\circ)$ (A44(4)), $P(\partial_t, L_q)u = 0 \iff P(\partial_t, L_0)Wu = 0$.

Why it works. the encoder is a function of x only, so it commutes with everything in t, and an intertwiner of L_q with L_0 is automatically an intertwiner of every polynomial in L_q.

■

Uses: A44.

8.2 What an encoder cannot do

Dim. A46 **ENC-P0b**

No time-independent encoder for a time-dependent potential [proof written, not audited]

(classical version, ENC-P0(b)^{cl}) Let J be an open interval, $V \in C(I \times J)$, $c \in C(I)$, and let $W : C^\infty(J) \to C^\infty(J)$ be linear with $\dim\ker W < \infty$ and

$$W\big(-\partial_x^2 + V(t,\cdot)\big)f = \big(-\partial_x^2 + c(t)\big)Wf \qquad \forall t \in I,\ f \in C_c^\infty(J).$$

Then $V(t,x) - V(s,x) = c(t) - c(s)$ for all t, s, x.

(L^2 version, ENC-P0(b)) The same conclusion holds a.e. for $W \in B(L^2(J))$ independent of t with $\dim\ker W < \infty$ on $L^2(J)$, $WD \subset D$, and $V(t) \in L^\infty$.

Stated in the paper as no. 14.3, p. 75

Idea. Subtracting the intertwining identities at two times gives $W(mf) = 0$ for a function m built from the difference of potentials; if $m \neq 0$ somewhere, the infinitely many independent functions $m\varphi_k$ with disjoint supports lie in $\ker W$.

Proof.

Step 1 **Subtract at two times**

Subtracting the identities at t and s gives $W(mf) = 0$ for all $f \in C_c^\infty(J)$, with

$$m = V(t,\cdot) - V(s,\cdot) - \big(c(t) - c(s)\big),$$

a continuous function.

Step 2 **Contradiction with the finite kernel**

If $m(x^*) \neq 0$ then $m \neq 0$ on an interval $U \ni x^*$. Choose non-zero $\varphi_1, \varphi_2, \dots \in C_c^\infty(U)$ with disjoint supports: the functions $m\varphi_k$ are linearly independent and all lie in $\ker W$, contradicting $\dim\ker W < \infty$. For the L^2 version use measurable sets $E_k \subset \{|m| > \varepsilon\}$ of finite positive measure.

Why it works. finiteness of the kernel is the whole content of "structured"; without it the zero map or the interaction picture $W(t) = U_0(t)U_V(t)^{-1}$ would be an encoder.

■

Uses: none.

Scope

The classical version covers the Volterra encoder of A44 (injective on C^0). Time-dependent encoders do exist: they are interaction pictures $W(t) = U_0(\theta(t))W(0)U_V(t)^{-1}$, and a pointwise representative plus gauge with $\theta = \mathrm{id}$ exists exactly for $V = \alpha(t)x^2 + \beta(t)x + \gamma(t)$ with $\alpha \equiv 0$ (ENC-P0′). The first version's synthesis "exceptions change the time" was false as written (Avron–Herbst).

Dim. A47 **ENC-N1p**

Blind encoders and identifiability: a reduction lemma (ENC-N1′) [proved]

Fix a family $\mathcal{W} = \{W_V\}$ of encoders. A *blind encoder* is a map E from classes of solutions to outputs with $E([u]) = [N_V W_V u']$ for all V and $u' \in \mathcal{S}_V$, $u' \sim u$. Let (C) be coherence: $u \in \mathcal{S}_V$, $u' \in \mathcal{S}_{V'}$, $u \sim u'$ implies $[N_V W_V u] = [N_{V'} W_{V'} u']$, and (S) separation. Then:

(i) a blind encoder exists iff (C) holds;

(ii) identifiability modulo $\sim$ implies (C), hence existence;

(iii) (C)$\wedge$(S) $\iff$ identifiability, by definition;

(iv) (witness: blind without identifiability) in model DM-φ with $\mathcal{V} = \{q_0 + c\}$ and W_{q_0+c} the Volterra encoder of $q_0 - q_0(x_0)$, (C) holds but V is identified only modulo constants;

(v) existence depends on the family: for the strict Volterra family normalised at x_0, (C) fails.

The correct formulation is: *blind encoder* $\iff$ *identifiability modulo the gauge group of the data model*, for a family that respects the gauge.

Stated in the paper as no. 14.3, p. 75

Proof.

Step 1 (i)–(iii)

(i) E exists iff $[N_V W_V u']$ does not depend on the choice of (V, u'), which is (C). (ii) If $u \sim u' \neq 0$ then $V = V'$ and by the single-V remark $u' = e^{i\theta_0} u$ with θ_0 constant; W_V commutes with constant phases. (iii) $\Leftarrow$ is (ii); $\Rightarrow$: if $V \neq V'$ with $u \sim u' \neq 0$, (S) contradicts (C).

Step 2 (iv) The witness

$W_{\hat{q}} L_{q_0+c} = (L_0 + q_0(x_0) + c) W_{\hat{q}}$ with $\hat{q} = q_0 - q_0(x_0)$, and $W_{\hat{q}}$ commutes with phases in t. For $u \in \mathcal{S}_{q_0}$, $u' = e^{-ict} u \in \mathcal{S}_{q_0+c}$ is equivalent to u, and $N_{V'} W_{\hat{q}} u' = e^{i(q_0(x_0)+c)t} e^{-ict} W_{\hat{q}} u \approx N_V W_{\hat{q}} u$.

Step 3 (v) Dependence on the family

For $q_0 = 0$ and $u = e^{-i\lambda t} \cos\sqrt{\lambda}(x - x_0) \in \mathcal{S}_0$, $u' = e^{-ict} u \in \mathcal{S}_c$: $W_0 u = u$ but $W_c u' = e^{-i(\lambda+c)t} \cos\sqrt{\lambda + c}\,(x - x_0)$, not equivalent modulo phases.

Why it works. "blind" just says the output does not remember which potential produced the data; that is coherence, and coherence is implied by, but strictly weaker than, identifiability.

■

Uses: A36, A37, A40, A44.

CHAPTER 9

The universal dictionary and the misfit certificate

In this chapter	
Starting from	$\Omega \subset \mathbb{R}^N$ open, $N = 1 + d$, $R = \mathbb{C}[\tau, \xi]$; a *filtered dictionary* $\mathcal{D}_0 \subset \mathcal{D}_1 \subset \cdots$ of finite sets of operators (atoms), $\mathcal{D}_\infty = \bigcup \mathcal{D}_n$; a *level* n is $\operatorname{span} \mathcal{D}_n = \operatorname{span}\{D_0, \ldots, D_{N_n}\}$; u *fits* at level n if some non-zero c gives $\sum c_i D_i u = 0$, otherwise it has a *misfit*.
Why	to prove what "a dictionary that takes everything into account" can and cannot be: it exists in the sense of approximation (Peetre's theorem), it cannot give exact identification of every operator (Baire), a misfit can be certified by finite data but a fit never can, and a stopping rule exists with a guarantee under a threshold that the data cannot supply.
Role	Dim. A48–A51. They prove Chapter 15 of the paper.

9.1 Universality

Dim. A48 — **DICT-U1**

Every local linear operator is a limit of dictionary elements [proved]

(U1-a) Let $T : C^\infty(\Omega) \to C^\infty(\Omega)$ be linear with $\operatorname{supp} Tu \subset \operatorname{supp} u$. Then $Tu = \sum_\alpha a_\alpha \partial^\alpha u$ with $a_\alpha \in C^\infty(\Omega)$ locally finite (on each compact only $|\alpha| \le r(K)$ occur).

(U1)(a) Let (χ_j) be an explicit locally finite smooth partition of unity subordinate to an exhaustion of Ω, and $\mathcal{D}_n^{\mathrm{loc}} = \{\chi_j(y) y^\beta \partial^\alpha : j, |\alpha|, |\beta| \le n\}$. Every linear local T is a limit, in the topology τ_1 (locally bounded order, coefficients converging in $C^\infty(K)$), of elements of $\operatorname{span}_{\mathbb{Q}(i)} \mathcal{D}_\infty^{\mathrm{loc}}$.

(N0) With $\mathcal{D}_n^{\mathrm{pol}} = \{y^\beta \partial^\alpha : |\alpha|, |\beta| \le n\}$, T is a pointwise limit of operators with polynomial coefficients iff T has globally finite order on Ω.

Stated in the paper as no. 15.1, p. 77

Idea. Peetre's theorem says a linear local operator has locally finite order; then the coefficients are read off by applying T to monomials, and they are approximated by polynomials. N0 uses Banach–Steinhaus: equicontinuity forces a uniform order.

Proof.

Step 1 (U1-a) Peetre's theorem (cited)

If $D : C^\infty \to C^\infty$ is linear with $\operatorname{supp} Ds \subset \operatorname{supp} s$, then for each compact K there is r with $j^r_y s = 0 \Rightarrow Ds(y) = 0$ for $y \in K$ (Peetre 1959–60; Kolář–Michor–Slovák, Thm 19.1; [from cited source]). So $j^r_y u \mapsto Tu(y)$ is well defined and linear on jets, hence $Tu(y) = \sum_{|\alpha|\le r} a_\alpha(y)\partial^\alpha u(y)$. For smoothness of a_α, apply T to $m_\beta = \chi z^\beta/\beta!$ near y: $T(m_\beta)(y) = \sum_{\alpha\le\beta} a_\alpha(y) y^{\beta-\alpha}/(\beta-\alpha)!$ is a triangular system for the a_α with smooth data.

Step 2 (U1)(a) Approximation

Polynomials are dense in $C^\ell(K)$: for g smooth near K, mollify χg with a Gaussian to get an entire function, whose Taylor series converges with all derivatives uniformly on K, then round the coefficients to $\mathbb{Q}(i)$. Let r_j be the order of T on $\operatorname{supp}\chi_j$; choose polynomials $p_{\alpha,j,n}$ with $\|a_\alpha - p_{\alpha,j,n}\|_{C^n(\operatorname{supp}\chi_j)} \le 2^{-n}/(1+\|\chi_j\|_{C^n})$ and put $T_n = \sum_{j\le n} \chi_j \sum_{|\alpha|\le r_j} p_{\alpha,j,n}\partial^\alpha$. On each compact only finitely many χ_j matter, and $T_n \to T$ in τ_1.

Step 3 (N0) Pointwise limits

($\Leftarrow$) If the order is $\le k$ globally, choose polynomials $p_{\alpha,n}$ with $\|a_\alpha - p_{\alpha,n}\|_{C^n(K_n)} \le 1/n$. ($\Rightarrow$) $C^\infty(\Omega)$ is Fréchet, so a pointwise convergent sequence $\{T_n\}$ is equicontinuous (Banach–Steinhaus): on a compact K with interior there are k, K', C with $\sup_K |T_n u| \le C\|u\|_{C^k(K')}$ for all n, u. A T_n with a non-zero coefficient of order $> k$ would contradict this bound; the contradiction is obtained from the principal symbol of the highest-order coefficient at a point of $\operatorname{int} K$ (*condensed*: the explicit test functions are in the package §1.1).

Why it works. a local operator sees a function only through finitely many derivatives at each point (Peetre), so it is a differential operator with smooth coefficients, and smooth coefficients are polynomial limits.

■

Uses: Peetre ([from cited source]), Banach–Steinhaus.

> **Example 9.1 An operator of infinite global order**
>
> On $\Omega = \mathbb{R}$ take $T = \sum_k \varphi_k(x)\partial_x^k$ with $\varphi_k \in C_c^\infty((k, k+1))$ non-zero. It is local and linear, of globally infinite order: it is not a pointwise limit of polynomial-coefficient operators, but it is a limit in $\mathcal{D}^{\mathrm{loc}}$ by (a).

9.2 The negative theorem

Dim. A49 **DICT-N1**

No countable dictionary identifies everything [proved]

(i) (Baire) Let X be an infinite-dimensional F-space and $\{a_k : P_k \to X\}$ countably many continuous maps from σ-compact $P_k \subset \mathbb{R}^{p_k}$. Then $\bigcup_k a_k(P_k)$ is meagre. In particular for any countable dictionary the exactly representable operators form a meagre set in each class of coefficients.

(ii) No finite set of data certifies the fit or the identification (A50, A7, A9).

(iii) (iii) (Non-monotonicity) If $N \geq 2$, $u \in \mathrm{EP}(\Omega)$, $u \not\equiv 0$, $D = \dim R \cdot u$, then for every n with $\binom{n+N-1}{N-1} > D$ and every $L \in \mathrm{Ann}(u)$ with $\deg L \geq 1$: $\mathrm{Ann}(u) \cap R_{\leq n} \supsetneq (L) \cap R_{\leq n}$.

(iv) (iv) A procedure with finite outputs over a finite alphabet (also as a stabilising limit) emits only countably many operators, so it never emits $\partial_t - c\partial_x$ for all but countably many c.

(v) (v) Let $V_0 \subset V_1 \subset \cdots$ be finite-dimensional spaces of differential polynomials in u and $B_n = \{u : \exists P \in V_n \setminus 0,\ P[u] = 0\}$. Then B_n is closed and nowhere dense (so $\bigcup B_n$ is meagre) and is *shy* (Christensen): the generic u satisfies no equation of the dictionary, in the topological sense and in the sense of measure.

Stated in the paper as no. 15.2, p. 78

Idea. (i) A compact set in an infinite-dimensional space has empty interior (Riesz), so countably many images of compacts are meagre. (v) Move u in a "direction of escape" w along which the determinant of the dictionary applied to $u + sw$ is a non-zero polynomial in s, so only finitely many s lie in B_n.

Proof.

Step 1 (i) Baire

Each $a_k(P_k)$ is a countable union of images of compacts, i.e. of compacts. A compact C in a Hausdorff topological vector space of infinite dimension has empty interior: if $C \supset x + V$ with V a neighbourhood of 0, V would lie in a compact neighbourhood of 0, and a locally compact Hausdorff TVS has finite dimension (Riesz). C is closed, hence nowhere dense; apply Baire. (With rational parameters each a_k has countable image, and the conclusion holds without any continuity.)

Step 2 (iii) A dimension count

$Q \mapsto Qu$ on $R_{\leq n}$ has kernel $\mathrm{Ann}(u) \cap R_{\leq n}$ and image of dimension $\leq D$, so $\dim(\mathrm{Ann}(u) \cap R_{\leq n}) \geq \binom{n+N}{N} - D$. Also $\dim((L) \cap R_{\leq n}) = \dim R_{\leq n-\deg L} \leq \binom{n-1+N}{N}$. Since $\binom{n+N}{N} - \binom{n+N-1}{N} = \binom{n+N-1}{N-1} > D$, the first dimension is larger, and $(L) \cap R_{\leq n} \subset \mathrm{Ann}(u) \cap R_{\leq n}$ because $L \in \mathrm{Ann}(u)$. (For $N = 1$ the statement is false: $\mathrm{Ann}(u)$ is principal.)

Step 3 (iv) Countability

Finite strings over a finite alphabet are countable; a stabilised sequence has a limit string; the image of a countable set is countable. The family $\{\partial_t - c\partial_x\}_{c\in\mathbb{R}}$ is uncountable and each member is identified by suitable solutions (C3-T4).

Step 4 (v) Escape directions

Take a degree-adapted basis $P_1, \dots, P_M$ of V_n (top-degree parts P_i^{top} linearly independent; possible by choosing complements in the filtration by degree), $e_i = \deg P_i$. *Closed:* if $u_k \to u$ with $\sum c_i^{(k)} P_i[u_k] = 0$, $|c^{(k)}| = 1$, pass to a convergent subsequence $c^{(k)} \to c$ and use continuity of $P_i[\cdot]$. *Escape:* by the non-degeneracy lemma (ND-NL: if $Q_1, \dots, Q_M$ are linearly independent as functions on $U \times J^r$, there is $w \in C_c^\infty(U)$ with $Q_i[w]$ independent) take w with $P_1^{\mathrm{top}}[w], \dots, P_M^{\mathrm{top}}[w]$ independent and test functionals ℓ_j with $\det(\ell_j(P_i^{\mathrm{top}}[w])) \neq 0$. For any u, $P_i[u + sw] = s^{e_i} P_i^{\mathrm{top}}[w] + (\text{lower in } s)$, so $p_u(s) = \det(\ell_j(P_i[u + sw]))$ is a polynomial in s with non-zero coefficient of $s^{\sum e_i}$. If $u + sw \in B_n$ the $P_i[u + sw]$ are dependent and $p_u(s) = 0$: at most finitely many s. So the complement of B_n is dense (meagre), and the probability measure of sw, $s \in [0, 1]$ uniform, gives $\mu(B_n - u) = 0$ for every u: B_n is shy; a countable union of shy sets is shy.

Why it works. a countable list of finite-dimensional families can only cover a "thin" set, and the thinness is of both kinds, topological and measure-theoretic, because every direction of escape is a polynomial identity in one real parameter.

■

Uses: A2, A7, A9, A50.

What the theorem does not say

"Everything" holds in the sense of approximation (A48) plus a misfit certificate (A50), not as exact identification. For a differential operator with C^∞ coefficients and one data function the statement "no operator is identified" (the first form of N1(vi)) is false; it was withdrawn and replaced by the rank theorem (Chapter 17).

9.3 Fit and misfit

Dim. A50 **DICT-A**

Misfit has a finite certificate; fit never does (Theorem A, MF1–MF3) [proved]

Let $\operatorname{span}\{D_0, \dots, D_N\}$ be a level of linear continuous operators $C^\infty(\Omega) \to \mathcal{D}'(\Omega)$, independent over $\mathbb{C}$; $U_n = \{u : D_0u, \dots, D_Nu \text{ independent}\}$ (misfit), $F_n = C^\infty \setminus U_n$ (fit).

(i) U_n is open, and every $u \in U_n$ has a finite certificate ($N+1$ test functions) valid in a neighbourhood.

(ii) F_n is closed with empty interior: no finite set of *approximate* data certifies a fit.

(iii) For differential atoms and any continuous functionals $\ell_1, \dots, \ell_M$ on $C^\infty(\Omega)$, every $u \in F_n$ has a $v \in U_n$, arbitrarily close, with $\ell_j(v) = \ell_j(u)$: a fit is not certifiable even from *exact* finite data.

Certificates: (MF1) $\operatorname{dist}(v, W) = \max\{|\varphi(v)| : \|\varphi\| \le 1, \varphi|_W = 0\}$; (MF2) in a Hilbert space $r_n^2 = \det G(D_0u, \dots, D_Nu) / \det G(D_1u, \dots, D_Nu)$; (MF3) weak form: independence is equivalent to $\det\langle f_i, \psi_j\rangle \neq 0$ for some test functions from a fixed countable dense set, with robustness $|\sigma_k(A) - \sigma_k(B)| \le \|A - B\|$.

Stated in the paper as no. 15.3, p. 78

Idea. Linear independence is an open condition witnessed by a non-vanishing determinant, so misfit is open and certified by finitely many test functions. A fit is a closed condition with empty interior because locality lets one perturb u on disjoint small balls to restore independence without changing finitely many functionals.

Proof.

Step 1 (MF1) The distance formula

For $w \in W$, $|\varphi(v)| = |\varphi(v - w)| \le \|v - w\|$. Conversely with $\delta = \operatorname{dist}(v, W) > 0$ define $\varphi_0(w + \lambda v) = \lambda\delta$ on $\operatorname{span}(W, v)$; since $\|\lambda v + w\| \ge |\lambda|\delta$, $\|\varphi_0\| \le 1$, and Hahn–Banach extends it with $\varphi(v) = \delta$.

Step 2 (MF2) The Gram quotient

Replacing the first row and column of $G(D_0u, \dots)$ by those of $\mathrm{res} = D_0u - P_W D_0u$ (elementary operations) gives $\operatorname{diag}(\|\mathrm{res}\|^2, G(D_1u, \dots))$, since $\mathrm{res} \perp W$; and $\varphi = \langle \cdot, \mathrm{res}\rangle / \|\mathrm{res}\|$ has $\varphi(D_iu) = 0$, $\varphi(D_0u) = \|\mathrm{res}\|$.

Step 3 (MF3) The weak form

$f_0, \dots, f_N \in \mathcal{D}'$ are independent iff some $\det\langle f_i, \psi_j\rangle \neq 0$: if $\sum c_i f_i = 0$ the columns are dependent; otherwise the image of $\psi \mapsto (\langle f_i, \psi\rangle)_i$ is a proper subspace orthogonal to some $c \neq 0$, giving $\sum \bar{c}_i f_i = 0$. For differential atoms $\langle D_iu, \psi\rangle = \langle u, D_i^t\psi\rangle$: no derivative of u is needed.

Step 4 (i)–(ii)

(i) By MF3, $\det A(u) \neq 0$ for some ψ_j; $u \mapsto A(u)$ is continuous on C^∞ (and in the $\mathcal{D}'$ topology when $D_i^t\psi_j \in C_c^\infty$), so $\det A(v) \neq 0$ nearby. (ii) F_n is the complement of (i); it has empty interior by A49(v) with the non-degeneracy lemma, and a set of approximate data contains a neighbourhood of u.

Step 5 (iii) Exact finite data

Let $Z = \bigcap_j \ker \ell_j$, closed of codimension $\leq M$. The idea is to use locality: perturb u by a function $w \in Z$ built from pieces supported in small disjoint balls, so that the perturbation does not change the finitely many data but restores independence of the $D_i v$, and $v = u + \varepsilon w$ lies in U_n for small $\varepsilon \neq 0$. (*Condensed*: the choice of the balls is in the package §3.3, where a flaw of the first version, "independence on an open set implies independence on each ball", was repaired; numerical check: $u = e^{-t}\sin x$ with fit in $\{\partial_t, \partial_x^2\}$, three functionals, relative $\lambda_{\min} = 4.5 \cdot 10^{-2}$.) The statement is open for non-local atoms.

Why it works. the asymmetry is topological: "$\neq$" is open, "$=$" is closed and thin; and locality lets one repair equality by a perturbation invisible to finitely many measurements.

■

Uses: A49.

Example 9.2 Certifying a misfit

Test whether $u = e^{-t}\sin x$ satisfies a law in the level $\{\partial_t, \partial_x\}$: the Gram matrix of (u_t, u_x) in L^2 is non-singular with $\lambda_{\min} > 0$, so $\varphi = \langle \cdot, \mathrm{res}\rangle / \|\mathrm{res}\|$ is a finite certificate that no first-order law holds. The same u *does* fit the level $\{\partial_t, \partial_x^2\}$ ($u_t = -u$, $u_{xx} = -u$), and no finite data can certify that.

Dim. A51 **DICT-GR**

Stopping rule (G-R) and identification in the limit with a dictionary (I-LIM) [proved]

(G-R) Let n_0 be the least level with an affine fit, $L \in \mathcal{A}_{n_0}$ with $Lu = 0$, and assume (H-ID): $\mathrm{Ann}(u) \cap \mathrm{span}\, \mathcal{D}_n = \langle L\rangle_n$ for all $n \geq n_0$. Then the test R1 ($r_n = \mathrm{dist}(D_0 u, W_n)$) stops exactly at n_0, and the rule R2 (coherence) signals failure when the kernel at a higher level is strictly larger than $\langle L\rangle_n$.

(I-LIM, approximate branch) Let Ω be bounded Lipschitz, $u \in H^m$, $\mathcal{K}_R = \{L = D_0 + \sum_{\alpha\in A} c_\alpha \partial^\alpha : \|c_\alpha\|_{C^{s,1}} \leq R\}$ (compact in C^s), (H-ID) uniqueness in $\mathcal{K}_{R'}$ and (H2) bounded approximability by $\tilde{L}_n \in \mathrm{span}\, \mathcal{D}_n \cap \mathcal{K}_{R'}$ with $\|\tilde{L}_n - L\|_{C^s} \leq \varepsilon_n \to 0$. Then the modulus $\omega(\eta) = \sup\{\|L' - L\|_{C^s} : L' \in \mathcal{K}_{R'},\ \|L'u\|_{L^2} \leq \eta\} \to 0$, every minimiser L_n of $\|L'u\|$ on $\mathrm{span}\, \mathcal{D}_n \cap \mathcal{K}_{R'}$ satisfies $\|L_n - L\| \leq \omega(C_u \varepsilon_n)$, $C_u = \sum_\alpha \|\partial^\alpha u\|_{L^2}$, and with data $\tilde{u}$, $\|\tilde{u} - u\|_{H^m} \leq \delta$: $\|L_n^\delta - L\| \leq \omega(C_u\varepsilon_n + 2C_{R'}\delta)$.

Stated in the paper as no. 15.4, p. 79

Idea. Compactness of $\mathcal{K}_{R'}$ plus uniqueness (H-ID) gives that small residual means close coefficients; the rest is the triangle inequality.

Proof.

Step 1 (a) The modulus tends to 0

Otherwise there are $\varepsilon_0 > 0$ and $L'_j \in \mathcal{K}_{R'}$ with $\|L'_j u\| \to 0$, $\|L'_j - L\| \geq \varepsilon_0$. By compactness $L'_j \to L^*$ in C^s with $L^* \in \mathcal{K}_{R'}$; the map (coefficients)$\mapsto L'u$ is continuous $C^0 \to L^2$ ($\|(L' - L'')u\| \leq \sum \|c'_\alpha - c''_\alpha\|_\infty \|\partial^\alpha u\|_{L^2}$), so $L^* u = 0$ and $\|L^* - L\| \geq \varepsilon_0$, against (H-ID).

Step 2 (b) and (c)

$\|L_n u\| \leq \|\tilde{L}_n u\| = \|(\tilde{L}_n - L)u\| \leq C_u \varepsilon_n$, then apply (a). For data $\tilde{u}$ and $L' \in \mathcal{K}_{R'}$, $\|L'\tilde{u} - L'u\| \leq C_{R'}\delta$ with $C_{R'} = \|D_0\|_{H^m \to L^2} + |A|R'$; hence $\|L^\delta_n u\| \leq \|L^\delta_n \tilde{u}\| + C_{R'}\delta \leq \|\tilde{L}_n \tilde{u}\| + C_{R'}\delta \leq C_u \varepsilon_n + 2C_{R'}\delta$.

Step 3 (G-R) The stopping test

By (H-ID), for $n < n_0$ there is no affine fit, so $r_n > 0$, and at n_0 the fit exists, so $r_{n_0} = 0$ (computed by MF2 on a maximal independent subfamily of W_n). For $n > n_0$ the coherence rule R2 compares $\mathrm{Ann}(u) \cap \mathrm{span}\, \mathcal{D}_n$ with $\langle L \rangle_n$; a strictly larger kernel means u is a closed-form datum or does not identify (A49(iii), A5).

Step 4 When (H-ID) holds

With a single coefficient and $\{\partial^{\alpha_1} u \neq 0\}$ dense, (H-ID) holds for every R' (rank theorem, Chapter 17). For $|A| = 1$, $D_0 = \partial_t$, $|\partial^{\alpha_1} u| \geq \gamma > 0$ and a cone condition on Ω, the modulus is Hölder of exponent $2/(N+2)$: $\|L^\delta_{n(\delta)} - L\|_{C^0} = O((\varepsilon_n + \delta)^{2/(N+2)})$.

Why it works. identification in the limit with a dictionary is a compactness argument; what the data can never supply is the margin: the rate depends on ε_n, the approximability of the true law.

■

Uses: A41, A43, A49, A50.

CHAPTER 10

Phase 10: corrections to the results of Phases 0–5

In this chapter	
Starting from	the setting of Phases 1–3: the commuting-terms class $\mathfrak{B}$ on a bounded rectangle (Chapter 3), and the factor decomposition $\ker P = \sum \ker Q_j$ in Sobolev spaces on bounded Lipschitz regions (Chapter 2).
Why	to record the proofs that changed what Phases 0–5 had claimed: a statement that turned out false as written (the continuity no-go for any operator with two terms), its correct form, and the counterexample showing that density of the sum of kernels fails on regions with holes, which also settles the question "closed but not dense".
Role	Dim. A52–A53. They prove Chapter 16 of the paper. They prevail over the statements of Phases 0–5 with the same identifiers (ledger sections F10-A, F10-C2, F10-TRIAGE).

10.1 The continuity no-go with constant terms

Setting

$\Omega = I \times J$ is a bounded rectangle, $I = (0, T)$, $\mathfrak{B}$ the class of Chapter 3, $t_0 \in \bar{I}$ the base time, $s = t - t_0$. Consider

$$Q = \tau - \sum_{n=1}^{N} a_n(\xi) - \sum_{k=1}^{K} c_k, \qquad a_n \in \mathbb{C}[\xi] \text{ non-constant}, \; c_k \in \mathbb{C},$$

$a = \sum a_n$, $\kappa = \sum c_k$, $Q' = \tau - a(\xi)$. A *term* is a monomial $\lambda_k \xi^{l_k}$; a monomial of degree 0 (linear reaction $\lambda_0 \,\mathrm{id}$) is a term. On $\mathfrak{B} \cap \ker Q$,

$$E^{\mathrm{add}}(u)(t) = u(t) - u_0 - \sum_{n} \left(e^{s a_n(\partial)} u_0 - u_0\right) - \sum_{k} \left(e^{s c_k} - 1\right) u_0,$$

and E' is the same object for Q'.

Dim. A52 **S10A-T1**

P1-Q6: the no-go for E_{int} holds with two non-constant terms, and fails with constants only (S10A-L1, T1, C1) [proved]

(a) (S10A-L1) $M : v \mapsto e^{\kappa(t-t_0)}v$ is a linear bijection $\mathfrak{B} \cap \ker Q' \to \mathfrak{B} \cap \ker Q$ with $c_m\|v\|_{H^m} \le \|Mv\|_{H^m} \le C_m\|v\|_{H^m}$, and for $u = Mv$

$$E^{\text{add}}(u) = E'(v) + B(v), \qquad B(v) = \big(e^{s\kappa} - 1\big)v(t) - \sum_k \big(e^{sc_k} - 1\big)v_0, \quad \|B(v)\|_{L^2} \le C\|v\|_{H^1}.$$

(b) (S10A-T1) For $m \ge 1$, E^{add} is bounded $H^m \to L^2$ iff E' is; if E' is unbounded in H^m for every $m \ge 0$, so is E^{add}; if $N \le 1$, E^{add} is bounded for every $m \ge 1$.

(c) (S10A-C1) For monomial terms of distinct degrees $0 \le l_1 < \cdots < l_M$: E^{add} is unbounded $H^m \to L^2$ for all m and t_0 iff $\#\{k : l_k \ge 1\} \ge 2$; otherwise bounded for every $m \ge 1$.

Consequently the literal statement of P1-Q6, "no continuity for any operator with at least two non-proportional terms", is **false**: reaction–diffusion $u_t = Du_{xx} + ru$ has E^{add} continuous from H^1, with

$$E^{\text{add}}(u) = (e^{sr} - 1)\big(v(t) - v_0\big), \qquad G_\zeta(s) = (e^{sD\zeta^2} - 1)(e^{sr} - 1) \text{ on modes } e^{a(\zeta)t + \zeta x}.$$

Stated in the paper as no. 16.1, p. 80

Idea. A constant term only multiplies the solution by $e^{\kappa s}$ and shifts the interaction by a bounded, lower-order piece; so it neither creates nor destroys the no-go, which comes from the non-constant terms.

Proof.

Step 1 (a) The constants separate

If $u = e^{\kappa s}v$ then $\partial_t u - a(\partial_x)u - \kappa u = e^{\kappa s}(\partial_t v - a(\partial_x)v)$, so $Q(\partial)u = 0 \iff Q'(\partial)v = 0$. The class $\mathfrak{B}$ is stable under multiplication by $e^{\gamma t}$: by Leibniz $e^{\gamma t}\mathfrak{B}_\rho \subset \mathfrak{B}_{\rho+|\gamma|}$, and the H^m estimates follow with $e^{\pm\kappa s}$ smooth and bounded with all derivatives on $\bar{I}$. With $u_0 = v_0$ and $u(t) = e^{\kappa s}v(t)$, substituting in the definition gives $E^{\text{add}}(u) - E'(v) = (e^{\kappa s} - 1)v(t) - \sum_k (e^{sc_k} - 1)v_0 = B(v)$. The bound $\|B(v)\|_{L^2(\Omega)} \le C\|v\|_{H^1(\Omega)}$ uses the trace $\|v_0\|_{L^2(J)} \le C_I\|v\|_{H^1(\Omega)}$.

Step 2 (b) Transfer of boundedness

$E^{\text{add}} \circ M = E' + B$ with B bounded $H^1 \to L^2$ and M an isomorphism of H^m: so E^{add} is bounded iff E' is. If E' is unbounded in H^{m+1}, then (applying this with $m+1 \ge 1$) E^{add} has $\|E^{\text{add}}u_j\|/\|u_j\|_{H^{m+1}} \to \infty$, and $\|u_j\|_{H^m} \le \|u_j\|_{H^{m+1}}$ gives the same blow-up in H^m. For $N = 1$, P1-T2(i) gives $e^{sa_1(\partial)}v_0 = v(t)$, so $E' = 0$ and $E^{\text{add}} \circ M = B$; for $N = 0$, $E'(v) = v(t) - v_0$ is bounded $H^1 \to L^2$ as in (a).

Step 3 (c) Monomial terms

If at least two monomials have degree ≥ 1, the no-go of Chapter 3 (OPA-T3(b), unbounded for every m and t_0) applies to Q', and (b) transfers it to Q. With at most one, (b) gives boundedness. For reaction–diffusion, (a) with $N = 1$ gives $E' = 0$ and the formula; the mode computation is the product $(e^{sD\zeta^2} - 1)(e^{sr} - 1)$ of the two single-term increments (A19).

Why it works. boundedness of the interaction is decided by the non-constant part of the operator; constants only multiply by a bounded exponential.

■

Uses: A19, A21, A22; P1-T2(i), OPA-T3 (Chapter 3).

Example 10.1 The numerical dichotomy

On $\Omega = (0,1)^2$ in H^1: for reaction–diffusion the ratio $R(\zeta)$ decreases, $0.031, 0.0084, 0.0021, 0.00054$ for $\zeta = 5, 10, 20, 40$; for advection–diffusion–reaction $u_t = u_{xx} - u_x + 2u$ it explodes, 0.79, 30.9, $1.7 \cdot 10^5$, $2.0 \cdot 10^{13}$, with the same dichotomy for $t_0 = \frac{1}{2}$ and $t_0 = 1$.

10.2 Holes: the sum of kernels is closed but not dense

Dim. A53 **S10A-P1**

On a region with a hole the sum of kernels is not dense (P2-Q1, C12-P7) [proved]

Let $P = \Delta = 4\partial_z\partial_{\bar{z}}$, $Q_1 = \partial_t + i\partial_x$, $Q_2 = \partial_t - i\partial_x$ (coprime, $\delta(P) = 1$); $\mathcal{H}_1$ are the holomorphic and $\mathcal{H}_2$ the anti-holomorphic functions in H^m.

(i) On the annulus $A = \{1 < |z| < 2\}$ (Lipschitz, not convex, $b_1 = 1$), for every $m \geq 0$, $\log|z| \in \ker_{H^m(A)} \Delta \setminus \overline{\mathcal{H}_1 + \mathcal{H}_2}$: the sum is *not dense*. For every $m \geq 1$, moreover,

$$\ker_{H^m(A)} \Delta = (\mathcal{H}_1 + \mathcal{H}_2) \oplus \mathbb{C} \log|z| :$$

the sum is *closed, of codimension* $1 = b_1$, *and not dense*.

(ii) If Ω is bounded Lipschitz with a cycle $\gamma \subset \Omega$ and a point $a \notin \Omega$ with $\mathrm{Ind}_\gamma(a) \neq 0$, then $\log|z - a| \notin \overline{\mathcal{H}_1 + \mathcal{H}_2}$ for every $m \geq 0$.

Consequently P2-Q1 is **false** on non-convex regions in general (density holds on convex regions, A18), and (i) is a *differential* example of "closed but not dense", the question left open in C12-P7.

Stated in the paper as no. 16.1, p. 80

Idea. A flux through a closed curve is a continuous functional on harmonic functions in L^2; it vanishes on holomorphic and anti-holomorphic functions (their derivative has a single-valued primitive) but not on $\log|z|$.

Proof.

Step 1 The flux functional is continuous

For a smooth cycle $\gamma \Subset \Omega$ and $h \in \ker_{L^2}\Delta$ put $\Phi_\gamma(h) = \int_\gamma \partial_n h\,ds = \int_\gamma *dh$. With $\delta = \operatorname{dist}(\gamma, \partial\Omega)/2$ the mean-value property and Cauchy's gradient estimate on a ball of radius $\delta/2$ give, for $y \in \gamma$,

$$|\nabla h(y)| \le C'\delta^{-2}\|h\|_{L^2(\Omega)}, \qquad \text{hence} \qquad |\Phi_\gamma(h)| \le C_\gamma\|h\|_{L^2} \le C_\gamma\|h\|_{H^m}.$$

(Numerically on the annulus, $\sup_{r=3/2}|\partial_r h|/\|h\|_{L^2} \le 0.78$ on $r^{\pm n}\cos n\theta$, $n \le 8$.)

Step 2 It vanishes on $\mathcal{H}_1 + \mathcal{H}_2$ and not on $\log|z|$

For f holomorphic, $df = f'dz$ and $*dz = -i\,dz$, so $*df = -if'dz$ and $\Phi_\gamma(f) = -i\oint_\gamma f'(z)\,dz = 0$ because f' has the single-valued primitive f; the anti-holomorphic case is conjugate. For $h = \log|z|$ on a circle, $\Phi(h) = 2\pi \ne 0$. By continuity Φ_γ vanishes on the closure of $\mathcal{H}_1 + \mathcal{H}_2$, so $\log|z|$ is not in it. This proves (i)'s first claim for every $m \ge 0$, and (ii) the same way with the cycle γ and the point a.

Step 3 Exact codimension on the annulus for $m \ge 1$

A harmonic h on A has a Laurent form $h = f + \bar g + c\log|z|$ with f, g holomorphic *single-valued* and $c = \Phi(h)/2\pi$. If $\Phi(h) = 0$ then $h = f + \bar g$, and one must check $f, \bar g \in H^m(A)$. Now $f' = \partial_z h \in H^{m-1}(A)$ (uses $m \ge 1$). In polar coordinates, for $1 < r < 2$ and $r_0 = 3/2$,

$$|f(re^{i\theta}) - f(r_0e^{i\theta})|^2 \le |r - r_0|\int_1^2 |f'(\varrho e^{i\theta})|^2\,d\varrho,$$

so $\|f - f(r_0e^{i\cdot})\|_{L^2(A)} \le C\|f'\|_{L^2(A)}$ and $f(r_0e^{i\theta})$ is bounded; the derivatives of f of order $1..m$ are derivatives of $f' \in H^{m-1}$, hence $f \in H^m(A)$; likewise for $\bar g$. So $\ker\Phi = \mathcal{H}_1 + \mathcal{H}_2$ and $\ker_{H^m}\Delta = \ker\Phi \oplus \mathbb{C}\log|z|$, with $\ker\Phi$ closed (kernel of a continuous functional).

Why it works. the obstruction is topological: a hole gives a period, and a period is a continuous functional that the sum of holomorphic and anti-holomorphic pieces cannot see.

■

Uses: A13, A14, A17, A18.

What remains open

(1) Non-Lipschitz regions (for example a disc with a slit, simply connected but not Lipschitz). (2) The exact codimension for general P on a multiply connected region: for the Laplacian the flux gives it; in general one would need continuous "period" functionals in H^m. (3) Closedness on the annulus for $m = 0$. In Phase 10 the Lipschitz simply connected case was proved modulo two cited facts [F1], [F2] (S10A-T3, A18 extension), and the extension of A18 to non-convex regions was shown to be *false* (S10B-C1).

Part III

Phases 0–5: the other results

CHAPTER 11

Phases 0–1: the ODE prototype, identification, decidability

In this chapter	
Starting from	the toolkit of Part I, Chapter 1 (A1–A9). For the companion matrix: $P \in \mathbb{C}[\xi][\tau]$ monic in τ of degree k, $C(\xi)$ its companion matrix (ones on the superdiagonal, last row $(-p_0, \dots, -p_{k-1})$), $v(\tau) = (1, \tau, \dots, \tau^{k-1})^{\mathsf{T}}$.
Why	to prove the remaining results of Chapters 3–5 of the paper: the one-variable prototype, the projectors of the companion matrix, the refinements of identification (evolutions, waves, symmetries, local data) and the decidability results beyond A8–A9.
Role	Dim. B1–B21. Two elementary facts are used throughout: (F0) $C(\xi)v(\tau) = \tau v(\tau) - P(\tau, \xi)e_k$, so $v(\tau_0)$ is an eigenvector when $P(\tau_0, \xi_0) = 0$; (F1′) e_k is a cyclic vector of C, so the minimal polynomial of C is P.

11.1 The one-variable prototype and the companion matrix

Dim. B1 R4-T1

ODE identification [proved]

Let $u \in \ker p(D)$, $D = d/dx$, $\deg p = m$. The minimal annihilator of u has degree $\dim \operatorname{span}\{u, u', u'', \dots\}$, the number of basis functions $x^j e^{\lambda x}$ actually present in u. It equals p iff the Wronskian of $u, \dots, u^{(m-1)}$ does not vanish.

Stated in the paper as no. 3.4.1, p. 15

Proof.

Step 1 The derivative space

In one variable every ideal of $\mathbb{C}[\xi]$ is principal, so $\operatorname{Ann}(u) = (q)$ with q the minimal annihilator, and $\mathbb{C}[\xi]u \cong \mathbb{C}[\xi]/(q)$ has dimension $\deg q$.

Step 2 Which basis functions are present

Write $u = \sum_j q_j(x)e^{\lambda_j x}$. By A2(c) in one variable, $(D - \lambda_j)^{\deg q_j + 1}$ kills the j-th summand and no lower power does, so $q = \prod_j (\xi - \lambda_j)^{\deg q_j + 1}$, of degree equal to the number of functions $x^l e^{\lambda_j x}$ present.

Step 3 The Wronskian

$q = p$ iff $\deg q = m$ iff $u, \dots, u^{(m-1)}$ are linearly independent iff their Wronskian is not identically zero.

Why it works. in one variable the derivative space of u is the quotient by its minimal annihilator.

■

Uses: A2.

Dim. B2 — **II-L1**

First-order reduction as a module equivalence [proof written, not audited]

(a) $\mathcal{R}^{1\times k}/\mathcal{R}^{1\times k}(\tau I - C(\xi)) \cong \mathcal{R}/(P)$, with $\bar{e}_1 \mapsto \bar{1}$. (b) The t-jet map $J : u \mapsto (u, \partial_t u, \dots, \partial_t^{k-1} u)$ is a bijection from $\ker P$ onto the solutions of $(\partial_t - C(\partial_x))U = 0$, with inverse $U \mapsto U_1$, and $J(f(\partial)w) = f(\partial_x, C(\partial_x))J(w)$.

Proof.

Step 1 (a) The presentation

Map $e_i \mapsto \tau^{i-1}$. The rows $\tau e_i - e_{i+1}$ go to 0, and the last row $\tau e_k + \sum_j p_j e_{j+1}$ goes to $P \equiv 0$, so the map passes to the quotient and is onto. In the quotient $e_{i+1} \equiv \tau e_i$, so the module is cyclic, generated by $\bar{e}_1$, and its annihilator is exactly (P).

Step 2 (b) The jet map

If $Pu = 0$, $U = J(u)$ satisfies $\partial_t U_i = U_{i+1}$ and $\partial_t U_k = -\sum_j p_j(\partial_x) U_{j+1}$: this is $(\partial_t - C(\partial_x))U = 0$. Conversely the first $k-1$ equations give $U_i = \partial_t^{i-1} U_1$ and the last gives $P(\partial)U_1 = 0$. Intertwining holds for $f = \tau$ by the equation itself, and for $f = \xi$ trivially.

Why it works. the companion system is a presentation of the same cyclic module $\mathcal{R}/(P)$.

■

Uses: none.

Dim. B3 — **II-T1**

When the companion form exists [proof written, not audited]

(i) $\mathcal{R}/(P)$ is a finitely generated $\mathbb{C}[\xi]$-module iff the leading coefficient of P in τ is a nonzero constant; then it is free with basis $1, \tau, \dots, \tau^{k-1}$ and τ acts by $C(\xi)^{\mathsf{T}}$. (ii) For P, Q monic in τ: $C_P \sim C_Q \iff P = Q \iff \mathcal{R}/(P) \cong \mathcal{R}/(Q)$. (iii) If the principal symbol does not vanish at $(1, 0)$, (i) holds with $k = \deg P$; the converse is false (heat).

Proof.

Step 1 (i)

For monic P, division by P in $\mathbb{C}[\xi][\tau]$ gives the basis. Conversely, if the module is generated by $g_1, \dots, g_n$ over $\mathbb{C}[\xi]$, then $\tau g_i = \sum a_{ij} g_j$, and the determinant trick gives a monic $h(\tau) = \det(\tau I - (a_{ij}))$ killing the module, so $h = Pq$ and $1 = \mathrm{lc}_\tau(P)\,\mathrm{lc}_\tau(q)$.

Step 2 (ii)

$\det(\tau I - C_P) = P$, similarity preserves it, and an isomorphism of modules preserves the annihilator (P).

Step 3 (iii)

The coefficient of τ^m is $\sigma_m(1,0)$. Counterexample to the converse: $P = \tau - \xi^2$ has $\sigma_2(1,0) = 0$ but a companion form of size one.

Why it works. a companion form is a choice of a non-characteristic time direction.

■

Uses: B2.

Dim. B4 — II-P2

The principal symbol sees only infinity [proof written, not audited]

(a) $\sigma_m(P+Q) = \sigma_m(P)$ if $\deg Q < m$: the principal symbol is not a complete invariant. (b) The projective closure of $V(P)$ meets the line at infinity exactly at the zeros of σ_m. (c) σ_m transforms as a binary form under linear changes of variables.

Proof.

Step 1 (a)

The top-degree part does not change. Example: $\partial_t - \partial_x^2$ and $\partial_t - \partial_x^2 + c$ have the same $\sigma_2 = -\xi^2$ but different kernels.

Step 2 (b), (c)

$P^h(\tau, \xi, \zeta) = \sum_d \zeta^{m-d} \sigma_d$, so at $\zeta = 0$ only σ_m remains; the top-degree part commutes with linear substitution.

Why it works. the principal symbol is the restriction of the equation to the line at infinity.

■

Uses: none.

Dim. B5 **II-T2**

Projectors of the companion matrix, and their poles [proved]

Let $P = Q_1 \cdots Q_s$ with Q_j monic in τ and pairwise coprime over $K = \mathbb{C}(\xi)$; $B_j = P/Q_j$, $r_j = \mathrm{Res}_\tau(Q_j, B_j) \in \mathbb{C}[\xi]$.

(a) $r_j \neq 0$, and $U_j Q_j + V_j B_j = r_j$ with $U_j, V_j \in \mathbb{C}[\xi][\tau]$.

(b) $\Pi_j = (V_j B_j)(\xi, C(\xi))/r_j(\xi)$ are orthogonal idempotents commuting with C, summing to I; Π_j projects onto $\ker Q_j(C)$ along $\ker B_j(C)$. At a point ξ_0 with $r_j(\xi_0) \neq 0$ the same holds with kernel $\ker B_j(\xi_0, C_0)$.

(c) The poles of Π_j are exactly the ξ-coordinates of the points of $V(Q_j) \cap V(B_j)$.

(d) On solutions: if $u = \sum_i u_i$ with $u_i \in \ker Q_i$, then $(V_j B_j)(\partial)u = r_j(\partial_x)u_j$.

Stated in the paper as no. 3.4.2, p. 16

Idea. This is the Chinese remainder theorem over the field $\mathbb{C}(\xi)$, evaluated at the companion matrix; the resultant is the common denominator, and it vanishes exactly where two factor curves meet.

Proof.

Step 1 (a) Resultant identity

Q_j and B_j are monic and coprime over K, so their resultant is nonzero, and Cramer's rule on the Sylvester matrix gives $U_j Q_j + V_j B_j = r_j$ with coefficients in $\mathbb{C}[\xi]$.

Step 2 (b) Idempotents

$e_j = V_j B_j / r_j \equiv 1 \pmod{Q_j}$ and $\equiv 0 \pmod{Q_i}$ for $i \neq j$. So $e_j^2 \equiv e_j$, $e_i e_j \equiv 0$, $\sum e_j \equiv 1$ modulo P. Since $P(C) = 0$ (F1′, Cayley–Hamilton), $f \mapsto f(C)$ is a ring homomorphism from $K[\tau]/(P)$, and the $\Pi_j = e_j(C)$ inherit these relations. If $Q_j(C)w = 0$ then $\Pi_j w = w$; and $Q_j(C)\Pi_j = (V_j P)(C)/r_j = 0$. So the image is $\ker Q_j(C)$; $B_j(C)(I - \Pi_j) = (U_j P)(C)/r_j = 0$ gives the kernel.

Step 3 (c) Poles

If no point of $V(Q_j) \cap V(B_j)$ has abscissa ξ_0, the specialised polynomials are coprime and $r_j(\xi_0) \neq 0$. Conversely, if Π_j were holomorphic at a common point (τ_0, ξ_0), its value Π_0 would be an idempotent commuting with the cyclic C_0, hence a polynomial in C_0, hence a sum of spectral projectors; but τ_0 is a root of both Q_j and B_j at ξ_0, and C_0 has a single Jordan block there, so Π_0 cannot both keep and kill it.

Step 4 (d) On solutions

For $i \neq j$, $(V_j B_j)(\partial)u_i = 0$ because $Q_i \mid B_j$. And $(V_j B_j)(\partial)u_j = (r_j - U_j Q_j)(\partial)u_j = r_j(\partial_x)u_j$.

Why it works. the resultant is the universal denominator of the Chinese remainder idempotents.

■

Uses: B2.

Example 11.1 Wave

For $(\tau-\xi)(\tau+\xi)$: $r=2\xi$ and

$$\Pi_1=\begin{pmatrix}\frac{1}{2} & \frac{1}{2\xi}\\ \frac{\xi}{2} & \frac{1}{2}\end{pmatrix},$$

with a pole at $\xi=0$, the abscissa of the crossing point of the two lines.

Dim. B6 **II-T3**

Polynomial projectors iff disjoint varieties [proved]

For $P=P_1P_2$ and any Ω, equivalent: (i) $V(P_1)\cap V(P_2)=\emptyset$; (ii) $(P_1,P_2)=\mathcal{R}$; (iii) $\mathcal{R}/(P)\cong\mathcal{R}/(P_1)\times\mathcal{R}/(P_2)$; (iv) some $E_1\in\mathcal{R}$ is the identity on $\ker P_1$ and zero on $\ker P_2$; (v) $\ker P_1\cap\ker P_2=\{0\}$. Under (ii), $\ker P=\ker P_1\oplus\ker P_2$ on every Ω. **(II-C2)** If (i) holds for nonconstant P_1,P_2, their principal symbols share a linear factor.

Stated in the paper as no. 6.5, p. 37

Proof.

Step 1 (i) $\Longleftrightarrow$ (ii) $\Longleftrightarrow$ (iii)

Weak Nullstellensatz; Chinese remainder theorem for comaximal ideals.

Step 2 (ii)⇒(iv),(v), and the splitting

From $aP_1+bP_2=1$ put $E_1=bP_2$: $E_1u_1=(1-aP_1)u_1=u_1$ and $E_1u_2=0$. Every $u\in\ker P$ splits as $(bP_2)(\partial)u+(aP_1)(\partial)u$, with the two pieces in $\ker P_1$ and $\ker P_2$.

Step 3 (iv),(v)⇒(i)

A common zero (τ_0,ξ_0) gives $w=e^{\tau_0 t+\xi_0 x}\in\ker P_1\cap\ker P_2$, $w\neq0$, contradicting (v) and, via $w=E_1w=0$, (iv).

Step 4 (II-C2)

Two projective plane curves always meet (Bézout). Under (i) they meet only at infinity, that is (B4(b)) at a common zero of the principal symbols.

Why it works. a polynomial projector is a Bézout identity, which exists exactly when the curves never meet.

■

Uses: B4.

Dim. B7 **II-T4**

Only factors give single-valued projectors [proved]

For P monic in τ and a set S of roots $\tau_a(\xi)$ (with full multiplicities), the spectral projector Π_S of $C(\xi)$ is rational in ξ iff $Q_S = \prod_{a\in S}(\tau-\tau_a)^{\mu_a}$ lies in $\mathbb{C}[\xi][\tau]$, that is, iff S is a union of irreducible factors of P. There are exactly 2^r rational projectors, r the number of distinct irreducible factors.

Proof.

Step 1 Rational projector ⇒ rational factor

$\Pi_S = g(C)$ with g of degree $< k$; since $I,\dots,C^{k-1}$ are independent with coefficients in K, Cramer gives $g \in K[\tau]$. Then $Q_S = \gcd(P, g-1)$ (with multiplicities) is computed by Euclid in $K[\tau]$, so $Q_S \in K[\tau]$, and by Gauss in $\mathbb{C}[\xi][\tau]$.

Step 2 Rational factor ⇒ rational projector

B5(b) with $s = 2$.

Why it works. the branches of an irreducible factor are permuted by monodromy, so no single-valued projector separates them.

■

Uses: B5.

Dim. B8 **II-P5**

Normality of the companion matrix [proved]

For a companion matrix C of $p(s) = s^k + \sum_{j<k} p_j s^j$, equivalent: (1) C normal; (2) C has an orthonormal eigenbasis; (3) the roots are simple and $\lambda_a\bar\lambda_b$ is a nontrivial k-th root of unity for all $a \neq b$; (4) $k = 1$; or $k = 2$ with $|p_0| = 1$ and $\bar p_0 p_1 = -\bar p_1$; or $k \geq 3$ with $p = s^k - c$, $|c| = 1$. **The earlier claim "normal iff $p = s^k - c$" is false for** $k = 2$ [false]: $s^2 + s - 1$ has the symmetric companion matrix $\begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}$.

Proof.

Step 1 Normal forces simple roots

A normal matrix is diagonalisable, and a cyclic one (F1′) is diagonalisable only with simple roots.

Step 2 Orthogonality of eigenvectors

With simple roots the eigenvectors are $v(\lambda_a)$, and

$$\langle v(\lambda_a), v(\lambda_b)\rangle = \sum_{j<k} z^j, \qquad z = \lambda_a\bar\lambda_b,$$

which vanishes iff $z \neq 1$ and $z^k = 1$. This proves (1) $\iff$ (3).

Step 3 $k \geq 3$

With three indices, $|\lambda_a|^2 = |\lambda_a\lambda_b||\lambda_a\lambda_c|/|\lambda_b\lambda_c| = 1$, so the roots are λ_1 times all k-th roots of unity and $p = s^k - \lambda_1^k$.

Step 4 $k = 2$

Direct computation:

$$CC^* - C^*C = \begin{pmatrix} 1 - |p_0|^2 & -p_1\bar{p}_0 - \bar{p}_1 \\ -p_0\bar{p}_1 - p_1 & |p_0|^2 - 1 \end{pmatrix}.$$

Why it works. the eigenvectors of a companion matrix are Vandermonde vectors, orthogonal only for roots on a rotated circle of roots of unity.

■

Uses: none.

11.2 Refinements of identification

Dim. B9 — **II-C1**

The t-jet is a cyclic vector [proved]

For P monic in τ of degree k: $\mathrm{Ann}(u) = (P)$ iff no nonzero $(b_0, \dots, b_{k-1}) \in \mathbb{C}[\xi]^k$ has $\sum_j b_j(\partial_x)\partial_t^j u = 0$.

Stated in the paper as no. 3.4.3, p. 16

Proof.

Step 1 Division by P

This is A3(d): every Q is $AP + \sum_{j<k} b_j(\xi)\tau^j$ uniquely, $Q \in \mathrm{Ann}(u)$ iff $\sum b_j(\partial_x)\partial_t^j u = 0$, and $Q \in (P)$ iff all $b_j = 0$.

Why it works. modulo P, every operator is a $\mathbb{C}[\xi]$-combination of the first k time derivatives.

■

Uses: A3.

Dim. B10 **C3-T4**

Evolutions: Hermite interpolation [proved]

Let $P = \tau - a(\xi)$. (a) $\mathrm{Ann}(u) = (P) \iff u \notin \mathrm{EP}$. (b) If $u = \sum_k q_k e^{\langle z_k, \cdot \rangle}$ (distinct z_k, $q_k \neq 0$), $\delta_k = \deg_x q_k$, $D = \sum_k (\delta_k + 1)$ and $\deg a \leq n$, then

$$\{b \in \mathbb{C}[\xi]_{\leq n} : \tau - b \in \mathrm{Ann}(u)\} = a + \prod_k (\xi - \xi_k)^{\delta_k + 1} \cdot \mathbb{C}[\xi]_{\leq n - D}.$$

Stated in the paper as no. 4.4.1, p. 23

Proof.

Step 1 (a)

A3(a) with one factor and $P/P_1 = 1$.

Step 2 (b) Reduction to an ODE in x

$\tau - b = P + (a - b)(\xi)$, so $\tau - b \in \mathrm{Ann}(u)$ iff $c(\partial_x)u = 0$ with $c = a - b$, $\deg c \leq n$.

Step 3 (b) Triangular conditions

By A2(a),(b), $c(\partial_x)u = 0$ iff for each k

$$c(\xi_k + \partial_x) q_k = \sum_j \frac{c^{(j)}(\xi_k)}{j!} \partial_x^j q_k = 0.$$

Since $\partial_x^j q_k$ has x-degree exactly $\delta_k - j$, the system is triangular and equivalent to $c^{(j)}(\xi_k) = 0$ for $j \leq \delta_k$, that is, $\prod_k (\xi - \xi_k)^{\delta_k + 1} \mid c$.

Why it works. each mode with a polynomial prefactor of degree δ fixes $\delta + 1$ Hermite data of the symbol.

■

Uses: A2, A3.

Dim. B11 **C3-T5**

Wave, Klein–Gordon, telegraph, beam [proved]

(a) Wave $P = (\tau - c\xi)(\tau + c\xi)$, $c \in \mathbb{R} \setminus \{0\}$, $u = f(x - ct) + g(x + ct)$: $\mathrm{Ann}(u) = (P) \iff f \notin \mathrm{EP}$ and $g \notin \mathrm{EP}$. (b) $\tau^2 - \xi^2 + \mu^2$ ($\mu \neq 0$) and $\tau^2 + \alpha\tau - c^2\xi^2$ ($\alpha, c \neq 0$) are irreducible: identified iff $u \notin \mathrm{EP}$. (c) Beam $\tau^2 + \xi^4$: identified iff $(\partial_t \pm i\partial_x^2)u \notin \mathrm{EP}$ for both signs.

Stated in the paper as no. 4.2, p. 20

Proof.

Step 1 (a) The partial quotients

$$w_1 = (\partial_t + c\partial_x)u = 2c\,g'(x+ct), \qquad w_2 = (\partial_t - c\partial_x)u = -2c\,f'(x-ct).$$

Step 2 (a) Pull-backs

$F(x-ct) \in \mathrm{EP}(\Omega)$ iff $F \in \mathrm{EP}(J_F)$: if an exponential-polynomial E equals $F(x-ct)$ on Ω, then $(\partial_t + c\partial_x)E = 0$ on Ω, hence everywhere (A2(b)), so $E = F_0(x-ct)$ with $F_0 \in \mathrm{EP}$, and $F = F_0$ on J_F. In one variable $f \in \mathrm{EP} \iff f' \in \mathrm{EP}$. Apply A3(a).

Step 3 (b), (c)

A nondegenerate conic is irreducible (bordered determinant $-\mu^2$, resp. $c^2\alpha^2/4$, nonzero). For the beam apply A3(a) to the two factors.

Why it works. each factor of the wave operator carries one profile, and each profile must be non-exponential.

■

Uses: A2, A3.

Dim. B12 — **C3-C1**

The spectrum of an expansion is not intrinsic [false] (counterexample proved)

The claim "if $u = \sum_{z\in Z} c_z e^{\langle z,\cdot\rangle}$ on a bounded Ω with all $c_z \neq 0$ and $Z \cap V(P_i)$ infinite for each i, then $\mathrm{Ann}(u) = (P)$" is false.

Stated in the paper as no. 4.4.2, p. 23

Proof.

Step 1 The counterexample

$P = \tau + \xi$, $\Omega \subset \{|x-t| < \pi\}$. On $(-\pi, \pi)$,

$$e^s = \sum_{k\in\mathbb{Z}} c_k e^{iks}, \qquad c_k = \frac{\sinh(\pi)\,(-1)^k}{\pi(1-ik)} \neq 0,$$

in L^2, hence in $\mathcal{D}'(\Omega)$. So $u = e^{x-t}$ has infinitely many exponents $(-ik, ik)$ on $\tau + \xi = 0$, but $\tau + 1$, $\xi - 1 \in \mathrm{Ann}(u)$.

Step 2 Diagnosis

The expansion is not unique: $(-1, 1)$ also lies on the line, giving a nontrivial relation. On a window of length $\geq 2\pi$, Fourier uniqueness restores the criterion (C3-P4).

Why it works. on a bounded domain a function has many exponential expansions, so their exponents are not invariants.

■

Uses: none.

Dim. B13 C3-C2

Exponential-polynomial data never identify [proof written, not audited]

If $u \in \mathrm{EP}(\Omega)$ and $P(\partial)u = 0$ with P nonconstant, then $\mathrm{Ann}(u) \neq (P)$.

Proof.

Step 1 Finite versus infinite

$V(\mathrm{Ann}\, u)$ is finite (A2(c)), while $V(P)$ is infinite (A1(F)).

Why it works. finitely many exponentials sit on finitely many points of a curve.

■

Uses: A1, A2.

Dim. B14 C3-P3

Translation symmetry [proof written, not audited]

Let $\ell = \alpha\tau + \beta\xi \neq 0$, $\ell(\partial)u = 0$ and $P(\partial)u = 0$, P nonconstant. (i) If $P \not\sim \ell$, $\mathrm{Ann}(u) \neq (P)$. (ii) If $\ell \nmid P$, $u \in \mathrm{EP}$. (iii) Every dictionary containing the monomials of P and ℓ has a Gram kernel of dimension ≥ 2.

Stated in the paper as no. 4.4.3, p. 24

Proof.

Step 1 (i)

$\ell \in \mathrm{Ann}(u)$; if $\mathrm{Ann}(u) = (P)$ then $P \mid \ell$, so $P \sim \ell$.

Step 2 (ii)

ℓ is irreducible and does not divide P, so they are coprime; $V(\mathrm{Ann}\, u) \subseteq V(P, \ell)$ is finite and $u \in \mathrm{EP}$ (A2).

Step 3 (iii)

P and ℓ are independent elements of the annihilator.

Why it works. a constant-coefficient symmetry is itself an equation, and two independent equations leave only finitely many exponentials.

■

Uses: A1, A2.

Dim. B15 **C3-P4**

Spectral criterion under uniqueness of the expansion

[proof written, not audited]

Let P be squarefree, $u = \sum_{z\in Z} c_z e^{\langle z,\cdot\rangle}$ in $\mathcal{D}'(\Omega)$, $Z \subset V(P)$ countable, $c_z \neq 0$, and assume uniqueness of such expansions (H-uniq). Then $\mathrm{Ann}(u) = \{Q : Q|_Z = 0\}$, and $\mathrm{Ann}(u) = (P)$ iff $Z \cap V(P_i)$ is infinite for every i.

Proof.

Step 1 **Annihilators**

$Q(\partial)u = \sum c_z Q(z) e^{\langle z,\cdot\rangle}$, zero iff $Q|_Z = 0$ by (H-uniq).

Step 2 **(⇐)**

If $Q|_Z = 0$ and $Z \cap V(P_i)$ is infinite, then Q and P_i have infinitely many common zeros, so $P_i \mid Q$ (A1(V)); hence $P \mid Q$.

Step 3 **(⇒)**

If $F = Z \cap V(P_i)$ is finite, $Q = (P/P_i)\prod_{z\in F} \ell_z$ with linear $\ell_z(z) = 0$ vanishes on Z and is not a multiple of P.

Why it works. with a unique expansion, the exponents are the points where annihilators must vanish.

■

Uses: A1.

Dim. B16 **C3-P5**

Local data

[proof written, not audited] (a) / [from cited source] (b)

(a) For $P = \tau + \xi$, $F(s) = e^s + \varphi(s)$ with a nonzero bump $\varphi \in C_c^\infty(2,3)$ and $u = F(x-t)$: on $(0,1)\times(-5,5)$, $\mathrm{Ann}(u) = (\tau+\xi)$; on $(0,1)^2$, $u = e^{x-t}$ and Ann is zero-dimensional. (b) For elliptic P, $\mathrm{Ann}(u|_{\Omega'}) = \mathrm{Ann}(u)$ for every connected $\Omega' \subset \Omega$ (Petrowsky analyticity) [from cited source].

Stated in the paper as no. 4.4.3, p. 24

Proof.

Step 1 **(a)**

If $u \in$ EP on the large rectangle, F would be an exponential-polynomial on $(-6,5)$, analytic, equal to e^s on $(-1,1)$ hence everywhere, but $F \neq e^s$ on $(2,3)$. On the small square $x - t \in (-1,1)$ and $u = e^{x-t}$.

Step 2 **(b)**

$Q(\partial)u$ solves the elliptic equation, so it is analytic; if it vanishes on Ω' it vanishes on Ω.

Why it works. a non-analytic bump can hide outside a smaller window; analyticity forbids it.

■

Uses: A2.

11.3 Decidability and variable coefficients

Dim. B17 **P4-C1**

Restriction does not identify in the Weyl algebra [proof written, not audited]

(a) For every nonzero smooth u of one variable, the action $A_1 \to \mathrm{End}(A_1 \cdot u)$ is injective.
(b) For $u = e^x$ and $\mathcal{L} = \mathrm{span}\{1, \partial\}$ the restriction to $\mathcal{L}$ is injective, yet $\partial - 1 \in \mathrm{Ann}(u) \cap \mathcal{L}$.
(c) The correct object is the evaluation $L \mapsto Lu$, whose kernel is $\mathrm{Ann}(u) \cap \mathcal{L}$.

Stated in the paper as no. 5.6, p. 29

Proof.

Step 1 (a)

The kernel of the action is a two-sided ideal not containing 1; A_1 is simple, so it is 0.

Step 2 (b)

$(\partial - 1)e^x = 0$. If $c_0 + c_1\partial$ acts as zero on $\mathbb{C}[x]e^x$, applying it to xe^x gives $c_0 x + c_1(1 + x) = 0$, so $c_0 = c_1 = 0$.

Why it works. in a noncommutative ring, killing the generator does not kill the cyclic module.

■

Uses: none.

Dim. B18 **OP-ALG-T1**

Decidability in the holonomic model [proved mod. (AC)]

In the analytic holonomic model M3 (and for analytic solutions with D-finite Cauchy data, (P4-T6)), under (AC): "$L_* u = 0$?" is decidable for every operator with polynomial coefficients; "$Q(\partial)u \in \mathrm{EP}$?" is decidable; "$\mathrm{Ann}(u) = (P)$?" and "$V(\mathrm{Ann}\, u) = V(P)$?" are decidable and G_u is computable; a bound B and generators of $\mathrm{Ann}(u)$ are computable.

Stated in the paper as no. 5.5, p. 28

Proof.

Step 1 Zero test

Holonomic functions are closed under the operators involved; a holonomic function given with enough initial values is zero iff those values are zero (OP-ALG-P1), which is a finite check in a computable field under (AC).

Step 2 EP test

For $w = Q(\partial)u$ compute eliminating operators in t alone and in x alone; $w \in \mathrm{EP}$ iff their constant-coefficient parts $a^*(\partial_t)$, $b^*(\partial_x)$ kill w (OP-ALG-L4, P2), two zero tests.

Step 3 Identification and G_u

A3 and A4 with at most $\sum_i e_i$ EP tests.

Step 4 Bound and generators

$w_0 = G_u(\partial)u \in \mathrm{EP}$ has $\mathcal{R}u \cap \mathrm{EP} = \mathcal{R}w_0$ (A5); its eliminants give a finite-dimensional stable space containing $\mathcal{R}w_0$, hence B; A6(ii) and A8(i) give generators.

Why it works. holonomic functions carry finite certificates of their own equations, which supply the bound that a black box lacks.

■

Uses: A3, A4, A5, A6, A8; research file `work/OP-ALG`.

Dim. B19 — **OP-ALG-T3**

The minimal sufficient datum

[proof written, not audited]

The minimal datum that makes "$\mathrm{Ann}(u) = (P)$" decidable in $\mathrm{M2}^+$ is equivalent to a degree D such that, if identification fails, a witness of degree $\leq D$ exists. The bound B of A8 is equivalent to knowing G_u, and it is strictly stronger than the minimal datum on some regions when $\sum_i e_i \geq 2$.

Proof.

Step 1 D suffices and is computable from any sufficient datum

Given D, compare annihilators in degree $\max(D, \deg P)$ (A8(i)). Conversely, from the answer, run the semi-decision A8(ii), which returns a witness degree when identification fails.

Step 2 $B \equiv G_u$

A8(iii) computes G_u from B; conversely, from G_u, $w_0 = G_u(\partial)u$ is an exponential-polynomial and searching annihilators $a(\tau)$, $b(\xi)$ of w_0 gives $B = \deg a \cdot \deg b$.

Why it works. deciding needs only the degree of a witness, while B describes the whole exponential part.

■

Uses: A5, A8.

Dim. B20 **OP-ALG-T4**

Saturated identification criterion

[proof written, not audited]

Let $\mathfrak{D}$ be a ring of operators $\mathfrak{A}[\partial_t]$ with $\mathfrak{A}$ a domain of operators in x, P monic in ∂_t of degree m, $Pu = 0$. Equivalent: (a) $\mathrm{Ann}_{\mathfrak{D}}(u) = \mathfrak{D}P$; (b) the t-jet is free over $\mathfrak{A}$; (c) $\mathrm{rank}_{\mathfrak{A}}(\mathfrak{D}u) = m$; (d) no $Q \notin \mathfrak{D}P$ has Qu in the $\mathfrak{A}$-torsion.

Stated in the paper as no. 5.6, p. 29

Proof.

Step 1 Division

$Q = AP + Q_0$ with $\deg_{\partial_t} Q_0 < m$, $Qu = Q_0 u$, and $Q \in \mathfrak{D}P$ iff $Q_0 = 0$.

Step 2 The equivalences

(a) $\Longleftrightarrow$ (b): a strictly larger annihilator is a nonzero Q_0 of degree $< m$ killing u. (b) $\Longleftrightarrow$ (c): $\mathfrak{D}u \cong \mathfrak{A}^m/N$ with N the module of relations, torsion-free. (b) $\Longleftrightarrow$ (d): a relation is a Q_0 with $Q_0 u = 0$; conversely $bQ_0 u = 0$ with $b \neq 0$ gives the relation bQ_0.

Why it works. as in B9, modulo P everything is a combination of the first m time derivatives.

■

Uses: B9.

Dim. B21 **OP-ALG-C1**

With variable coefficients the finite test fails

[proof written, not audited]

With $\theta = x\partial_x$, $P_1 = \partial_t - \theta$, $P_2 = \partial_t - \theta + 1$, $P = P_1 P_2$, $u_2 = e^{-t}\exp(x^2 e^{2t})$ and $u^{(\lambda)} = (1 + \lambda\partial_x)u_2$: $\mathrm{Ann}(u^{(\lambda)}) \neq \mathfrak{D}P$ for every λ, yet neither partial quotient $P_1 u^{(\lambda)} = -u_2$ nor $P_2 u^{(\lambda)} = \lambda\partial_x u_2$ is torsion. So the literal transfer of A3(a) is false. **(OP-ALG-P5)** In the full Weyl algebra A_2, a holonomic solution never identifies its equation.

Proof.

Step 1 The identities

$P_1 P_2 = P_2 P_1$, $P_2 - P_1 = 1$, $P_1\partial_x = \partial_x P_2$, so $(1 + \lambda\partial_x)P = P_1 Q_\lambda$ with $Q_\lambda = P_2 + \lambda\partial_x P_1$, and $Q_\lambda u^{(\lambda)} = 0$ with $Q_\lambda \notin \mathfrak{D}P$.

Step 2 (P5)

A holonomic u is killed by a nonzero L in x alone, so u is torsion and B20(d) fails with $Q = 1$.

Why it works. without a gcd, annihilators of noncommutative operators are not controlled by finitely many quotients.

■

Uses: B20.

11.4 Results used without proof

C3-P1[gen] Malgrange density: on convex Ω exponential solutions are dense in $\ker P$; finitely many continuous functionals never certify identification. [from cited source]

Malgrange, Ann. Inst. Fourier 6, Chap. I §2 Thm 2 (primary source checked).

II-T9 With variable coefficients in $\mathbb{C}(x)[\partial]$, identification needs the pair (module, cyclic vector). [from cited source]

Ore 1933; van der Put–Singer.

R6-T3 "The companion matrix is normal iff $p = s^m - c$, $|c| = 1$" for every m. [false]

Counterexample $s^2 + s - 1$ in B8; true for $m \geq 3$.

CHAPTER 12

Phase 2: decomposition and regularity

In this chapter	
Starting from	Part I, Chapter 2 (A10–A18): the finite quotient, the twisted Poincaré lemma, the factor decomposition, the regularity criterion and density.
Why	to prove the remaining decomposition results: the one-dimensional lemmas on rectangles, constructive projectors, the Vandermonde formula, interior regularity, the cases where the H^m split holds, the cohomological criterion for regions with holes, and the closedness results for families of transports. The chapter ends with the complements on variable-coefficient identification proved in the same research package.
Role	Dim. B22–B42.

12.1 Rectangles and convex sets

Dim. B22 — **C12-L1**

One-dimensional lemmas on a rectangle [proved]

On $\Omega = I \times J$: (a) $\{\varphi \in \mathcal{D}'(\Omega) : \partial_x \varphi = 0\} = \{c \otimes 1 : c \in \mathcal{D}'(I)\}$; (b) there is a linear $\mathcal{J}$ with $\partial_x \mathcal{J} = \mathrm{id}$ and $[\partial_t, \mathcal{J}] = 0$; (c) the twisted Poincaré lemma holds on Ω; (d) $\{w : D_t w = D_x w = 0\} = \mathbb{C}\, e_{(p,q)}$.

Proof.

Step 1 (a), (b)

Fix $\theta \in C_c^\infty(J)$ with $\int \theta = 1$. Every test function splits as $\psi = \partial_x K\psi + \Psi \otimes \theta$ with $\Psi(t) = \int \psi(t,x)\,\mathrm{d}x$ and $K\psi$ compactly supported. If $\partial_x \varphi = 0$, then $\langle \varphi, \psi \rangle = \langle c, \Psi \rangle$ with $\langle c, \chi \rangle = \langle \varphi, \chi \otimes \theta \rangle$. Define $\langle \mathcal{J}\varphi, \psi \rangle = -\langle \varphi, K\psi \rangle$.

Step 2 (c), (d)

Remove the twist by $e = e_{(p,q)}$. With $w_0 = \mathcal{J}\tilde{\psi}$, $\partial_x(\partial_t w_0 - \tilde{\varphi}) = 0$, so $\partial_t w_0 - \tilde{\varphi} = c \otimes 1$ by (a); subtract a primitive $C \otimes 1$ of $c \otimes 1$. For (d), $\tilde{w} = c \otimes 1$ with $c' = 0$.

Why it works. on a rectangle every equation in one variable can be integrated with the other variable as a parameter.

■

Uses: none.

Dim. B23 **C12-L2**

Finite-dimensional modules on a rectangle [proved]

For a finite-dimensional $\mathcal{R}$-module M and $\mathcal{A}$ on a rectangle: $\operatorname{Ext}^1_{\mathcal{R}}(M,\mathcal{A}) = 0$, $\dim \operatorname{Sol}_{\mathcal{A}}(M) = \dim M$, and every solution is an exponential-polynomial.

Proof.

Step 1 Simple modules

For $M = \mathcal{R}/\mathfrak{m}_z$ the Koszul resolution reduces both claims to B22(c), (d).

Step 2 Induction

A common eigenvector of τ, ξ on M gives a simple submodule S; the long exact sequence for $0 \to S \to M \to M/S \to 0$ and induction give the two claims.

Step 3 Exponential-polynomials

The minimal polynomials $s(\tau)$, $r(\xi)$ of τ, ξ on M kill every solution value, and A2, Steps 5–7, applies.

Why it works. finite modules are built from points, and each point contributes one exponential.

■

Uses: B22, A2.

Dim. B24 **C12-L5**

Functions constant along a direction on a convex set [proved]

If $D \subset \mathbb{R}^2_{(s,r)}$ is open and convex and $\varphi \in \mathcal{D}'(D)$ has $\partial_r \varphi = 0$, then $\varphi = F(s)$ for a unique $F \in \mathcal{D}'(\mathrm{pr}_s D)$; if $\varphi \in L^2_{loc}$, so is F.

Proof.

Step 1 Local form and gluing

On each rectangle $Q \subset D$, B22(a) gives $\varphi = F_Q \otimes 1$. Two rectangles whose projections overlap are joined by a thin rectangle along a segment inside D (convexity), so the F_Q agree and glue.

Step 2 L^2

$F = \int \varphi(\cdot, r)\theta(r)\,dr$ locally, and Cauchy–Schwarz.

Why it works. convexity makes every fibre of the projection connected.

■

Uses: B22.

Dim. B25 **C12-T1**

Factor decomposition on rectangles and convex sets [proved]

On a rectangle (and on every convex set), statements (i)–(iv) of A13 hold.

Proof.

Step 1 Reduction

Rectangles and convex sets are simply connected, so this is a special case of A13. The original proof replaces A12 by B23.

Why it works. as for A13.

■

Uses: A13, B23.

Dim. B26 **C12-P1**

Constructive projectors [proved]

(a) For $s = 2$, $Q_2(\partial)u \in \ker Q_1$, and the components u_1 are exactly the solutions of $\{Q_1 w = 0,\ Q_2 w = Q_2 u\}$, unique up to $\ker Q_1 \cap \ker Q_2$. (b) If all factor curves are disjoint, $u_j = E_j(\partial)u$ with Chinese remainder idempotents $E_j \in \mathcal{R}$, and $u \in H^m \Rightarrow u_j \in H^{m-\deg E_j}$. (c) If $r = AQ_1 + BQ_2 \in \mathbb{C}[\xi]$ and $s = A'Q_1 + B'Q_2 \in \mathbb{C}[\tau]$, every component satisfies

$$r(\partial_x)u_1 = B(\partial)Q_2(\partial)u, \qquad s(\partial_t)u_1 = B'(\partial)Q_2(\partial)u.$$

Proof.

Step 1 (a)

$Q_1 Q_2 u = 0$; if $u = u_1 + u_2$ then $Q_2 u_1 = Q_2 u$, and two solutions differ by an element of $\ker Q_1 \cap \ker Q_2$.

Step 2 (b)

From $A_i Q_i + B_i \prod_{j \neq i} Q_j = 1$ put $E_i = B_i \prod_{j \neq i} Q_j$; then $Q_i E_i = B_i P$ and $\sum_i E_i \equiv 1 \pmod P$. Each derivative costs one Sobolev order.

Step 3 (c)

$r(\partial_x)u_1 = (AQ_1 + BQ_2)u_1 = BQ_2 u_1 = BQ_2 u$.

Why it works. every element of the ideal (Q_1, Q_2) turns into an equation that a component must satisfy.

■

Uses: A11.

Dim. B27 **C12-T2**

Vandermonde formula [proved]

For $P = \prod_{j=1}^{k}(\tau - a_j(\xi))$ with distinct a_j, $V(\xi) = [a_j^l]$, $\Delta = \det V = \prod_{i<j}(a_j - a_i)$:

$$\Delta(\partial_x)\, u_j = \sum_{l=0}^{k-1} \operatorname{adj}(V)_{jl}(\partial_x)\, \partial_t^l u, \qquad \delta(P) = \deg \Delta.$$

The split is unique iff Δ is a nonzero constant, and then $u_j = \Delta^{-1} \sum_l \operatorname{adj}(V)_{jl}(\partial_x)\partial_t^l u$.

Proof.

Step 1 The jet in terms of the components

$\partial_t u_j = a_j(\partial_x)u_j$, so $\partial_t^l u = \sum_j a_j(\partial_x)^l u_j$, that is $U = V(\partial_x)\mathbf{u}$.

Step 2 Inverting

Multiply by $\operatorname{adj} V$ and use $\operatorname{adj}(V)V = \Delta I$.

Step 3 The ambiguity

$\mathcal{R}/(\tau - a_i, \tau - a_j) \cong \mathbb{C}[\xi]/(a_i - a_j)$, so $\delta(P) = \sum_{i<j} \deg(a_i - a_j) = \deg \Delta$.

Why it works. the time derivatives of u are the components weighted by a Vandermonde matrix in the symbols.

■

Uses: A13.

> **Example 12.1** Wave
>
> For $(\tau - \xi)(\tau + \xi)$, $\Delta = -2\xi$ and the formula is d'Alembert's $\partial_x u_+ = \frac{1}{2}(\partial_x u + \partial_t u)$.

Dim. B28 **C12-P2**

Reducibility in second-order time; first order [proved] (Lean-checked algebra)

(C12-P2) $P = \tau^2 + b(\xi)\tau + c(\xi)$ is reducible iff $D = b^2 - 4c$ is a square d^2 in $\mathbb{C}[\xi]$; then $P = (\tau - a_+)(\tau - a_-)$, $a_\pm = (-b \pm d)/2$, and $\delta(P) = \deg d$. (C12-P3) An evolution $\tau - a(\xi)$ has exactly one component.

Proof.

Step 1 (P2)

A factor of degree 0 in τ divides the leading coefficient 1, so a nontrivial factorisation has two monic linear factors $\tau - \alpha$, $\tau - \beta$ with $\alpha + \beta = -b$, $\alpha\beta = c$, $(\alpha - \beta)^2 = D$. And $\delta = \deg(a_+ - a_-)$ by B27.

Step 2 (P3)

A primitive polynomial of degree one in τ is irreducible (Gauss).

Why it works. a quadratic in τ factors exactly when its discriminant is a square.

■

Uses: B27.

> **Example 12.2 Klein–Gordon and telegraph**
>
> For $\tau^2 - \xi^2 + m^2$, $D = 4(\xi^2 - m^2)$ is a square iff $m = 0$; for $\tau^2 + a\tau - \xi^2$, $D = a^2 + 4\xi^2$ is a square iff $a = 0$. So both are single components.

Dim. B29 **C12-T4**

Interior regularity of the components [proved]

(C12-L4) If $p(\partial_x)w$ and $q(\partial_t)w$ lie in H^k_{loc}, with $\deg p = \deg q = D \geq 1$, then $w \in H^{k+D}_{loc}$.
(C12-T4) For coprime, non-comaximal Q_1, Q_2, with r, s as in B26(c), $D = \deg r \deg s$ and e the degree of the operators BQ_2, $B'Q_2$: every component of $u \in H^m_{loc} \cap \ker Q_1 Q_2$ lies in H^{m-e+D}_{loc}.

Proof.

Step 1 (L4) A Fourier estimate

$|p(i\kappa)| \geq c|\kappa|^D - C(1 + |\kappa|)^{D-1}$, and likewise for q, so

$$(1 + \omega^2 + \kappa^2)^{D/2} \leq C\big(1 + |p(i\kappa)| + |q(i\omega)|\big).$$

Localise with cutoffs; each commutator costs one order less, so finitely many steps reach H^{k+D}_{loc}.

Step 2 (T4)

Apply B26(c) with $p = r^{\deg s}$, $q = s^{\deg r}$ in the ideal, then (L4).

Why it works. an ODE in x and an ODE in t together form a quasi-elliptic pair, which regularises.

■

Uses: B26.

Interior regularity is never lost; A15 shows the loss happens only at the boundary.

Dim. B30 **C12-P4**

Two cases where the H^m split holds [proved]

(a) $P = \tau\xi$ on a bounded rectangle, every m: $c_F = 0$ after removing the constants. (b) The wave $\tau^2 - \xi^2$ on the square $(0,1)^2$, every m: $\|u_+\|^2 + \|u_-\|^2 \le C\|u\|^2$ after normalisation, and $c_F < 1$.

Proof.

Step 1 (a)

By B25, $u = A(t) \otimes 1 + 1 \otimes B(x)$. Integrating against $\theta(x)$ shows $A \in H^m$, hence $B \in H^m$. If $\int A = 0$, the H^m product of $A \otimes 1$ and $1 \otimes B$ reduces to the $\alpha = 0$ term $\int A \int B = 0$.

Step 2 (b) Characteristic coordinates

With $s = x + t$, $r = x - t$ the square becomes the rhombus $|r| < s < 2 - |r|$, and $u = F(s) + G(r)$ (B24).

Step 3 (b) Five pieces

On the central square $(\frac{1}{2}, \frac{3}{2}) \times (-\frac{1}{2}, \frac{1}{2})$, averaging against test functions controls F and G by u. On each of the four corner triangles one of the two profiles is already controlled (its argument stays in the central range), so the other is u minus it. The five pieces cover the rhombus.

Why it works. when the two families of characteristics are transversal, every point sees one profile on a long chord.

■

Uses: B24, B25.

Dim. B31 **C12-P5**

Variable coefficients: LCLM instead of the product [proved mod. Ore 1933]

For $L_1, L_2 \in \mathbb{C}(x)[\partial]$ on an interval J without singular points: $\ker \mathrm{LCLM}(L_1, L_2) = \ker L_1 + \ker L_2$ and $\ker \mathrm{GCRD}(L_1, L_2) = \ker L_1 \cap \ker L_2$. The product $L_1 L_2$ does not work: $\ker((x\partial - 1)\partial) \neq \ker(x\partial - 1) + \ker \partial$.

Proof.

Step 1 Inclusions

$G = UL_1 + VL_2$, $L_i = W_i G$, $M = AL_1 = BL_2$ hold as identities of operators, and continuity extends them across the poles of the rational coefficients. So $\ker G = \ker L_1 \cap \ker L_2$ and $\ker L_i \subseteq \ker M$.

Step 2 Dimensions

By Ore's identity $\operatorname{ord} M + \operatorname{ord} G = \operatorname{ord} L_1 + \operatorname{ord} L_2$ [from cited source], so $\dim(\ker L_1 + \ker L_2) = \operatorname{ord} M = \dim \ker M$.

Why it works. in a noncommutative ring the least common left multiple plays the role of the product.

■

Uses: Ore 1933.

12.2 Regions with holes

Dim. B32 **OPF-T4**

Cohomological criterion [proof written, not audited] ((i), (ii) need only sheaf theory)

Let Ω be connected, $\mathcal{N}$ the $\delta(P)$-dimensional space of global exponential tuples with sum 0, $\mathcal{K}_j$, $\mathcal{K}_P$ the solution sheaves. (i) $0 \to \underline{\mathcal{N}}_\Omega \to \bigoplus_j \mathcal{K}_j \xrightarrow{\Sigma} \mathcal{K}_P \to 0$ is exact. (ii) $\operatorname{coker} \Sigma_\Omega \cong \ker\big[\iota : H^1(\Omega;\mathbb{C}) \otimes \mathcal{N} \to \bigoplus_j H^1(\Omega;\mathcal{K}_j)\big]$; so the decomposition holds on Ω iff ι is injective, and $\dim \operatorname{coker} \le \delta(P)\, b_1(\Omega)$. (iii) For the Laplacian on a disc with k holes the cokernel has dimension exactly k [from cited source].

Stated in the paper as no. 6.4, p. 36

Proof.

Step 1 (i) Exactness on small discs

On every disc A13 gives surjectivity and kernel $\mathcal{N}$; restriction between discs is the identity on $\mathcal{N}$, so the kernel sheaf is the constant sheaf $\underline{\mathcal{N}}$.

Step 2 (ii) The long exact sequence

$$0 \to \mathcal{N} \to \bigoplus_j \ker Q_j \xrightarrow{\Sigma_\Omega} \ker P \xrightarrow{\partial} H^1(\Omega;\underline{\mathcal{N}}) \xrightarrow{\iota} \bigoplus_j H^1(\Omega;\mathcal{K}_j).$$

So $\operatorname{coker} \Sigma_\Omega \cong \operatorname{im} \partial = \ker \iota$, and $H^1(\Omega;\underline{\mathcal{N}}) \cong H^1(\Omega;\mathbb{C}) \otimes \mathcal{N}$ has dimension $\delta(P)\, b_1(\Omega)$.

Step 3 (iii)

Elliptic factors are surjective on every planar open set [from cited source], so $H^1(\Omega;\mathcal{K}_j) = 0$ and the cokernel is all of $H^1 \otimes \mathcal{N}$; for the Laplacian $\delta = 1$, with basis $\log|z - a_l|^2$.

Why it works. the obstruction to splitting is the monodromy of the shared exponentials around the holes.

■

Uses: A13; sheaf cohomology.

12.3 Closedness of sums of kernels

Dim. B33 **P2-T3**

Two subspaces: exact constants [proved]

For closed $\mathcal{H}_1, \mathcal{H}_2$ with intersection $\mathcal{I}$: $\ker S = \{(z, -z) : z \in \mathcal{I}\}$; on $(\ker S)^\perp$ the spectrum of S^*S lies in $[1 - c_F, 1 + c_F] \cup \{2\}$ and contains $1 \pm c_F$. The minimal-energy split is stable iff $c_F < 1$, with optimal constant $(1 - c_F)^{-1}$. **(P2-C2)** For $N \geq 2$ subspaces the same holds with γ_N; Tikhonov regularisation converges to the minimal-energy split, and its norm blows up when $\gamma_N = 0$.

Proof.

Step 1 Spectrum

This is A17, Steps 2–3: $S^*S = I + K$ on $A \oplus B$ with $\|K\| = c_F$ and symmetric spectrum, and $S^*S = 2$ on the diagonal of $\mathcal{I}$.

Step 2 Tikhonov

$\omega_\alpha = (S^*S + \alpha)^{-1} S^* u$; in the spectral resolution of S^*S it converges to $S^\dagger u$ when $u \in \operatorname{im} S$, and $\|\omega_\alpha\| \to \infty$ for $u \in \overline{\operatorname{im} S} \setminus \operatorname{im} S$.

Why it works. the sum map is a block operator whose off-diagonal part has norm equal to the cosine of the angle.

■

Uses: A17.

Dim. B34 **P2-C1**

Where regularising the factor split fails [proved]

(a) For two parallel transports on a convex Ω in L^2 the sum is dense in $\ker fg$; if the chords shrink, some u has no L^2 components, and for every such u the Tikhonov split diverges while its residual tends to 0. (b) On the square, for the function of A15, either every regularisation leaves a residual, or $c_F = 1$ and the regularised components diverge.

Proof.

Step 1 (a) Density

In coordinates (t, s), $u = e^{d_1 t} F(s) + e^{d_2 t} G(s)$; truncating F, G to compact intervals of s gives elements of $\mathcal{H}_1 + \mathcal{H}_2$ converging to u.

Step 2 Divergence

B33 with $c_F = 1$ (A16).

Why it works. dense but not closed means: approximable, but only by pieces of unbounded size.

■

Uses: A16, B33.

Dim. B35 **OPF-T5**

Density via contracting symmetries

[proof written, not audited]

Let Ω be bounded and convex, and suppose affine maps $\Phi_\lambda(y) = o + A_\lambda(y-o)$, $A_\lambda \to I$, satisfy $P(A_\lambda^{\mathsf{T}} z) = \kappa_\lambda P(z)$ and $\Phi_\lambda(\bar{\Omega}) \subset \Omega$. Then $\sum_j \ker_{H^m} Q_j$ is dense in $\ker_{H^m} P$ for every m. This covers homogeneous P (the wave on every convex set) and quasi-homogeneous P on axis-parallel rectangles (the beam).

Proof.

Step 1 Shrink, mollify, decompose

$u \circ \Phi_\lambda \to u$ in H^m and solves P (the symmetry). Mollify on a convex neighbourhood of $\Phi_\lambda(\bar{\Omega})$, pull back: a smooth solution on a convex neighbourhood of $\bar{\Omega}$, which splits by A13 into components smooth up to the boundary.

Why it works. a symmetry of P pulls a solution slightly inward without leaving the solution space.

■

Uses: A13.

This is superseded on convex sets by A18, which needs no symmetry.

Dim. B36 **OPF-P6**

Parallel families: pairwise closed iff totally closed

[proof written, not audited]

For $Q_j = \partial_t + c\partial_x - d_j$ (d_j distinct) on a bounded convex Ω in L^2, equivalent: (a) the chords have $\inf \ell > 0$; (b) $\sum_j \mathcal{H}_j$ closed; (c) every pairwise sum closed; (d) some pairwise sum closed.

Proof.

Step 1 Structure

The factors are comaximal, so the split is unique, $u = \sum_j e^{d_j t} F_j(s)$.

Step 2 (a)⇒(b)

The $N \times N$ Gram matrix of $\{e^{d_j t}\}$ on chords of length $\geq \ell_0$ is uniformly positive definite, as in A16, Step 2.

Step 3 (d)⇒(a)

If $\inf \ell = 0$, A16, Step 4, applied to any pair shows that pair is not closed.

Why it works. for parallel transports everything happens chord by chord, and all pairs see the same chords.

■

Uses: A16.

Dim. B37 **OPF-P7**

Separable families: the sum is always closed

[proof written, not audited]

If every factor depends on t alone or on x alone, $P = A(\tau)B(\xi)$ on a bounded rectangle, then $\sum_j \ker_{H^m} Q_j$ is closed in H^m for every grouping.

Proof.

Step 1 **A bounded projector**

With a basis e_l of $\ker A(\partial_t)$ and biorthogonal $\psi_l \in C_c^\infty(I)$, $\pi_A u = \sum_l e_l(t) \int u(t', x)\psi_l(t')\,\mathrm{d}t'$ is a bounded idempotent on H^m with image $\ker_{H^m} A(\partial_t)$; likewise π_B.

Step 2 **Closed sum**

The t-factors sum to $\operatorname{im}\pi_A$ and the x-factors to $\operatorname{im}\pi_B$; two ranges of commuting bounded projectors have a closed sum, $\operatorname{im}(\pi_A + \pi_B - \pi_A\pi_B)$.

Why it works. separated variables give commuting projectors.

■

Uses: B23.

Dim. B38 **OPR2-T2**

Real transports in L^2

[proof written, not audited] *(condensed)*

For real transports $Q_j = v_j \cdot \nabla - d_j$ on a bounded convex Ω, grouped in classes of parallel directions, assume that of any two classes at least one has chords bounded below. Then $\sum_j \ker_{L^2} Q_j$ is closed iff every class with at least two factors has chords bounded below; in particular pairwise closed implies totally closed.

Stated in the paper as no. 7.3, p. 41

Proof.

Step 1 **Class projectors**

For a class with chords bounded below, B36 gives a closed sum with a chord-by-chord projector whose kernel is bounded.

Step 2 **Compact cross products**

For two transversal classes, the product of their projectors has a kernel integrable over the transversal chords, hence is compact, under the hypothesis on the chords.

Step 3 Closed sum

A finite sum of closed subspaces whose projectors have compact pairwise products and trivial intersections is closed (Fredholm argument). If a class with two factors has shrinking chords, B36 shows the class sum itself is not closed.

Why it works. transversal transports interact compactly, so only parallel ones can destroy closedness.

■

Uses: B36; research file `work/OP-R2`.

Dim. B39 **OPR2-T1**

Heat times transport: density [proof written, not audited]

For $P = (\tau - \xi^2)(\tau + c\xi)$ on a rectangle, $\ker_{H^m}(\partial_t - \partial_x^2) + \ker_{H^m}(\partial_t + c\partial_x)$ is dense in $\ker_{H^m} P$ for every m and c; so the components lie in H^m iff $c_F < 1$.

Proof.

Step 1 From A18

A rectangle is bounded and convex, so A18 applies directly. (The original proof used local inward translations at the corners and the hypoellipticity of the heat operator.)

Why it works. as for A18.

■

Uses: A18.

Dim. B40 **OPF5B-P8**

The tangent model of a wedge is closed [proof written, not audited]

On the infinite wedge $C = \{r\omega(\varphi) : 0 < \varphi < \beta\}$, $\beta < \pi$, with pairwise non-parallel homogeneous transports whose lines all cut C in bounded chords, there is $\gamma_* > 0$ with $\|\sum_j f_j\|^2 \geq \gamma_* \sum_j \|f_j\|^2$ for $f_j \in \mathcal{H}_j$.

Proof.

Step 1 Structure

$f \in \mathcal{H}_j$ iff $f(y) = F(n_j \cdot y)$ with $F \in L^2(s\,\mathrm{d}s)$, since the chords have length proportional to s.

Step 2 Mellin transform

The Mellin transform in r is unitary and maps $\mathcal{H}_j$ to $\{F^\sharp(\sigma)\, a_j(\varphi)^{-1+i\sigma}\}$, with $\|a_j^{-1+i\sigma}\|_{L^2(0,\beta)}$ independent of σ.

Step 3 Fibrewise Gram matrices

The norm of a sum becomes an integral over σ of quadratic forms with the Gram matrix $G(\sigma)$ of the functions $a_j^{-1+i\sigma}$, which is positive definite and has a positive infimum of its smallest eigenvalue (the limits $\sigma \to \pm\infty$ decouple by oscillation).

Why it works. dilation invariance diagonalises the problem, leaving one finite Gram matrix per frequency.

■

Uses: none.

12.4 Complements: variable-coefficient identification

Dim. B41 **OPF5B-T2**

Saturated criterion for non-monic P [proof written, not audited]

Let $P = \sum_{j\le m} p_j \partial_t^j$ with $p_m \in \mathfrak{A} \setminus 0$, $\mathfrak{K}$ the Ore localisation. For $Pu = 0$, equivalent: (a′) $\mathrm{Ann}_{\mathfrak{D}}(u) \subseteq \mathfrak{K}P$; (b) the t-jet is free over $\mathfrak{A}$; (c) $\mathrm{rank}_{\mathfrak{A}}\mathfrak{D}u = m$; (d′) no $Q \notin \mathfrak{K}P$ has Qu torsion. **(OPF5B-P3)** If $p_m = \pi(x)$ is a function and $\mathfrak{D}/\mathfrak{D}P$ has no π-torsion, B20 holds verbatim. **(OPF5B-C4)** The radial heat kernel $t^{-1}e^{-x^2/4t}$ identifies the radial heat operator $x\partial_t - \partial_x x \partial_x$; the Bessel modes $e^{-k^2 t}J_0(kx)$ do not.

Proof.

Step 1 Localise

In $\mathfrak{K}$, $P = p_m \hat{P}$ with $\hat{P}$ monic; divide by $\hat{P}$ and clear denominators with the Ore condition to turn a non-multiple annihilator into a nontrivial relation of degree $< m$, as in B20.

Step 2 Rank

Localisation is exact: $\mathrm{rank}\,\mathfrak{D}u = \dim_K \mathfrak{K}/\mathfrak{K}\,\mathrm{Ann}(u) \le m$, with equality iff $\mathrm{Ann}(u) \subseteq \mathfrak{K}P$.

Step 3 (C4)

The right action of P_r on $\mathfrak{D}/x\mathfrak{D} \cong \mathbb{C}[\eta, T]$ is $f \mapsto Tf' - \eta f - \eta^2 f'$, injective (leading coefficient $-(d+1)c_d$), so there is no x-torsion. The heat kernel is not killed by any nonzero operator in x alone (its derivatives are polynomials of growing degree times the Gaussian); the Bessel modes satisfy Bessel's ODE.

Why it works. a non-monic operator behaves like a monic one after inverting its leading coefficient, provided nothing is lost to torsion.

■

Uses: B20.

Dim. B42 **OPF5B-T6**

Repeated socle: no finite test

[proof written, not audited]

Assume some solution u^* identifies P (monic) and $\mathrm{End}_{\mathfrak{D}}(\mathfrak{D}/\mathfrak{D}P)$ contains infinitely many non-injective endomorphisms E_λ with pairwise trivially intersecting kernels. Then for every finite set of tests $\mathcal{T}_f$ the criterion "$\mathrm{Ann}(u) = \mathfrak{D}P$ iff Tu is not torsion for all $T \in \mathcal{T}_f$" fails for some solution. The hypothesis holds for $P = P_1P_2$, $P_2 = P_1 + c$, whenever a nonzero J with $P_1J \in \mathfrak{D}P_2$ exists (example: $\theta = x\partial_x$, $J = \partial_x$).

Proof.

Step 1 **Solutions from endomorphisms (OPF5B-L5)**

$E(\bar{Q}) = \overline{QR}$ with $PR \in \mathfrak{D}P$; $u_R = Ru^*$ solves P, and u_R identifies iff E_R is injective.

Step 2 **Beating a finite test**

Each test T lies in at most one kernel $\ker E_\lambda$; choose λ outside the finitely many used. Then u_λ passes every test but does not identify.

Step 3 **The example (OPF5B-P7)**

$\mathfrak{D}/\mathfrak{D}P \cong \mathfrak{D}/\mathfrak{D}P_1 \oplus \mathfrak{D}/\mathfrak{D}P_2$ via $\bar{Q} \mapsto (\bar{Q}, \bar{Q})$, and $E_\lambda(x_1, x_2) = (0, x_2 + \lambda J x_1)$ has kernel $\{(x_1, -\lambda J x_1)\}$, pairwise trivially intersecting.

Why it works. a repeated simple factor offers infinitely many ways to fail, more than any finite list of tests can detect.

■

Uses: B20.

12.5 Results used without proof

C12-P6	No orthogonality of components at fixed time in general.	[from cited source]

Shohat, Amer. J. Math. (checked, CIT-AUDIT-2).

C12-Q2	Closedness of the sum for $N \geq 3$ general differential factors.	[open]

Solved for real transports (B38), separable families (B37), wedges in the tangent model (B40).

OPF-Q3	Density for quasi-homogeneous P on non-rectangular convex sets.	closed by A18

A18 covers every bounded convex set.

CHAPTER 13

Phase 3: the interaction term

In this chapter	
Starting from	Part I, Chapter 3 (A19–A24): the class $\mathfrak{B}$, the closed form, the cost, the instability, Dyson–Phillips and the no-go for infinite adjoint orbits.
Why	to prove the supporting lemmas of the class $\mathfrak{B}$, the results on factors and second-order evolutions inside $\mathfrak{B}$, the solvable and non-solvable Lie cases, the interaction created by a boundary, the exact frontier between continuous and discontinuous interaction, the balance decomposition, and the residual-based no-go results.
Role	Dim. B43–B60.

13.1 The class $\mathfrak{B}$

Dim. B43 P1-L1

Properties of $\mathfrak{B}$ [proved]

(a) B_ρ, $\mathfrak{B}_\rho$ are Banach spaces and $\|\partial_x\|_{B_\rho \to B_\rho} \le \rho$. (b) $\mathfrak{B}_\rho \hookrightarrow H^m$ for every m. (c) Every exponential-polynomial lies in $\mathfrak{B}$. (d) For $u \in \mathfrak{B}_\rho$, $t \mapsto u(t,\cdot)$ extends to $\bar{I}$ and lies in $C(\bar{I}; B_{\rho'})$ for $\rho' > \rho$.

Stated in the paper as no. 8.2, p. 44

Proof.

Step 1 (a)

A Cauchy sequence has all derivatives Cauchy in L^2; the limits satisfy $g_{j+1} = g_j'$, so $g_0 \in H^\infty$ with $g_0^{(j)} = g_j$, uniformly in j. Shifting the index gives $\|f'\|_{B_\rho} \le \rho \|f\|_{B_\rho}$.

Step 2 (b), (c)

$\|u\|_{H^m}^2 \le \binom{m+2}{2} \max(1,\rho)^{2m} \|u\|_{\mathfrak{B}_\rho}^2$. For $x^l e^{\zeta x}$ the j-th derivative is $e^{\zeta x}$ times a polynomial with coefficients $O(j^l |\zeta|^j)$, and $j^l (|\zeta|/\rho)^j$ is bounded for $\rho > |\zeta|$.

Step 3 (d)

$H^1(I; L^2(J)) \hookrightarrow C(\bar{I}; L^2(J))$ applied to $\partial_x^j u$, with the geometric decay $(\rho/\rho')^j$.

Why it works. $\mathfrak{B}$ is the space where derivatives grow at most geometrically, so exponentials of polynomials in ∂_x converge.

■

Uses: none.

Dim. B44 P1-L2

Flows of terms in $\mathfrak{B}$ [proved]

On B_ρ: $\|A(\partial_x)\| \le |A|(\rho)$; $e^{sA(\partial_x)}$ converges in norm with $\|e^{sA}\| \le e^{|s||A|(\rho)}$; the family is entire in s, a group, independent of ρ; exponentials of different polynomials commute and $e^{s(A+A')} = e^{sA}e^{sA'}$; and $e^{sA(\partial_x)}(x^l e^{\zeta x}) = \partial_\zeta^l\big(e^{sA(\zeta)}e^{\zeta x}\big)$.

Proof.

Step 1 Norm and commutation

$\|A(\partial)\| \le \sum |\alpha_l|\|\partial\|^l$. Polynomials in one operator commute, and in a Banach algebra $e^{X+Y} = e^X e^Y$ for commuting X, Y (Cauchy product).

Step 2 Action on modes

$\zeta \mapsto e^{\zeta x} \in B_\rho$ is holomorphic for $|\zeta| < \rho$ and $A(\partial)^k e^{\zeta x} = A(\zeta)^k e^{\zeta x}$; a bounded operator commutes with ∂_ζ^l.

Why it works. on $\mathfrak{B}$ the derivative is a bounded operator, so its polynomials have exponentials.

■

Uses: B43.

Dim. B45 P1-L3

The Cauchy problem in the class [proved]

(P1-L3) A continuous curve v in B_ρ with $(\partial_t - A(\partial_x))v = 0$ is $v(t) = e^{(t-t_0)A}v(t_0)$; conversely these are solutions in $\mathfrak{B}$. (P1-L4) For $r \in \mathbb{C}[\xi]$ the Volterra primitive $w = \int_{x_0}^{x} G(x-y)F(t,y)\,dy$ (G the fundamental solution of $r(\partial)$) satisfies $r(\partial_x)w = F$ and maps $\mathfrak{B}_\rho$ into $\mathfrak{B}_R$. (P1-L5) Formal lemmas in $\mathcal{A}[[t]]$: $e^{gX}Ye^{-gX} = \sum_k g^k \mathrm{ad}_X^k Y/k!$, and for $U = \prod_i e^{g_i X_i}$, $U'U^{-1} = \sum_i g_i' \,\mathrm{Ad}(P_{i-1})X_i$.

Proof.

Step 1 (L3)

Testing the equation against $\varphi \otimes \psi$ shows that the distributional time derivative of v, as a B_ρ-valued distribution, is $A(\partial)v$; a Banach-valued distribution with zero derivative is constant, so $\frac{d}{dt}\big(e^{-(t-t_0)A}v\big) = 0$.

Step 2 (L4)

Variation of constants gives $r(\partial_x)w = F$. For $j \ge \delta = \deg r$, $\partial_x^j w$ is expressed through $\partial_x^{j-\delta}F$ and lower derivatives of w; induction gives $\sup_i \rho^{-i}\|\partial_t^i \partial_x^j w\| \le KR^j$.

Step 3 (L5)

Both sides of each identity solve the same formal ODE with the same initial value.

Why it works. uniqueness in $\mathfrak{B}$ is what excludes Tychonoff and boundary-fed solutions.

■

Uses: B43, B44.

Dim. B46 P1-P1

Single flows are not determined without a class [proved] / [from cited source]
(Tychonoff)

(a) For $a_n = \xi$ the solutions with a given datum on J are $F(x+t)$ for any extension F: infinitely many. (b) For $a_n = \xi^2$ there are nonzero heat solutions with zero datum; those with datum u_0 form an infinite-dimensional affine space. (c) So without a class E_{int} is not a function of u; in $\mathfrak{B}$ it is unique.

Stated in the paper as no. 8.4, p. 47

Proof.

Step 1 (a)

In coordinates $(t, y = x + t)$ the equation is $\partial_t w = 0$ on a parallelogram, so $w = F(y)$, fixed on J and free elsewhere.

Step 2 (b)

Tychonoff's $v(t,x) = \sum_k g^{(k)}(t)\, x^{2k}/(2k)!$ with $g(t) = e^{-t^{-2}}$ for $t > 0$, 0 otherwise, is smooth and vanishes for $t \le 0$ [from cited source]. Its time shifts $v(t-\sigma, x)$ are independent: on the first interval only the earliest survives, and $v(s,0) = g(s) \neq 0$.

Step 3 (c)

B45(L3).

Why it works. on a bounded domain both the transport and the heat flow need information that the datum does not contain.

■

Uses: B45.

13.2 Factors and second-order evolutions in $\mathfrak{B}$

Dim. B47 **P1-P2**

Factors in $\mathfrak{B}$ [proved]

On a bounded rectangle, $u \in \mathfrak{B} \cap \ker P$: (a) if the factor curves are disjoint, the components $E_j(\partial)u$ lie in $\mathfrak{B}$; (b) if one decomposition lies in $\mathfrak{B}$, all do; (c) for $P = \prod_j(\tau - a_j(\xi))$, every decomposition lies in $\mathfrak{B}$. **(P1-P3)** The interaction per term inside each factor is then defined modulo a finite-dimensional space of exponential-polynomials, of dimension at most $\delta(P)$.

Proof.

Step 1 (a), (b)

$\mathfrak{B}$ is closed under derivatives; the ambiguity consists of exponential-polynomials, which lie in $\mathfrak{B}$.

Step 2 (c)

By B27, $\Delta(\partial_x)u_j = F_j \in \mathfrak{B}$. B45(L4) gives $w \in \mathfrak{B}$ with $\Delta(\partial_x)w = F_j$; then $h = u_j - w \in \ker\Delta(\partial_x) = \mathcal{D}'(I) \otimes V_\Delta$, and $(\partial_t - a_j)h \in \mathfrak{B}$ turns the coefficients into solutions of a linear ODE system in t with $\mathfrak{B}$ data, hence of geometric growth.

Step 3 (P3)

Changing decomposition adds $(w_j) \in \mathcal{N}$, which changes each interaction term linearly by an exponential-polynomial.

Why it works. the Vandermonde formula expresses each component through derivatives of u, up to an ODE in x that preserves geometric growth.

■

Uses: B27, B45.

Dim. B48 **P1-P4**

Second-order factors: propagator and Gauss form [proved]

For $Q = \tau^2 + b(\xi)\tau + c(\xi)$ and $u \in \mathfrak{B}_\rho \cap \ker Q$: (a) $U = (u, \partial_t u)$ satisfies $U(t) = e^{(t-t_0)C(\partial_x)}U(t_0)$, with e^{tC} given by the canonical fundamental system φ_1, φ_2 of $\varphi'' + b(\zeta)\varphi' + c(\zeta)\varphi = 0$; (b) for $|t| < T^*(\rho')$, with $T^* \geq \log 2/K(\rho')$, the Gauss factorisation

$$e^{tC} = \begin{pmatrix} 1 & 0 \\ \varphi_1'/\varphi_1 & 1 \end{pmatrix} \begin{pmatrix} \varphi_1 & 0 \\ 0 & e^{-tb}/\varphi_1 \end{pmatrix} \begin{pmatrix} 1 & \varphi_2/\varphi_1 \\ 0 & 1 \end{pmatrix}$$

holds. **(II-T7)** Mode by mode, the Wei–Norman (Gauss) form is global iff $\varphi_1(t;\zeta)$ has no zero.

Proof.

Step 1 (a)

B45(L3) for systems; the entries of $e^{tC(\zeta)}$ solve the scalar ODE in t with the canonical initial values.

Step 2 (b)

Multiply out the three matrices and use the Wronskian $\varphi_1\varphi_2' - \varphi_1'\varphi_2 = e^{-tb}$. An entire function without zeros on $|\zeta| \le \rho'$ is invertible in the closed algebra generated by ∂_x; $\varphi_1(t;\zeta) = 1 + O(t^2)$ has no zero for $|t| \le \log 2/K$.

Why it works. a 2×2 matrix factors into lower, diagonal and upper parts exactly where its top-left entry does not vanish.

■

Uses: B45.

13.3 Non-commuting terms

Dim. B49 **P1-T4**

Solvable Lie algebra: exact global formula [proved] (Lie's theorem [from cited source])

If the terms generate a solvable Lie algebra $\mathfrak{g}$ of dimension d, then for every $A \in \mathfrak{g}$ there are entire functions $g_1, \dots, g_d$, obtained by d quadratures, with $e^{tA} = \prod_{i=1}^{d} e^{g_i(t)X_i}$; hence $E_{\mathrm{int}}^{\mathrm{mult}}$ has a closed form with d scalar functions.

Stated in the paper as no. 8.7, p. 50

Proof.

Step 1 A flag of ideals

By Lie's theorem there is a basis with $\mathfrak{i}_k = \mathrm{span}\{X_k, \dots, X_d\}$ ideals [from cited source]; ad_{X_j} acts on $\mathfrak{i}_k/\mathfrak{i}_{k+1}$ by a scalar λ_{jk}.

Step 2 A triangular Wei–Norman system

By B45(L5), $\mathrm{Ad}(P_{i-1})X_i = \sum_{l \ge i} M_{il} X_l$ with entire M_{il} and $M_{ii} = \exp(\sum_{j<i} \lambda_{ji} g_j) \neq 0$. The system $\sum_{i \le l} g_i' M_{il} = c_l$ is triangular: $g_1 = c_1 t$, and each g_l is a quadrature of an entire function.

Why it works. a solvable algebra can be climbed one ideal at a time, each step a scalar ODE.

■

Uses: B45.

Dim. B50 **P1-P5**

Non-solvable: local only [proved]

For every finite-dimensional $\mathfrak{g}$ the Wei–Norman factorisation exists with analytic germs g_i near $t = 0$; globality can fail: in $\mathfrak{sl}_2$ with $A = E - F$ in Gauss order, $g_{21}(t) = -\tan t$. **(P1-C1)** The earlier claim "infinite Lie algebra ⇒ no finite factorisation of $E_{\text{int}}^{\text{mult}}$" is false as stated [false]: $E_{\text{int}}^{\text{mult}} = e^{-tT_2}e^{-tT_1}e^{t(T_1+T_2)}$ is always a product of three exponentials.

Proof.

Step 1 **Local existence**

The map $g' \mapsto \sum_i g_i' \mathrm{Ad}(P_{i-1}(g))X_i$ is given by an entire matrix $\Xi(g)$ with $\Xi(0) = I$; the analytic ODE $g' = \Xi(g)^{-1}c$ has a local solution.

Step 2 **The example**

Direct computation in Mat_2: the Gauss coordinates of $e^{t(E-F)}$, a rotation, involve $\tan t$.

Step 3 **(C1)**

Definition of $E_{\text{int}}^{\text{mult}}$; the right question is A24.

Why it works. coordinates of the second kind exist near the identity but can run into singularities.

■

Uses: B45.

13.4 Boundaries

Dim. B51 **P1-P6**

A boundary creates an interaction of order one [proved]

On $L^2(0, L)$ with Dirichlet diffusion $A_1 = D\partial^2$, inflow transport $A_2 = -c\partial$ and the advection–diffusion generator A: $S_2(t) = 0$ for $t \geq L/c$, while $S(t)$ is injective. So for $t \geq L/c$, $E_{\text{int}}^{\text{mult}}(t) = S(t)$: the relative splitting error is 1 for every nonzero datum.

Stated in the paper as no. 8.6, p. 49

Proof.

Step 1 **The transport empties the interval**

$S_2(t)f(x) = f(x - ct)\mathbf{1}_{x>ct}$, which is 0 for $ct \geq L$.

Step 2 The full flow is injective

A is similar, by $e^{-cx/2D}$, to a self-adjoint operator, so $S(t)$ is an analytic semigroup of self-adjoint type, injective.

Why it works. the boundary condition belongs to the sum of the terms and cannot be shared between them.

■

Uses: none.

Dim. B52 **OPA-T2**

Boundary interaction: sharp rate [proved]

For advection–diffusion with Dirichlet conditions on $(0, L)$ and a datum $u_0 \in C^{k+1}$ vanishing at the ends, with contact order k (first nonzero derivative at an endpoint):

$$\|E(t)\| = K_k\, t^{k/2+3/4}\big(1 + O(t^{1/2})\big), \qquad K_k = |c|\, N_{k-1}\, D^{k/2-1/4}\, \frac{\sqrt{u_0^{(k)}(0)^2 + u_0^{(k)}(L)^2}}{(k-1)!},$$

for the first-order interaction (OPA-T2) and, with $\tilde{N}_k = kN_{k-1}$, for the full $E_{\mathrm{int}}^{\mathrm{mult}}$ (OPF5A-T5). Uniformly: $\|E(t)\| \le 1.301\,|c|D^{1/4}\|u_0'\|_\infty t^{5/4}$. Flat data give $O(t^p)$ for every p. **(P3-T3)** The first-order matrix elements are $V_{mn} = -4c\,mn/(L(m^2-n^2))$ for $m+n$ odd and 0 otherwise (a reflection symmetry).

Proof.

Step 1 A commutator identity

The first-order interaction is $\int_0^t S_1(t-s)[\,\cdot\,]S_1(s)\,ds$ applied to Vu_0; the commutator of $V = -c\partial$ with the Dirichlet heat flow is a boundary term involving u_0' at the ends (OPA-L2).

Step 2 Localisation at the ends

Near each end the Dirichlet heat kernel is that of the half-line, and $u_0(y) = \gamma y^k/k! + O(y^{k+1})$.

Step 3 Scaling

With $x = \sqrt{Dt}\,w$ the boundary term becomes $t^{k/2+3/4}$ times a universal profile, whose L^2 norm is N_{k-1}; the corrections are $O(t^{1/2})$ relative.

Step 4 Full interaction

A conjugation reduces $E_{\mathrm{int}}^{\mathrm{mult}}$ to the same boundary terms plus a remainder $O(t^{k+5/4})$ and an exponentially small image term (OPF5A-L5).

Why it works. the boundary interaction is born where the datum meets the wall, and the heat flow sees the wall at distance $\sqrt{Dt}$.

■

Uses: A23; research files `work/OP-ANA` §3, `work/OP-F5A` (L5, T5).

13.5 Where the interaction is continuous

Dim. B53 **OPA-T3**

The exact frontier of discontinuity [proved]

For an evolution $\tau - \sum_n a_n(\xi)$ on a bounded rectangle: (b) if the terms are at least two distinct non-constant monomials, $u_0 \mapsto E_{\mathrm{int}}^{\mathrm{add}}$ is unbounded from $(\mathfrak{B} \cap \ker Q, H^m)$ to L^2, for every m and every base time. **(OPA-P3)** If the terms are affinely proportional, $a_n = \theta_n a + \kappa_n$ with real θ_n, and $t_0 + \theta_n(t - t_0) \in \bar{I}$ (TC), then

$$E_{\mathrm{int}}^{\mathrm{add}}(u)(t) = u(t) + (N-1)u(t_0) - \sum_n e^{(t-t_0)\kappa_n} u\big(t_0 + \theta_n(t-t_0)\big),$$

which is continuous in H^1; without (TC) it is unbounded. The general non-proportional case (e) is proved modulo the Turán–Nazarov inequality [from cited source].

Proof.

Step 1 Exponential test data

On $u = e^{a(\xi)t + \xi x}$ with real ξ the interaction is $e_{\geq 2}(z)\, u$, $z_n = e^{s a_n(\xi)} - 1$ (A19). If some term grows faster than the full symbol along a ray of ξ (condition (S)), the ratio to any H^m norm grows exponentially, as in A21.

Step 2 Proportional terms

$e^{s a_n(\partial)} = e^{s\kappa_n} e^{s\theta_n a(\partial)}$, and $e^{\tau a(\partial)} u_0 = u(t_0 + \tau)$ for $t_0 + \tau \in \bar{I}$: the flows of the terms are time-rescaled copies of the true solution, controlled by $\|u\|_{H^1}$.

Why it works. the interaction is continuous only when every term is a rescaled copy of the whole equation.

■

Uses: A19, A21, B43.

Dim. B54 **P3-T6**

Balance decomposition [proof written, not audited]

For $u \in H^m(\Omega)$ with $(\partial_t - \sum_i a_i(\partial_x))u = 0$, $m \geq \max(\deg a_i, 1)$, put $u^{(i)} = \int_{t_0}^{t} a_i(\partial_x)u(s)\,\mathrm{d}s$. Then

$$u = u_0 \otimes 1 + \sum_i u^{(i)}, \qquad \|u^{(i)}\|_{H^{m-d_i}} \leq C\|u\|_{H^m}.$$

In $\mathfrak{B}$ the $u^{(i)}$ solve a coupled Faddeev system whose Neumann series converges on every bounded time interval, with $\|\mathbb{K}^n\| \leq (\kappa_0|I|)^n/n!$.

Proof.

Step 1 Balance

Integrate the equation in time: $u(t) - u_0 = \sum_i \int_{t_0}^{t} a_i(\partial_x)u$. Each term loses d_i derivatives in x and gains one in t.

Step 2 Faddeev system

$\partial_t u^{(i)} = a_i(\partial)u = a_i(\partial)(u_0 + \sum_j u^{(j)})$, so $u^{(i)} = G_i a_i(\partial)(u_0 + \sum_{j\neq i} u^{(j)})$ with the Green operator G_i of $\partial_t - a_i$; the off-diagonal Volterra operators have the factorial bound.

Why it works. a split that uses the true solution, not the single-term flows, inherits its continuity.

■

Uses: B44.

13.6 Regularisation of the term split

Dim. B55 **P2-T1**

Every regulariser has the same limit residual [proved]

Let $S : W \to H$ be linear with closure of the range M, R any regulariser with $S(\operatorname{dom} R)$ dense in M, and ω_α near-minimisers of $\frac{1}{2}\|S\omega - u\|^2 + \alpha R(\omega)$ (errors $\varepsilon_\alpha \to 0$). Then $u - S\omega_\alpha \to P_M^{\perp} u$, so the limit residual is $\operatorname{dist}(u, M)$ for every regulariser. **(P2-L1)** If Q kills M, then $\|E^{\perp}(u)\|_{H^m} \geq \|Q(\partial)u\|_{H^{m-k}}/C_Q$.

Proof.

Step 1 Upper bound

For any $v \in \operatorname{dom} R$, $\frac{1}{2}\|S\omega_\alpha - u\|^2 \leq \frac{1}{2}\|Sv - u\|^2 + \alpha R(v) + \varepsilon_\alpha$; let $\alpha \to 0$, then choose Sv close to $P_M u$.

Step 2 Lower bound

$\|u - S\omega\| \geq \operatorname{dist}(u, M)$ always; Pythagoras in $u = P_M u + P_M^{\perp} u$ gives norm convergence.

Step 3 (L1)

$Q(\partial)u = Q(\partial)E^{\perp}(u)$ and $Q(\partial)$ is bounded $H^m \to H^{m-k}$.

Why it works. a regulariser only chooses among representations; it cannot reach what lies outside their closure.

■

Uses: none.

Dim. B56 **P2-P3**

The regulariser acts only on the ambiguity [proved]

The exact representations of u form $\omega^\dagger + \ker S$. For convex, lower semicontinuous, coercive R the regularised solutions converge (along subsequences) to an R-minimal exact representation, to $S^\dagger u$ for $R = \frac{1}{2}\|\cdot\|^2$. On a rectangle $\ker S$ has dimension $\delta(P)$ for factors and $\sum_{n<m} \deg(a_n - a_m)$ for terms. **(P2-P2)** For ODEs, the sum of the per-term kernels is $\ker \operatorname{lcm}(p_n)(D)$.

Proof.

Step 1 Bounded minimisers

$R(\omega_\alpha) \le R(\omega^\dagger) + \varepsilon_\alpha/\alpha$ and coercivity bound ω_α; weak limits are exact and R-minimal by weak lower semicontinuity (Mazur).

Step 2 Dimensions

A13(ii) for factors; for terms $\mathcal{R}/(\tau - a_n, \tau - a_m) \cong \mathbb{C}[\xi]/(a_n - a_m)$.

Step 3 (P2)

The roots of the lcm, with maximal multiplicities, give a basis contained in the union of the per-term kernels.

Why it works. all methods agree up to a finite-dimensional gauge, where the regulariser picks a representative.

■

Uses: A13.

Dim. B57 **P2-P4**

Linear encoders do not help [proved]

(a) A bounded, boundedly invertible linear encoder E preserves closedness of sums and distances up to the constants m_E, $\|E\|$. (b) An injective compact encoder can make a non-dense M have dense image, but then the encoded components diverge.

Proof.

Step 1 (a)

$m_E\|u - v\| \le \|Eu - Ev\| \le \|E\|\|u - v\|$; isomorphisms preserve closed subspaces.

Step 2 (b)

With $Ee_k = e_k/k$ and $M = w^\perp$, $w \propto \sum_k e_k/k$: $y \perp EM$ forces $Ey \in \mathbb{C}w$, impossible in ℓ^2 unless $y = 0$. If $Em_k \to Eu$ with $u \notin M$ and m_k bounded, a weak limit would give $u \in M$.

Why it works. stable encodings preserve geometry; unstable ones hide it by sending pieces to infinity.

■

Uses: none.

Dim. B58 P2-P5

Where the interaction is stable [proved]

(a) In $\mathfrak{B}_\rho$: $\sup_t \|E_{\mathrm{int}}(t)\|_{B_{\rho'}} \le e_{\ge 2}(\beta)\|u_0\|_{B_{\rho'}}$ with $\beta_n = e^{|I||a_n|(\rho')} - 1$. (b) On $\mathbb{R}$ with L^2 data: $\|E_{\mathrm{int}}(t)\|_{L^2} \le e_{\ge 2}(\beta)\|u_0\|$ with $\beta_n = \sup|e^{ta_n(i\kappa)} - 1|$; for advection–diffusion $\|E_{\mathrm{int}}\| \le 2\|u_0\|$.

Proof.

Step 1 (a)

A19(ii) and $\|e^{sa_n} - I\| \le e^{|s||a_n|(\rho')} - 1$; $e_{\ge 2}$ has nonnegative coefficients.

Step 2 (b)

Plancherel: $\widehat{E_{\mathrm{int}}} = e_{\ge 2}(z)\hat{u}_0$ and $|e_{\ge 2}(z)| \le e_{\ge 2}(|z|)$. For advection–diffusion $|z_1| \le 2$, $|z_2| \le 1$.

Why it works. stability of the interaction is a property of the data class, not of the method.

■

Uses: A19, B44.

Dim. B59 P3-C1

The hypotheses of Dyson–Phillips cannot be dropped [proved]

(a) Same order ($A_0 = \partial^2$, $V = -2\partial^2$): the Dyson terms have norm $(2n)^n e^{-n}/n! \sim 2^n/\sqrt{2\pi n}$ and $A_0 + V$ generates no semigroup. (b) Dispersive principal term ($A_0 = i\partial^2$, $V = i\partial$): $A_0 + V$ generates no semigroup. (c) $A_0 = i\partial^2$, $V = \partial$: the sum generates a group but the Dyson terms are unbounded.

Proof.

Step 1 Multipliers

All operators are Fourier multipliers; in (a) $\sup_y (2ty)^n e^{-ty}/n! = (2n)^n e^{-n}/n!$.

Step 2 Non-generation

The spectrum of a multiplier is the closure of its symbol's range; in (a), (b) the real part is unbounded above, so by Hille–Yosida there is no generator.

Why it works. without smoothing by the principal term, each perturbation step costs a full derivative.

■

Uses: A23.

Dim. B60 **OPF-T2**

Repeated factors: the multiplicity term

[proof written, not audited]

For $Q = (\tau - a(\xi))^e$ and $u \in \mathfrak{B}$, $U = (u^{(0)}, \dots, u^{(e-1)})$ with $u^{(k)} = (\partial_t - a)^k u$ satisfies $U(t) = e^{s(a(\partial_x)I + N_e)}F$ with the nilpotent shift N_e; the multiplicity term commutes with the others, its flow is polynomial in s (secular growth), and $E_{\text{int}}^{\text{mult}} = 0$. **(OPF-T1)** For every P, every factor decomposition of $u \in \mathfrak{B}$ lies in $\mathfrak{B}$.

Proof.

Step 1 (T2)

$\partial_t U = (a(\partial_x)I + N_e)U$, a system with commuting constant-coefficient generator; B45(L3) for systems.

Step 2 (T1)

For each j, B26(c) gives $r(\partial_x)u_j$ and $s(\partial_t)u_j$ in $\mathfrak{B}$; B45(L4) inverts $r(\partial_x)$ inside $\mathfrak{B}$, and the remaining finite-dimensional part is controlled as in B47.

Why it works. a repeated factor adds a nilpotent term, which commutes with everything and grows polynomially.

■

Uses: B26, B45, B47.

13.7 Results used without proof here

TROP-3′, TROP-4	Uniform-in-time bound of the interaction on dissipative faces of the Newton polygon; no uniform bound on conservative faces.	[proved]

Proofs in `work/SCOUT-TROP/VERDETTO.md` (audited); not reproduced. TROP-5..8: [proof written, not audited].

R5-P6, R5-T4	Global Wei–Norman for Ornstein–Uhlenbeck; Duhamel with bound.	[proved]

Proofs in `work/CHECK/DECLASSAMENTI.md` §B; not reproduced.

P3-T2, P3-T4, P3-T5	Path formula for Dyson terms; closed form for an εx perturbation; Appell and Rosenbloom–Widder duals.	[proved] / [proof written, not audited]

Proofs in `work/P3`; not reproduced.

OPA-P4	Splitting error at full coupling equals the first-order term to relative $O(t^{1/2})$.	[proved]

Now contained in B52 (OPF5A-T5).

P1-Q6	No-go of continuity for every operator with at least two terms.	closed

By B53 (monomials) and B53(e) mod Turán–Nazarov.

P1-Q3, **P1-Q7**	No finite product of flows for ∂^2 and x^k, $k \geq 3$; A24 without the switching-off hypothesis.	[open]

CHAPTER 14

Phase 4: nonlinear equations

In this chapter	
Starting from	Part I, Chapter 4 (A25–A31), and its notation: Burgers flows S, S_B, S_A, the Lagrangian map X, J, the Hopf–Lax functional Φ_x, and the dictionary setting.
Why	to prove the tools behind A26–A28 (comparison principle, pre-shock bounds), the blow-up at the shock time, the jumps after the shock, the $\sqrt{\nu}$ splitting error, the KdV–Burgers remainder, and the remaining identification results (the negative theorem for complete dictionaries, closed forms, the degree bound for Burgers).
Role	Dim. B61–B72, followed by the list of nonlinear results whose long proofs are not reproduced here.

14.1 Tools before the shock

Dim. B61 — **S4T-L0**

Comparison principle [proved]

Let b, c be continuous and real on $[0,T]\times\mathbb{T}$, $|c|\le K$, $\nu\ge 0$, and $Lg = g_t + bg_x + cg - \nu g_{xx}$. (a) If $Lg \ge 0$ and $g(0)\ge 0$ then $g\ge 0$. (b) If $|Lg|\le F(t)$ then

$$\|g(t)\|_\infty \le e^{Kt}\Big(\|g(0)\|_\infty + \int_0^t F\Big).$$

Proof.

Step 1 (a) First touching point

For $\theta>0$ put $z = g + \theta e^{(K+1)t}$, so $Lz>0$ and $z(0)>0$. At a first point (t_1,x_1) with $z=0$: $z_t\le 0$, $z_x=0$, $z_{xx}\ge 0$, so $Lz\le 0$, a contradiction. Let $\theta\to 0$.

Step 2 (b)

$Y(t) = e^{Kt}(\|g(0)\| + \int_0^t F)$ satisfies $Y' \ge KY + F$; apply (a) to $Y \mp g$.

Why it works. a parabolic operator cannot create a new minimum.

■

Uses: none.

Dim. B62 **S4T-L1**

Derivative bounds before the shock

[proved]

For real nonconstant u_0, $M_0 = \max u_0'$, $0 < T < t^*$, and w the viscous or inviscid Burgers solution: $-1/(t^* - t) \le \partial_x w \le M_0$, $\|w\|_\infty \le \|u_0\|_\infty$, and $\|\partial_x^j w(t)\|_\infty \le Y_j(t)$ with Y_j independent of ν.

Proof.

Step 1 The slope

$p = \partial_x w$ solves $p_t + wp_x + p^2 - \nu p_{xx} = 0$. Apply B61(a) to $M_0 - p$ (source $p^2 \ge 0$) and to $p - \vartheta$ with $\vartheta = -1/(t^* - t)$, which solves $\vartheta' = -\vartheta^2$.

Step 2 Higher derivatives

$p_j = \partial^j w$ solves $p_{j,t} + wp_{j,x} + (j+1)p_1 p_j - \nu p_{j,xx} = -\sum_{i=2}^{j-1} \binom{j}{i} p_i p_{j+1-i}$; B61(b) and induction give ν-independent bounds.

Why it works. before the shock every derivative obeys a transport equation with bounded coefficients.

■

Uses: B61.

Dim. B63 **S4T-P3**

The closed form blows up at the shock time

[proof written, not audited]

If u_0' has a unique minimum a^* with $\kappa = u_0'''(a^*) > 0$ and $\eta = 1 - t/t^*$, the first-order coefficients of A26 satisfy, as $t \uparrow t^*$,

$$\eta^{3/2} \sup_x |E_1^{\mathrm{add}}(t)| \to \frac{3\sqrt{3}}{16}\sqrt{2t^*\kappa}.$$

Proof.

Step 1 Local form

With $y = a - a^*$: $u_0''(a) = \kappa y(1 + O(y))$ and $J = \eta + \frac{t\kappa}{2} y^2(1 + O(y))$.

Step 2 Scaling

With $y = z\sqrt{\eta/c_t}$, $c_t = t\kappa/2$, the quantity $|u_0''|/J^2$ equals $\frac{\kappa}{\sqrt{c_t}} \eta^{-3/2} z/(1+z^2)^2$ up to $1 + O(\eta^{1/4})$; the maximum of $z/(1+z^2)^2$ is $3\sqrt{3}/16$. Away from the window the function is smaller.

Why it works. the Jacobian vanishes quadratically at the cusp, and the coefficient is a ratio of a linear to a squared quadratic quantity.

■

Uses: A26.

14.2 After the shock

Dim. B64 **S4T-T3**

After the shock: jumps and the splitting order [proved mod. [F1]]

For smooth nonconstant u_0, $t > t^*$ and every $\nu > 0$: (i) $E_{\mathrm{int}}^{\mathrm{add}}(t)$ has a jump $+D$ at each shock, independent of ν, so $\|E_{\mathrm{int}}^{\mathrm{add}}\|_\infty \geq D/2$ and $E_{\mathrm{int}}^{\mathrm{add}} \notin H^{1/2}$; (ii) the same for the splitting error B then A; (iii) the splitting error A then B is smooth but $\liminf_\nu \nu^{-1/2}\|E_{AB}^{\mathrm{mult}}\|_{L^1} \geq 2D\sqrt{t/\pi}$.

Proof.

Step 1 (i)

$E_{\mathrm{int}}^{\mathrm{add}} = S(t)u_0 - S_B(t)u_0 - S_A(t)u_0 + u_0$: the viscous, heat and initial terms are smooth, while $S_B(t)u_0$ jumps by $-D$ at the shock (Lax–Oleinik).

Step 2 (iii)

$E_{AB}^{\mathrm{mult}} = S(t)u_0 - S_A(t)S_B(t)u_0$; heat-smoothing a jump of size D during time t leaves an error of L^1 size $\geq 2D\sqrt{\nu t/\pi}$, and the viscous solution is $O(\nu|\log\nu|)$-close to the shock profile.

Why it works. the inviscid term alone produces a discontinuity that no amount of viscosity in the other term can remove.

■

Uses: [F1], [F2].

Dim. B65 **OPF5D-T6**

The vanishing-viscosity L^1 bound [proved] (mod [F1] for the corollary)

For every $t, \nu > 0$,

$$\|S(t)u_0 - v(t)\|_{L^1(\mathbb{T})} \leq 8\pi\sqrt{\frac{2\nu\log(1 + t\|u_0'\|_\infty)}{t}}.$$

Consequently $\|E_{AB}^{\mathrm{mult}}(t)\|_{L^1} = \Theta(\sqrt{\nu})$ for $t > t^*$, and $\|E_{\mathrm{int}}^{\mathrm{add}}(t)\|_{L^1} = O(\sqrt{\nu})$.

Stated in the paper as no. 9.2.2, p. 54

Proof.

Step 1 **Convex potentials**

$G_\nu(x) = 2\nu \log \int \exp\left((xy/t - h(y))/2\nu\right) \mathrm{d}y$ and $G_0(x) = \sup_y(xy/t - h(y))$, $h = U_0 + y^2/2t$, are convex, and $u - v = G_0' - G_\nu'$.

Step 2 **Laplace gap**

$G_\nu - G_0 = 2\nu \log I(x)$ with $I \geq c_0 = \sqrt{4\pi\nu/M}$, $M = \|u_0'\|_\infty + 1/t$; put $\varepsilon = G_\nu - c - G_0 \geq 0$.

Step 3 **Strong convexity and averaging**

G_0 is $(1/t^2M)$-strongly convex, so each $D_x(y)$ is quadratic-from-below in x; Fubini over a period and Jensen give $\int_{\mathbb{T}} \varepsilon \leq 4\pi\nu \log(tM)$.

Step 4 **Derivatives of close convex functions**

With $f = G_\nu - c$, $g = G_0$, $f - g = \varepsilon \geq 0$ and $h > 0$, convexity gives

$$|f' - g'|(x) \leq g'(x+h) - g'(x-h) + \frac{\varepsilon(x+h) + \varepsilon(x-h)}{h}.$$

Over one period $\int (g'(x+h) - g'(x-h))\,\mathrm{d}x = 4\pi h/t$, so

$$\|u - v\|_{L^1} \leq \frac{4\pi h}{t} + \frac{8\pi\nu \log(tM)}{h},$$

and $h = \sqrt{2\nu t \log(tM)}$ gives the bound.

Step 5 **The splitting error**

$E_{AB}^{\mathrm{mult}} = (u - v) - (S_A v - v)$ with $\|S_A v - v\|_{L^1} \leq 2\sqrt{\nu t/\pi}\,\mathrm{TV}(v(t))$; the lower bound is B64(iii).

Why it works. the viscous and inviscid solutions are derivatives of two convex potentials that differ by a Laplace correction of size ν log.

■

Uses: [F1], [F2].

Dim. B66 **OPR2-T5**

KdV–Burgers before the shock

[proved mod. local well-posedness]

For $u_t + uu_x = \nu u_{xx} - d u_{xxx}$, $k \geq 2$, $T < t^*$, and small $\nu + |d|$:

$$\sup_{t \leq T} \left\| u(t) - v(t) - \nu\delta(t) - d\,\delta_d(t) \right\|_{H^k} \leq C_k(\nu + |d|)^2,$$

so $E_{\mathrm{int}}^{\mathrm{add}}$ has an explicit first-order form in (ν, d).

Proof.

Step 1 Energy estimate for $e = u - v$

$e_t + (ve)_x + ee_x - \nu e_{xx} + de_{xxx} = \nu v_{xx} - dv_{xxx}$. In H^k the dispersive term is antisymmetric, the viscous term dissipates, and the transport terms are bounded by $C\|e\|_{H^k}^2$.

Step 2 Bootstrap

$(\sqrt{Y})' \le C\sqrt{Y} + G(\nu + |d|)$ keeps $\|e\|_{H^k} \le C(\nu + |d|)$ on $[0, T]$; the same for $e - \nu\delta - d\delta_d$ gives the square.

Why it works. before the shock both small terms are regular perturbations of the inviscid flow.

■

Uses: B62.

Dim. B67 **S4T-P5**

Fisher–KPP: the jet [proved]

For $u_t = \nu u_{xx} + ru(1-u)$ with terms $A = \nu u_{xx}$, $B = ru(1-u)$:

$$[s^2]\,\hat{E}^{\mathrm{add}} = \tfrac{1}{2}\big(A'[B] + B'[A]\big) = r\nu\big(u_0'' - u_0'^2 - 2u_0u_0''\big);$$

the estimate of the remainder is not uniform in time.

Proof.

Step 1 The two derivatives

$A'[w] = \nu w_{xx}$, $B'[w] = r(1-2u_0)w$. So $A'[B] = r\nu(u_0 - u_0^2)'' = r\nu(u_0'' - 2u_0'^2 - 2u_0u_0'')$ and $B'[A] = r\nu(1-2u_0)u_0''$.

Step 2 Half the sum

$\frac{1}{2}\big(A'[B] + B'[A]\big) = r\nu\big(u_0'' - u_0'^2 - 2u_0u_0''\big)$.

Why it works. A25 at order two.

■

Uses: A25.

Example 14.1 The reference datum $u_0 = \varepsilon \sin x$ (OPA-C1)

Here $t^* = 1/\varepsilon$, there is one shock at $x = \pi$ for $t > t^*$, symmetric ($\rho = 0$), and A27 gives

$$\Lambda(t) = 2\log\frac{1}{\varepsilon t - 1} + 4\log 2 + o(1) \qquad (t \downarrow t^*),$$

consistent with the uniform rate of A28.

14.3 Identification in dictionaries

Dim. B68 **S4ID-P1**

The literal statement holds only in the minimal dictionary [proved]

For $F = y_t - \sigma y_n - N'$ with $\sigma \neq 0$ and $\deg N = 2$ (Burgers, Fisher–KPP, KdV, KdV–Burgers): (a) $F \in V_{d,k}$ iff $d \geq 2$ and $k \geq n$; (b) if $(d,k) \neq (2,n)$ then $\dim([F] \cap V_{d,k}) \geq 2$; (c) $[F] \cap V_{2,n} = \mathbb{C}F$.

Proof.

Step 1 (b) Two independent consequences

If $k \geq n+1$: $\partial_x F \in V_{d,k}$ contains y_{n+1}, which F does not. If $d \geq 3$: yF contains $y\,y_t$, which F does not.

Step 2 (c) Elimination

In $V_{2,n}$, the generators involving time derivatives other than y_t have normal forms of distinct orders $na+b \geq n+1$ with nonzero leading coefficients σ^a, so they must cancel; what remains is a multiple of F by degree count.

Why it works. a complete dictionary always contains derivatives or multiples of the equation.

■

Uses: A29.

Dim. B69 **S4ID-T4**

Closed forms do not identify; identifying solutions exist [proof written, not audited] ((iii) mod Hölder, Bona–Smith [from cited source])

(i) If u lies in a finite tower of field extensions of $\mathbb{C}(t,x)$ (closed forms), then $u, u_x, \dots, \partial_x^r u$ are algebraically dependent, so $\mathcal{J}_u \neq 0$ and u does not identify its equation in any dictionary containing that relation. (ii) For $u = P/Q$ in t, x and exponentials, an explicit degree $D^* = (n+1)c^n - n$ of such a relation exists. (iii) For a datum g that satisfies no algebraic ODE (for example Γ on an interval), the Burgers or KdV solution has $\mathcal{J}_u = 0$ and identifies its equation modulo consequences in every dictionary.

Proof.

Step 1 (i) Transcendence degree

Each step of the tower adds at most a bounded transcendence degree, so the field generated by u and its x-derivatives has finite transcendence degree r; $r+1$ derivatives are dependent. A29(c) concludes.

Step 2 (iii) A transcendental datum

At $t = 0$ any relation in the x-jet would be an algebraic ODE for g; by continuity a relation for u restricts to one for g. Hölder's theorem says Γ satisfies none [from cited source].

Why it works. closed forms live on finite-dimensional varieties of jets; differentially transcendental data do not.

■

Uses: A29.

Dim. B70 **OP-T7**

Burgers N-shocks: no relation of degree $\leq N$ [proved]

Under the hypotheses of A30, no nonzero polynomial of degree $\leq N$ in the jet of u (of any order) vanishes on u. Hence, for every order $K \geq N$, u identifies Burgers in the evolutionary dictionary iff $d \leq N$.

Stated in the paper as no. 9.3, p. 55

Proof.

Step 1 Cumulants

By A30, Steps 1–2, the jet of w is the sequence of cumulants κ_n of the moments r_j, polynomial in $r_1, \dots, r_N$, which are algebraically independent.

Step 2 Weights

Give r_j weight j. The top-weight part of κ_n is a truncated cumulant; a relation $P(\kappa) = 0$ of degree $\leq N$ has a nonzero top-weight part that must vanish identically.

Step 3 Confluent limit

The top-weight parts are cumulants of a measure with N atoms; in the confluent limit they become power sums of N variables, and power sums $p_1, \dots, p_N$ are algebraically independent (fundamental theorem of symmetric polynomials), which excludes any relation of degree $\leq N$.

Why it works. N atoms leave N independent power sums, so the first relation needs degree $N + 1$.

■

Uses: A30.

Dim. B71 **F4L-P3**

Analyticity radius of Burgers in the amplitude

[proof written, not audited]

For Burgers with datum $\varepsilon U_0'$, the solution is analytic in ε for $|\operatorname{Im}\varepsilon| < 2\pi\nu/\operatorname{osc}(U_0)$, uniformly in time, and this is optimal. **(KIEFFER-C)** The "infinite radius" claimed in a cited preprint is not proved there: its own formula has denominator zeros at finite ε.

Proof.

Step 1 A moment generating function

By Cole–Hopf, $\varphi_\varepsilon(t,x) = \mathbb{E}\, e^{sX}$ with $X = U_0(Y)$, Y distributed by the heat kernel, $s = -\varepsilon/2\nu$; X takes values in an interval of length $w = \operatorname{osc} U_0$.

Step 2 The strip lemma

For a random variable with values in an interval of length w, $M(s) = \mathbb{E}e^{sX}$ has no zeros for $|\operatorname{Im} s| < \pi/w$ (L-S), and $u = -2\nu\partial_x \log\varphi$ is analytic there. The symmetric Bernoulli law attains the bound (the tanh case).

Why it works. Cole–Hopf turns Burgers into the logarithm of a moment generating function, whose zeros are kept away by the bounded range.

■

Uses: [F2].

Dim. B72 **S4ID-L4**

Burgers in Hammerstein form

[proof written, not audited]

With $\mathcal{R}$ acting on differential polynomials, and $F_B = (\tau - \nu\xi^2)y + \frac{\xi}{2}y^2$: $[F_B] \cap (\mathcal{R}y \oplus \mathcal{R}y^2) = \mathcal{R}F_B$.

Proof.

Step 1 Degree splitting

Elements of $[F_B]$ are $\sum_j L_j\, \partial^j F_B$ with differential-polynomial coefficients; those of degree ≤ 2 in y with linear–quadratic structure must have constant-coefficient multipliers, because any y-dependent coefficient raises the degree.

Why it works. in the Hammerstein form the linear and quadratic parts cannot be mixed without leaving the space.

■

Uses: A29.

14.4 Nonlinear results whose proofs are not reproduced

The following results are proved in the research files named in the last column. Their proofs rely on long computations (scattering theory, Hirota bilinear forms, computer-assisted syzygies). They are listed with their labels so that the catalogue is complete as a list; the proofs are not reproduced in this edition.

INT-T1	KdV in the class $\mathcal{H}_M$: identification by the scattering data; (A) and unique continuation [proved]; (B), (C), INT-T2 [conditional] / [proved] at fixed time (OPF5D).	see label

`work/S4-INTEG` §1.1, `work/OP-F5D`.

INT-P5, INT-P6	Hirota N-solitons: the interaction is pairwise, integrates to 0, costs $O(N^3)$.	[proved]

`work/S4-INTEG`.

OP-P17, OP-P18	Lax and zero-curvature relations for all n (sign at $n = 0$ corrected).	[proved]

`work/OP-F5D` (OPF5D-T1, T2).

OPF5D-R2	NLS identification by scattering data without multiplicity.	[false] mod [DP]

`work/OP-F5D`.

OPF5D-T7	Lax–Oleinik equals Kružkov entropy solution ([F1]).	[proved mod. Kružkov]

`work/OP-F5D`.

OPF5A-T3, T4, C1	OPA-T1 with multiple points; genericity of (H_t); global uniform rate for generic data (several births [conditional]).	[proved] mod [F1]

`work/OP-F5A`.

OPF5C-T1, L1–L8, P7, P9, T6	Structure of Novikov's variety; upper bounds for all N; nine exact syzygy degrees $\delta_N(K)$ (computer-assisted).	[proved] mod (Nov′)

`work/OP-F5C`; the main line is in A31.

S4ID-L2, L5, C1, C2, P5–P7	Normal forms and elimination lemmas; evolutionary versus complete dictionaries.	[proved]

`work/S4-ID`.

OP-C11, **OP-T1**, **OP-E3**, **OP-C3′**, **OP-L2..P14**	Complete criteria for the Burgers family; KdV examples and obstructions.	[proved]

`work/OP-NL`.

ALG-T1, **MODREL-T1**, **F4LIN-T2**	Identification in the full ring for analytic u; moment relations; tropical interaction $4\nu \ln 2$.	[proved]

`work/S4-INTEG`.

OPR2-P4, **P6**, **O8**, **O9′**	Novikov relation, obstructions to INT-Q1.	[proved] / mod Novikov

`work/OP-R2`.

S4ID-P8, **OPR2-P7**, **OPF5D-P14**, **P15**	Nonlinear asymptotics with derivatives (Deift–Zhou).	[conditional]

`work/S4-ID`, `work/OP-F5D`.

INT-Q1, **OPF5C-Q1**	Analytic KdV data with reflection; general $\delta_N(K)$.	[open]

Part IV

Phases 6–10: the other results

CHAPTER 15

Phases 6–8: systems, gauge and variable coefficients (other results)

In this chapter	
Starting from	the notation of Chapters 5 and 6.
Why	to give the remaining results of the system and variable-coefficient packages: short proofs in full, and the others as labelled statements with a pointer to the research file.
Role	Dim. B73–B80, and the list cards of this chapter. They prove the corresponding parts of Chapters 11 and 12 of the paper.

15.1 Systems

Dim. B73 — **SYS-R1**

Finite data and ODE systems [proved]

(SYS-R1) If $u \in \mathrm{EP}^k$ then M_u has finite length. Hence from exponential-polynomial data one identifies only systems with finite-dimensional solution space.

(SYS-T0) For $n = 1$, $R = \mathbb{C}[s]$, $P \in R^{l\times k}$ of rank r with invariant factors $e_1 \mid \cdots \mid e_r$ on an open interval: an identifying u exists iff $e_1 = \cdots = e_{r-1} = 1$, i.e. iff the gcd of the $(r-1)\times(r-1)$ minors of P is 1. (Known.)

Stated in the paper as no. 11.5, p. 67

Proof.

Step 1 SYS-R1

Every component of u lies in a finite-dimensional derivative-invariant space of EP (A2), so $M_u \cong \sum Ru_i \subseteq \mathcal{A}$ has finite dimension over $\mathbb{C}$.

Step 2 SYS-T0

By the Smith form $M_P \cong \bigoplus_{i\le r} \mathbb{C}[s]/(e_i) \oplus \mathbb{C}[s]^{k-r}$, and $\dim \operatorname{soc}_{(s-\lambda)}(M_P) = \#\{i : e_i(\lambda) = 0\}$ (a cyclic torsion summand with root λ has a one-dimensional socle there; free summands have none). Since $e_i \mid e_r$, "≤ 1 for every λ" means e_{r-1} has no root, i.e. $e_1 = \cdots = e_{r-1} = 1$; the gcd of the $(r-1)$-minors is $e_1 \cdots e_{r-1}$. Apply A32.

Why it works. in one variable the socle count is read off the Smith form.

■

Dim. B74 **SYS-E**

Dirac $1+1$ as an evolution: what survives of the no-product theorem [proved]

Write $\psi_t = A(\partial_x)\psi$, $A(\xi) = X + mY$, $X = i\xi\sigma_z$ (transport), $Y = -i\sigma_x$ (mass), in the algebra $\mathrm{Mat}_2(\mathbb{C}[\xi])$.

(E1) $X, Y_0 = Y, Y_k = \mathrm{ad}_X^k Y$ are linearly independent over $\mathbb{C}$. Hence (P1-T5, A24) no finite word $(Z_1, \dots, Z_d) \in \{X, Y\}^d$ and holomorphic $g_i(t, m)$ with $g_i(0, m) = 0$, $g_i(t, 0) \equiv 0$ for $Z_i = Y$, gives $e^{t(X+mY)} = \prod e^{g_i Z_i}$ in $\mathrm{Mat}_2(\mathbb{C}[\xi])[[t]]$ for m near 0.

(E2) $\mathrm{Lie}_{\mathbb{C}}\langle X, Y\rangle = \mathfrak{L}^\theta = \{Z \in \mathfrak{sl}_2 \otimes \mathbb{C}[\xi] : \theta(Z) = Z\}$, $\theta(Z)(\xi) = \sigma_x Z(-\xi)\sigma_x$, of infinite dimension.

(E3) With $\omega = \sqrt{\xi^2 + m^2}$, $b = -\arcsin(m \sin(\omega t)/\omega)$ and $a = \frac{1}{2}\arg(\cos \omega t + i\xi \sin(\omega t)/\omega)$ one has $e^{tA} = e^{g_1 X} e^{g_2 mY} e^{g_1 X}$ with $g_1 = a/\xi$, $g_2 = -b/m$: explicit, analytic in ξ for $|t| < \pi/(2|m|)$, but g_2 is not a polynomial in ξ (a bounded polynomial would be constant, while $b \to 0$ at infinity and $b(t, 0) = -mt \neq 0$), and $e^{g_2(t,\partial_x)mY}$ is a non-local Fourier multiplier. At $t = \pi/(2|m|)$, $g_1 \to \pm\infty$ as $\xi \to 0^{\pm}$.

Stated in the paper as no. 11.5, p. 67

Proof.

Step 1 (E1) Independence

With $[\sigma_z, \sigma_x] = 2i\sigma_y$, $[\sigma_z, \sigma_y] = -2i\sigma_x$: $\mathrm{ad}_X \sigma_x = -2\xi\sigma_y$, $\mathrm{ad}_X \sigma_y = 2\xi\sigma_x$, so $Y_k = c_k \xi^k \sigma_x$ (k even) or $c_k \xi^k \sigma_y$ (k odd), $c_k \neq 0$; distinct degrees in ξ make them independent, and X has a σ_z component absent from all Y_k.

Step 2 (E2) The algebra

θ is a Lie algebra automorphism, so $\mathfrak{L}^\theta$ is a subalgebra containing X, Y. Conversely $\mathrm{Lie}\langle X, Y\rangle$ contains $\xi^{2k}\sigma_x$, $\xi^{2k+1}\sigma_y$ (E1) and $[\sigma_x, \xi^{2k+1}\sigma_y] = 2i\xi^{2k+1}\sigma_z$, a basis of $\mathfrak{L}^\theta$. So P1-T4 (global Wei–Norman with $\dim \mathfrak{g}$ functions) does not apply over $\mathbb{C}$.

Step 3 (E3) Euler angles

$A^2 = -\omega^2 I$ gives $e^{tA} = \cos \omega t\, I + \frac{\sin \omega t}{\omega} A$. The product $e^{ia\sigma_z} e^{ib\sigma_x} e^{ia\sigma_z}$ has entries $(11) = \cos b\, e^{2ia}$, $(12) = (21) = i \sin b$, $(22) = \cos b\, e^{-2ia}$; matching $(12) = -im \sin(\omega t)/\omega$ gives $\sin b = -m \sin(\omega t)/\omega$ and $|\cos \omega t + i\xi \sin(\omega t)/\omega|^2 = \cos^2 b$ gives a. The multiplier $\cos b\, I + i \sin b\, \sigma_x$ is smooth, bounded, tends to I at infinity and is not constant; a convolution operator is local iff its symbol is a polynomial (Schwartz).

Why it works. the Lie algebra generated by transport and mass is infinite-dimensional, so a uniform finite product with polynomial times is impossible; the explicit product that exists uses non-local multipliers.

■

Uses: A24, A32.

Open (resolved in Phase 10 for real times)

Whether finite products with times polynomial in ξ exist (Q-S8SYS-1) was closed in Phase 10: no finite word at polynomial real times (SYS-DIRAC-POL); complex times remain open (Chapter 18).

Dim. B75 **LAME-3**

Lamé in space: an identifying displacement field and its test [proved]

Let $P = \rho\tau^2 I_3 - \mu|\xi|^2 I_3 - (\lambda+\mu)\xi\xi^\top$ with $\rho, \mu \neq 0$, $\lambda+\mu \neq 0$, $\lambda+2\mu \neq 0$, $q_P = \rho\tau^2 - (\lambda+2\mu)|\xi|^2$, $q_S = \rho\tau^2 - \mu|\xi|^2$. Then $\det P = q_P q_S^2$, $\mathrm{Ass}(M_P) = \{(q_P), (q_S)\}$ with Bass multiplicities $(1,2)$; an identifying field always exists (A33); and u identifies Lamé iff (a) $\mathrm{Ann}(\operatorname{div} u) = (q_P)$ and (b) every relation $\sum r_i(\partial)(\operatorname{rot} u)_i = 0$ has $r \in q_S R^3 + R\xi$ up to a factor outside (q_S).

Stated in the paper as no. 11.5, p. 67

Proof.

Step 1 Structure

$\det P = q_P q_S^2$; q_P, q_S are irreducible quadrics of rank 4; $\bar{P}$ has rank 1 mod q_S and rank 2 mod q_P, so over the DVR $R_{(q_S)}$ the Smith form gives $M_{(q_S)} \cong \kappa(q_S)^2$ and $M_{(q_P)} \cong \kappa(q_P)$.

Step 2 An essential submodule

Let E be generated by the row $d = \xi^\top$ (div) and the rows c_i of $[\xi]_\times$ (rot). Then $\xi^\top P = q_P \xi^\top$, $[\xi]_\times P = q_S [\xi]_\times$, and Helmholtz $|\xi|^2 I = \xi\xi^\top - [\xi]_\times^2$ gives $|\xi|^2 M_P \subseteq E$; $|\xi|^2 \notin (q_P) \cup (q_S)$ is a non-zero-divisor, so E is essential. $Rd \cap \sum Rc_i = 0$ (a common element is killed by (q_P, q_S) of height 2).

Step 3 The test

By SYS-L0 on E, u identifies iff (a) and (b) hold. For (b), $M_{(q_S)} = \kappa^{1\times 3}/\kappa\xi^\top$ and $\bar{r}^\top[\xi]_\times \in \kappa\xi \iff \bar{r}\times\xi = 0 \iff \bar{r} \in \kappa\xi$.

Why it works. one field identifies Lamé iff its P-part (div) and its S-part (the two transverse polarisations of rot) satisfy no extra equation. A pure P-wave ($\operatorname{rot} u = 0$) never identifies.

■

Uses: A33, A32.

Dim. B76 **G-T2**

Gauge covariance, and what gauge-invariant data can reveal [proved]

(G-L0) $e^{-i\chi}(\partial_\mu - i(A_\mu + \partial_\mu\chi))e^{i\chi} = \partial_\mu - iA_\mu$; so $L_{A+d\chi}(e^{i\chi}u) = e^{i\chi}L_A u$ for every operator polynomial in $\nabla_\mu = \partial_\mu - iA_\mu$ with constant matrix coefficients. For $\chi = c\cdot x$, $\mathrm{Ann}(e^{ic\cdot x}u) = \sigma_c\,\mathrm{Ann}(u)$ with $\sigma_c(r)(\zeta) = r(\zeta - ic)$.

(G-T2) Let $\Omega \subseteq \mathbb{R}^n$ be open connected and gauge group the smooth maps $g : \Omega \to U(1)$ acting as $A \mapsto A + \frac{1}{i}g^{-1}dg$. (a) $A' \sim A \iff d(A'-A) = 0$ and $\oint_\gamma (A'-A) \in 2\pi\mathbb{Z}$ for every cycle γ; on simply connected Ω, $A' \sim A \iff dA' = dA$. (b) If an observation O satisfies $O(gu) = O(u)$, any procedure reconstructing A from $O(u)$ recovers at most the class $[A]$.

Stated in the paper as no. 11.5, p. 66

Proof.

Step 1 (G-L0)

$(\partial - iA - i\partial\chi)(e^{i\chi}f) = e^{i\chi}(\partial f - iAf)$.

Step 2 (G-T2)(a)

($\Rightarrow$) Locally $g = e^{i\chi}$ and $A' - A = d\chi$ is closed; along γ, $\frac{1}{i}\oint g^{-1}dg = 2\pi\deg(g\circ\gamma) \in 2\pi\mathbb{Z}$. ($\Leftarrow$) $B = A' - A$ is closed with periods in $2\pi\mathbb{Z}$, so $g(x) = \exp(i\int_{x_0}^{x} B)$ is path-independent, smooth, and $\frac{1}{i}g^{-1}dg = B$.

Step 3 (G-T2)(b)

If $L_A u = 0$ and $A' = A + \frac{1}{i}g^{-1}dg$ then $L_{A'}(gu) = gL_A u = 0$ and $O(gu) = O(u)$: the data do not distinguish A from A'. On a non-simply connected Ω the holonomies are the extra datum (Aharonov–Bohm; [from cited source]).

Why it works. gauge acts on A and on u compatibly; an observation insensitive to the phase cannot see the compensating change in A.

■

Uses: none.

Dim. B77 **SYS-N3**

Magnetic Schrödinger: one field does not identify (A, V), even modulo gauge

[proved]

Let $n \geq 2$, $u \in C^\infty(J \times \Omega)$ nowhere zero with $iu_t = -(\nabla - iA)^2 u + Vu$, A, V real smooth; $\rho = |u|$, $\nabla S = \mathrm{Im}(\nabla u/u)$, $S_t = \mathrm{Im}(u_t/u)$, $w = \nabla S - A$, $j = \rho^2 w$. Then:

(a) a real smooth pair (A', V') satisfies the same equation with the same u iff $A' = \nabla S - j'/\rho^2$, $V' = -S_t + \Delta\rho/\rho - |j'|^2/\rho^4$, $\mathrm{div}_x j' = -\frac{1}{2}\partial_t(\rho^2)$;

(b) the compatible pairs form the affine space $(A, V) + \{(-k/\rho^2, (|j|^2 - |j+k|^2)/\rho^4) : \mathrm{div}_x k = 0\}$, of infinite dimension;

(c) there are k with $d(k/\rho^2) \neq 0$: the magnetic fields differ, the pairs are not gauge-equivalent, and a gauge fixing u is trivial ($gu = u \Rightarrow g = 1$).

The variations $B = A' - A = -k/\rho^2$ satisfy $\mathrm{div}(\rho^2 B) = 0$ (not $\mathrm{div} B = 0$).

Stated in the paper as no. 11.5, p. 66

Proof.

Step 1 (a) Madelung form

With $u = \rho e^{iS}$, $(\nabla - iA)^2 u = e^{iS}[\Delta\rho - \rho|w|^2 + \frac{i}{\rho}\mathrm{div}(\rho^2 w)]$ and $iu_t = e^{iS}(i\rho_t - \rho S_t)$. Real and imaginary parts give $V = -S_t + \Delta\rho/\rho - |w|^2$ and $\partial_t \rho^2 + 2\mathrm{div}(\rho^2 w) = 0$; A' enters only through $w' = \nabla S - A'$.

Step 2 (b), (c)

Two admissible j, j' differ by k with $\mathrm{div}_x k = 0$, and every such k gives an admissible pair. For $n = 2$ take $k = \nabla^\perp \psi$, $\psi \in C_c^\infty \setminus 0$; $\mathrm{rot}(k/\rho^2) = \mathrm{div}(\rho^{-2}\nabla\psi)$ is not identically zero, otherwise ψ would be a compactly supported solution of an elliptic divergence-form equation, so $\psi \equiv 0$ by the maximum principle. For $n \geq 3$ take $k = \mathrm{rot}(\psi e_3)$.

Why it works. one complex scalar equation cannot determine $n+1$ real unknowns: the count, not gauge, is the obstruction.

■

Uses: B76.

Dim. B78 **VAR-P5**

Dirac $1+1$ with a potential: finite speed leaves places the data never visited

[proved]

Let $\Omega = I \times J$, $\psi \in C^1(\Omega;\mathbb{C}^2)$, $i\psi_t = -i\sigma_1\psi_x + m\sigma_3\psi + V(x)\psi$, residual $r = i\psi_t + i\sigma_1\psi_x - m\sigma_3\psi = V\psi$, $Y_0 = \{x : \psi(t,x) = 0\ \forall t\}$.

(a) (scalar continuous V) $V = \langle\psi, r\rangle / |\psi|^2$ where $\psi \neq 0$, with compatibility $r \parallel \psi$; V is identified iff $\operatorname{int} Y_0 = \emptyset$.

(b) Unlike Schrödinger, $\operatorname{int} Y_0 \neq \emptyset$ with $\psi \not\equiv 0$ is possible: for $m = V = 0$, $\Omega = (0,1)\times(0,2)$, $\psi = f(x-t)v_+$ with $\sigma_1 v_+ = v_+$ and $f \in C_c^\infty((0.5,0.7)) \neq 0$, $Y_0 \supset (0,0.5] \cup [1.7,2)$ and every V supported in $(0,0.5)$ is compatible.

(c) (matrix V, pointwise class) $V(x_0)$ is determined iff $\operatorname{span}\{\psi(t,x_0) : t \in I\} = \mathbb{C}^2$; if the span is a line $\mathbb{C}v$ the ambiguity is $\{W : Wv = 0\}$.

Stated in the paper as no. 12.4, p. 70

Proof.

Step 1 (a)

As in A36(a): $(V-W)\psi = 0 \iff V = W$ off Y_0; for continuous V, iff on the dense set $J \setminus Y_0$, iff $\operatorname{int} Y_0 = \emptyset$.

Step 2 (b)

With $m = V = 0$ the equation is $\psi_t = -\sigma_1\psi_x$, i.e. $\psi_{\pm,t} = \mp\psi_{\pm,x}$ on the components $\sigma_1 = \pm 1$, solved by $f(x-t)$. Then $\psi \neq 0$ only if $x - t \in (0.5, 0.7)$, which never occurs for $x \leq 0.5$ or $x \geq 1.7$, $t \in (0,1)$.

Step 3 (c)

$W \mapsto (W\psi(t,x_0))_t$ is injective iff no $W \neq 0$ vanishes on the span. A Hermitian W with $Wv = 0$ preserves $v^\perp$, so $W = cP_{v^\perp}$ with c real.

Why it works. finite propagation speed creates regions of space-time the solution never visits, where the potential cannot be read; Schrödinger's infinite speed, through A37, rules this out.

■

Uses: A36, A37.

Dim. B79 **ENC-K0**

A constant potential has no strict differential intertwiner [proved]

Let J be an open interval, $c \in \mathbb{C} \setminus 0$ and $W = \sum_{j=0}^{n} w_j \partial^j$, $w_j \in C^\infty(J;\mathbb{C})$, $w_n \not\equiv 0$. Then $WL_c \neq L_0 W$. (With the relaxed requirement $WL_c = (L_0 + c)W$, $W = I$ works.)

Stated in the paper as no. 14.1, p. 74

Proof.

Step 1 **Read the coefficients**

$WL_c - L_0 W = [\partial^2, W] + cW = \sum_j (w_j'' \partial^j + 2w_j' \partial^{j+1}) + cW$. If it vanishes: the coefficient of ∂^{n+1} gives $2w_n' = 0$, so w_n is a non-zero constant; for $1 \le j \le n$, $w_j'' + 2w_{j-1}' + cw_j = 0$; and $w_0'' + cw_0 = 0$.

Step 2 **Induction on $k = 0, \dots, n$**

w_{n-k} is a polynomial of degree exactly k with leading coefficient $(-c/2)^k w_n / k!$: if w_j has degree k and leading coefficient ℓ, then $w_{j-1}' = -(w_j'' + cw_j)/2$ has degree k and leading coefficient $-c\ell/2$, so w_{j-1} has degree $k+1$ and leading coefficient $-c\ell/(2(k+1))$.

Step 3 **Contradiction**

For $k = n$, w_0 has degree n and non-zero leading coefficient, so $w_0'' + cw_0$ has degree n and leading coefficient $c(-c/2)^n w_n / n! \neq 0$, against the last equation.

Why it works. the strict equation forces a polynomial tower that cannot close; the constant c must be moved to the right-hand side.

■

Uses: none.

Dim. B149 **SYS-T1**

An identification criterion on an essential family [proof written, not audited] (a tool, not a structural theorem)

Let M be a finitely generated R-module and $\{x_{\mathfrak{p},j}\}$ a finite family with $Rx_{\mathfrak{p},j} \cong R/\mathfrak{p}$, $\mathfrak{p} \in \operatorname{Ass}(M)$, whose sum E is direct and essential in M. For $u \in \operatorname{Sol}_{\mathcal{A}}(P)$ put $w_{\mathfrak{p},j} = \varphi_u(x_{\mathfrak{p},j})$. Then u identifies P iff for every $\mathfrak{q} \in \operatorname{Ass}(M_P)$ there is no relation $\sum_{\mathfrak{p} \supseteq \mathfrak{q}} \sum_j r_{\mathfrak{p},j}(\partial) w_{\mathfrak{p},j} = 0$ with some $r_{\mathfrak{q},j} \notin \mathfrak{q}$.

Stated in the paper as no. 11, p. 64

Proof.

Step 1 **Injectivity can be tested on an essential submodule (SYS-L0)**

Let $\varphi : M \to F$ be a homomorphism and $E \subseteq M$ essential. Then $\ker \varphi \cap E = \ker(\varphi|_E)$; if this is 0, then $\ker \varphi$ meets E trivially, hence is zero by essentiality. So φ is injective iff $\varphi|_E$ is.

Step 2 A non-zero kernel element has prime annihilator

A non-zero element of $\ker \varphi_u \cap E$ can be chosen with annihilator a prime $\mathfrak{q} \in \operatorname{Ass}(E) = \operatorname{Ass}(M_P)$. Its expression involves only summands with $\mathfrak{p} \supseteq \mathfrak{q}$ and at least one coefficient $r_{\mathfrak{q},j} \notin \mathfrak{q}$: a finite intersection of primes equal to a prime $\mathfrak{q}$ contains the product of those primes, hence $\mathfrak{q}$ contains one of them.

Step 3 Reading

So φ_u fails to be injective iff such a relation exists. For $k = 1$ this is exactly C3-T1.

Why it works. a module map that kills nothing on an essential submodule kills nothing at all.

■

Uses: Matlis (essential extensions), cited.

Status and prior art

SYS-T1 is not a structural theorem: it restates SYS-L0 and is no more decidable than computing $\operatorname{Ann}(u)$ directly for non-EP u. It shortens the test to $\mu_{\mathfrak{p}}$ scalar functions per associated prime (Dirac 3+1, Lamé). The structural content is SYS-EX. For exponential-polynomial data M_u has finite length (SYS-R1), the setting of Schindelar–Levandovskyy–Zerz 2011 (abstract read only).

Dim. B150 **SYS-N1**

The socle obstruction

[proof written, not audited]

Let Ω be connected and M an R-module. If $\dim_{\mathbb{C}} \operatorname{soc}_{\mathfrak{m}_\lambda}(M) \geq 2$ for some maximal ideal $\mathfrak{m}_\lambda$, no homomorphism $M \to \mathcal{A}$ is injective. In particular if this holds for M_P, no $u \in \operatorname{Sol}_{\mathcal{A}}(P)$ identifies P.

Stated in the paper as no. 11, p. 64

Proof.

Step 1 The image of the socle

A homomorphism φ sends $\operatorname{soc}_{\mathfrak{m}_\lambda}(M)$ into $\{f \in \mathcal{A} : (\partial_j - \lambda_j) f = 0 \ \forall j\}$.

Step 2 That space is a line

On a connected Ω, write $f = e^{\lambda \cdot x} g$: then $\nabla g = 0$ in $\mathcal{D}'(\Omega)$, so g is constant, and the space is $\mathbb{C} e^{\lambda \cdot x}$.

Step 3 No injectivity

A linear map from a space of dimension ≥ 2 to a line is not injective.

Why it works. a connected domain carries only one exponential per eigenvalue, so a socle with two copies cannot embed.

■

Uses: none.

Prior art

On $\Omega = \mathbb{R}^n$ SYS-N1, SYS-EX and SYS-T0 are known (Oberst 1993 for the minimal number of trajectories, Pommaret, Macaulay for the length-finite case); the only new point is arbitrary connected Ω (A32), where $C^\infty(\Omega)$ is not injective. The texts of Oberst 1990/1993 were not read.

Dim. B151 **VAR-T6**

Order of the minimal operator, socle and cyclicity (Picard–Vessiot) [from cited source]

Let V be the solution space of L in a Picard–Vessiot extension $E \supset K$ with group G, $u \in V \setminus 0$, $W_u = \operatorname{span}_{\mathbb{C}}(G \cdot u)$. (a) G-stable subspaces $W \subset V$ correspond to monic right factors $L_W = \operatorname{Wr}(y_1..y_k, Y)/\operatorname{Wr}(y_1..y_k)$ of L, with $\operatorname{Sol}_E(L_W) = W$. (b) $\operatorname{Sol}_E(L_u) = W_u$, so $\operatorname{ord} L_u = \dim \operatorname{span}(G \cdot u)$ and u identifies L iff u is a cyclic vector of V over $\mathbb{C}[G]$. (c) An identifying u exists iff V is cyclic over $\mathcal{A} = \operatorname{span}(G)$; if $V \cong \bigoplus S_i^{m_i}$ is semisimple, iff $m_i \le \dim S_i$. (d) The right factor C_1 attached to $\operatorname{soc}(V)$ induces $V/\operatorname{soc} V \cong \operatorname{Sol}(L_1)$ for $L = L_1 C_1$; iterating gives the Loewy series.

Stated in the paper as no. 12, p. 68

Proof.

Step 1 (a)

For W G-stable with basis y_i, $g(y_1..y_k) = (y_1..y_k)A_g$, both Wronskians are multiplied by $\det A_g$, so the coefficients of L_W are G-invariant, i.e. in K ($E^G = K$, cited). $\operatorname{Sol}(L_W) \supset W$ and has dimension $\le k$. Division $L = QL_W + R$ with $\operatorname{ord} R < k$ gives $R|_W = 0$, so $R = 0$. Conversely, if $L = QR$, $\operatorname{Sol}_E(R) \subset V$ is G-stable of dimension $\operatorname{ord} R$ (since R maps V into $\operatorname{Sol}(Q)$).

Step 2 (b)

$\operatorname{Sol}(L_u)$ is G-stable and contains u, hence $\supset W_u$; conversely L_{W_u} kills u, so $L_{W_u} \in \mathcal{D}L_u$ and $\operatorname{Sol}(L_u) \subset W_u$. Thus u identifies iff $W_u = V$.

Step 3 (c)

The first statement is (b). If V is semisimple, $\mathcal{A}$ acts through $\bigoplus \operatorname{End}(S_i)$ (Jacobson density); $S_i^m \cong S_i \otimes \mathbb{C}^m$ is cyclic over $\operatorname{End}(S_i)$ iff $m \le \dim S_i$ (a vector is cyclic iff its tensor rank is m), and components at distinct factors are independent.

Step 4 (d)

C_1 commutes with G (coefficients in K), so $C_1 : V \to E$ is a G-morphism with kernel $\operatorname{Sol}(C_1) = \operatorname{soc} V$ by (a); its image lies in $\operatorname{Sol}(L_1)$ and has dimension $n - \operatorname{ord} C_1 = \operatorname{ord} L_1$.

Why it works. a solution identifies the operator exactly when its Galois orbit spans the whole solution space.

■

Uses: van der Put–Singer ($E^G = K$, numbering to check).

Examples that follow

Airy $\partial^2 - x$ has no rational Riccati solution, so every non-zero solution identifies it. For $L_E = -\partial^2 + x^2 - E$ the rational solutions are $p(x)e^{\mp x^2/2}$ with p Hermite, existing iff $E = \pm(2n+1)$; for $E \notin \pm(2\mathbb{N}+1)$ every $u \neq 0$ identifies L_E, and for $E = 2n+1$ one has u identifying iff $u \notin \mathbb{C}h_n$ (Loewy length 2, socle $\mathbb{C}h_n$).

15.2 Statements recorded without proof here

SYS-T1	Identification criterion on an essential family: proved in B149 (as a tool).	[proof written, not audited] (a tool, not a structural theorem)

S8-SYS §1. Reduces to C3-T1 for $k = 1$; the structural content is SYS-EX.

SYS-N1, SYS-EX, SYS-T0	Socle obstruction (proved in B150); existence criterion; ODE systems on $\Omega = \mathbb{R}^n$.	known (Oberst, Pommaret, Macaulay)

The only new point is arbitrary Ω (A32).

SYS-S1-C	Failure of (i⇐) on a simply connected non-convex Ω, in C^∞ and $\mathcal{D}'$.	[false]

Proved in Chapter 5, after A34.

SYS-II-T2-C	"Crossing of eigenvalues implies a pole of the projector" is false for a derogatory A ($A = \mathrm{diag}(\xi, -\xi)$ has constant projectors); see B138.	[false]

one-line counterexample, S8-SYS §8.

Q-S8SYS-4	"The pole depends only on $A(\xi_0)$ and the factorisation"; the counterexample pair A_1, A_2 is in B138.	[false]

Phase 10 (F10-C).

G-T0	A gauge-fixed Maxwell potential never identifies Maxwell by itself; if $\mathrm{Ann}(A) = U_A + Rg$ with $g \cdot \partial \neq 0$ then $U_A = \{r \in \mathrm{Ann}(A) : r \cdot \partial = 0\}$.	observation

S8-SYS §6.

15.3 Variable coefficients

VAR-T6	Order of L_u is $\dim \operatorname{span}(Gu)$; socle/Loewy structure; existence iff V is cyclic ($m_i \le \dim S_i$). Proved from $E^G = K$ in B151.	from cited source

Galois correspondence $E^G = K$; S8-VAR §2.2.

VAR-P3 (Lipschitz, no continuity from C^1, Hölder $1/3$ sharp)	See A39.	[proof written, not audited]

Chapter 6.

Proofs moved. The results VAR-T2a, VAR-E1/E2, VAR-P8, VAR-P9, VAR-P11, VAR-T10, VAR-P12 and VAR-C3Q2 have a proof card in Chapter 19 (Dim. B80–B86). VAR-T6 (cited) and VAR-P3 stay as above.

CHAPTER 16

Phases 7–8: encoder and convergence (other results)

In this chapter	
Starting from	the notation of Chapters 7 and 8: $H_0 = -\partial_x^2$, $H(t) = H_0 + V(t)$, the propagators $U_0(t) = e^{-itH_0}$ and $U_V(t)$, and the encoder W with the intertwining relation $WL_q = (L_0 + c)W$.
Why	to prove the remaining results on the encoder: the classification of time-dependent intertwiners, the normal form with reparametrised time, the blind encoder, the equivalence on a bounded domain, the Faddeev decomposition and the Foldy–Wouthuysen transform. Definitions and statements from cited sources stay as list cards.
Role	Dim. B110–B116. They prove the corresponding parts of Chapter 14 of the paper.

16.1 The encoder

ENC-D Definition of a *structured encoder*: (E1) $\dim\ker W < \infty$ on the domain of the encoder; (E2′) $WL_q = (L_0 + c)W$ for a constant c; (E3) explicit. — definition

The strict version $c = 0$ excludes the constant case (B79).

Dim. B110 — **ENC-P0(a)**

Time-dependent intertwiners are interaction pictures

[proof written, not audited]

Let H_0 be self-adjoint on a Hilbert space with domain D, $V \in C(\mathbb{R}; B(\mathcal{H}))$ norm-continuous and self-adjoint, $H(t) = H_0 + V(t)$, and assume a unitary propagator $U_V(t,s)$ strongly continuous in both arguments, with $U_V(t,s)D \subset D$ and the usual differentiation rules (Dyson–Phillips; [from cited source]). Let $W(t)$ be strongly continuous, $W(t)D \subset D$, C^1 on D, with

$$iW'(t)\psi = H_0W(t)\psi - W(t)H(t)\psi \quad (\psi \in D). \tag{INT}$$

Then (INT) holds iff $W(t) = U_0(t)W(0)U_V(t)^{-1}$ for all t.

Stated in the paper as no. 14, p. 74

Idea. Along a solution of the true equation, $W(t)u(t)$ solves the free equation; and the free flow is unique. Conversely, compose the three flows and differentiate.

Proof.

Step 1 ($\Rightarrow$)

Let $u_0 \in D$, $u(t) = U_V(t)u_0$, $g(t) = W(t)u(t)$. Write $[g(t+h) - g(t)]/h = W(t+h)\frac{u(t+h)-u(t)}{h} + \frac{W(t+h)-W(t)}{h}u(t)$. The first term tends to $W(t)u'(t)$ (W locally bounded by Banach–Steinhaus and strongly continuous), the second to $W'(t)u(t)$ because $u(t) \in D$. So $ig' = H_0 g$ with $g(t) \in D$, whence $\frac{d}{dt}[U_0(-t)g(t)] = 0$ and $g(t) = U_0(t)g(0)$: $W(t)U_V(t)u_0 = U_0(t)W(0)u_0$ on a dense set, hence on $\mathcal{H}$ by boundedness.

Step 2 ($\Leftarrow$)

Put $M = W(0)$; only $MD \subset D$ is used. M is closed as a map $(D, \|\cdot\|_{\mathrm{graph}}) \to (D, \|\cdot\|_{\mathrm{graph}})$: if $\psi_n \to \psi$ and $M\psi_n \to \varphi$ in the graph norm, then $M\psi_n \to M\psi$ in $\mathcal{H}$, so $\varphi = M\psi$. By the closed graph theorem M is graph-bounded. For $\psi \in D$ the map $t \mapsto U_V(0,t)\psi$ is C^1 in $\mathcal{H}$ and continuous in the graph norm, so $U_0(t)MU_V(0,t)\psi$ is differentiable and the computation with $iU_V(0,t)H(t)\psi$ gives (INT).

Why it works. the intertwiner is the interaction picture; the free flow is the only thing that can come out.

■

Uses: none.

Dim. B111 — **ENC-P0b-loc, POLE-N-cl**

An encoder independent of time sees only the autonomous region [proof written, not audited]

Let $U \subset J$ be open and $W : D_W \to \mathcal{D}'(U)$ linear with $D_W \supset C(U) \cdot C_c^\infty(U)$ and $\dim\ker W < \infty$ on all of D_W, or $V(t,\cdot) \in C^\infty(U)$ and $W : C^\infty(U) \to C^\infty(U)$. If $W(-\partial_x^2 + V(t,\cdot))f = (-\partial_x^2 + c(t))Wf$ for all t and $f \in C_c^\infty(U)$, then $V(t,x) - V(s,x) = c(t) - c(s)$ on U.

(POLE-N^{cl}) If $W(t) = \sum_j w_j(t,x)\partial_x^j$ is an instantaneous intertwiner that also intertwines $i\partial_t - L_{V(t)}$ with $i\partial_t - L_0 - c$, then $\partial_t W = 0$ on any rectangle free of poles, and $V - c$ is autonomous there.

Stated in the paper as no. 14, p. 74

Proof.

Step 1 Subtract the identities

Taking the identity at t and at s and subtracting gives $W(mf) = 0$ for $f \in C_c^\infty(U)$, with $m = V(t,\cdot) - V(s,\cdot) - (c(t) - c(s))$ continuous.

Step 2 An infinite independent family in the kernel

If $m(x^*) \neq 0$, then $m \neq 0$ on an interval $U' \ni x^*$. Pick non-zero $\varphi_1, \varphi_2, \dots \in C_c^\infty(U')$ with disjoint supports: the functions $m\varphi_k$ are linearly independent and lie in $D_W \cap \ker W$, against $\dim \ker W < \infty$.

Step 3 Why the kernel must be finite on all of D_W

Requiring it only on $C_c^\infty(U)$ is false: for $X = C_c^\infty(U)$, w continuous and smooth on no interval, $W(g + wf) := g$ on $X \oplus wX$, $V = tw$, $c = 0$ satisfies the identity with $\ker W|_X = 0$ and V not autonomous.

Step 4 POLE-N

If W also intertwines the evolutions, $\partial_t W = 0$, so $\partial_t w_j = 0$ on $[s,t] \times U$ and $w_j(t,\cdot) = w_j(s,\cdot)$ there. The leading coefficient w_n is a non-zero constant (ENC-K1′), so $\dim \ker W = n \geq 1$, and the first part applies. A pole-free interval at the two times is not enough: with the pole at $x = z(t) = t$ and $U = (\frac{1}{2}, \frac{3}{2})$ the interval is free at $t = 0, 2$ but crossed at $t = 1$.

Why it works. an infinite family of functions supported where V moves would all be in the kernel of the encoder.

■

Uses: A44.

Dim. B112 — **ENC-P0p, ENC-P0p-DIC**

Normal form of the encoder, and when the time change can be avoided [proof written, not audited]

Let $U_V(t)$ be unitary with $U_V(0) = I$, $W(t) \in B(\mathcal{H})$, $\theta \in C^1(I)$ with $\theta' > 0$, $\theta(0) = 0$, and $D_0 \subset \mathcal{H}$ dense. If for every $u_0 \in D_0$ the function $v(\tau) = W(\theta^{-1}(\tau))U_V(\theta^{-1}(\tau))u_0$ is C^1 with values in $D(H_0)$ and $iv' = H_0 v$, then $W(t) = U_0(\theta(t))W(0)U_V(t)^{-1}$; in particular θ is not an invariant of the encoder.

(DIC) For $V = \alpha(t)x^2 + \beta(t)x + \gamma(t)$ real, a pointwise-plus-gauge representative has equations $T' = 1/\rho^2$, $\rho' = 4a\rho$, $a' + 4a^2 = -\alpha$. There is a representative with $T = \mathrm{id}$ iff $\alpha \equiv 0$; for $\alpha \not\equiv 0$, T is affine in no representative.

Stated in the paper as no. 14, p. 74

Proof.

Step 1 Normal form

$\frac{d}{d\tau}[U_0(-\tau)v(\tau)] = U_0(-\tau)(v' + iH_0 v) = 0$, so $v(\tau) = U_0(\tau)v(0)$ with $v(0) = W(0)u_0$. Conclude by density of D_0 and boundedness of $W(t)$ and $U_V(t)$. The proof uses neither the propagator hypothesis of B110 nor the class of encoders.

Step 2 Examples

The lens ($H_0 = -\frac{1}{2}\partial^2$, $\theta = \tan$), Avron–Herbst ($V = Fx$, $\theta = \mathrm{id}$, $u = e^{-iFtx - iF^2t^3/3}v(t, x + Ft^2)$) and the class $V = \beta(t)x + \gamma(t)$ (with $\theta = \mathrm{id}$) are instances on $D_0 = \mathcal{S}(\mathbb{R})$.

Step 3 (DIC) $\Leftarrow$

If $\alpha \equiv 0$ take $a \equiv 0$, $\rho \equiv 1$, $T = \mathrm{id}$.

Step 4 (DIC) $\Rightarrow$

If T is affine, $T' = 1/\rho^2$ is constant so ρ is constant; $\rho' = 4a\rho$ forces $a \equiv 0$ and $a' + 4a^2 = -\alpha$ forces $\alpha \equiv 0$.

Step 5 The earlier synthesis is false

"The exceptions change the time" fails: for $V = 0$ the pseudo-conformal representative $a = a_0/(1 + 4a_0t)$, $\rho = 1 + 4a_0t$, $\theta(t) = t/(1 + 4a_0t)$ solves the system with $\alpha \equiv 0$ and has $\theta \neq \mathrm{id}$. Also $a \equiv 0$ gives only ρ constant, not $\rho \equiv 1$. The time change is needed for the quadratic part in the existential sense only.

Why it works. the free flow is unique, so any reparametrisation is absorbed; whether it can be avoided is decided by one Riccati equation.

■

Uses: B110.

Dim. B113 **ENC-N1-F6, ENC-N1p**

The blind encoder: why the claimed equivalence is false [proof written, not audited]

Fix a family of encoders $\mathcal{W} = \{W_V\}$ with $W_V L_V = (L_0 + c_V)W_V$, W_V injective on the solution space $\mathcal{S}_V$, $N_V(v) = e^{ic_V t}v$. A *blind encoder* is a map $E : \bigcup_V [\mathcal{S}_V] \to \mathcal{O}$ with $E([u]) = [N_V W_V u']$ for every V and $u' \in \mathcal{S}_V$, $u' \sim u$. Let (C) be coherence ($u \sim u'$ implies $[N_V W_V u] = [N_{V'} W_{V'} u']$) and (S) separation. (i) A blind encoder exists iff (C). (ii) Identifiability modulo $\sim$ implies (C). (iii) (C)$\wedge$(S) iff identifiability, by definition. (iv) Witness: with DM-φ (phase $e^{i\theta(t)}$ allowed) and $\mathcal{V} = \{q_0 + c\}$, $W_{q_0+c} := W_{\hat{q}}$, $\hat{q} = q_0 - q_0(x_0)$, a blind encoder exists but V is identified only modulo constants. (v) With the strict family (the Volterra encoder of $q_0 + c$ normalised at x_0) no blind encoder exists.

Stated in the paper as no. 14, p. 74

Proof.

Step 1 (i)–(iii)

(i) If E exists, (C) is its well-definedness; conversely (C) says $[N_V W_V u']$ does not depend on the pair (V, u'). (ii) If $u \sim u' \neq 0$ then $V = V'$, and for a single V, $u' = e^{i\theta_0}u$ with θ_0 constant (since $\theta' u = 0$), and W_V commutes with a constant phase. (iii) ($\Leftarrow$) is (ii); ($\Rightarrow$) if $V \neq V'$ with $u \sim u' \neq 0$ existed, (S) would contradict (C).

Step 2 (iv) Blind without identifiability

One checks $W_{\hat{q}} L_{q_0+c} = (L_0 + q_0(x_0) + c) W_{\hat{q}}$ and $W_{\hat{q}}$ commutes with phases in t. For $u \in \mathcal{S}_{q_0}$, $u' = e^{-ict}u \in \mathcal{S}_{q_0+c}$ has $N_{V'} W_{\hat{q}} u' = e^{i(q_0(x_0)+c)t} e^{-ict} W_{\hat{q}} u \approx N_V W_{\hat{q}} u$. So (C) holds, but q_0 and $q_0 + c$ are not distinguished.

Step 3 (v) Dependence on the family

For $q_0 = 0$, $u = e^{-i\lambda t} \cos \sqrt{\lambda}(x - x_0) \in \mathcal{S}_0$ and $u' = e^{-ict}u \in \mathcal{S}_c$: $W_0 u = u$ while $W_c u' = e^{-i(\lambda+c)t} \cos \sqrt{\lambda + c}(x - x_0)$, outputs not equivalent modulo phases, so (C) fails.

Step 4 Gauge reading

With DM-φ, for any regular non-zero $w = e^{i\theta(t)}u$ one has $(iw_t + w_{xx})/w = q(x) - \theta'(t)$: q is identified modulo constants, which is the correct statement: *blind encoder* $\iff$ *identifiability modulo the gauge group of the data model, for a fixed gauge-respecting family*.

Why it works. the original equivalence compared a map on potentials with a map on classes; the constant shift of V is invisible on the output and breaks the injectivity of Φ.

■

Uses: A47.

Dim. B114 **NS-EQ-Omega**

Avron–Herbst and the lens on a bounded domain

[proof written, not audited]

Let $\Omega = I \times J$ be a bounded rectangle and Φ the pointwise-plus-gauge encoder: Avron–Herbst $v(t, y) = e^{iFt(y - Ft^2) + iF^2 t^3/3} u(t, y - Ft^2)$; lens $v(\tan t, x/\cos t) = (\cos t)^{1/2} e^{ix^2 \tan t/2} u(t, x)$, $I \subset (-\frac{\pi}{2}, \frac{\pi}{2})$. Then (a) $\Phi : \mathrm{Sol}_V(\Omega) \to \mathrm{Sol}_0(\Omega')$ is a bijection with no boundary condition, where Ω' is a curved strip (Avron–Herbst; a rectangle iff $F = 0$) or $\{|y|/\sqrt{1+\tau^2} \in J\}$ (lens: non-convex, star-shaped with respect to 0 if $0 \in I$), both simply connected. (b) Dirichlet goes to Dirichlet on the moving boundary; Neumann goes to a time-dependent Robin condition ($v_y = iFt\, v$; $v_y = iy \sin t \cos t\, v$). (c) (I) and (S) hold on Ω', but backwards transport takes place in the conjugated ring $\Phi^{-1}\mathbb{C}[\partial_\tau, \partial_y]\Phi$; (E) fails. (d) For $H = -\frac{1}{2}\partial^2 + \frac{1}{2}x^2$, $U(\pi) = -iP$ (P the parity), and gluing across the caustics $t = \frac{\pi}{2} + k\pi$ is exact on $\mathbb{R}$.

Stated in the paper as no. 14, p. 74

Proof.

Step 1 (a) Domain

Direct computation of the images. Non-convexity of the lens: $(\pm T, b\sqrt{1+T^2} - \varepsilon) \in \Omega'$ but $(0, b\sqrt{1+T^2} - \varepsilon) \notin \Omega'$. Star-shapedness: $s\sqrt{1+\tau^2} \le \sqrt{1+s^2\tau^2}$ for $s \in [0,1]$. Simple connectedness: Ω' is a homeomorphic image of a rectangle.

Step 2 (b) Boundary

The prefactor does not vanish, so Dirichlet is preserved. Differentiating the product gives the Robin conditions.

Step 3 (c) Identification, splitting, factorisation

C3-T1 needs Ω connected and C12-T1′ needs $H^1_{dR} = 0$, both true on Ω'. For Avron–Herbst $\Phi^{-1}\partial_y\Phi = \partial_x + iFt$ and $\Phi^{-1}\partial_t\Phi = \partial_t - 2Ft\partial_x + iFx - iF^2t^2$ (polynomial in (t,x)); for the lens the coefficients are trigonometric in t and polynomial in x. The potential is suppressed, not split.

Step 4 (d) Gluing

$e^{-i\pi(n+1/2)} = -i(-1)^n$ on the Hermite functions h_n, so $U(\pi) = -iP$ and $u(s+k\pi) = (-iP)^k u(s)$; $W_0(s)$ commutes with P (even chirp, even dilation), so $W_0(s)u(s+k\pi) = (-i)^k P^k W_0(s)u(s)$.

Why it works. the transformations are pointwise, so the boundary only moves; the conjugated ring has non-constant coefficients.

■

Uses: A44, A46.

Dim. B115 **FAD-1**

Faddeev decomposition at fixed energy [proof written, not audited]

Let $J = [x_0, x_0+\ell]$, $\lambda = k^2 \in \mathbb{C}$, $V = \sum_{i=1}^N V_i$, $V_i \in C(J)$, $(G_0 f)(x) = \int_{x_0}^x \frac{\sin k(x-y)}{k} f(y)dy$, u the solution of $-u'' + Vu = \lambda u$ with the Cauchy data at x_0 of the free solution $u^{(0)}$. Then (i) $u = u^{(0)} + G_0(Vu)$ and $\|(G_0V)^n\| \le (C\|V\|_\infty \ell)^n/n!$, $C = \sup|\sin(kz)/k|$, with no smallness; (ii) the Faddeev components $u^{(i)} = G_0(V_i u)$ satisfy $u = u^{(0)} + \sum u^{(i)}$ and the $N \times N$ system has a unique solution; (iii) with u_i the solution with V_i alone and $\mathbf{V} = \sum \|V_i\|_\infty$, $E^{\text{add}} := u - \sum u_i + (N-1)u^{(0)} = \sum_{i\neq j} G_0V_iG_0V_ju^{(0)} + R$, $\|R\| \le (C\ell\mathbf{V})^3 e^{C\ell\mathbf{V}}\|u^{(0)}\|/3$, uniformly in N.

Stated in the paper as no. 14, p. 74

Proof.

Step 1 (i)

$(-\partial^2 - \lambda)(u^{(0)} + G_0(Vu)) = -Vu$ and the Cauchy data agree, so uniqueness of the Cauchy problem gives the identity. G_0V is a Volterra operator with kernel bounded by $C\|V\|$, which gives the factorial bound.

Step 2 (ii)

Summing the components gives (i). Uniqueness follows from the Volterra estimate for the system on $C(J)^N$.

Step 3 (iii)

Expand $u = \sum_n (G_0 V)^n u^{(0)}$ and $u_i = \sum_n (G_0 V_i)^n u^{(0)}$. The terms of order 0 and 1 cancel. At order 2 what remains is $(G_0 V)^2 - \sum_i (G_0 V_i)^2 = \sum_{i \neq j} G_0 V_i G_0 V_j$. The remainder is $\sum_{n \geq 3} [(G_0 V)^n - \sum_i (G_0 V_i)^n] u^{(0)}$; since $\|V\| \leq \mathbf{V}$ and $\sum_i \|V_i\|^n \leq \mathbf{V}^n$, it is at most $2 \sum_{n \geq 3} (C\ell \mathbf{V})^n / n! \, \|u^{(0)}\|$. With $\|\sum V_i\|_\infty$ in place of $\mathbf{V}$ the remainder is not controlled, since centres may cancel.

Why it works. a Volterra operator has a factorially convergent Neumann series, so the exact decomposition by centres costs no smallness; only the cross terms are of second order.

■

Uses: A44.

Dim. B116 **FW-1**

Foldy–Wouthuysen for the free Dirac equation

[proof written, not audited]

Let $H = \alpha p + \beta m$ with $\alpha = \sigma_x$, $\beta = \sigma_z$, $m > 0$, and $U(p) = \cos\theta + \beta\alpha \sin\theta$, $\tan 2\theta = p/m$. Then U is unitary, $UHU^\dagger = \beta E(p)$ with $E = \sqrt{p^2 + m^2}$, and U is a Fourier multiplier with non-rational entries ($\sin^2\theta = (1 - m/E)/2$), hence non-local. So FW is a decomposition (S) by sheets, not an encoder (E2′) towards free Dirac.

Stated in the paper as no. 14, p. 74

Proof.

Step 1 Unitarity and diagonalisation

$\beta\alpha = i\sigma_y$ (up to sign) is anti-Hermitian with square -1, so $U = \cos\theta + \beta\alpha \sin\theta$ is a rotation, unitary. Using $\beta\alpha\beta = -\alpha$, one computes $UHU^\dagger = \beta(m\cos 2\theta + p \sin 2\theta) + \alpha(p\cos 2\theta - m \sin 2\theta)$; $\tan 2\theta = p/m$ kills the second bracket and makes the first equal to $E(p)$.

Step 2 Non-locality

A convolution operator is local iff its symbol is a polynomial. The entries $\cos\theta, \sin\theta$ are algebraic non-rational functions of p that tend to constants at infinity without being constant, so U is not a differential operator, consistent with "rational projectors $\iff$ factors". With a potential $V(x)I$ the exact decomposition into sheets $\pm E$ exists abstractly ($\Lambda = \operatorname{sign} H$, Eriksen; Thaller 1992, ch. 5, [from cited source]); the FW series in $1/m$ is only asymptotic.

Why it works. the diagonalising rotation depends on the momentum through $\sqrt{p^2 + m^2}$, which no polynomial symbol can reproduce.

■

Uses: B74.

Dim. B152 **ENC-POLE-N**

The classical form of ENC-P0(b), and the failure of the POLE route [proof written, not audited]

Let J be an interval, $V \in C(I \times J)$, $c \in C(I)$, and let W be linear, defined on $D_W \supset C(J) \cdot C_c^\infty(J)$ (or on $C^\infty(J)$ if $V(t,\cdot) \in C^\infty$), with $\dim\ker W < \infty$ on all of D_W. If $W(-\partial_x^2 + V(t,\cdot))f = (-\partial_x^2 + c(t))Wf$ for all $t \in I$ and $f \in C_c^\infty(J)$, then $V(t,x) - V(s,x) = c(t) - c(s)$ for all t, s, x. Consequence (POLE-N): an *instantaneous* intertwiner $WL_{V(t)} = (L_0 + c)W$ that is also an encoder for $iu_t = L_{V(t)}u$ forces $\partial_t W = 0$, hence V autonomous modulo $c(t)$.

Stated in the paper as no. 14, p. 74

Proof.

Step 1 Subtract two times

Subtracting the identity at t and s gives $W(mf) = 0$ for all $f \in C_c^\infty(J)$, with $m = V(t,\cdot) - V(s,\cdot) - (c(t) - c(s))$ continuous.

Step 2 Infinitely many independent kernel elements

If $m(x^*) \neq 0$, then $m \neq 0$ on an interval $U \ni x^*$. Choose non-zero $\varphi_1, \varphi_2, \dots \in C_c^\infty(U)$ with disjoint supports: the $m\varphi_k$ are linearly independent, lie in D_W and in $\ker W$, against $\dim\ker W < \infty$.

Step 3 POLE-N

If $W(t)$ also satisfies the encoder relation $iW' = (L_0 + c)W - WL_{V(t)}$, the instantaneous intertwining gives $\partial_t W = 0$ on a rectangle $[s,t] \times U$ free of poles (for a polar potential such as Adler–Moser one cannot just ask a pole-free interval at both times: with a pole $z(t) = t$ and $U = (\frac{1}{2}, \frac{3}{2})$ the pole crosses U at $t = 1$). Then the coefficients w_j do not depend on t, $\dim\ker W = n$ because w_n is a non-zero constant, and Steps 1–2 apply: V is autonomous modulo c on U.

Why it works. a time-independent map with a small kernel cannot absorb a potential difference that is non-zero on an interval: it would have to kill infinitely many independent functions.

■

Uses: ENC-K1′(i), ENC-1.

Scope

In $L^2(J)$ with bounded J and a self-adjoint H_0 the boundary conditions are not respected by the Volterra encoder, so the Hilbert version of P0(b) does not exclude an ENC-1 type encoder for $V(t,x)$. The localised version needs finite kernel on all of D_W: with only $C_c^\infty(U)$ it is false (counterexample $X = C_c^\infty(U)$, w smooth nowhere, $V = tw$). Sub-case: W commuting with ∂_t rather than independent of t (e.g. convolutions in t) with non-constant $c(t)$ is open.

Dim. B153 **ENC-P0p**

Time change is not an invariant of the encoder; "the exceptions change the time" is false [proof written, not audited]

(i) Let $U_V(t)$ be unitary with $U_V(0) = I$, $\theta \in C^1(I)$, $\theta' > 0$, $\theta(0) = 0$, D_0 dense, and suppose that for $u_0 \in D_0$, $v(\tau) = W(\theta^{-1}\tau)U_V(\theta^{-1}\tau)u_0$ is C^1 with values in $D(H_0)$ and $iv' = H_0 v$. Then $W(t) = U_0(\theta(t))W(0)U_V(t)^{-1}$; so every such encoder is equivalent to one with $\theta = \mathrm{id}$. (ii) For $V = \alpha(t)x^2 + \beta(t)x + \gamma(t)$ there is a point representative plus gauge with $\theta = \mathrm{id}$ iff $\alpha \equiv 0$; for $\alpha \not\equiv 0$ no representative has θ affine. (iii) The rev. 1 summary "the exceptions change the time" is false as written: the Avron–Herbst potential $V = Fx$ and the whole class $V = \beta(t)x + \gamma(t)$ have $\theta = \mathrm{id}$; and for $V = 0$ the pseudo-conformal representative has $\theta \neq \mathrm{id}$.

Stated in the paper as no. 14, p. 74

Proof.

Step 1 (i)

$\frac{d}{d\tau}[U_0(-\tau)v(\tau)] = U_0(-\tau)(v' + iH_0 v) = 0$, so $v(\tau) = U_0(\tau)v(0)$ with $v(0) = W(0)u_0$. Density of D_0 and boundedness of W, U_V conclude.

Step 2 (ii) The point ansatz

In the ansatz (chirp $a(t)$, dilation $\rho(t) > 0$, translation, phase, time $T(t)$) the equations are $T' = 1/\rho^2$, $\rho' = 4a\rho$, $a' + 4a^2 = -\alpha$. If T is affine, T' is constant so ρ is constant, then $a \equiv 0$ by $\rho' = 4a\rho$, and $\alpha \equiv 0$ by the third equation. Conversely $\alpha \equiv 0$ allows $a = 0$, $\rho = 1$, $T = \mathrm{id}$.

Step 3 (iii) Counterexamples

For $V = 0$, take $a = a_0/(1 + 4a_0 t)$, $\rho = 1 + 4a_0 t$, $\theta(t) = t/(1 + 4a_0 t)$: $u = \rho^{-1/2} e^{iax^2} v(\theta(t), x/\rho)$ maps free solutions to free solutions, with $\theta \neq \mathrm{id}$. For Avron–Herbst, $u = e^{-iFtx - iF^2t^3/3} v(t, x + Ft^2)$ with $\theta = \mathrm{id}$; for $V = \beta(t)x + \gamma(t)$, $u = e^{-iB(t)x - i\Phi(t) - i\int\gamma} v(t, x - \Xi(t))$ with $B' = \beta$, $\Xi' = -2B$, $\Phi' = B^2$.

Why it works. the time change belongs to the local representative, not to the encoder, since two representatives differ by a free-flow unitary.

■

Uses: B152, Bluman–Shtelen 1996 (classification, cited).

POLE-N (Hilbert version)	Instantaneous intertwiners with an evolving $V(t)$ are not encoders for Schrödinger; they serve only the Lax flow. Classical version in B152.	conditional on the propagator hypothesis of B110

S8-ENC §4. Classical version in B111.

ENC-P0-sintesi (rev. 1)	"The exceptions change the time"; refuted in B153.	[false] as written

Avron–Herbst and the whole class $V = \beta(t)x + \gamma(t)$ have $\theta = \mathrm{id}$ (B112, Step 5).

ENC-N1 (F6)	"A blind encoder exists iff $\Phi(V,v) = W_V^{-1}v$ is injective."	[false] as written

witness ENC-N1′(iv) in B113.

GAM-O1	Gamblets: an L_q-orthogonal decomposition (S), not an encoder: the bases depend on q, so there is no intertwiner $WL_q = L_0W$.	observation, from cited source

Owhadi 2017 (theorem numbers to be checked).

ENC-K1	Rational $V \to 0$: a differential W exists iff V is an iterated Darboux transform of 0 (Adler–Moser).	from cited source, to be checked

Duistermaat–Grünbaum 1986.

16.2 Convergence

Dim. B154 **CONV-D2**

On a convex set, $\ker(Q_1Q_2) = \ker Q_1 + \ker Q_2$ in $C^\infty(\Omega)$ [from cited source]

Let $\Omega \subset \mathbb{R}^n$ be convex, $\mathcal{A} = C^\infty(\Omega)$, $Q_1, Q_2 \in R_n = \mathbb{C}[\xi_1..\xi_n]$ coprime. Then $\ker_{\mathcal{A}}(Q_1Q_2) = \ker_{\mathcal{A}} Q_1 + \ker_{\mathcal{A}} Q_2$, and the ambiguity of the sum is isomorphic to $\mathrm{Sol}(R/(Q_1,Q_2))$.

Stated in the paper as no. 13, p. 71

Proof.

Step 1 The source (cited)

For Ω convex $C^\infty(\Omega)$ is an injective R_n-module (Malgrange, Sém. Bourbaki exp. 246, Thm 3.2(2), verified; the convention $X_j \mapsto -i\partial_j$ differs by a ring automorphism preserving injectivity).

Step 2 The homological step

In a UFD, $(Q_1)\cap(Q_2) = (Q_1Q_2)$, so $0 \to R/(Q_1Q_2) \to R/(Q_1)\oplus R/(Q_2) \to R/(Q_1,Q_2) \to 0$ is exact. Applying the exact functor $\mathrm{Hom}(\cdot,\mathcal{A})$ yields $0 \to \mathrm{Sol}(R/(Q_1,Q_2)) \to \ker Q_1 \oplus \ker Q_2 \to \ker(Q_1Q_2) \to 0$. By induction the decomposition exists for any product of pairwise coprime factors.

Why it works. injectivity of the solution module makes the Hom functor exact on the sequence that expresses the intersection of two principal ideals.

■

Uses: none.

Restriction

Only C^∞ is claimed: for $\mathcal{D}'(\Omega)$ injectivity on convex Ω was not found in the sources (CONV-Q17), and non-convex Ω is open (CONV-Q5). CONV-D1 (the ambiguity) does not use injectivity and holds in $\mathcal{D}'(\Omega)$.

ALG-TK	Trotter–Kato convergence of splitting only for V bounded or relatively bounded in the sense of Trotter 1959, Kato 1978, Engel–Nagel III.5.8.	from cited source

F10-G9.

Proofs moved. ENC-K1′, ENC-A9, CONV-T6, P6, P7, P8, D1, B1, B2, R1, R1-F, MZ1–3, MZ3-F, T7 and T8 have a proof card in Chapters 20 and 21 (Dim. B88–B97). ENC-K1 (cited), CONV-D2 (cited) and CONV-B3 (statement only) stay in Chapter 20 or above.

CHAPTER 17

Phase 8: the dictionary (other results)

In this chapter	
Starting from	the notation of Chapter 9: levels span $\mathcal{D}_n$, atoms D_i, the Gram matrix $G_n(u)$ and the residual r_n.
Why	to give the proofs of the remaining universality, negative, rank, misfit and exponential-family results of the dictionary package. The few statements that are heuristic, from a cited source or only checked on examples stay as list cards at the end.
Role	Dim. B98–B109. They prove the corresponding parts of Chapter 15 of the paper.

17.1 Universality by class

Dim. B98 — **DICT-U1b, U4**

Singular coefficients and systems

[proof written, not audited]

Let $K \subset \mathbb{R}^N$ be compact and $T = \sum_{|\alpha| \le m} a_\alpha \partial^\alpha$ with $a_\alpha \in L^p(K)$, $1 \le p < \infty$. If q_n are polynomials with $\|a_\alpha - q_{n,\alpha}\|_{L^p(K)} \to 0$, then $\|(T - T_n)u\|_{L^p(K)} \le \sum_\alpha \|a_\alpha - q_{n,\alpha}\|_{L^p} \|u\|_{C^m(K)} \to 0$ for every $u \in C^m(K)$, with no uniform rate. For systems, $\mathcal{D}_n^{(k)} = \{E_{ij} \otimes D : D \in \mathcal{D}_n\}$ and every density statement holds componentwise.

Stated in the paper as no. 15, p. 77

Proof.

Step 1 Singular coefficients

Polynomials are dense in $L^p(K)$ for $p < \infty$. Hölder with the bounded factor $\partial^\alpha u$ gives $\|(a_\alpha - q_{n,\alpha})\partial^\alpha u\|_{L^p} \le \|a_\alpha - q_{n,\alpha}\|_{L^p} \|\partial^\alpha u\|_\infty$. Such a T maps $C^\infty(\Omega)$ into $C^\infty(\Omega)$ only outside the singular set, so it is outside Peetre's framework on Ω; exactness needs atoms $|x - x_0|^{-j}$, $\log|x - x_0|$ with a continuous parameter x_0 (with $x_0 \in \mathbb{Q}^d$ exact only in the limit).

Step 2 Systems

The topology on $k \times k$ matrix operators is the finite product; convergence is convergence of each entry. The relative annihilator $\{Q \in \operatorname{span} \mathcal{D}_n^{(1\times k)} : Qu = 0\}$ is the kernel of the block Gram matrix of $(De_c \otimes u_c)_{D,c}$, because $Qu = \sum q_{D,c} D u_c = 0$ iff q annihilates the quadratic

form $\|\sum q_{D,c} D u_c\|^2$. Rational symbols and sums of exponential memory are augmented local systems ($Q(D)v = P(D)u$; $w_j' = -\lambda_j w_j + f$).

Why it works. density in L^p costs nothing against a bounded factor; systems are finite products.

■

Uses: A48.

Dim. B99 **DICT-U2**

Non-linear local operators: the function F is smooth [proof written, not audited]

Let $D : C^\infty(\Omega, \mathbb{R}^q) \to C^\infty(\Omega, \mathbb{R}^p)$ be local ($Dg(y)$ depends on the germ of g at y) and regular (smooth families go to smooth families). For $f \in C^\infty$ and compact K there are an order r, a neighbourhood $\Omega_1 \supset K$, $\varepsilon_0 > 0$ and $F \in C^\infty(\mathcal{W}, \mathbb{R}^p)$, $\mathcal{W} = \{(y, j) \in \Omega_1 \times J^r : |j - j^r_y f| < \varepsilon_0\}$, with $Dg(y) = F(y, j^r_y g)$ for g in a C^∞-neighbourhood $\mathcal{V}$ of f and $(y, j^r_y g) \in \mathcal{W}$. Hence for every compact $\mathcal{C} \subset \mathcal{W}$, ℓ, M, ε there is a differential polynomial P in the dictionary $\mathcal{D}^{NL}_n$ with $\|Dg - P[g]\|_{C^\ell(K)} \le \varepsilon$ for all $g \in \mathcal{V}$ with jets in $\mathcal{C}$ and $\|g\|_{C^{r+\ell}(K)} \le M$.

Stated in the paper as no. 15, p. 77

Idea. A non-linear Peetre theorem says $Dg(y)$ depends only on a finite jet, uniformly near f. Realise each jet by an explicit smooth family of functions; regularity of D then makes the jet-to-value map smooth. Smooth functions are approximated by polynomials.

Proof.

Step 1 Finite jet dependence (cited)

By the regular version of the non-linear Peetre theorem (Slovák 1986/1988; Kolář–Michor–Slovák §19.10; [from cited source]) there are r and a neighbourhood $\mathcal{V} \ni f$ in the compact-open C^∞ topology such that $j^r_y g_1 = j^r_y g_2 \Rightarrow Dg_1(y) = Dg_2(y)$ for $g_1, g_2 \in \mathcal{V}$ and $y \in K_1 \ni K$.

Step 2 Smooth realisation of jets

Take $\chi \in C^\infty_c(B_\rho(0))$, $\chi \equiv 1$ near 0, and $g_{y,j}(z) = f(z) + \chi(z-y)\sum_{|\alpha| \le r} (j_\alpha - \partial^\alpha f(y)) \frac{(z-y)^\alpha}{\alpha!}$. Then $j^r_y g_{y,j} = j$, the perturbation has C^k norm $\le C_1 |j - j^r_y f|$, so $g_{y,j} \in \mathcal{V}$ for $|j - j^r_y f| < \varepsilon_0 = \eta / (2C_1)$, and $(y, j, z) \mapsto g_{y,j}(z)$ is C^∞.

Step 3 F is well defined and smooth

Put $F(y, j) = D(g_{y,j})(y)$. If $g \in \mathcal{V}$ and $(y, j^r_y g) \in \mathcal{W}$, then g and $g_{y, j^r_y g}$ both lie in $\mathcal{V}$ and share the r-jet at y, so $Dg(y) = F(y, j^r_y g)$. Smoothness: along a smooth curve $c(s) = (y(s), j(s))$ the family $G(s, z) = g_{c(s)}(z)$ is smooth, so by regularity $s \mapsto D(G(s, \cdot))(y(s)) = F(c(s))$ is smooth; Boman's theorem makes F smooth.

Step 4 Approximation

Take $\psi \in C_c^\infty(\mathcal{W})$, $\psi \equiv 1$ near $\mathcal{C}$. As in B98 there is a polynomial P_0 with coefficients in $\mathbb{Q}(i)$ and $\|F - P_0\|_{C^\ell(\mathcal{C})} \le \varepsilon'$. By the chain rule (Faà di Bruno) $\partial_y^\gamma[(F - P_0)(y, j_y^r g)]$, $|\gamma| \le \ell$, is a sum of derivatives of $F - P_0$ of order $\le |\gamma|$ times products of derivatives of g of order $\le r + |\gamma|$, bounded by M. Hence the claim with $\varepsilon = C(\ell, r, M)\varepsilon'$.

Why it works. locality gives finite order, regularity gives a smooth dependence on the jet, and a smooth function of finitely many variables is a limit of polynomials.

■

Uses: A48.

Where it stops

(a) If the order is only locally finite in the jet, $D(u) = \sum_k \psi_k(u)\partial_x^k u$ with $\psi_k \in C_c^\infty((k, k+1))$, no fixed level approximates D uniformly on all bounded g. (b) $\sin u$ is not in the global uniform closure: for $g \equiv c$ every polynomial P has $\sup_c |P(y_0, c, 0, \dots) - \sin c| \ge 1$. It becomes exact with the atoms e^{iku}, $e^{\lambda u}$ (B109).

Dim. B100 **DICT-N2**

The space of symbols is not separable

[proof written, not audited]

Let $S^m = S^m_{1,0}(\mathbb{R}^d \times \mathbb{R}^d)$. There is a family $\{m_\varepsilon\}_{\varepsilon \subset \mathbb{N}_{\ge 1}}$, bounded in S^m uniformly in ε, with $\|m_\varepsilon(D) - m_{\varepsilon'}(D)\|_{H^s \to H^{s-m}} \ge 1$ for $\varepsilon \ne \varepsilon'$. Hence no countable set is norm-dense in a bounded set of $\mathrm{Op}(S^m)$.

Stated in the paper as no. 15, p. 77

Proof.

Step 1 A dyadic family

Let φ be smooth, non-increasing, $= 1$ on $[0,1]$, $= 0$ on $[2,\infty)$; $\psi(r) = \varphi(r) - \varphi(2r) \ge 0$ is supported in $[\frac{1}{2}, 2]$; $\psi_k(\xi) = \psi(2^{-k}|\xi|)$ and $m_\varepsilon(\xi) = \langle\xi\rangle^m \sum_{k \in \varepsilon} \psi_k(\xi)$.

Step 2 Bounded in S^m

On $\operatorname{supp} \psi_k$ one has $|\xi| \asymp 2^k$, each ξ-derivative of ψ_k brings a factor $2^{-k} \asymp \langle\xi\rangle^{-1}$, and at most two ψ_k are non-zero at a point. So all seminorms are bounded independently of ε.

Step 3 Separation

For a multiplier, $\|p(D)\|_{H^s \to H^{s-m}} = \sup |p|\langle\xi\rangle^{-m}$ (Plancherel). If $k \in \varepsilon \setminus \varepsilon'$, at $|\xi| = 2^k$ one has $\psi_k = 1$ and all other $\psi_j = 0$ (since $\psi \ge 0$, $\sum \psi_j \le 1$), so $|m_\varepsilon - m_{\varepsilon'}|\langle\xi\rangle^{-m} = 1$ there.

Why it works. the continuum of subsets of dyadic shells gives a continuum of 1-separated symbols of the same size.

■

Uses: B98.

Dim. B101 **DICT-U3c, U3d**

Symbols with loss, completely monotone memory

[proof written, not audited]

(c) Let $m < m'$. Every x-independent $p \in S^m$ is a limit of p_n in the explicit dictionary $\mathcal{S}_n = \{\xi^\gamma \chi(2^{-j}\xi) : |\gamma|, j \le n\}$ in $\|\cdot\|_{H^s \to H^{s-m'}}$; the same holds for symbols with compact x-support (continuity of $\mathrm{Op} : S^{m'} \to \mathcal{L}(H^s, H^{s-m'})$ cited).

(d) Let k be completely monotone and locally integrable, $k(t) = \int e^{-\lambda t} d\mu(\lambda)$. (i) $(k * f)(t) = \int w(t,\lambda) d\mu(\lambda)$ with $w_t = -\lambda w + f$, $w(0) = 0$: an exact local system with a continuous parameter λ. (ii) There are finite sums $k_J = \sum_{i \le J} \mu_i e^{-\lambda_i t}$ with $\|k - k_J\|_{L^1(0,T)} \to 0$, hence $\|(k - k_J) * f\|_\infty \to 0$. (iii) For $k = t^{-\beta}$, $0 < \beta < 1$, nodes $\lambda_j = e^{jh}$ give a relative error $\le 2(\cos a)^{-\beta}/(e^{2\pi a/h} - 1)$ uniformly in $t > 0$, $a \in (0, \frac{\pi}{2})$.

Stated in the paper as no. 15, p. 77

Proof.

Step 1 (c) Tail and head

Tail: $(1 - \chi(\xi/R))p$ has $S^{m'}$ seminorms $\le C_j R^{m-m'} |p|_{m,j}$ since $\langle\xi\rangle \gtrsim R$ on its support. Head: $\chi(\xi/R)p$ has compact support; choose $2^j \ge 2R$ and approximate by a polynomial in C^J (Weierstrass). For multipliers the operator norm is $\sup |\cdot| \langle\xi\rangle^{-m'}$.

Step 2 (d)(i) Exact representation

$w(t,\lambda) = \int_0^t e^{-\lambda(t-s)} f(s) ds$ with $|w| \le \|f\|_\infty \min(t, 1/\lambda)$, integrable in $d\mu$ because $k \in L^1(0,T)$; Fubini gives $\int w \, d\mu = (k * f)(t)$.

Step 3 (d)(ii) Convergence

Write $\mu = \mu(\{0\})\delta_0 + \mu'$. The atom at 0 is kept exactly (the constant $\mu(\{0\})$). For μ' choose Λ with $\int_{\lambda > \Lambda} d\mu'/\lambda < \eta$ and ε_0 with $T\mu'((0,\varepsilon_0)) < \eta$, and split $[\varepsilon_0, \Lambda]$ into cells I_i with $|e^{-\lambda t} - e^{-\lambda_i t}| \le T \operatorname{diam} I_i$. The three errors are $< \eta$ each.

Step 4 (d)(iii) Rate for $t^{-\beta}$

$\Gamma(\beta) t^{-\beta} = \int_{\mathbb{R}} \exp(-e^s t + \beta s) ds = t^{-\beta} \int G(\sigma) d\sigma$ with $G(\sigma) = \exp(-e^\sigma + \beta\sigma)$, analytic in $|\operatorname{Im}\sigma| < \frac{\pi}{2}$ with $\int |G(x+iy)| dx = \Gamma(\beta)(\cos y)^{-\beta}$. The trapezoid rule on $h\mathbb{Z} + \log t$ has error $\le 2M(a)/(e^{2\pi a/h} - 1)$ uniformly in the shift (cited: Trefethen–Weideman 2014, theorem number to be checked). Truncation costs a geometric tail on each side.

Why it works. a completely monotone kernel is a mixture of exponentials, so memory becomes a continuum of ordinary equations; a quadrature in $\log\lambda$ converges geometrically for $t^{-\beta}$.

■

Uses: B98.

Dim. B102 **DICT-RIT1, N3**

A delay is a transport, and not a finite ODE system

[proof written, not audited]

(RIT-1) For $u \in C^1([-\tau, T])$ the function $w(t,\theta) = u(t-\theta)$ is the unique C^1 solution of $w_t + w_\theta = 0$, $w(t,0) = u(t)$, $w(0,\theta) = u(-\theta)$, and $u(t-\tau) = w(t,\tau)$. So $F(u, u', u(\cdot - \tau)) = 0$ is a local system in (u, w) with the history as datum.

(N3) Let $\mathcal{R}$ be the input–output maps $H = D\delta + h*$, $h(t) = Ce^{At}B$, of finite-dimensional constant-coefficient systems. (a) On $L^2(\mathbb{R})$: $\operatorname{dist}(S_\tau, \mathcal{R}) = 1$. (b) On $L^2(0,T)$, $T > \tau$: $\operatorname{dist}(S_\tau, \mathcal{R}) \geq 1$.

Stated in the paper as no. 15, p. 77

Proof.

Step 1 RIT-1

$w(t,\theta) = u(t-\theta)$ satisfies the three conditions. If $\tilde{w}$ does too, $z = w - \tilde{w}$ has $z_t + z_\theta = 0$, vanishes on $\{\theta = 0\} \cup \{t = 0\}$, and each characteristic $s \mapsto (t_0 + s, \theta_0 + s)$ run backwards meets that set; so $z \equiv 0$.

Step 2 N3(a) on the line

The symbol of $H \in \mathcal{R}$ is a proper rational $R(i\omega)$, with $R(i\omega) \to c$ as $|\omega| \to \infty$. The norm is $\sup_\omega |e^{-i\tau\omega} - R(i\omega)|$. For $\omega \geq \omega_0$, $|R - c| < \eta$, and on $[\omega_0, \omega_0 + 2\pi/\tau]$ the phase $e^{-i\tau\omega}$ takes every value of the unit circle, in particular $-c/|c|$ where $|e^{-i\tau\omega} - c| = 1 + |c|$. So the sup is $\geq 1 - \eta$ for every η. Equality holds for $H = 0$.

Step 3 N3(b) on an interval

Take $\varphi \in C_c^\infty((0, T-\tau))$ with support of diameter $< \tau$ and $f_\omega = \varphi e^{i\omega t}$. Then $S_\tau f_\omega$ has support disjoint from that of f_ω, and $\|h * f_\omega\|_{L^2} \leq C/\omega \to 0$ by integration by parts. So $\|(S_\tau - H) f_\omega\|^2 \to \|\varphi\|^2(1 + |D|^2) \geq \|f_\omega\|^2$.

Why it works. a delay is a pure phase of modulus 1 at every frequency, while a finite-dimensional system tends to a constant at infinity.

■

Uses: none.

Dim. B103 **DICT-CS1**

The fractional Laplacian as a local degenerate system [proof written, not audited]

Let $0 < s < 1$, $f \in H^s(\mathbb{R}^d)$, $\varphi_s(r) = \frac{2^{1-s}}{\Gamma(s)} r^s K_s(r)$ (K_s the modified Bessel function) and $\hat{U}(\xi, y) = \hat{f}(\xi)\varphi_s(|\xi|y)$. Then $\operatorname{div}(y^{1-2s}\nabla U) = 0$ in $\mathbb{R}^d \times (0,\infty)$, $U(\cdot, y) \to f$ in H^s, and $-y^{1-2s}\partial_y U(\cdot, y) \to c_s(-\Delta)^s f$ in H^{-s} with $c_s = 2^{1-2s}\Gamma(1-s)/\Gamma(s)$.

Stated in the paper as no. 15, p. 77

Proof.

Step 1 The ODE in y

In Fourier, $-|\xi|^2\hat{U} + \hat{U}_{yy} + (1-2s)y^{-1}\hat{U}_y = 0$; with $r = |\xi|y$, $\phi'' + (1-2s)r^{-1}\phi' - \phi = 0$. Putting $\phi = r^s g$ gives the modified Bessel equation of order s, whose bounded decaying solution is $r^s K_s$. Since $K_s(r) \sim \frac{\Gamma(s)}{2}(r/2)^{-s}$ as $r \to 0$, $\varphi_s(0^+) = 1$.

Step 2 Trace

$\varphi_s' = -\frac{2^{1-s}}{\Gamma(s)} r^s K_{1-s} < 0$, so $0 \le \varphi_s \le 1$ decreases to 1 at 0; dominated convergence gives $U(\cdot, y) \to f$ in H^s.

Step 3 Neumann data

$-y^{1-2s}\partial_y \hat{U} = \frac{2^{1-s}}{\Gamma(s)}|\xi|^{2s} h(|\xi|y)\hat{f}$ with $h(r) = r^{1-s}K_{1-s}(r)$, decreasing, $h(0^+) = 2^{-s}\Gamma(1-s)$. Hence the limit is $\frac{2^{1-s}}{\Gamma(s)} h(0^+)|\xi|^{2s}\hat{f} = c_s|\xi|^{2s}\hat{f}$; dominated convergence in H^{-s}.

Why it works. the non-local symbol $|\xi|^{2s}$ is the Dirichlet-to-Neumann map of a local, degenerate, elliptic equation in one more variable.

■

Uses: none.

17.2 The negative theorem and the rank theorem

Dim. B104 **DICT-N1(iv)(a), N1(v)**

Finite outputs cannot reach every operator, and the generic function has no law

[proof written, not audited]

(iv)(a) A procedure whose outputs are finite strings over a finite alphabet, even as a stabilised limit, emits a countable set of operators; so for all but countably many c it never emits $\partial_t - c\partial_x$ (each identified by suitable solutions, C3-T4).

(v) Let $V_0 \subset V_1 \subset \cdots$ be finite-dimensional spaces of differential polynomials in u with smooth coefficients, and $B_n = \{u : \exists P \in V_n \setminus 0,\ P[u] = 0\}$. Then B_n is closed and nowhere dense, $\bigcup B_n$ is meagre, and each B_n (so the union) is Haar-null in the sense of prevalence (shy). Taking V_n = all differential polynomials of order, degree and coefficient degree $\le n$, the set of differentially algebraic functions over $\mathbb{C}(y)$ is meagre and shy: the generic u is differentially transcendental.

Stated in the paper as no. 15, p. 77

Idea. The linear case needs a function w for which the atoms become independent. For the non-linear case, add sw with s large: the highest-degree parts dominate and a determinant becomes a non-zero polynomial in s, so only finitely many s can give a law.

Proof.

Step 1 (iv)(a)

Finite strings are countable, a stabilised sequence has a limit string, and the image of a countable set is countable.

Step 2 Lemma ND

Let $D_1, \dots, D_M$ be linear differential operators, independent over $\mathbb{C}$ on an open set U. Then there is $w \in C_c^\infty(U)$ with $D_1 w, \dots, D_M w$ independent. Indeed, let $L_i(y) : J_y^r \to \mathbb{C}$ be the jet form of D_i at y and $S = \{(L_i(y)(j))_i\} \subset \mathbb{C}^M$. If $\operatorname{span} S \neq \mathbb{C}^M$ some $c \neq 0$ has $\sum c_i D_i = 0$ on U. So there are distinct y_k and polynomial jets with an invertible evaluation matrix; $w = \sum_k \chi(\cdot - y_k) p_k$ with disjoint supports works. The same proof gives Lemma ND-NL for differential polynomials independent on $U \times J^r$.

Step 3 Closed and nowhere dense

Take a degree-adapted basis $P_1, \dots, P_M$ of V_n (top-degree parts P_i^{top} independent), $e_i = \deg P_i$. Closedness: if $u_k \to u$ with $\sum c_i^{(k)} P_i[u_k] = 0$, $|c^{(k)}| = 1$, pass to a convergent subsequence of $c^{(k)}$. By ND-NL choose w with $P_i^{\mathrm{top}}[w]$ independent and functionals ℓ_j with $\det(\ell_j(P_i^{\mathrm{top}}[w])) \neq 0$. For any u, $p_u(s) = \det(\ell_j(P_i[u + sw]))$ is a polynomial in s of degree $\sum e_i$ with non-zero leading coefficient. If $u + sw \in B_n$ then $p_u(s) = 0$: at most $\sum e_i$ values of s.

Step 4 Meagre and shy

So $u + sw \notin B_n$ for small $s \neq 0$: B_n is nowhere dense, and Baire gives meagre. The probability measure of sw, s uniform in $[0, 1]$, has $\mu(B_n - u) = 0$ for every u, so B_n is shy; countable unions of shy sets are shy (Christensen 1972).

Step 5 The algebraic corollary

The differentially algebraic functions are the union over n of the B_n above. They are dense (they contain polynomials) but meagre and shy. Combined with ALG-T1 this is the non-linear form of the trichotomy: a non-linear law exists only for exceptional u, and for those exact identification of the equation needs transcendence degree ∞.

Why it works. one direction w makes the dependence between atoms a polynomial identity in s, which can vanish only finitely often.

■

Uses: A49.

Dim. B105 **DICT-RANGO**

The rank theorem and its corollaries [proof written, not audited]

Let $L_\theta = L_0 + \sum_{k=1}^{K} \theta_k T_k$, $\theta \in C^\infty(\Omega, \mathbb{F}^K)$, $A_u(y)\vartheta = \sum \vartheta_k T_k u(y)$ and $\Phi_u(\theta) = \sum \theta_k T_k u$. A *law* is a θ with $L_\theta u = 0$; the class is *identifiable* (ID-ker) if $\ker \Phi_u = 0$. (I) The set of laws is empty or $\theta^0 + \ker \Phi_u$. (II) $\ker \Phi_u = 0$ iff $A_u(y)$ is injective on a dense set. (III) If $A_u(y)$ is surjective at every y, a smooth law exists; if $-L_0 u(y_0) \notin \operatorname{im} A_u(y_0)$, none exists. (IV) If $K \dim_{\mathbb{R}} \mathbb{F} > \dim_{\mathbb{R}}(\text{values})$ the class is never identifiable.

Stated in the paper as no. 15, p. 77

Proof.

Step 1 (I)

Two laws differ by an element of $\ker \Phi_u$, and conversely.

Step 2 (II)

($\Leftarrow$) $\Phi_u(\theta) = 0$ gives $\theta(y) \in \ker A_u(y) = 0$ on a dense set, so $\theta = 0$. ($\Rightarrow$) The set where $A_u(y)$ has full rank is open (lower semicontinuity of rank). If it is not dense, on some open U the rank is $\leq K \dim \mathbb{F} - 1$; where the rank is maximal and constant on an open U', $\ker A_u$ is a smooth vector bundle of rank ≥ 1, with a non-zero local section; a cut-off times it lies in $\ker \Phi_u$.

Step 3 (III) and (IV)

If $A_u(y)$ is onto, $\theta(y) = -A_u^T (A_u A_u^T)^{-1} L_0 u$ is smooth and solves $L_\theta u = 0$. If $-L_0 u(y_0) \notin \operatorname{im} A_u(y_0)$ no value of $\theta(y_0)$ works. (IV) is linear algebra on the dimension of the target.

Step 4 Corollaries

(a′) All operators of order $\leq m$ ($d \geq 1$ or $m \geq 2$, so $K \geq 2$): $K \cdot 2 > 2$, never identifiable; if u has no zeros the coefficient of the term $\alpha = 0$ is $u(y) \neq 0$, A_u is onto, so every such u has a law: neither identifiable nor falsifiable. (b′) Schrödinger with complex V: $A_u \vartheta = -\vartheta u$ is injective iff $u(y) \neq 0$, so identifiable iff $\operatorname{int} Z(u) = \emptyset$. With real V one has $\mathbb{F} = \mathbb{R}$, values in $\mathbb{C} = \mathbb{R}^2$, identifiable and falsifiable: for u without zeros a law exists iff $\operatorname{Im}((iu_t + u_{xx})/u) \equiv 0$; in general only ($\Rightarrow$) holds ($u = x^3$ gives $6/x^2$, real but singular). Transport $u_t = a(t,x) u_x$: identifiable iff $\{u_x \neq 0\}$ is dense, $a = u_t / u_x$.

Step 5 The retracted claim

The original N1(vi) ("with $C^\infty(t,x)$ coefficients no operator is identified") is false: $u = t + x$ in the class $u_t = a(t,x)u_x$ has $A_u = -u_x = -1$, so $a \equiv 1$ is the unique law. In the normalised class $(\partial_x, 1)$, $u = t - x^2$ has $A_u(0,0) = (0,0)$ and $L_0 u(0,0) = 1$: no law, so "every u has a law" fails too.

Why it works. identifiability is injectivity of a pointwise linear map; the right axis is the count of unknowns against the dimension of the values, not the size of the class.

■

Uses: B77.

Dim. B106 **DICT-MAG, MAG-NU**

Magnetic case: outside the rank theorem [proof written, not audited]

The magnetic Schrödinger operator $(\nabla - iA)^2$ is not affine in A ($\operatorname{div} A$ contains derivatives of the unknown and $|A|^2$ is quadratic). Let (A, V) be a real law for u, and B real with $\operatorname{supp} B \subset \{u \neq 0\}$. Then $(A + B, V + \delta V)$ with $\delta V = -2iB \cdot \nabla u/u - i \operatorname{div} B - 2A \cdot B - |B|^2$ is a real law iff $\operatorname{div}_x(|u|^2 B) = 0$. Given $\chi \in C_c^\infty(U)$, $U \subset \{u \neq 0\}$, $B = |u|^{-2}(\partial_{x_2}\chi, -\partial_{x_1}\chi, 0, \dots)$ gives a second law with $dB \not\equiv 0$.

Stated in the paper as no. 15, p. 77

Proof.

Step 1 The difference identity

Substituting $A + B$ in $(\nabla - iA)^2 u$ and subtracting gives $\operatorname{Im}(\delta V)|u|^2 = -\operatorname{div}_x(|u|^2 B)$. So δV is real iff $\operatorname{div}_x(|u|^2 B) = 0$.

Step 2 The construction

With $k = -|u|^2 B = -\nabla^\perp \chi$, $\operatorname{div} k = 0$: this is B77(a)(b) on U. Outside $\operatorname{supp} \chi$ one has $B = 0$ and $\delta V = 0$, so the pair is smooth on all of $J \times \Omega$ even if u has zeros elsewhere.

Step 3 Different magnetic field

The $(1,2)$ component of dB is $-\operatorname{div}_{12}(|u|^{-2}\nabla_{12}\chi)$, not identically zero by the maximum principle for a compactly supported $\chi \neq 0$; so the two laws are not gauge-equivalent (B76).

Why it works. the counterexample to rank-type uniqueness is the current: only $\operatorname{div}(\rho^2 B) = 0$ is needed, not $\operatorname{div} B = 0$.

■

Uses: B76, B77.

17.3 Misfit, rank of an atom, change of frame

Dim. B107 **DICT-MF, B2, R3**

Weak-form certificates, coefficient estimate, exact gain of an atom [proof written, not audited]

(MF3) (a) $f_0, \dots, f_N \in \mathcal{D}'(\Omega)$ are independent iff some $\det(\langle f_i, \psi_j\rangle) \neq 0$ with ψ_j in a fixed countable dense set $\Psi \subset C_c^\infty$. (b) $\langle D_i u, \psi\rangle = \langle u, D_i^t \psi\rangle$ uses no derivative of u. (c) If $|\langle \tilde{u} - u, D_i^t \psi_j\rangle| \le \delta_{ij}$, then $\sigma_{N+1}(A(u)) \ge \sigma_{N+1}(A(\tilde{u})) - \|\delta\|_F$. (d) With an oracle for the integrals, the search over Ψ terminates iff there is homogeneous misfit.

(MF-C) For a compact convex $B \subset \mathbb{C}^N$: $\min_{c\in B} \|D_0 u + \sum c_i D_i u\| = \max_{\|\phi\|\le 1} \min_{c\in B} \operatorname{Re}\phi(\cdot)$; a cone constraint $Ac = b$, $c \ge 0$ is infeasible iff $A^T z \ge 0$, $b^T z < 0$ for some z.

(B2) If $\lambda_{\min}(G_n') > 0$, every $\tilde{L}_n = D_0 + \sum \tilde{c}_i D_i$ with $\eta = \|\tilde{L}_n u\|$ has $|c_n - \tilde{c}_n| \le 2\eta/\sqrt{\lambda_{\min}}$.

(R3) $r_{n+1}^2 = r_n^2 - |\langle \mathrm{res}_n, Du\rangle|^2 / \|P_W^\perp Du\|^2$.

Stated in the paper as no. 15, p. 77

Proof.

Step 1 MF3

(a) ($\Leftarrow$) a relation makes the columns dependent. ($\Rightarrow$) otherwise the image of $\psi \mapsto (\langle f_i, \psi\rangle)_i$ is a proper subspace, orthogonal to some $c \neq 0$, and $\sum \bar{c}_i f_i = 0$. Continuity of det in ψ allows $\psi_j \in \Psi$. (b) is the definition of distributional derivative. (c) is Weyl's inequality $|\sigma_k(A) - \sigma_k(B)| \le \|A - B\|_2 \le \|A - B\|_F$. (d) With misfit some determinant is non-zero and interval arithmetic excludes 0 at finite precision; without misfit all vanish and no interval excludes 0.

Step 2 MF-C

The functional is affine real in c and in ϕ, B is compact convex and the dual ball is weak-$*$ compact convex: Sion's minimax theorem gives the equality. The cone case is Farkas' lemma.

Step 3 B2

$\sum (c_n - \tilde{c}_n)_i D_i u = L_n u - \tilde{L}_n u$ has norm $\le r_n + \eta \le 2\eta$, and $\|\sum \delta c_i D_i u\|^2 \ge \lambda_{\min} |\delta c|^2$.

Step 4 R3

Put $e = P_W^\perp Du / \|P_W^\perp Du\|$. The projector onto $W' = W \oplus \mathbb{C}e$ is $P_W + e\,e^*$, so the new residual is $\mathrm{res}_n - \langle \mathrm{res}_n, e\rangle e$, and $\langle \mathrm{res}_n, P_W^\perp Du\rangle = \langle \mathrm{res}_n, Du\rangle$ because $\mathrm{res}_n \perp W$.

Why it works. each statement is a finite-dimensional linear-algebra identity applied to the vectors $D_i u$.

■

Uses: A50.

Dim. B108 **DICT-R6**

Changing frame with the Volterra encoder

[proof written, not audited]

Let $\Omega = I \times J^\circ$, $J = [x_0 - a, x_0 + a]$, $q \in C^1(J)$ given, $Q = \|q\|_\infty$, $W = T^{-1}$ the Volterra encoder of ENC-1, $C_W = 1 + a^2 Q e^{a\sqrt{Q}}$. On the even subring: (a) $W(Lu) = \hat{L}(Wu)$ for $L = P(\partial_t, L_q)$, $\hat{L} = P(\partial_t, L_0)$; so $Lu = 0 \iff \hat{L}(Wu) = 0$. (b) $\|W\|, \|W^{-1}\| \le C_W$ on $C(\bar{\Omega})$ and $L^2(\Omega)$. (c) $r_n^{\mathrm{even}}(Wu)/\|W\| \le r_n^q(u) \le \|W^{-1}\| r_n^{\mathrm{even}}(Wu)$, and the same for $\lambda_{\min}$ with squared constants. (d) A misfit certificate ψ for Wu becomes $\psi \circ W$ for u with $\|\psi \circ W\| \le \|W\| \|\psi\|$.

Stated in the paper as no. 15, p. 77

Proof.

Step 1 (a) Exact intertwining

This is ENC-C1 applied to each monomial $\partial_t^j L_q^k$; W commutes with ∂_t.

Step 2 (b) Norms

$W = I + V$ with $Vf(x) = \int_{2x_0 - x}^{x} H(x,s) f(s) ds$ and $\|H\|_\infty \le \frac{aQ}{2} e^{a\sqrt{Q}}$ (ENC-1). On $C(J)$, $|Vf| \le 2a\|H\|_\infty \|f\|_\infty$; on $L^2(J)$, Cauchy–Schwarz gives $|Vf(x)| \le \|H\|_\infty \sqrt{2a} \|f\|_{L^2}$ and integrating over J (length $2a$) $\|Vf\|_{L^2} \le 2a\|H\|_\infty \|f\|_{L^2}$. The same for the kernel of T. In t take the sup or integrate (Fubini).

Step 3 (c) Distances

From (a), $\|\hat{L}Wu\| = \|WLu\| \le \|W\| \|Lu\|$ and $\|Lu\| = \|W^{-1}\hat{L}Wu\| \le \|W^{-1}\| \|\hat{L}Wu\|$. Take the infimum over L; $L \mapsto \hat{L}$ is a bijection of the affine levels. For Gram matrices compare $\|\sum c_i L_i u\|$ and $\|W \sum c_i L_i u\|$ and take $\min_{|c|=1}$.

Step 4 (d) Certificates

$(\psi \circ W)(L_i u) = \psi(W L_i u) = \psi(\hat{L}_i W u)$ by (a).

Why it works. the encoder intertwines the true and the free operator exactly on the even subring, with a controlled distortion of distances.

■

Uses: B88, B107.

What R6 does not give

Odd atoms in ξ have no counterpart at level q; a free odd law such as $\xi - ik$ transports only through its minimal even multiple. Finite jets never certify a fit, in any frame: for every n and every finite datum there is v with the same data and no law at level n (B107 together with Theorem A, and a Baire step for no level at all).

17.4 Limits, exponential families

Dim. B109 **DICT-ILIM-H0, NS**

I-LIM with a Hölder modulus; polynomialisation of exponential families [proof written, not audited]

(I-LIM) Let $\mathcal{K}_R = \{D_0 + \sum_{\alpha\in A} c_\alpha \partial^\alpha : \|c_\alpha\|_{C^{s,1}} \le R\}$ (compact in C^s), with (H-ID): there is $L \in \mathcal{K}_R$, $Lu = 0$, unique in $\mathcal{K}_{R'}$; (H2): approximants $\tilde L_n \in \operatorname{span}\mathcal{D}_n \cap \mathcal{K}_{R'}$ with $\|\tilde L_n - L\|_{C^s} \le \varepsilon_n \to 0$. Then $\omega(\eta) = \sup\{\|L'-L\|_{C^s} : \|L'u\| \le \eta\} \to 0$ and every minimiser L_n has $\|L_n - L\| \le \omega(C_u\varepsilon_n)$; with data $\|\tilde u - u\|_{H^m} \le \delta$ the bound is $\omega(C_u\varepsilon_n + 2C_{R'}\delta)$. For one coefficient with $|\partial^{\alpha_1}u| \ge \gamma$ and a cone condition, $s = 0$: $\omega(\eta) = O(\eta^{2/(N+2)})$.

(NS-L1) If $E[u]$ is a polynomial in the derivatives $\partial^\alpha u$ ($|\alpha| \ge 1$) and in $e^{n_j\lambda u}$, then $E[u] = w^{-k}\tilde E[w]$ with $w = e^{\lambda u}$; on a simply connected Ω, u solves $E = 0$ iff w solves $\tilde E = 0$. **(NS-T1)** $\operatorname{trdeg}\mathbb{C}\langle w\rangle = \infty \iff \operatorname{trdeg}\mathbb{C}\langle u\rangle = \infty$. **(Kink)** For $u = 4\arctan e^\theta$, $\theta = ax + t/a$, the dictionary $\{u_{xt}, \sin u, u_{xx}, u_{tt}, u\}$ has kernel of dimension 3 and the affine kernel has dimension 2: the kink does not identify sine-Gordon.

Stated in the paper as no. 15, p. 77

Idea. I-LIM is a compactness argument: if the modulus did not tend to zero, a limit of near-solutions would be a second solution. The Hölder rate comes from a ball argument for a Lipschitz coefficient. Exponential families become polynomial after substituting $w = e^{\lambda u}$.

Proof.

Step 1 (a) The modulus tends to 0

Otherwise there are ε_0 and $L_j' \in \mathcal{K}_{R'}$ with $\|L_j'u\| \to 0$, $\|L_j' - L\| \ge \varepsilon_0$. By compactness $L_j' \to L^*$ in C^s; since $(c) \mapsto L'u$ is continuous $C^0 \to L^2$, $L^*u = 0$ and $\|L^* - L\| \ge \varepsilon_0$, against (H-ID).

Step 2 (b), (c) Stability

$\|L_nu\| \le \|\tilde L_nu\| = \|(\tilde L_n - L)u\| \le C_u\varepsilon_n$, then (a). With noise, $\|L'\tilde u - L'u\| \le C_{R'}\delta$ gives $\|L_n^\delta u\| \le \|\tilde L_n u\| + 2C_{R'}\delta$.

Step 3 Hölder rate

For $L'u - Lu = (c'-c)\partial^{\alpha_1}u$, $\|c'-c\|_{L^2} \le \eta/\gamma$. With $f = c'-c$, $\operatorname{Lip} f \le 2R'$, $M = |f(y_0)|$, on $B_\rho(y_0)\cap\Omega$ with $\rho = \min(M/(4R'), r_0)$ one has $|f| \ge M/2$ so $(\eta/\gamma)^2 \ge (M/2)^2\kappa\rho^N$, which gives $M^{N+2} \lesssim (\eta/\gamma)^2$ or $M \lesssim \eta/(\gamma\sqrt{\kappa r_0^N})$. For $s \ge 1$ the interpolation inequality is cited (case closed in Phase 10 by an elementary cone lemma).

Step 4 NS-L1

$\partial_i u = \lambda^{-1}\partial_i w/w$; by induction $\partial^\alpha u = \lambda^{-1}P_\alpha[w]/w^{|\alpha|}$ with P_α polynomial, and $e^{n_j\lambda u} = w^{n_j}$. Multiplying by a power of w clears the denominators. Examples: sine-Gordon $u_{xt} = \sin u$ ($\lambda = i$) gives $2(ww_{xt} - w_xw_t) - (w^3 - w) = 2iw^2(u_{xt} - \sin u)$; Liouville $u_{xt} = e^u$ gives $vv_{xt} - v_xv_t - v^3 = v^2(u_{xt} - e^u)$. A bare u cannot occur, since $u = \lambda^{-1}\log w$.

Step 5 NS-T1

With $F_1 = \mathbb{C}(\partial^\alpha u : |\alpha| \geq 1)$, NS-L1 gives $\mathbb{C}\langle w\rangle = F_1(w)$ and $\mathbb{C}\langle u\rangle = F_1(u)$; adjoining one element changes the transcendence degree by at most 1.

Step 6 Kink

Writing $U'' = \sin U$, the five atoms equal $U'', U'', a^2U'', a^{-2}U'', U$. U'' and U are independent (if $\alpha U'' + \beta U = 0$, then $\theta \to +\infty$ gives $U'' \to 0$, $U \to 2\pi$, so $\beta = 0$, then $\alpha = 0$). So the span has dimension 2, the kernel 3: $u_{xt} - \sin u$, $u_{xx} - a^2 u_{xt}$, $u_{tt} - a^{-2}u_{xt}$. With $D_0 = u_{xt}$ the affine kernel has dimension 2.

Why it works. stability from compactness plus uniqueness; exact polynomialisation because the exponential is a unit of the differential ring.

■

Uses: B107, A51.

Form of the equation

The transfer of ALG-T1 through NS-T1 is valid for *evolutionary* forms (for example sine-Gordon in laboratory coordinates $u_{tt} = u_{xx} + \sin u$); it does not apply to the light-cone form $u_{xt} = \sin u$. The logarithm in the direction $w \mapsto u$ needs Ω simply connected, or u read modulo $2\pi i/\lambda$.

17.5 Statements without a written proof

DICT-N1(iv)(b,c)	(b) exact stabilisation in the BSS model (CONV-T1); (c) "Ann $\neq 0$" is verifiable but not decidable in the limit.	[proof written, not audited]

Imported from S8-CONV (Chapter 7, CONV-T1/T2), §2.4.

DICT-2KINK	The 2-kink has kernel 1.	checked on examples

numerical rank 4 for $a_1 = \frac{1}{2}$, $a_2 = 2$; a proof needs exact independence of $u_{xt}, u_{xx}, u_{tt}, u$ (open question Q6).

DICT-AX	Differential Ax–Schanuel.	from cited source, to be read

used only in a tool, and only to treat e^u as an independent variable next to the jet.

DICT-R4, DICT-PAINLEVE	Heuristic family-change signatures (power law in the residual spectrum $\Rightarrow |D|^s$; exponential correlation in time $\Rightarrow$ memory; phase $e^{-i\tau\omega} \Rightarrow$ delay; concentration near $x_0 \Rightarrow$ singular atoms); Painlevé as a diagnostic.	tool

no theorem, no guarantee depends on it.

DICT-Q3 (c), (a)	Test of formal non-membership is insufficient (b, proved); (a) from Malgrange; (c) reformulated as C^∞ division, open.	(b) [proof written, not audited]; (a) from cited source; (c) [open]

F10-C.

DICT-Q4-C	(1) $\Longleftrightarrow$ (2) for finite rank [proof written, not audited]; infinite rank implies A(iii) from Amitsur 1965 / Brešar–Šemrl 1999, then replaced by DICT-LOCDEP in the triage.	[proof written, not audited]; replaced

F10-C, F10-TRIAGE.

DM-δ-MINIMAX = VAR-Q19	Under $M > \delta_0$ the minimax rate is $\Theta(\varepsilon^{1/3})$ (matching the sharp exponent $1/3$ of A39).	[proof written, not audited]

F10-C.

CHAPTER 18

Phase 10: closures, reductions and examples (other results)

In this chapter	
Starting from	the rows of the ledger sections F10-A, F10-B2, F10-C2, F10-COL, F10-GTU, F10-INT, F10-LEAN and F10-TRIAGE. Where two sections give the same identifier, the later one prevails (F10-B2 over F10-B, F10-C2 over F10-C, F10-TRIAGE over all).
Why	to prove, in full, the Phase 10 results whose argument is self-contained and short enough to write out (the dilation theorem, the closedness lemma, the annulus counterexample, the Dyson–Phillips regularity, the rank-2 counterexample and the Cantor-type zeros), and to record the others with the status exactly as in the ledger.
Role	Dim. B117–B123, then list cards. The two proofs that change statements of Phases 0–5 are A52 and A53 (Chapter 10). The summary of the results is Chapter 16 of the paper.

18.1 Non-commuting flows and the factor decomposition

Dim. B117 — **S10A-T2, P1-Q7**

No finite word of flows at fixed coupling, under a weighted dilation [proof written, not audited]

Let $\mathcal{A}$ be a unital associative $\mathbb{C}$-algebra, $X, Y \in \mathcal{A}$ satisfying hypothesis (H) of P1-T5 (the elements $\mathrm{ad}_X^k Y$ are linearly independent and $X \neq 0$), and $\mathcal{A}_0$ the unital subalgebra they generate. Assume **(Dil)**: there are integers $\alpha \geq 0$, $\beta < \alpha$ and an infinite $\Lambda \subset \mathbb{C}^*$ such that for each $\mu \in \Lambda$ a unital endomorphism σ_μ of $\mathcal{A}_0$ has $\sigma_\mu X = \mu^\alpha X$, $\sigma_\mu Y = \mu^\beta Y$. Then for every $\varepsilon_0 \in \mathbb{C}^*$ there are no word $(Z_1, \dots, Z_d) \in \{X, Y\}^d$ and series $g_i \in t\mathbb{C}[[t]]$ with $e^{t(X+\varepsilon_0 Y)} = \prod_{i=1}^{d} e^{g_i(t) Z_i}$ in $\mathcal{A}[[t]]$. Hence P1-T5 holds without the hypothesis "the flows of Y vanish at $\varepsilon = 0$".

Stated in the paper as no. 16, p. 80

Idea. Rescale by σ_μ and substitute $t = \mu^{-\alpha} s$: the fixed coupling ε_0 becomes $\varepsilon_0 \nu^{\mathfrak{a}} \to 0$ and the flows of Y become $1 + O(\nu^{\mathfrak{a}})$. The old proof of P1-T5 then works in $\mathbb{C}[\nu]$.

Proof.

Step 1 Rescaling

Applying σ_μ coefficientwise gives $e^{t(\mu^\alpha X+\varepsilon_0\mu^\beta Y)} = \prod_i e^{g_i(t)\mu^{w_i}Z_i}$ with $w_i = \alpha$ for $Z_i = X$ and β for $Z_i = Y$. Put $t = \nu^\alpha s$, $\nu = \mu^{-1}$ and $\mathfrak{a} = \alpha - \beta \geq 1$:

$$e^{s(X+\varepsilon_0\nu^{\mathfrak{a}}Y)} = \prod_i e^{h_i(s,\nu)Z_i}, \qquad h_i = \sum_k g_{i,k}\nu^{\alpha k - w_i}s^k.$$

For $Z_i = X$ the exponents are $\alpha(k-1) \geq 0$; for $Z_i = Y$ they are $\geq \alpha - \beta = \mathfrak{a}$.

Step 2 A polynomial identity in ν

The coefficient of s^m lies in the finite-dimensional span of words of length $\leq m$ and has coordinates polynomial in ν on both sides. They agree on the infinite set Λ^{-1}, hence as polynomials.

Step 3 Orders below $\nu^{\mathfrak{a}}$

Modulo $\nu^{\mathfrak{a}}$ the Y-factors are the identity and the left side is e^{sX}; the X-factors commute, so $e^{sX} \equiv e^{HX}$ with $H = \sum_{Z_i=X} h_i$. If $H - s \not\equiv 0$, its lowest term $c\,s^p\nu^q$ gives $cX = 0$ in $e^{(H-s)X} = 1$, impossible. So $H \equiv s \bmod \nu^{\mathfrak{a}}$.

Step 4 Order $\nu^{\mathfrak{a}}$

On the left the coefficient is $\varepsilon_0 e^{sX}\sum_k(-1)^k\frac{s^{k+1}}{(k+1)!}Y_k$ (formal Duhamel), $Y_k = \mathrm{ad}_X^k Y$. On the right only two terms contribute: all Y-factors at order 0 (giving $r(s)Xe^{sX}$, r the $\nu^{\mathfrak{a}}$-coefficient of H), or exactly one Y-factor at order $\nu^{\mathfrak{a}}$ with coefficient $q_i(s)Y$. Multiplying by e^{-sX} and using $e^{-\tau X}Ye^{\tau X} = \sum(-1)^k\frac{\tau^k}{k!}Y_k$:

$$rX + \sum_k(-1)^k Y_k \sum_{i:Z_i=Y} q_i\frac{\tau_i^k}{k!} = \varepsilon_0\sum_k(-1)^k\frac{s^{k+1}}{(k+1)!}Y_k.$$

Step 5 The Hankel contradiction

By (H), $r = 0$ and $\sum_{i:Z_i=Y} q_i\tau_i^k = \varepsilon_0 s^{k+1}/(k+1)$ for all $k \geq 0$. With $p = \#\{i : Z_i = Y\}$ and $n = p+1$, the Hankel matrix $[m_{i+j}]_{i,j<n}$ of the left sides is $W^{\mathsf{T}}\mathrm{diag}(q)W$ with W of size $p \times n$, so its determinant vanishes identically (Cauchy–Binet). That of the right sides is $\varepsilon_0^n s^{n^2}\det \mathrm{Hilb}_n \neq 0$. For $p = 0$, $m_0 = \varepsilon_0 s \neq 0$. Contradiction.

Why it works. dilation turns a fixed coupling into one that tends to 0, recovering by symmetry the hypothesis that was to be removed.

■

Uses: A24.

What remains open

Algebras with (H) but without weighted dilations. The candidate $X = (x^3+x)\partial$, $Y = \partial$ in the Weyl algebra A_1 is not homogeneous; it was settled in the triage, but the general case stays open (P1-Q7).

Dim. B118 **S10A-L2, C2**

A factor with compact interaction closes the sum

[proof written, not audited]

Let $\mathcal{H}_1, \dots, \mathcal{H}_r$ be closed subspaces of a Hilbert space with orthogonal projections Π_i. If $M = \mathcal{H}_1 + \cdots + \mathcal{H}_{r-1}$ is closed and $\Pi_r \Pi_i$ is compact for each $i < r$, then $\mathcal{H}_1 + \cdots + \mathcal{H}_r$ is closed. For $r = 3$, put an edge $\{i, j\}$ when $\Pi_i \Pi_j$ is not compact: if the three pairwise sums are closed and the triple sum is not, every vertex has degree ≥ 1, so there are at least two edges.

Stated in the paper as no. 16, p. 80

Proof.

Step 1 **The lemma**

The sum map $S : \bigoplus_{i<r} \mathcal{H}_i \to M$ is bounded and onto the Banach space M. By the open mapping theorem each $m \in M$ with $\|m\| \leq 1$ is $\sum_{i<r} h_i$ with $\sum \|h_i\| \leq C$. Then $\Pi_r m = \sum_i \Pi_r \Pi_i h_i$, so $\Pi_r(B_M) \subset \sum_{i<r} \Pi_r \Pi_i(B_C)$, relatively compact. Hence $\Pi_r|_M$ and $\Pi_r \Pi_M$ are compact, and by OPR2-L1 for two closed subspaces $\mathcal{H}_r$ and M the sum $\mathcal{H}_r + M$ is closed.

Step 2 **The corollary**

The relation is symmetric, $(\Pi_i \Pi_j)^* = \Pi_j \Pi_i$. If a vertex k were isolated, apply the lemma with $\mathcal{H}_r = \mathcal{H}_k$ and $M = \mathcal{H}_i + \mathcal{H}_j$ (closed by hypothesis): the triple sum is closed, a contradiction. Three vertices of degree ≥ 1 need at least two edges.

Why it works. compactness of the interaction lets the new subspace be handled as a finite-rank perturbation of an orthogonal complement.

■

Uses: none.

Consequence for C12-Q2

A counterexample to "pairwise closed implies triple closed" for $P = Q_1 Q_2 Q_3$ must have at least two pairs of factors with non-compact interaction. Real transports are excluded (OPR2-T2), and no candidate family has been analysed: the question stays open, reduced to such triples.

Dim. B119 **S10A-P1-annulus**

The annulus: closed but not dense

[proof written, not audited]

Let $P = \Delta$, factors $Q_1 = \partial_t + i\partial_x$, $Q_2 = \partial_t - i\partial_x$ ($\delta(P) = 1$), so $\mathcal{H}_1$ are holomorphic and $\mathcal{H}_2$ anti-holomorphic functions in H^m. (i) On the annulus $A = \{1 < |z| < 2\}$ (Lipschitz, non-convex, $b_1 = 1$), $\log|z| \in \ker_{H^m(A)} \Delta \setminus \overline{\mathcal{H}_1 + \mathcal{H}_2}$ for every $m \geq 0$, and for $m \geq 1$ $\ker_{H^m(A)} \Delta = (\mathcal{H}_1 + \mathcal{H}_2) \oplus \mathbb{C} \log|z|$: the sum is closed of codimension $1 = b_1$ and not dense. (ii) If a bounded Lipschitz Ω has a cycle γ and a point $a \notin \Omega$ with non-zero winding number, then $\log|z-a| \notin \overline{\mathcal{H}_1 + \mathcal{H}_2}$. So P2-Q1 in its universal form is false, and the differential example "closed but not dense" of C12-P7 exists.

Stated in the paper as no. 16, p. 80

Proof.

Step 1 The flux functional

For a smooth cycle $\gamma \Subset \Omega$ and h harmonic put $\Phi_\gamma(h) = \int_\gamma *dh$. With $\delta = \operatorname{dist}(\gamma, \partial\Omega)/2$, the mean-value property and Cauchy's gradient estimate give $|\nabla h(y)| \leq C\delta^{-2}\|h\|_{L^2(\Omega)}$ on γ, so $|\Phi_\gamma(h)| \leq C_\gamma \|h\|_{H^m}$: Φ_γ is continuous.

Step 2 It vanishes on $\mathcal{H}_1 + \mathcal{H}_2$

For f holomorphic, $*df = -if'dz$ and $\Phi_\gamma(f) = -i\oint_\gamma f'dz = 0$ since f is a single-valued primitive; for $\bar{g}$ anti-holomorphic, $\Phi_\gamma(\bar{g}) = \overline{\Phi_\gamma(g)} = 0$. By continuity Φ_γ vanishes on the closure.

Step 3 It does not vanish on the logarithm

$*d\log|z-a| = d\arg(z-a)$, so $\Phi_\gamma(\log|z-a|) = 2\pi \operatorname{Ind}_\gamma(a) \neq 0$. For a hole one takes a in the interior of the bounded component of $\mathbb{C} \setminus \Omega$ (the index is constant on it); such a cycle exists when $b_1 \geq 1$ (Rudin RCA 13.11, [from cited source]). This proves (ii) and the first part of (i) with $\gamma = \{|z| = \frac{3}{2}\}$.

Step 4 Exact codimension on the annulus, $m \geq 1$

A harmonic h on A is $f + \bar{g} + c\log|z|$ with f, g holomorphic single-valued (Laurent) and $c = \Phi(h)/2\pi$. If $\Phi(h) = 0$, then $h = f + \bar{g}$ and $f' = \partial_z h \in H^{m-1}$. In polar coordinates $|f(re^{i\theta}) - f(\frac{3}{2}e^{i\theta})|^2 \leq \int_1^2 |f'|^2 \varrho\, d\varrho$, so $\|f - f(\frac{3}{2}e^{i\cdot})\|_{L^2(A)} \leq C\|f'\|_{L^2}$, while $f(\frac{3}{2}e^{i\theta})$ is bounded. Derivatives of order $1, \dots, m$ of f are derivatives of $f' \in H^{m-1}$, so $f \in H^m$; likewise $\bar{g}$. Thus $\ker\Phi = \mathcal{H}_1 + \mathcal{H}_2$, closed as the kernel of a continuous functional.

Why it works. a hole carries a flux that no holomorphic or anti-holomorphic function can have, and that flux is a continuous functional.

■

Uses: A52, A53.

Dim. B120 **S10A-T3**

Density up to the first Betti number, on Lipschitz regions [proved mod. [F1], [F2]]

Let $\Omega \subset \mathbb{R}^2$ be bounded, connected and Lipschitz, $P = c\prod_{j=1}^{N} Q_j$ with pairwise coprime primary Q_j, $\mathcal{H}_j = \ker_{H^m(\Omega)} Q_j$, $\mathcal{K} = \ker_{H^m(\Omega)} P$, $S = \overline{\sum \mathcal{H}_j}$. Then (a) $\operatorname{codim}_{\mathcal{K}} S \le \delta(P)\, b_1(\Omega)$; (b) if Ω is simply connected, $\sum \mathcal{H}_j$ is dense in $\mathcal{K}$, and the consequences of OPF5B-T1 hold.

Stated in the paper as no. 16, p. 80

Idea. Approximate a solution in Ω by one on a neighbourhood $U \supset \bar{\Omega}$ with the same topology; there the cokernel of the sum map is bounded by $\delta(P)b_1$ (OPF-T4); a finite-dimensional argument transfers the bound back to Ω.

Proof.

Step 1 A uniform segment condition replaces convexity

Cover $\partial\Omega$ by finitely many cylinders C_i in which Ω is the supergraph of an L_i-Lipschitz function in direction e_i; choose $C_i' \Subset C_i$ still covering, a partition of unity θ_0, θ_i and translations $\tau_i = \varepsilon e_i$. For $y_0 \in \bar{\Omega} \cap \bar{C}_i'$ one has $B(y_0 + \tau_i, r_\varepsilon) \subset \Omega$ with $r_\varepsilon = \varepsilon/(2\sqrt{1+L^2})$, because $y_2 - \varphi_i(y_1) \ge \varepsilon - \sqrt{1+L^2}|q| > \varepsilon/2$. This is the only place where OPF5B-T1 used convexity.

Step 2 Approximation by solutions on a neighbourhood

The remaining steps of OPF5B-T1 (Stein extension [F2], continuity of translations, the facts F1 and L1, mollification) do not use convexity. For $u \in \mathcal{K}$ and $\epsilon > 0$ there are an open $V_\varepsilon' \supset \bar{\Omega}$ and $v_\eta \in C^\infty$ with $P(\partial)v_\eta = 0$ on V_ε' and $\|v_\eta - u\|_{H^m(\Omega)} < 2\epsilon$.

Step 3 A neighbourhood with the same topology

By the collar fact [COL] there is a flow Φ_σ of a transversal field with $\bar{\Omega} \subset \Phi_\sigma(\Omega) \subset \{\operatorname{dist} < C\sigma\}$; take σ with $C\sigma < r_\varepsilon/4$ and $U = \Phi_\sigma(\Omega)$, diffeomorphic to Ω, so $b_1(U) = b_1(\Omega)$.

Step 4 The cokernel on U

OPF-T4(ii) for C^∞ gives $\dim \operatorname{coker} \Sigma_U \le \delta(P) b_1(U) =: \Delta$ with $\Sigma_U : \bigoplus \ker_{C^\infty(U)} Q_j \to \ker_{C^\infty(U)} P$. If $\Delta = 0$, $v_\eta|_U = \sum w_j$ and, $\bar{\Omega} \subset U$ being compact, $w_j|_\Omega \in \mathcal{H}_j$: this is (b).

Step 5 Finite codimension

Suppose $\operatorname{codim}_{\mathcal{K}} S > \Delta$: there are $u_0, \dots, u_\Delta \in \mathcal{K}$ with independent images in the Hilbert space $\mathcal{K}/S$. Independence is open (Gram determinant), so there is ϵ with the same property for every family ϵ-close. Steps 2–3 with one U give $v_k \in \ker_{C^\infty(U)} P$ with $\|v_k|_\Omega - u_k\| < \epsilon$. As $\dim \operatorname{coker} \Sigma_U \le \Delta$, the $\Delta + 1$ classes of v_k are dependent: $\sum c_k v_k = \sum w_j$ on U with $c \neq 0$. Restricting to Ω, $\sum c_k v_k|_\Omega \in \sum \mathcal{H}_j \subset S$, so the images in $\mathcal{K}/S$ are dependent. Contradiction.

Why it works. a Lipschitz boundary supplies a uniform inward segment, a collar supplies a neighbourhood with the same holes, and the cokernel of the sum map there is bounded by the number of holes times $\delta(P)$.

■

Uses: A53.

18.2 Variable coefficients: regularity, counterexample and zeros

Dim. B121 **VAR-P11(e)**

Classical solutions for a C^1 perturbation

[proof written, not audited]

Let A_0 generate a C_0-semigroup $T(t)$ on a Banach space X, $\|T(t)\| \le M_0$ on $[0,T]$, $B \in C^1([0,T];\mathcal{L}(X))$ with $\beta_0 = \sup\|B\|$, $\beta_1 = \sup\|B'\|$, $U(t,s)$ the Dyson–Phillips propagator (the mild solution $u(t) = T(t-s)x + \int_s^t T(t-r)B(r)u(r)dr$). If $x \in D(A_0)$ then $u = U(\cdot,s)x \in C^1([s,T];X) \cap C([s,T];D(A_0))$ and $u' = (A_0 + B(t))u$; the solution is unique.

Stated in the paper as no. 12, p. 68

Proof.

Step 1 Convolution lemma

If $f \in C^1$ and $F(t) = \int_s^t T(t-r)f(r)dr$, then $F \in C^1$, $F'(t) = T(t-s)f(s) + \int_s^t T(t-r)f'(r)dr$, and $F(t) \in D(A_0)$ with $A_0F = F' - f$. Write $F(t) = \int_0^{t-s} T(\sigma)f(t-\sigma)d\sigma$; the difference quotient has a boundary term tending to $T(t-s)f(s)$ and an integral term tending to $\int T(\sigma)f'(t-\sigma)d\sigma$ by dominated convergence. For the domain, $h^{-1}(T(h)-I)F(t) = h^{-1}[F(t+h)-F(t)] - h^{-1}\int_t^{t+h} T(t+h-r)f(r)dr \to F'(t) - f(t)$.

Step 2 A candidate derivative

With $y_0 = A_0x + B(s)x$ let v be the unique solution of $v(t) = T(t-s)y_0 + \int_s^t T(t-r)[B'(r)u(r) + B(r)v(r)]dr$ (Picard, kernel bounded by $M_0\beta_0$). Put $w(t) = x + \int_s^t v$, $f = Bw$ (so $f' = B'w + Bv$) and $z(t) = T(t-s)x + \int_s^t T(t-r)f(r)dr$.

Step 3 Gronwall

By Step 1, $z' = T(t-s)(A_0x + B(s)x) + \int_s^t T(t-r)[B'w + Bv]dr$, so with $e = u - w$: $u - z = \int_s^t T(t-r)Be\,dr$ and $(w-z)' = v - z' = \int_s^t T(t-r)B'e\,dr$ with $(w-z)(s) = 0$. Hence $\|e(t)\| \le (M_0\beta_0 + (t-s)M_0\beta_1)\int_s^t \|e\|$, so $e \equiv 0$: $u = w \in C^1$, $u' = v$.

Step 4 Classical

$u(t) = T(t-s)x + F(t)$ with $f = Bu$; both terms are in $D(A_0)$ and $A_0u = T(t-s)A_0x + F' - Bu = u' - Bu$.

Why it works. differentiating the Duhamel formula once more shows that the mild solution is already the derivative of itself, by Gronwall.

■

Uses: none.

Dim. B122 **VAR-Q18-CE**

A norm-continuous perturbation with no classical solution

[proof written, not audited]

There are a Hilbert space X, A_0 skew-adjoint and $B \in C([0,T];\mathcal{L}(X))$ skew-adjoint at every t such that for some $x_* \in D(A_0)$ and every t with $\sin(nt) \neq 0$, $U(t,0)x_* \notin D(A_0)$. So VAR-P11(e) is false without C^1. The sub-case of multiplication operators $B = -iV(t)\cdot$ with V norm-continuous in L^∞ stays open.

Stated in the paper as no. 12, p. 68

Proof.

Step 1 Construction

Let A be skew-adjoint with unitary group $T(t)$, $x \in \mathcal{H} \setminus D(A)$, $n = \|x\|$, $X = \mathcal{H} \oplus \mathbb{C}$, $A_0 = A \oplus 0$, $f(t) = T(t)x$ and $B(t)(y,c) = (cf(t), -\langle y, f(t)\rangle)$.

Step 2 Properties of B

$\langle B(y,c),(y',c')\rangle = c\langle f,y'\rangle - \langle y,f\rangle\bar{c}' = -\langle (y,c),B(y',c')\rangle$: skew-adjoint. $\|B(t)-B(t')\| \leq 2\|T(t)x - T(t')x\| \to 0$: continuous. It is not C^1, else $t \mapsto B(t)(0,1) = (T(t)x,0)$ would be differentiable, giving $x \in D(A)$. The datum is $x_* = (0,1) \in D(A_0)$.

Step 3 Explicit solution

Put $C(t) = \sin(nt)/n$, $y(t) = C(t)T(t)x$, $c(t) = C'(t) = \cos nt$. The first component of the mild equation is $\int_0^t T(t-r)c(r)f(r)dr = (\int_0^t c)T(t)x = y(t)$; the second is $c(t) = 1 - \int_0^t \langle y,f\rangle = 1 - n^2\int_0^t C$, which is the identity $\cos nt = 1 - n^2\int_0^t \sin(nr)/n\,dr$.

Step 4 Conclusion

By uniqueness $U(t,0)x_* = (C(t)T(t)x, \cos nt)$. For $0 < t < \pi/n$, $C(t) \neq 0$ and $T(t)x \notin D(A)$ since $D(A)$ is invariant under $T(\pm t)$.

Why it works. the perturbation moves the datum along the very direction that is not in the domain, at the speed of a rank-2 rotation.

■

Uses: B121.

Dim. B123 **VAR-Q10-EX**

Zeros of positive measure with empty interior

[proof written, not audited]

There is $u \in C^\infty([0,1]\times\mathbb{R})$ with $i\partial_t u + \partial_x^2 u = 0$ ($V = 0$), $u(t,\cdot)$ entire in x for each t, $\operatorname{int} Z(u) = \varnothing$ and $|Z(u)| > 0$. So in the class L^∞, $V = 0$ and $V' = \mathbf{1}_{Z(u)}$ are both laws ($V'u \equiv 0$), while in the class C the potential is identified ($\operatorname{int} Z = \varnothing$).

Stated in the paper as no. 12, p. 68

Proof.

Step 1 A flat function

Let $K \subset [0,1]$ be a fat Cantor set (closed, nowhere dense, $|K| > 0$) with gaps $(\alpha_j, \alpha_j + \ell_j)$, $\sum \ell_j \le 1$. Take $g_0(s) = e^{-s^{-2}}$ for $s > 0$, $B(s) = g_0(s)g_0(1-s)$, with $|B^{(k)}| \le (2C_1)^k (k!)^{3/2}$, and $g = \sum_j g_j$, $g_j(t) = e^{-\ell_j^{-4}} B((t-\alpha_j)/\ell_j)$. Since $\sup_{\ell\in(0,1]} \ell^{-k} e^{-\ell^{-4}/2} \le C_2^k (k!)^{1/4}$ and $e^{-\ell^{-4}/2} \le \ell$, the series of derivatives converge uniformly and $|g^{(k)}| \le A^k (k!)^{7/4}$. Every $g_j^{(k)}$ vanishes on K, so $g^{(k)} = 0$ on K, and $g > 0$ on the gaps.

Step 2 The Tychonoff series

$u(t,x) = \sum_k (-i)^k g^{(k)}(t) x^{2k}/(2k)!$. As $(2k)! \ge (k!)^2$ the k-th term is $\le A^k |x|^{2k}/(k!)^{1/4}$, so the series converges for all x, as do those of the derivatives; u is smooth, entire in x, and $i\partial_t u + \partial_x^2 u = 0$ termwise (telescoping).

Step 3 Zeros

For $t \in K$ all $g^{(k)}(t) = 0$, so $u(t,\cdot) \equiv 0$ and $Z(u) \supset K \times \mathbb{R}$, of positive measure. If an open $W \subset Z(u)$ existed it would contain a point with t in a gap (K is nowhere dense); there $u(t,\cdot)$ is entire and vanishes on an interval, hence $\equiv 0$, against $u(t,0) = g(t) > 0$.

Why it works. Tychonoff's non-uniqueness for the heat-type equation gives a solution flat on a fat Cantor set but analytic in x.

■

Uses: none.

Dim. B147 **S10COL-ii**

An inward vector field on a Lipschitz domain: $\bar\Omega \subset \Phi_\sigma(\Omega)$

[proof written, not audited]

Let $\Omega \subset \mathbb{R}^n$ be bounded and Lipschitz with charts (C_i, R_i, g_i, L_i) covering $\partial\Omega$ and vertical directions e_i; let $\theta_i \in C_c^\infty(C_i)$, $\theta_i \ge 0$, $\sum \theta_i > 0$ on $\partial\Omega$, $W = \sum \theta_i e_i$, $V = -W$ with flow Φ_s and $M = \sup|V|$. Then for every $\sigma > 0$: $\bar\Omega \subset \Phi_\sigma(\Omega) \subset \{\operatorname{dist}(\cdot,\bar\Omega) \le M\sigma\}$, and $\Phi_\sigma(\Omega)$ is open and diffeomorphic to Ω (same b_1). This is (COL-ii); it replaces the cited Grisvard/Hofmann–Mitrea–Taylor source in S10A-T3.

Stated in the paper as no. 16, p. 80

Proof.

Step 1 The cone of inward directions

For $p \in \partial\Omega$, $D(p)$ is the set of w for which there are $r, a > 0$ with $\operatorname{dist}(x + tu, \complement\Omega) \geq at$ for all $x \in \bar{\Omega} \cap B(p,r)$, $u \in B(w,r)$, $t \in (0,r]$. It is open and invariant under positive scaling (witness $(\rho, \lambda a)$, $\rho = \min(r, \lambda r, r/\lambda)$) and closed under sums: with witnesses (r_k, a_k) for w_k, $\rho = \min(1, r_1, r_2, r_2/(2 + |w_1|))$ and $u_k = w_k + \frac{1}{2}(u - w_1 - w_2)$, the point $x_1 = x + tu_1$ lies in Ω and then $x_1 + tu_2 = x + tu$ has distance $\geq a_2 t$. So $D(p)$ is a convex open cone, and $0 \notin D(p)$.

Step 2 Each chart cone is inward

For a chart, $\Gamma = \{z : z_n > L|z'|\} \subset D(p)$ for $p \in \partial\Omega \cap C$. With $z \in \Gamma$, $\delta = z_n - L|z'|$, $d = \operatorname{dist}(p, \complement C)$, take $r \leq 1$ with $\sqrt{1 + L^2}\, r \leq \delta/2$ and $r(2 + |z|) \leq d/2$. For $x \in \bar{\Omega} \cap B(p,r)$ one has $x_n \geq g(x')$ (limit of points of Ω), $y = x + tu \in C$, $u_n - L|u'| \geq \delta/2$ and $y_n - g(y') \geq t\delta/2$. Since $\operatorname{dist}(x, \complement\Omega) \geq \min(b/\sqrt{1 + L^2}, \operatorname{dist}(x, \complement C))$ for $b = x_n - g(x')$, one gets $\operatorname{dist}(y, \complement\Omega) \geq at$. In particular $e \in D(p)$.

Step 3 The chart field is inward at every point

$W(p) = \sum_{i \in I(p)} \theta_i(p) e_i$ with $e_i \in D(p)$ for $i \in I(p)$, so $W(p) \in D(p)$ by Step 1. No compatibility between overlapping charts is needed: e_j may lie outside Γ_i and both are in the intrinsic convex cone $D(p)$.

Step 4 From a direction to trajectories

If $W(p) \in D(p)$ with witness (r, a), continuity of W gives η with $|W(y) - W(p)| \leq r/2$ on $B(p, \eta)$; with $\rho_p = \min(r, \eta/(1 + M))$, $\psi_t(x) = x + tu$, $u = \frac{1}{t}\int_0^t W(\psi_s x)\, ds \in B(W(p), r)$, so $\operatorname{dist}(\psi_t(x), \complement\Omega) \geq at$, where $\psi_t = \Phi_{-t}$.

Step 5 Uniform inclusion

Cover the compact $\partial\Omega$ by balls $B(p_j, \rho_{p_j}/2)$; let $\sigma_1 = \min \rho_{p_j}/2$, $c = \min a_{p_j}$. Points of $\bar{\Omega}$ within σ_1 of $\partial\Omega$ are handled by Step 4; farther points have $\operatorname{dist}(x, \complement\Omega) \geq \sigma_1$ and move at most Mt. With $\sigma_0 = \min(\sigma_1, \sigma_1/(M + c))$ one has $\operatorname{dist}(\Phi_{-\sigma}(x), \complement\Omega) \geq c\sigma$ for $\sigma \leq \sigma_0$.

Step 6 All $s > 0$

$\psi_t(\Omega) \subset \Omega$ for $t \geq 0$: otherwise take the first exit time $\tau > 0$, $y = \psi_\tau(x) \in \partial\Omega$; for small h, $z = \psi_{\tau - h}(x) \in \Omega$ and Step 5 gives $y = \psi_h(z) \in \Omega$, contradiction. Then for $s > \sigma_0$ compose with $\psi_{\sigma_0}(x) \in \Omega$. Finally $|\Phi_\sigma(x) - x| \leq M\sigma$ and Φ_σ is a diffeomorphism of $\mathbb{R}^n$.

Why it works. the inward cone at each boundary point is convex and open, so a positive combination of chart directions is inward; compactness of the boundary turns this local fact into a uniform margin.

■

Uses: none.

Where it stops

For a non-Lipschitz domain $D(p)$ can be empty: in the disc with a slit, at a point p of the slit a transversal direction crosses the slit at time $h/|w \cdot n|$ and the estimate $\geq at$ fails, a tangent one keeps p on the slit. This is why the slit disc is outside S10A-T3.

Dim. B148 **P2-Q1-NL-model**

On a simply connected slit disc, harmonic functions are holomorphic plus anti-holomorphic [proof written, not audited]

Let $P = \Delta$ ($Q_1 = \partial_{\bar z}$, $Q_2 = \partial_z$), $m \geq 1$, and Ω bounded, simply connected, satisfying a Poincaré inequality (for instance the slit disc $D \setminus [0,1)$, which is not Lipschitz). Then $\ker_{H^m} \Delta = \mathcal{H}_1 + \mathcal{H}_2$ as sets, stronger than density.

Stated in the paper as no. 16, p. 80

Proof.

Step 1 **A holomorphic derivative**

For $h \in \ker_{H^m} \Delta$, h_z is holomorphic.

Step 2 **A primitive**

Ω is simply connected, so h_z has a primitive f, and $h - f$ is anti-holomorphic ($\partial_z(h - f) = 0$).

Step 3 **Regularity**

$f' = h_z \in H^{m-1}$, and by Poincaré $\|f - f_\Omega\|_{L^2} \leq C\|\nabla f\|_{L^2}$; hence $f \in H^m$ and also $h - f \in H^m$.

Step 4 **Poincaré on the slit disc**

Cover $D \setminus [0,1)$ by three convex sets (upper half-disc, lower half-disc, $D \cap \{\operatorname{Re} z < 0\}$) with pairwise overlaps of positive measure, and chain the means.

Why it works. the obstruction of the annulus is the flux, which vanishes on a simply connected set; the only thing needed from the geometry is Poincaré.

■

Uses: B119.

What is left

General P (elliptic factors with $\lambda \neq 0$, hyperbolic or transport ones, where the translations of Step 2′ of S10A-T3 fail), $m = 0$, and domains without Poincaré ("rooms and corridors") are open; there is no route for general P. The exact codimension for general P (P2-Q1-COD) lacks continuous "period" functionals in H^m.

18.3 Statements recorded without proof here

S10A-C2, C12-Q2 = OPF-Q2	The question reduces to triples with at least two non-compact pairs $\Pi_i\Pi_j$, outside real transports (OPR2-T2); proved in B118.	reduction [proof written, not audited]; the question [open]

S10-OPEN-A §4.

S10A-P1 (ring, (i))	Ring structure; (ii) holds with the topological premise from Rudin, RCA 13.11 (cited) or an elementary argument.	[proof written, not audited]

S10-OPEN-A §3; the annulus is B119.

S10COL-L1, L2, L3, TA, TB, TC = (COL-ii)	(COL-ii) proved in B147; the step B of S10A-T3 is replaced by a chart field $X = -\sum \theta_i e_i$, $\theta_i \geq 0$. Theorem B (a field with $V \cdot \nu \geq \kappa$ a.e. is inward at every point, by integrating along almost every segment) is not rewritten here.	[proof written, not audited]

S10-COL; (COL-i) from cited source: Grisvard Lemma 1.5.1.9 (verified), Hofmann–Mitrea–Taylor Cor. 2.13.

S10B-C1	The extension of OPF5B-T1 (density) to non-convex regions is **false** (a hole with an interior is enough); proved in B146.	[false] (extension)

S10-OPEN-B; F10-B2 prevails; see B119.

P2-Q1 (universal form)	False for $P = \Delta$ on every Lipschitz region with $b_1 \geq 1$ (A53, B119).	[false]

S10A-P1.

P2-Q1-NL	Model case: Δ, $m \geq 1$, with $L^{1,2} \subset L^2$ proved in B148; general non-Lipschitz case open.	model [proof written, not audited]; general [open]

F10-TRIAGE.

C12-P7 ("closed but not dense")	Differential example on the annulus, $m \geq 1$.	[proof written, not audited]

A53, B119.

S10B-P3	Density of (SB) proved in B126; "open and dense" after the fix P3-c (bumps also at minima of value 0; counterexample $u_0' = -\cos x - \cos 2x$) proved in the triage.	[proof written, not audited]

F10-B2, F10-TRIAGE.

18.4 Nonlinear and scattering closures

Dim. B124 **S10B-T1**

Hirota N-solitons are reflectionless potentials

[proof written, not audited]

Let $0<\kappa_1<\cdots<\kappa_N$, $c_j>0$, $\varepsilon_j=(-1)^{j-1}$, $k_j=2\kappa_j$, and

$$f(x)=\sum_{S\subseteq\{1..N\}}\prod_{j\in S}e^{\eta_j}\prod_{\substack{i<j\\ i,j\in S}}A_{ij},\qquad A_{ij}=\left(\frac{k_i-k_j}{k_i+k_j}\right)^2,\quad e^{\eta_j}=c_jd_je^{k_jx},$$

with $d_j=\prod_{l\neq j}\frac{\kappa_j+\kappa_l}{|\kappa_j-\kappa_l|}$ and $u=2(\log f)''$. Then $u\in\mathcal{S}(\mathbb{R})$, $b\equiv0$, $a(k)=\prod_j\frac{k-i\kappa_j}{k+i\kappa_j}$, and the discrete spectrum of $L_u=-\partial^2-u$ is $\{-\kappa_1^2,\dots,-\kappa_N^2\}$. The same holds at every time with $c_j(t)=c_j(0)e^{-8\kappa_j^3t}$. That $u(t)$ solves KdV is Hirota's theorem, taken as part of the definition of the multi-soliton class.

Stated in the paper as no. 16, p. 80

Proof.

Step 1 The Wronskian of the half-solutions

Put $\varphi_j=e^{-\kappa_jx}+\varepsilon_jc_je^{\kappa_jx}$, so $-\varphi_j''=-\kappa_j^2\varphi_j$, and $W=W(\varphi_1,\dots,\varphi_N)$. Multilinearity in the columns, choosing in column j the term $e^{s_j\kappa_jx}$ with $s_j=+1$ iff $j\in S$, gives

$$W=V_0\,e^{-\sum\kappa_jx}\sum_S\prod_{j\in S}(\varepsilon_jc_j)\,\rho(S)\,e^{2\sum_{j\in S}\kappa_jx},\qquad\rho(S)=\frac{V(s_1\kappa_1,\dots,s_N\kappa_N)}{V_0},$$

where V is the Vandermonde determinant and $V_0=V(-\kappa_1,\dots,-\kappa_N)\neq0$.

Step 2 Sign and modulus of $\rho(S)$

Factor by factor, for $i<j$: the pairs outside S give 1, the pairs inside S give -1, a mixed pair gives $\pm(\kappa_i+\kappa_j)/(\kappa_j-\kappa_i)$. The number of negative factors is $\binom{|S|}{2}+\#\{i<j:\ i\notin S,\ j\in S\}\equiv\sum_{j\in S}(j-1)$, so $\operatorname{sign}\rho(S)=\prod_{j\in S}\varepsilon_j$. For the modulus, $|\rho(S)|=\prod_{i\in S,l\notin S}\frac{\kappa_i+\kappa_l}{|\kappa_i-\kappa_l|}=\prod_{j\in S}d_j\prod_{i<j\in S}\left(\frac{\kappa_i-\kappa_j}{\kappa_i+\kappa_j}\right)^2$. Hence $W=V_0e^{-\sum\kappa_jx}f(x)$ with $f>0$, so W never vanishes and $2(\log W)''=u$.

Step 3 Intertwining without quoting Crum

Let $Ag=W(\varphi_1,\dots,\varphi_N,g)/W$: a monic operator of order N with $\ker A=\operatorname{span}\{\varphi_j\}$ and $a_{N-1}=-W'/W$. Divide: $A(-\partial^2)=QA+R$ with $\operatorname{ord}R<N$. Both $A(-\partial^2)$ and QA kill each φ_j, so $R\varphi_j=0$ for N independent functions, hence $R=0$. Comparing the coefficients of $\partial^{N+2},\partial^{N+1},\partial^N$ gives $q_2=-1$, $q_1=0$, $q_0=2a_{N-1}'=-2(\log W)''$. So $Q=-\partial^2-u=L_u$ and $L_uA=A(-\partial^2)$.

Step 4 Jost solutions and scattering data

For real $k \neq 0$ the function $\psi = Ae^{ikx}$ solves $L_u\psi = k^2\psi$. As $x \to +\infty$ the terms $S = \{1..N\}$ dominate and $A \to \prod_j(\partial - \kappa_j)$, so $\psi e^{-ikx} \to \prod_j(ik - \kappa_j)$. As $x \to -\infty$ the terms $S = \emptyset$ dominate and $\psi e^{-ikx} \to \prod_j(ik + \kappa_j)$. Thus $\psi_+ = \psi / \prod_j(ik - \kappa_j)$ is the Jost solution at $+\infty$ and at $-\infty$ it has *no* e^{-ikx} component. With the convention $\psi_+(x,k) = a(k)\psi_-(x,-k) - b(-k)\psi_-(x,k)$ this means $b(-k) = 0$ for all real $k \neq 0$, so $b \equiv 0$, and $a(k) = \prod_j \frac{k-i\kappa_j}{k+i\kappa_j}$ on $\mathbb{R} \setminus 0$.

Step 5 Zeros of a and the spectrum

Let $R(k) = \prod_j \frac{k-i\kappa_j}{k+i\kappa_j}$, analytic in $\operatorname{Im} k > 0$ and continuous to $\mathbb{R}$. Then $h = a - R$ is analytic in the upper half-plane, continuous to $\mathbb{R} \setminus 0$ and zero there. By Schwarz reflection h extends across $\mathbb{R} \setminus 0$ and vanishes on an interval, so $h \equiv 0$. The zeros of a in the upper half-plane are exactly $i\kappa_j$, and the discrete spectrum is $\{-\kappa_j^2\}$.

Step 6 Schwartz decay

At each end, $\log f$ equals a linear exponent plus $\log(1 + \text{exponentially small})$. Hence $u = 2(\log f)''$ and all its derivatives are $O(e^{-2\kappa_1|x|})$.

Why it works. the Wronskian of the half-solutions is a positive sum of exponentials, so the Darboux–Crum operator A is smooth, and its asymptotic form at the two ends is a constant-coefficient operator that only multiplies the plane wave: nothing is reflected.

■

Uses: OPF5D-L3, OPF5D-L4, OPF5D-L9.

Checks and convention

SymPy/mpmath: Cauchy minors for $N \leq 4$ (4/4), sign and modulus of ρ for $N \leq 6$ (exact, 12/12), Wronskian relative error $\leq 1.8 \cdot 10^{-41}$, residual of $(-\partial^2 - u)\psi = k^2\psi$ at 10^{-196}, asymptotics at $x = \pm 40$ of order e^{-40}. The class of multi-solitons is *defined* by Hirota's formula, as in INT-P5/P6; with this convention INT-T1(B) has no external source.

Dim. B125 **S10B-T2**

A prime differential ideal in one indeterminate is $[A] : H_A^\infty$ [proof written, not audited]

Let F be a differential field of characteristic 0 and $J \subsetneq F\{y\}$ a non-zero prime differential ideal. Let $A \in J$ have minimal rank (order n first, then degree d in y_n), with initial I_A, separant $S_A = \partial A/\partial y_n$ and $H_A = I_A S_A$. Then $J = [A] : H_A^\infty$.

Stated in the paper as no. 16, p. 80

Proof.

Step 1 A is irreducible

If $A = BC$ with $B, C \notin F$, each factor has strictly smaller rank (order $< n$, or order n and degree $< d$). By primality one of them lies in J, against minimality. A factor in $F \setminus 0$ is impossible since J is proper.

Step 2 $H_A \notin J$

$I_A \neq 0$ has order $< n$ or lies in $F \setminus 0$. $S_A \neq 0$ because $d \geq 1$ and $\operatorname{char} F = 0$, and it has degree $d-1$ in y_n. Both have rank smaller than A, so they are not in J; by primality no power H_A^k is in J.

Step 3 $\supseteq$

If $H_A^k f \in [A] \subseteq J$, then $f \in J$, since J is prime and $H_A^k \notin J$.

Step 4 $\subseteq$

Let $f \in J$. Ritt's reduction (differential pseudo-division by the derivatives $A^{(i)}$, whose initial is S_A, then algebraic pseudo-division by A) gives $a, b \geq 0$ and R reduced with respect to A such that $S_A^a I_A^b f \equiv R \pmod{[A]}$. Then $R \in J$ and has smaller rank than A unless $R = 0$; minimality forces $R = 0$. So $H_A^{\max(a,b)} f \in [A]$.

Why it works. minimality of the rank leaves no room for a reduced element in J, so reduction by A must end at zero.

■

Uses: none.

Application to a solution

For u real-analytic on an interval, $J_u = \{P : P[u] \equiv 0\}$ is prime (the ring of analytic functions on a connected set is a domain) and differential. If u solves $u_t = F[u]$ then $(D_F P)[u] = \partial_t(P[u]) \equiv 0$, so J_u is D_F-stable, and the theorem applies: this removes the dependence of OPF5C-T5 on a numbered citation of Ritt or Kolchin.

Dim. B126 **S10B-P3**

Morse and (SB) is dense

[proof written, not audited]

Let $r \geq 3$. The class of $u_0 \in C^r(\mathbb{T})$ with u_0' Morse and satisfying (SB) (at each birth time t_b there is at most one degenerate minimiser (x_b, a_b), and $M_{t_b}(x_b) = \{a_b\}$) is dense. Together with openness and the density of Morse functions it is open and dense, so "generic" in OPA-Q2 is no longer conditional.

Stated in the paper as no. 16, p. 80

Proof.

Step 1 Candidates and distinct critical values

Let $m_1, \dots, m_p$ be the local minima of u_0' with negative critical value; the candidate times are $t_i = -1/u_0'(m_i)$ and $x_i = m_i + t_i u_0(m_i)$. Take $\psi_i(x) = (x - m_i)\chi(x - m_i)$ with χ even, $\chi = 1$ on $[-\rho, \rho]$, supported in $(-2\rho, 2\rho)$. Then $\psi_i' = 1$ near m_i and ψ_i'' lives in an annulus without critical points where $|u_0''| \geq c$. For small ε the perturbation $u_0 + \varepsilon\psi_i$ moves only the critical value $u_0'(m_i)$. Choosing the ε_i small and generic makes the critical values pairwise distinct and non-zero; so at each t_i there is only one candidate. The set of candidates is unchanged by all further steps.

Step 2 **Uniqueness of the minimiser at t_i**

If m_i is not a minimiser of Φ_{x_i}, or $M_{t_i}(x_i) = \{m_i\}$, do nothing. Otherwise m_i is an isolated strict local minimum of Φ_{x_i}; choose r_i so that the annulus $C_i = \{r_i \le |a - m_i| \le 2r_i\}$ contains neither critical points of u_0' nor other minimisers, and $\Phi_{x_i} \ge \Phi_{x_i}(m_i) + \eta_i$ on C_i. With $\Theta_i \in C^\infty$, $0 \le \Theta_i \le 1$, $\Theta_i = 0$ on $|a - m_i| \le r_i$ and $= 1$ outside $|a - m_i| < 2r_i$, set $\theta_i = \Theta_i'$ (periodic, zero mean, supported in C_i). Replacing u_0 by $u_0 + \varepsilon\theta_i$ changes U_0 by $\varepsilon\Theta_i$, leaves the jet at m_i (hence t_i, x_i) untouched, and raises Φ_{x_i} away from m_i. Then $M_{t_i}(x_i) = \{m_i\}$.

Step 3 **Earlier conditions are open**

Each condition SB_j is open in C^3: the first alternative is a strict inequality between continuous quantities; in the second the margin outside a neighbourhood of m_j is stable, and inside it Φ'_{x_j} is strictly increasing except at one point because m_j remains a non-degenerate minimum of u_0' with value $-1/t_j$. Choose each ε_i smaller than half the margin of every SB_j, $j < i$.

Step 4 **Conclusion**

After p steps $\tilde{u}_0 = u_0 + \sum \varepsilon_i \psi_i + \sum \varepsilon_i' \theta_i$ is Morse with (SB), and $\|\tilde{u}_0 - u_0\|_{C^r} \le \sum |\varepsilon_i| \|\psi_i\|_{C^r} + \sum |\varepsilon_i'| \|\theta_i\|_{C^r}$ is as small as desired. Translates $m_i + 2\pi n$ do not interfere: a minimiser satisfies $X_{t_i}(a') = x_i$ while $X(m_i + 2\pi n) = x_i + 2\pi n$.

Why it works. the candidate times are finitely many and are not moved by the bumps; each bump kills the competitor at its own time and, being chosen last-and-small, cannot undo what was done before.

■

Uses: OPF5A-L4, OPF5A-11, OPF5A-12.

The case of minima of value zero (triage fix P3-c)

A local minimum of u_0' with critical value 0 would correspond to an infinite time. For "open and dense" one also adds bumps at such minima: for $u_0' = -\cos x - \cos 2x$ the value is attained at degenerate points, and perturbing it away is what the triage proved.

Dim. B127 **S10B-L4**

The uniform shock rate with several births

[proved mod. [F1]]

Under Morse+(SB), $u_0 \in C^5$: for every birth time t_b there are $\tau_b, r_b > 0$ such that, uniformly for $|t - t_b| \le \tau_b$,

$$\int_{|x - x_b| < r_b} |u - v|\, dx = 2\nu \log \frac{1}{\max(|t - t_b|, \sqrt{\nu})} + O(\nu), \qquad \int_{|x - x_b| \ge r_b} |u - v|\, dx = O(\nu).$$

Hence on $(0, T]$: $\|u - v\|_{L^1} = 2\nu \log \frac{1}{\max(d(t), \sqrt{\nu})} + O(\nu)$, with $d(t) = \min_b |t - t_b|$, and $\sup_t \|u - v\|_{L^1} = \nu |\log \nu| + O(\nu)$, modulo the Oleinik $\Longleftrightarrow$ Kružkov equivalence [F1].

Stated in the paper as no. 16, p. 80

Proof.

Step 1 Margin instead of global convexity

Fix z_0 below the Taylor radius and the distance from a_b to the other critical points of u_0'. By (SB), $M_{t_b}(x_b) = \{a_b\}$; upper semicontinuity of $(t,x) \mapsto M_t(x)$ gives $r_b, \tau_b, \eta_0 > 0$ such that for $|t - t_b| \le \tau_b$, $|x - x_b| < 2r_b$ every minimiser has $|\alpha| < z_0/2$ ($\alpha = a - a_b$) and $\Phi_x \ge \min \Phi_x + \eta_0$ for $|\alpha| \ge z_0$. The region $|\alpha| \ge z_0$ weighs $O(e^{-\eta_0/2\nu})$ in the Hopf integral.

Step 2 Local geometry

With $\eta = 1 - t/t_b$ and $\kappa_3 = u_0'''(a_b) > 0$, $J_t = \frac{t}{t_b} J_{t_b} + \eta$, $J_{t_b}(a_b + \alpha) = \frac{t_b \kappa_3}{2}\alpha^2(1 + O(\alpha))$, hence $c(\eta + \alpha^2) \le J_t \le C(\eta + \alpha^2)$ for $\eta \ge 0$; Φ_x is convex in the window for $t \le t_b$. The estimates (a)–(c) of OPF5A-L2 follow with the tail replaced by the margin η_0.

Step 3 Before the birth, $t \in [t_b - \tau_b, t_b]$

Inside the window, Steps 1–4 of OPF5A-T1 repeat. Step 2 ("$dx = J_t\, da$ with X_t bijective") is false globally after the first birth; it is replaced by the local bijection $X_t|_{W_b}$, strictly increasing because $J_t > 0$ on $W_b \setminus \{a_b\}$. Since $t u_0'' = \partial_a J_t$ has constant sign on each side of a_b, $\int |\partial_a J_t| / J_t \, da = 2 \log \frac{1}{\max(\sqrt{\nu}, \eta)} + O(1)$. Outside, the minimisers are non-degenerate: on the compact set $\mathcal{K}_b$ one has $J \ge 2j_0 > 0$, and the box argument of OPF5A-L3 gives $O(\nu)$.

Step 4 After the birth, $t \in [t_b, t_b + \tau_b]$

This is the delicate point. With $\tau = t - t_b$, $a = a_b + \sqrt{\tau} A$, $x = X_t(a_b) - t\tau^{3/2}\xi$, the normal form $\Psi_\tau(\xi, A) = \tau^{-2}[\Phi_x(a_b + \sqrt{\tau}A) - \Phi_x(a_b)]$ converges in C^4 on compacts to $\xi A + \frac{\kappa_3}{24}A^4 - \frac{A^2}{2t_b^2}$. Lemma U uses only the Taylor expansion of u_0 at a_b to order 5, hence is local. Large $|A|$ inside the window is controlled by the A^4 growth, outside by η_0. In the intermediate region ($|\xi| > \Xi$, $|x - x_b| < r_b$) the convexity of 4.2 fails for $t > t_b$; one uses instead the local estimates $\psi(z) \ge \max(0, c_1(z^4 + \alpha^2 z^2) - C\tau z^2)$, $\psi(z) \le C(z^4 + \alpha^2 z^2 + \tau z^2)$ for $|z| \le z_0$.

Step 5 Gluing

$T_1' \cap (0, T] = \{t_1 < \cdots < t_B\}$ is finite. Take τ_b below half the gaps between births: at most one window is active at any t. Away from the windows OPF5A-L3 applies.

Why it works. near a birth only a bounded window of labels a matters; the margin η_0 replaces global convexity there, and away from births the minimisers are non-degenerate.

■

Uses: OPF5A-L2, OPF5A-L3, OPF5A-T1, OPF5A-T2, S10B-P3.

Checked on an example

For $u_0 = \sin x + 0.45 \sin(2x + 1)$ (second birth at $t_2 = 1.8370$) the ratio $(\|u - v\|_{L^1} - \text{pred})/\nu$ is stable at $-1.086, -1.085, -1.087$ for $\nu = 10^{-3}, 3 \cdot 10^{-4}, 10^{-4}$, and bounded at $|t - t_2| = 0.02$. The point of risk is Step 4: uniformity of Lemma U with the margin in place of global convexity.

Dim. B128 **S10B-T3**

Uniform-in-time H^{n-1} bounds from the Gardner laws [proof written, not audited]

Let $u \in C(\mathbb{R}; \mathcal{S}(\mathbb{R}))$ solve $u_t + u_{xxx} + 6uu_x = 0$. For each $n \geq 1$, $\sup_{t\in\mathbb{R}} \|u(t)\|_{H^{n-1}} \leq C_n(\|u(0)\|_{H^{n-1}}) < \infty$. Global existence in $\mathcal{S}$ for Schwartz data is the hypothesis (WP), taken from the literature.

Stated in the paper as no. 16, p. 80

Proof.

Step 1 Gardner's identity

If $u = w + \varepsilon w_x - \varepsilon^2 w^2$ then

$$u_t + 6uu_x + u_{xxx} = (1 + \varepsilon\partial - 2\varepsilon^2 w)\big(w_t + 6(w - \varepsilon^2 w^2)w_x + w_{xxx}\big).$$

The formal inverse series $w = \sum \varepsilon^n w_n[u]$, with $w_0 = u$, $w_n = -\partial w_{n-1} + \sum_{i+j=n-2} w_i w_j$, solves Gardner's equation, which is in conservation form $w_t = -\partial(3w^2 - 2\varepsilon^2 w^3 + w_{xx})$. Order by order $D_F w_n$ is a total derivative, so $H_n = \int w_n[u]\,dx$ is conserved for $u \in C(\mathbb{R}; \mathcal{S})$ (checked symbolically for $n \leq 8$).

Step 2 Weights

With $\operatorname{wt}\partial = 1$, $\operatorname{wt} u = 2$, $\operatorname{wt}\varepsilon = -1$, w_n is homogeneous of weight $n+2$: a monomial $\prod y_{j_i}$ ($y_j = \partial^j u$) satisfies $\sum(j_i + 2) = n + 2$.

Step 3 The quadratic part

The linear part of w_n is $(-\partial)^n u$, a total derivative. The quadratic part obeys $Q_n = -\partial Q_{n-1} + \sum_{i+j=n-2} L_i L_j$, and modulo total derivatives

$$\int Q_{2n} = \sum_{i=0}^{2n-2} \int \partial^i u\, \partial^{2n-2-i} u = (-1)^{n+1} \int (\partial^{n-1} u)^2.$$

Step 4 Lowering the order of higher-degree terms

A monomial of degree $d \geq 3$ in w_{2n} carries $D = 2n + 2 - 2d \leq 2n - 4$ derivatives. If $j_1 > j_2 + 1$ an integration by parts moves one derivative off the largest-order factor, and $\sum j_i^2$ drops strictly; the process ends with $j_1 \leq (D+1)/2$, i.e. $j_1 \leq n - 2$.

Step 5 Gagliardo–Nirenberg

Put $m = n - 1$, $X = \|\partial^m u\|_{L^2}$, $M_0 = \|u\|_{L^2}$ (conserved). Then $\|\partial^j u\|_2 \leq C M_0^{1-j/m} X^{j/m}$ and $\|\partial^j u\|_\infty \leq C M_0^{1-\theta_j} X^{\theta_j}$, $\theta_j = \frac{j+1/2}{m} < 1$, so for a reduced monomial

$$\Big|\int \prod y_{j_i}\Big| \leq \|y_{j_1}\|_2 \|y_{j_2}\|_2 \prod_{i\geq 3} \|y_{j_i}\|_\infty \leq C(M_0) X^\gamma, \qquad \gamma = \frac{2n + 1 - 3d/2}{n-1} < 2.$$

Step 6 Conclusion

$X(t)^2 = \pm H_{2n} + O(1 + X(t)^\gamma)$ with $\gamma < 2$ and H_{2n} constant, hence $\sup_t X(t) < \infty$; no induction is needed.

Why it works. each Gardner law has the square of the top derivative as its only quadratic part, and every other term is of strictly lower growth.

■

Uses: OP-F5D (Class (CI)), (WP).

Dim. B129 **S10B-L8p**

The graded bound extends to $K = 2N$ [proof written, not audited]

Let Z_k be the variety of order-$(2N-1)$ jets of N-solitons and I_k its ideal, which is radical. The ideal of order-$2N$ jets is

$$P_k^{(2N)} = I_k\,\mathbb{C}[y_0..y_{2N}] + \big(y_{2N} - \Phi(k; y_0, \dots, y_{2N-1})\big),$$

and the initial form of the added generator, $y_{2N} - \Phi(0; y)$, is monic in the new variable y_{2N}. Hence the proof of OPF5C-L8 applies verbatim and the graded lower bound at $k = 0$ transfers to every k for $K = 2N$.

Stated in the paper as no. 16, p. 80

Proof.

Step 1 The flow gives Φ

By OPF5C-L1 the map Ψ is a weighted-homogeneous polynomial automorphism, and Mumford's flow $V' = (V^2 + f)/U - (y_0 + \lambda)U$ (exact division) is polynomial and weighted. On Z_k it expresses $y_{2N} = \Phi(k; y_0..y_{2N-1})$, homogeneous of weight $2N + 2$ in (k, y).

Step 2 The ideal

($\supseteq$) both kinds of generator vanish on $2N$-jets of N-solitons. ($\subseteq$) the set of such jets is the graph of Φ over the dense subset of order-$(2N-1)$ jets, so its Zariski closure is the graph Γ of $\Phi|_{Z_k}$. The projection $\Gamma \to Z_k$ is an isomorphism, so $\mathbb{C}[y_0..y_{2N}]/(I_k, y_{2N} - \Phi) \cong \mathbb{C}[Z_k]$ is reduced and $(I_k, y_{2N} - \Phi)$ is the ideal of Γ.

Step 3 Regular sequence of initial forms

The initial form of $y_{2N} - \Phi$ (maximal y-weight component) is $y_{2N} - \Phi(0; y)$; it is monic in y_{2N}, a variable not occurring in the other generators, so it is a non-zero-divisor modulo them and the sequence of initial forms stays regular. This is the only property of the generators used in L8.

Why it works. adding one monic equation in a fresh variable cannot spoil regularity of the initial forms.

■

Uses: OPF5C-L1, OPF5C-L8, OPF5C-T1.

Dim. B130 **S10B-P10**

Finite-test criteria exist exactly when (I) fails or a finite hitting set exists [proof written, not audited]

Let $\mathcal{L} = \{L_u : Pu = 0,\ \mathrm{Ann}(u) \neq \mathfrak{D}P\}$ and call $\{T_1, \dots, T_s\} \subset \mathfrak{D}$ a finite-test criterion if, for every solution u of $Pu = 0$, $\mathrm{Ann}(u) = \mathfrak{D}P$ iff $T_i u \notin \mathcal{T}$ for all i. A finite-test criterion exists iff (I) *no solution is identifying*, or $\mathcal{L}$ has a finite hitting set in $\mathfrak{D} \setminus \mathfrak{D}P$. In particular (I) is necessary for the theorem "no finite test".

Stated in the paper as no. 16, p. 80

Proof.

Step 1 $\Leftarrow$, first case

If (I) holds, take $T_1 \in \mathfrak{D}P$: then $T_1 u = 0 \in \mathcal{T}$ for all u, so the criterion declares every u non-identifying, which is correct.

Step 2 $\Leftarrow$, second case

Let $T_1..T_s \notin \mathfrak{D}P$ hit every $L \in \mathcal{L}$. If u is not identifying, some $T_i \in L_u$, i.e. $T_i u \in \mathcal{T}$. If u is identifying, $L_u = \mathfrak{D}P$ and no $T_i \in \mathfrak{D}P$ gives $T_i u \notin \mathcal{T}$.

Step 3 $\Rightarrow$

Let $\{T_i\}$ be a correct criterion and suppose (I) fails, i.e. an identifying u^* exists. If some $T_i \in \mathfrak{D}P$ the criterion fails on u^*. So all $T_i \notin \mathfrak{D}P$, and for each non-identifying u some $T_i u \in \mathcal{T}$, i.e. $T_i \in L_u$: a finite hitting set.

Why it works. a test either detects every non-identifying solution or it is useless; the detection set is exactly a hitting set of the annihilator ideals.

■

Uses: OP-ALG-T4, OPF5B-T6.

What it does and does not give

A condition such as "infinitely many u_λ with pairwise intersecting L_{u_λ} in $\mathfrak{D}P$" (any $T \notin \mathfrak{D}P$ is in at most one of them) is only *sufficient* for the absence of a finite hitting set; realisability of the atoms of a repeated socle is not shown to be equivalent. OPR2-Q3 stays open in the exact form: with repeated socle, is (I) true and does $\mathcal{L}$ have no finite hitting set?

Dim. B131 **S10B-P11**

Localised compact interaction closes the sum on a polygon [proof written, not audited]

Let $\Omega \subset \mathbb{R}^2$ be a convex bounded polygon with vertices $v_1..v_V$, $Q_j = n_j \cdot \nabla + d_j$ with pairwise non-parallel n_j, $K_j = \ker_{L^2} Q_j$ with orthogonal projections P_j. Let $\chi_0, \dots, \chi_V$ be smooth with $\sum \chi_\alpha^2 = 1$ on $\bar{\Omega}$, $\chi_v \equiv 1$ near v, $\chi_0 \equiv 0$ near all vertices. Assume (L-int): $P_i \chi_0^2 P_j$ is compact for $i \neq j$; and (L-v): at each vertex, $\|\sum_j \chi_v f_j\|^2 \geq c_v \sum_j \|\chi_v f_j\|^2 - \langle \mathcal{K}_v f, f\rangle$ for some compact self-adjoint $\mathcal{K}_v$ on $\mathcal{H} = K_1 \times \cdots \times K_N$. Then $K_1 + \cdots + K_N$ is closed.

Stated in the paper as no. 16, p. 80

Proof.

Step 1 Partition of the norm

For $f = (f_j) \in \mathcal{H}$ and $Sf = \sum f_j$, $\|Sf\|^2 = \sum_\alpha \|\sum_j \chi_\alpha f_j\|^2$.

Step 2 Interior term

$\|\sum_j \chi_0 f_j\|^2 = \sum_j \|\chi_0 f_j\|^2 + \sum_{i \neq j} \langle f_i, P_i \chi_0^2 P_j f_j\rangle = \sum_j \|\chi_0 f_j\|^2 - \langle \mathcal{K}_0 f, f\rangle$ with $(\mathcal{K}_0 f)_i = -\sum_{j \neq i} P_i \chi_0^2 P_j f_j$, compact by (L-int) and self-adjoint since $(P_i \chi_0^2 P_j)^* = P_j \chi_0^2 P_i$.

Step 3 Vertex terms and sum

Use (L-v). With $c = \min(1, c_v)$ and $\mathcal{K} = \mathcal{K}_0 + \sum_v \mathcal{K}_v$ one gets $\|Sf\|^2 \geq c\|f\|_{\mathcal{H}}^2 - \langle \mathcal{K} f, f\rangle$.

Step 4 Fredholm

Hence $\mathcal{G} + \mathcal{K}$ with $\mathcal{G} = S^*S$ is coercive, thus invertible, and $\mathcal{G}$ is Fredholm. On $(\ker S)^\perp$, $\|Sf\|^2 \geq c'\|f\|^2$, so $\operatorname{ran} S$ is closed.

Why it works. a compact error in a coercive inequality does not change the closedness of the range: it can only add a finite-dimensional kernel.

■

Uses: OPF5B-P8, OPR2-L1.

What is not proved

The passage from the tangent-cone model (OPF5B-P8: a direct closed sum for homogeneous transports) to (L-v) on the polygon needs $\chi_v f_j$ to lie in the model kernel modulo compact errors; the commutator $(n_j \cdot \nabla \chi_v) f_j$ and the zeroth-order terms d_j are not controlled. (L-int) is also not proved in this generality. So OPR2-Q2′ ($m = 0$, polygon) stays open, reduced to (L-int)∧(L-v).

Dim. B132 **S10B-E8**

No autonomous ODE in x iff relation (R) fails (modulo S4ID-P8) [conditional]

Assume S4ID-P8 (hence [GT-u$_m$] and (WP)): for a solution $u \in C(\mathbb{R}; \mathcal{S})$ of KdV with $u(0)$ analytic and $r \not\equiv 0$, every relation $G[u(t)] \equiv 0$ has zero linear part and $\mathrm{in}(G) \in I_{PW}$, where $\mathrm{in}(G)$ is the homogeneous component of minimal degree and I_{PW} the ideal of jets of $v'' = cv$. Let (R) say: no $G \neq 0$ with zero linear part and $\mathrm{in}(G) \in I_{PW}$ satisfies $G[u(t)] \equiv 0$ for all t. Then INT-Q1 has the answer "no ODE" iff (R) holds.

Stated in the paper as no. 16, p. 80

Proof.

Step 1 $\Rightarrow$

If no relation exists, none exists in the class of (R).

Step 2 $\Leftarrow$

Let $G \neq 0$ with $G[u(t)] \equiv 0$. By S4ID-P8 its linear part vanishes and $\mathrm{in}(G) \in I_{PW}$; so G is in the class of (R), contradicting (R).

Why it works. S4ID-P8 already confines every possible relation to the class of (R).

■

Uses: S4ID-P8, OPF5C-T6.

It is an equivalence, not a reduction

The case "G with non-zero linear part" is excluded by S4ID-P8 itself, not by OPF5C-T6 (which for order ≥ 4 gives only exponential tails, compatible with $r \not\equiv 0$). Deciding (R) needs the sub-leading terms of the radiative asymptotics ($t^{-1/2}$ log corrections, the Deift–Zhou phase); Airy's cubic relation (OP-O5) lies in this class. INT-Q1 stays open.

Dim. B133 **S10B-V9**

Values of $\delta_N(K)$: the computed table and what it proves computer-assisted [proof written, not audited] mod (Nov)

Let $\delta_N(K)$ be the minimal degree of a homogeneous relation among the graded functions of OP-F5C at $k = 0$ (the lower bound transferred to every k by L8, L8′). Computed: $\delta_N(2N) = N+1$ for $N = 1, \ldots, 5$; $\delta_5(8) = 9$; and, in the triage with an independent mod-p implementation on Adler–Moser jets, $\delta_5(10) = 6$, $\delta_5(9) = 7$ (reproduced) and $\delta_6(12) = 7$ (new), the latter depending on the hypothesis that Adler–Moser jets lie in the closure of the principal stratum. Together with the upper bounds $\delta_N(K) \leq N+1$ for $K \geq 2N$ (OPR2-P4) and $\delta_N(2N-2) \leq N+4$ (OPF5C-P7(b)), these give equalities for the values listed. The case $N = 6$ in general is not executed.

Stated in the paper as no. 16, p. 80

Proof.

Step 1 Reduction to a graded problem

By L8 and L8′ the lower bound at $k = 0$ holds for all $k \in U_0$. So the conjectures $\delta_N(2N) = N + 1$, $\delta_N(2N-1) = N+2$ and $\delta_N(2N-2) = N+4$ are equivalent to the absence of homogeneous relations of degree $\leq N$, $\leq N+1$, $\leq N+3$ respectively among the graded functions.

Step 2 Computation

N	K	minimal degree	time
1	2	2	0.0 s
2	4	3	0.1 s
3	6	4	0.7 s
4	8	5	21.5 s
5	10	6	3231 s

Each row gives $\delta_N(2N)(k) \geq N+1$ for every k; with OPR2-P4, equality.

Step 3 $\delta_5(8)$

The completed run at $N = 5$, $K = 8$ gives $\delta_5(8) \geq 9$ at $k = 0$; L8 transfers it, and OPF5C-P7(b) gives ≤ 9.

Why it works. the parameter k disappears after L8: what remains is a finite Gröbner computation in a graded ring.

■

Uses: OPF5C-L8, B129, OPR2-P4, OPF5C-P7(b).

Status of the computations

$N \leq 4$ was reproduced by the audit; $N = 5$ and $\delta_5(8) = 9$ were computed once in the package (the first $(6,12)$ run was stopped after 3 h 28 min), and then reproduced in the triage in 26–161 s by the independent implementation (timings: $\delta_5(10)$ 26 s, $\delta_5(9)$ 88 s, $\delta_5(8)$ 161 s, $\delta_6(12)$ 195 s). The triage runs with Hirota jets for $N = 5$ and with $(6,11)$ were not concluded; the value $\delta_6(11) = 8$ is recorded in the ledger with a written proof not reproduced here. The proofs are computer-assisted modulo L8/L8′ and the hypothesis on Adler–Moser jets. The general question is OPF5C-Q1, open.

S4ID-P8, OPR2-P7 Conditional on [GT-u$_m$] and (WP): the radiative asymptotics with derivatives of the KdV solution, from Grunert–Teschl (arXiv:0807.5041, Thm 5.4) with a uniform error for x/t in compacts. [conditional]

F10-GTU. The use made of it is in B132.

[GT-u_0] Boundedness of the Hölder constants of GT (5.20)–(5.21) and of the decoupling ε for $k_0 = \sqrt{-x/12t}$ in compacts. Every step of GT §5 is continuous in k_0, so this is plausible, but it needs §5 rewritten with explicit constants; the earlier "reduction" (U) is the same statement, not a simpler one, and was withdrawn. [open]

F10-GTU; blocked on sources.

[GT-u_m] Justification of the differentiation in GT Rem. 5.5 (Deift–Zhou, CMP 165, not read). [open]

blocked on a source not available offline.

18.5 Systems, variable coefficients, encoder and dictionary closures

Dim. B134 **VAR-Q17a**

Lateral observability for $V = V_0(x) + v(t)$ [proof written, not audited]

Let $V = V_0(x) + v(t)$ with $V_0 \in L^\infty(J;\mathbb{R})$, $v \in L^\infty(0,T;\mathbb{R})$, in the frame of VAR-L2-cor. If the mild solution ψ vanishes a.e. on $(0,T)\times\omega$, $\omega \subset J$ open, then $\psi_0 = 0$.

Stated in the paper as no. 12, p. 68

Proof.

Step 1 Remove the time-dependent part

$\psi = e^{-i\int_0^t v}\tilde{\psi}$ where $\tilde{\psi}$ solves with V_0 only; the zero sets coincide, so one may take $V = V_0(x)$.

Step 2 Mollify in time

$H = H_0 + V_0$ is self-adjoint on $D(H_0)$ and $\psi(t) = e^{-itH}\psi_0$. For $\varphi \in C_c^\infty(-\eta,\eta)$ put $\psi_\varphi(t) = \int \varphi(s)\psi(t+s)\,ds = e^{-itH}\check{\varphi}(H)\psi_0$. Since $\check{\varphi}$ decays faster than any power, $\check{\varphi}(H)\psi_0 \in D(H^\infty) \subset D(H_0)$.

Step 3 Apply VAR-L2-cor

ψ_φ vanishes on $(\eta, T-\eta)\times\omega$ and V_0 does not depend on t, hence is C^1 in t; VAR-L2-cor on the shifted interval gives $\check{\varphi}(H)\psi_0 = 0$.

Step 4 Limit

With φ_n an approximate identity, $\check{\varphi}_n(H)\psi_0 \to \psi_0$ strongly, so $\psi_0 = 0$.

Why it works. a time-independent potential commutes with time mollification, which moves the datum into the domain where the classical unique-continuation theorem applies.

■

Uses: VAR-L2-cor, B121.

Part (b): $V(t,x)$ stays open

For genuinely time-dependent V, ψ_φ solves $i\partial_t\psi_\varphi = (H_0 + V(t))\psi_\varphi + R_\varphi$ with a source $R_\varphi = \int \varphi(s)[V(t+s) - V(t)]\psi(t+s)ds$. It vanishes on ω where $\psi = 0$, but the source lives outside the observation set, and VAR-L2 is a *homogeneous* unique-continuation statement. What would suffice is a Carleman estimate in H_t^{-1} or a mollification commuting with $V(t)$. The only regularity lemma proved is: for $h \in C_c^1(0,T)$ and mild ψ, $\int h(s)U(s,0)\psi_0 ds \in D(H_0)$. VAR-Q17(b) is reformulated, not reduced.

Dim. B135 **VAR-P12d-quad**

No quadratic differential logarithm for a switched oscillator

[proof written, not audited]

Let $\tau_1 = \tau_2 = \pi/(4\omega_1)$, $T = \tau_1 + \tau_2$, $H(t) = -\partial_x^2 + \omega(t)^2x^2$ on $L^2(\mathbb{R})$ with $\omega = \omega_1$ on $[0,\tau_1)$ and $\omega = \omega_2 = 5\omega_1$ on $[\tau_1, T)$. There is no self-adjoint $K' = \frac{1}{2}(ap^2 + b(xp+px) + cx^2) + dx + ep + f$ ($a,\dots,f \in \mathbb{R}$, $p = -i\partial_x$) with $e^{-iTK'} = U(T,0)$ up to a phase. The same holds for a C^∞ regularisation of ω close in L^1.

Stated in the paper as no. 12, p. 68

Proof.

Step 1 Monodromy

For quadratic Hamiltonians $U^*(x,p)U = M(x,p)$ with $M \in SL_2(\mathbb{R})$ solving $\dot{x} = 2p$, $\dot{p} = -2\omega^2 x$.

Step 2 Constant segments

A segment of length τ at constant ω contributes $\begin{pmatrix} \cos\theta & \sin\theta/\omega \\ -\omega\sin\theta & \cos\theta \end{pmatrix}$, $\theta = 2\omega\tau$. Here $\theta_1 = \pi/2$ and $\theta_2 = 5\pi/2$.

Step 3 Trace

$\operatorname{tr} M = -(\omega_1/\omega_2 + \omega_2/\omega_1) = -26/5 < -2$.

Step 4 An affine quadratic K' would give an exponential

For such K', $e^{iTK'}(x,p)e^{-iTK'} = e^{T\Xi}(x,p) + v$ with $\Xi \in \mathfrak{sl}_2(\mathbb{R})$, and equality of the propagators up to phase forces $v = 0$, $e^{T\Xi} = M$.

Step 5 Contradiction

$\exp(\mathfrak{sl}_2(\mathbb{R})) \subset \{\operatorname{tr} > -2\} \cup \{-I\}$: elliptic Ξ gives $\operatorname{tr} = 2\cos$, parabolic 2, hyperbolic $2\cosh$. The trace is continuous in $\omega \in L^1$, so the regularised problem fails too.

Why it works. a real symplectic matrix with trace below -2 is not an exponential in $\mathfrak{sl}_2(\mathbb{R})$.

■

Uses: none.

What it does not say

At period $2T$ one has $\operatorname{tr} M^2 = (26/5)^2 - 2 > 2$ and positive eigenvalues, so a quadratic logarithm exists for $U(2T, 0)$ (up to phase and metaplectic sign). The potential $\omega^2 x^2$ is unbounded, outside the L^∞ class. The register must *not* contain "(d) false": for bounded $V(t, x)$ and for non-quadratic logarithms (d) is open; the sub-question is whether a metaplectic operator with monodromy trace < -2 can be $e^{-iTK'}$ with K' non-quadratic differential.

Dim. B136 **VAR-C3Q2-R**

A rank criterion for the annihilator, and data that do not identify [proof written, not audited]

Let $A_2 = \mathbb{C}[t, x]\langle \partial_t, \partial_x \rangle$, $B = \mathbb{C}[t, x]\langle \partial_x \rangle$ (a left Ore domain with fraction field D_B), $L = \partial_t^m + \sum_{j<m} b_j \partial_t^j$ monic with $b_j \in B$, $\operatorname{rank}_B M = \dim_{D_B}(D_B \otimes_B M)$. (a) If $Lu = 0$, then $\operatorname{Ann}_{A_2}(u) = A_2 L \iff \operatorname{rank}_B(A_2 u) = m$. (b) $u(t, x) = f(x + t)$ with $f(y) = \exp(e^y)$ is not D-finite, and $(\partial_t - \partial_x)u = 0$, so u does not identify $\Box = \partial_t^2 - \partial_x^2$. (a′) If $A_2 u$ is holonomic no $L \neq 0$ is identified.

Stated in the paper as no. 12, p. 68

Proof.

Step 1 Division

$A_2 = A_2 L \oplus \bigoplus_{j<m} B\partial_t^j$ (S8-VAR §5(a)).

Step 2 (a) The map π

$\pi : B^m \to A_2 u$, $(\beta_j) \mapsto \sum \beta_j \partial_t^j u$, is onto since $A_2 L u = 0$. For $Q \in \operatorname{Ann}(u)$ write $Q = Q' L + \sum \beta_j \partial_t^j$ with $\sum \beta_j \partial_t^j u = 0$: $Q \in A_2 L \iff \beta = 0$. So $\operatorname{Ann}(u) = A_2 L \iff \ker \pi = 0$. If $\ker \pi \neq 0$ it is a non-zero submodule of the torsion-free B^m, of rank ≥ 1, hence $\operatorname{rank} A_2 u < m$; if $\ker \pi = 0$ the rank is m.

Step 3 (b) The example

u is not D-finite: otherwise $f = u(0, \cdot)$ would be D-finite in x, and entire solutions of linear ODEs with polynomial coefficients have finite order of growth, while $\exp(e^y)$ has infinite order. (The elementary proof via Touchard polynomials replaces the Wiman–Valiron citation.) And $(\partial_t - \partial_x)u = 0$ with $\partial_t - \partial_x \notin A_2 \Box$ by ∂_t-order, so $\operatorname{Ann}(u) \supsetneq A_2 \Box$: the criterion "not D-finite $\Rightarrow$ identifies", true for $m = 1$, fails for $m \geq 2$.

Step 4 (a′) Holonomic data

$A_2 / A_2 L$ has Gelfand–Kirillov dimension 3 while a holonomic module over A_2 has dimension 2 (Bernstein's inequality, Björk, cited), so $A_2 u \neq A_2 / A_2 L$.

Why it works. identifying L means that $A_2 u$ is as large as the free module on m generators over B; any extra relation lowers the rank.

■

Uses: S8-VAR §5.

Open

The non-monic case (VAR-Q13) stays open: without monicity the decomposition of Step 1 fails, and the equivalence "$\mathrm{Ann}(u) \subset A_2 \cap D_B L \iff \ker \pi = 0$" stated first is withdrawn. Part (a′) rests on Bernstein–Björk (cited), later replaced for the specific claim by the rank criterion.

Dim. B137 **SYS-DIRAC-POL**

Euler product for the Dirac 1+1 flow: formal yes, fixed time no [proof written, not audited]

Let $A(\xi) = X + mY$, $X = i\xi\sigma_z$, $Y = -i\sigma_x$, $\omega = \sqrt{\xi^2 + m^2}$, so $e^{tA} = e^{g_1 X} e^{g_2 m Y} e^{g_1 X}$ with $g_1 = a/\xi$, $g_2 = -b/m$, $b = -\arcsin(m \sin \omega t/\omega)$, $2a = \arctan(\xi \tan \omega t/\omega)$. (a) $g_1, g_2 \in \mathbb{Q}[\xi, m][[t]]$ and the identity holds in $\mathrm{Mat}_2(\mathbb{Q}[\xi, m][[t]])$. (b) For real $t \neq 0$, $m \neq 0$ there are no k and real polynomials $p_j, q_j \in \mathbb{R}[\xi]$ with $\prod_{j=1}^{k} e^{p_j X} e^{q_j m Y} = e^{tA(\xi)}$ for all $\xi \in \mathbb{R}$.

Stated in the paper as no. 11, p. 64

Proof.

Step 1 (a) Formal polynomiality

$\sin \omega t/\omega$ and $\tan \omega t/\omega$ are series in t with coefficients polynomial in $\omega^2 = \xi^2 + m^2$; $\arctan(\xi\,\cdot)$ and $\arcsin(m\,\cdot)$ are odd, so a/ξ and b/m are series with polynomial coefficients. The identity is analytic for small real t, hence holds between formal series. Each truncation is a product of flows of differential operators ($\xi \leftrightarrow -i\partial_x$).

Step 2 Lemma E

Let $R_i \in \mathbb{R}[\xi]$ with $R_i - R_j$ non-constant for $i \neq j$, $d_j \in \mathbb{C}$. If $S = \sum d_j e^{iR_j} \to 0$ as $\xi \to +\infty$, then all $d_j = 0$. Indeed $\frac{1}{X}\int_X^{2X} |S|^2 \to 0$ and expanding gives $\sum |d_j|^2 + \sum_{i \neq j} d_i \bar{d}_j \frac{1}{X}\int_X^{2X} e^{i(R_i - R_j)}$; for non-constant Q, Q' is eventually monotone and non-zero, and integrating by parts, $|\int_X^{2X} e^{iQ}| \leq 1/|Q'(X)| + 1/|Q'(2X)| + |1/Q'(2X) - 1/Q'(X)| = O(1)$, so the mixed terms tend to 0.

Step 3 (b) Expansion

$e^{pX} = \mathrm{diag}(e^{ip\xi}, e^{-ip\xi})$ and $e^{qmY} = \cos(qm)I - i\sin(qm)\sigma_x$ is a combination of $e^{\pm iqm}$. So $W_{11}(\xi) = \sum c_j e^{iP_j(\xi)}$ with $P_j \in \mathbb{R}[\xi]$ (group equal P_j up to constants).

Step 4 Asymptotics of the target

As $\xi \to +\infty$, $\omega = \xi + m^2/2\xi + O(\xi^{-3})$ and $(e^{tA})_{11} = \cos \omega t + i\xi \sin \omega t/\omega = e^{i\xi t} + \frac{im^2 t}{2\xi} e^{i\xi t} + O(\xi^{-2})$.

Step 5 Contradiction

If $W = e^{tA}$, $G = W_{11} - e^{i\xi t}$ has the form of Lemma E and tends to 0, so $G \equiv 0$; but $G = \frac{im^2 t}{2\xi} e^{i\xi t} + O(\xi^{-2}) \neq 0$ for large ξ.

Why it works. finite words in polynomial times produce only oscillating exponentials with polynomial phases, and the $1/\xi$ correction of the exact flow cannot be produced by them.

■

Uses: SYS-E3 (S8-SYS).

Scope

The ZXZ decomposition is a product of *local* flows only as a formal series in t. At fixed time the factors are non-local Fourier multipliers. Complex polynomial times are not covered: Lemma E does not treat e^{iP} with complex P; an ordering by growth would be needed. Open.

Dim. B138 **II-T2bp-CRIT**

The pole of a spectral projector, and why the Jordan form does not decide it [proof written, not audited]

Let $\chi_{A(\xi)} = Q \cdot B$ with Q, B coprime over $\mathbb{C}(\xi)$, $E(\xi) = \ker Q(A)$, $F(\xi) = \ker B(A)$ complementary near ξ_0, $P(\xi)$ the projector onto E along F. (a) P has a pole at ξ_0 iff $E_0 \cap F_0 \neq 0$, where $E_0 = \lim E$, $F_0 = \lim F$ in the Grassmannian; equivalently $\det[e(\xi_0)\ f(\xi_0)] = 0$ for saturated local bases. (b) The Jordan form of $A(\xi_0)$ does not decide: A_1, A_2 below have the same Jordan form at 0, P_1 has no pole and P_2 has one. (c) $A = \mathrm{diag}(\xi, -\xi)$ has constant projectors despite an eigenvalue crossing at 0.

Stated in the paper as no. 11, p. 64

Proof.

Step 1 Saturated bases

Columns of rational bases of E and F can be divided by powers of $(\xi - \xi_0)$ (Smith form over the DVR $\mathbb{C}[\xi]_{(\xi-\xi_0)}$), giving $e(\xi), f(\xi)$ analytic at ξ_0 with full-rank values; then $E_0 = \mathrm{im}\, e(\xi_0)$, $F_0 = \mathrm{im}\, f(\xi_0)$ and $P = [e\ 0][e\ f]^{-1}$.

Step 2 (a)

If $\det[e\ f](\xi_0) \neq 0$, P is analytic. If it vanishes, let $0 \neq v \in E_0 \cap F_0$. A bounded P would have a limit P_0 along a sequence with $P_0 e(\xi_0) = e(\xi_0)$ and $P_0 f(\xi_0) = 0$, so $P_0 v = v$ and $P_0 v = 0$: contradiction.

Step 3 (b) Counterexample

$$A_1 = \begin{pmatrix} \xi & 0 & 0 \\ 0 & -\xi & 1 \\ 0 & 0 & -2\xi \end{pmatrix}, \qquad A_2 = \begin{pmatrix} \xi & 1 & 0 \\ 0 & -\xi & 0 \\ 0 & 0 & -2\xi \end{pmatrix},$$

with $Q = \tau - \xi$, $B = (\tau + \xi)(\tau + 2\xi)$. Both have Jordan form $J_2 \oplus J_1$ at $\xi_0 = 0$. For A_1, $E = \langle e_1 \rangle$, $F = \langle e_2, e_3 \rangle$ for all ξ, so $P = \mathrm{diag}(1,0,0)$. For A_2, $P_{12} = 1/(2\xi)$ and $F \ni (1, -2\xi, 0)$, so $F_0 \ni e_1 \in E_0$.

Step 4 (c)

For $\mathrm{diag}(\xi, -\xi)$ the eigenvectors are e_1, e_2 for every ξ: the projectors are constant.

Why it works. the pole is a statement about the limit position of two subspaces, which depends on the first-order jet of A at ξ_0, not on $A(\xi_0)$ alone.

■

Uses: none.

Statements that are false

"Crossing of eigenvalues implies a pole" is false for a derogatory A (case (c)); and "the pole depends only on $A(\xi_0)$ and the factorisation" (Q-S8SYS-4) is false by (b). Both counterexamples are verified symbolically (idempotency, commutation).

Dim. B139 **SYS-MAXWELL**

An identifying Maxwell field exists on every connected Ω [proof written, not audited]

Let $R = \mathbb{C}[\tau, \xi_1, \xi_2, \xi_3]$, $P \in R^{8\times 6}$ the Maxwell matrix (rows $\tau E - \xi \times B$, $\tau B + \xi \times E$, $\xi \cdot E$, $\xi \cdot B$) and $M_P = R^{1\times 6}/R^{1\times 8}P$. Then $\mathrm{soc}_{\mathfrak{m}}(M_P) = 0$ for every maximal ideal $\mathfrak{m}$; by SYS-EX an identifying field (E, B) exists on every connected Ω. More generally, $\mathrm{pd}\, M_P \le n - 2$ implies $\mathrm{soc} = 0$.

Stated in the paper as no. 11, p. 64

Proof.

Step 1 Syzygies

$S_1 = (\xi, 0, -\tau, 0)$ and $S_2 = (0, \xi, 0, -\tau)$, in blocks $(3,3,1,1)$, satisfy $S_i P = 0$; e.g. $\xi \cdot (\tau E - \xi \times B) - \tau\, \xi \cdot E = 0$.

Step 2 Rank and kernel

A 6×6 minor is non-zero, so $\mathrm{rank}\, P = 6$ and the left kernel over $\mathrm{Frac}\, R$ is generated by S_1, S_2. If $sP = 0$ over R, then $s = f_1 S_1 + f_2 S_2$; the supports of S_1, S_2 are disjoint, so $f_1 \xi_i, f_1 \tau \in R$; writing $f_1 = a/b$ reduced, b divides $\xi_1, \xi_2, \xi_3, \tau$, hence is a unit. So the kernel is free of rank 2.

Step 3 Resolution

$0 \to R^2 \to R^8 \to R^6 \to M_P \to 0$ is exact: $\operatorname{pd} M_P \leq 2$.

Step 4 Socle

Let $k = R/\mathfrak{m}$ and $N = R^{1\times 8}P$. $R_{\mathfrak{m}}$ is regular of dimension 4, so $\operatorname{Ext}^i(k,R) = 0$ for $i < 4$. From $0 \to N \to R^6 \to M_P \to 0$: $\operatorname{Hom}(k, M_P) \hookrightarrow \operatorname{Ext}^1(k,N)$. From $0 \to R^2 \to R^8 \to N \to 0$: $\operatorname{Ext}^1(k,R^8) \to \operatorname{Ext}^1(k,N) \to \operatorname{Ext}^2(k,R^2)$, with both ends zero. Hence $\operatorname{Hom}(k,M_P) = 0$.

Why it works. a module of projective dimension at most $n-2$ over a regular ring of dimension n has no socle: depth is too high.

■

Uses: SYS-EX (S8-SYS), injectivity of $C^\infty(\mathbb{R}^n)$ (cited).

Dim. B140 **ENC-T2cp-P**

No finite jet certifies a principal annihilator [proof written, not audited]

Let $S = \mathbb{C}[\tau,\sigma]$, $\operatorname{Ann}_q(u) = \{P : P(\partial_t, L_q)u = 0\}$, $u \in C^\infty(\Omega)$ with $\operatorname{Ann}_q(u) = (P_*)$, $P_* \neq 0$. For every (t_1,x_1) and every η such that $J^\circ \setminus [x_1-\eta, x_1+\eta]$ has non-empty interior there is $u' \in C^\infty(\Omega)$ with $u' = u$ on $I \times (x_1-\eta, x_1+\eta)$ and $\operatorname{Ann}_q(u') = (P_*(\tau - \alpha)) \neq (P_*)$. So no finite jet, nor data on a strip around x_1, certifies $\operatorname{Ann}_q(u) = (P_*)$, even among principal annihilators.

Stated in the paper as no. 14, p. 74

Proof.

Step 1 Choice of α

$P_*(\alpha,\sigma) \not\equiv 0$ for all but finitely many α; take such an α, so that $(\tau-\alpha) \nmid P_*$.

Step 2 The perturbation

Take $0 \not\equiv f \in C_c^\infty(J^\circ \setminus [x_1-\eta, x_1+\eta])$, $h = e^{\alpha t} f$, $u' = u + h$. Then $P(\partial_t, L_q)h = e^{\alpha t} P(\alpha, L_q) f$. If $p(\sigma) = P(\alpha,\sigma) \neq 0$, then $p(L_q)f = 0$ is a linear ODE with constant non-zero leading coefficient and f has compact support, so $f \equiv 0$ by Cauchy uniqueness: contradiction. Hence $\operatorname{Ann}(h) = (\tau - \alpha)$.

Step 3 $\supseteq$

Obvious, the operators of S commute.

Step 4 $\subseteq$

Let $R \in \operatorname{Ann}(u')$, $g = Ru = -Rh$. Then $(\tau-\alpha)g = 0$, so $(\tau-\alpha)R \in (P_*)$, and since $\tau - \alpha$ is irreducible and does not divide P_*, $P_* \mid R$. Also $P_* g = RP_* u = 0$, so $P_* Rh = 0$ and $(\tau-\alpha) \mid R$. Coprimality in the factorial ring S gives $R \in (P_*(\tau-\alpha))$.

Why it works. a compactly supported exponential in time is invisible to a jet at x_1 but adds exactly one linear factor to the annihilator.

■

Uses: none.

Dim. B141 **ENC-Q4-S**

Non-vacuous separation: different constants are not equivalent after encoding [proof written, not audited]

Take $\mathcal{V} = \{q_0 + c : c \in \mathbb{R}\}$, $q_0 \in C^1(J)$ real, $J = [-a, a]$, and W_c the Volterra encoder of $q_0 + c$ normalised at $x_0 = 0$ ($c_V = 0$, $N_V = I$). Write $u \sim u'$ iff $u' = e^{i\theta(t)}u$. (a) Intersections are non-empty: if $u \in \mathcal{S}_{q_0}$ then $e^{-ict}u \in \mathcal{S}_{q_0+c}$ and $u \sim e^{-ict}u$. (S): for $u \sim u' \neq 0$ solutions with $V = q_0 + c \neq V' = q_0 + c'$, $[W_V u] \neq [W_{V'} u']$ (classes modulo phases).

Stated in the paper as no. 14, p. 74

Proof.

Step 1 A good time interval

Let $G = \{t : u(t, \cdot) \not\equiv 0\}$: it is open and non-empty. Fix a component G_0; outside G both u, u' vanish and give no information.

Step 2 The phase of u' is linear on G_0

Near $t_1 \in G_0$ choose x_1 with $u(t_1, x_1) \neq 0$; then $e^{i\theta_1} = u'/u$ at x_1 is C^1, and lifts to a C^1 θ_1 on the interval G_0. Subtracting the two equations on $G_0 \times J$: $0 = (-\theta_1' - \delta)e^{i\theta_1}u$ with $\delta = c' - c \neq 0$, so $\theta_1 = \kappa - \delta t$.

Step 3 Assume $W_{V'}u' = e^{i\theta}W_V u$

Put $w = W_V u$, $w' = W_{V'}u'$: free classical solutions since $WL_q = L_0 W$. As in Step 2, θ lifts to C^1 on G_0, and $0 = iw_t' + w_{xx}' = -\theta' e^{i\theta}w$, so $\theta \equiv \theta_0$ on G_0.

Step 4 An eigenvalue problem

$W_{V'}$ acts only in x: $e^{i\theta_0}w = W_{V'}u' = e^{i(\kappa-\delta t)}Zw$, $Z = W_{V'}W_V^{-1}$. So $Zw(t) = \mu(t)w(t)$, $\mu = e^{i(\theta_0 - \kappa + \delta t)}$.

Step 5 $Z = I + \mathcal{V}$ with $\mathcal{V}$ quasinilpotent

By ENC-1 $W_V = I + \int_{-x}^{x} H(x,s)\,ds$ with bounded kernel, and the composition of two such operators has the same form since $|r| \leq |s| \leq |x|$. Thus $\mathcal{V}f(x) = \int_{-|x|}^{|x|} k(x,r)f(r)dr$. With $G_n(\rho) = \max(|\mathcal{V}^n f(\rho)|, |\mathcal{V}^n f(-\rho)|)$, $G_{n+1} \leq 2\|k\| \int_0^\rho G_n$, so $\|\mathcal{V}^n\| \leq (2\|k\|a)^n/n!$ and $\sigma(\mathcal{V}) = \{0\}$.

Step 6 Conclusion

$Zw = \mu w$ with $w \neq 0$ gives $\mu - 1 \in \sigma(\mathcal{V}) = \{0\}$, so $\mu \equiv 1$ on G_0 and $\delta = 0$: contradiction.

Why it works. the encoder is the identity plus a Volterra operator, so it has only the eigenvalue 1; a shift in the potential would need a different eigenvalue.

■

Uses: ENC-1, ENC-N1′(v).

Literal DM3 reading

For $V = V(x)$ the intersection $\mathcal{S}_V \cap \mathcal{S}_{V'}$ is $\{0\}$ (VAR-T1(b)): (S) is vacuous. For $V(t,x)$ non-trivial intersections exist (B123: $u \in \mathcal{S}_0 \cap \mathcal{S}_{\mathbf{1}_Z}$) but $V' = \mathbf{1}_Z$ is not regular and time-independent encoders do not exist (ENC-P0(b)). ENC-Q4 in this reading stays open.

Dim. B142 **ENC-Q10-D**

A non-local explicit Darboux encoder for a time-dependent potential [proof written, not audited]

Let u be a free solution ($iu_t + u_{xx} = 0$) without zeros on $\Omega = I \times J$, $U = u_x/u$, $V_1 = -2\partial_x^2 \log u = -2U_x$, $\Lambda f = f_x - Uf = u\,\partial_x(f/u)$. (i) Λ maps free solutions to solutions of $L_0\chi = V_1\chi$, and $\ker\Lambda = \mathbb{C}u$ on each slice. (ii) For $\chi = \Lambda\varphi$ with φ free, put $\kappa(t) = (\chi_x + U\chi)(t,x_0)/u(t,x_0)$ and

$$\psi = u\Big[\int_{x_0}^{x} \frac{\chi}{u}\,dx' + c(t)\Big], \qquad c' = i\kappa.$$

Then ψ is free, $\Lambda\psi = \chi$, and ψ is unique modulo $\mathbb{C}u$. The encoder $\chi \mapsto \psi$ is non-local in x and in t.

Stated in the paper as no. 14, p. 74

Proof.

Step 1 (i) by hand

From $iu_t = -u_{xx}$ and $u_{xx}/u = U_x + U^2$: $iU_t = -U_{xx} - 2UU_x$. For $\chi = \varphi_x - U\varphi$ with $i\varphi_t = -\varphi_{xx}$: $i\chi_t = -\varphi_{xxx} - iU_t\varphi + U\varphi_{xx}$ and $\chi_{xx} = \varphi_{xxx} - U_{xx}\varphi - 2U_x\varphi_x - U\varphi_{xx}$. Summing, $L_0\chi = (U_{xx} + 2UU_x)\varphi - U_{xx}\varphi - 2U_x\varphi_x = -2U_x\chi = V_1\chi$. $\Lambda f = 0 \iff \partial_x(f/u) = 0$ and J is connected.

Step 2 The value of κ

With $W = u\varphi_x - u_x\varphi = u\chi$ and $\varphi_t = i\varphi_{xx}$, $u_t = iu_{xx}$: $\partial_t(\varphi/u) = iW_x/u^2 = i(u_x\chi + u\chi_x)/u^2 = i(\chi_x + U\chi)/u$, which depends only on χ and u.

Step 3 ψ is free exactly when $c' = i\kappa$

$\Lambda\psi = u\partial_x(\psi/u) = \chi$. Then $\psi - \varphi = a(t)u$, $a = c - (\varphi/u)(\cdot,x_0)$, and $L_0(au) = ia'u$; since $u \neq 0$, ψ is free iff $a' = 0$ iff $c' = \partial_t(\varphi/u)(\cdot,x_0) = i\kappa$.

Step 4 Uniqueness

Two free $\tilde\psi, \psi$ with the same Λ differ by $b(t)u$ with $ib'u = 0$, so b is constant.

Why it works. Darboux transformations relate free solutions to solutions of a Schrödinger equation with a time-dependent potential V_1; inverting Λ requires an integral and a time integration constant fixed by freeness.

■

Uses: none.

A real, regular, time-dependent case and what stays open

The Galilean boost $u_G = (x - 2kt + a)e^{i(kx-k^2t)}$ of the free solution $x + a$ gives $V_1 = 2/(x - 2kt + a)^2$: real, time-dependent and regular on $I \times J$ for $J \subset (2k \sup I - a, \infty)$. It is trivial (a symmetry of a stationary case), so what remains open is the existence of real time-dependent Darboux examples *not* reducible by symmetry to stationary ones, and whether history dependence is admitted in the class $\mathfrak{E}$. Also, $1/u$ does *not* solve the V_1-model when $u_{xx} \neq 0$ (residual $2u_{xx}/u^2$).

Dim. B143 **DICT-Q3**

The pointwise and formal tests do not decide in C^∞

[proof written, not audited]

Let the law be $A_u(y)\theta(y) = -L_0u(y)$ (RANGO). Necessary: (N1) $-L_0u(y) \in \operatorname{im} A_u(y)$ for all y; (N2) a formal solution at every point. (b) These do not suffice in C^∞: for Schrödinger with real V and $u(t,x) = e^{-1/x^2}$ $(x > 0)$, 0 $(x \leq 0)$, (N1) and (N2) hold and no continuous V exists. (a) For real-analytic u, (N2) at every point implies a C^∞ solution (Malgrange, cited).

Stated in the paper as no. 15, p. 77

Proof.

Step 1 (N1)

For $x \leq 0$: $u = u_{xx} = 0$, so $0 \in \operatorname{im} A_u$. For $x > 0$: $A_u = u \neq 0$ is bijective.

Step 2 (N2)

At $x = 0$ every series of u vanishes (u is flat), so $\theta = 0$ is a formal solution; elsewhere the case is trivial.

Step 3 No continuous V

On $x > 0$ the law is forced: $V = u_{xx}/u = 4/x^6 - 6/x^4 \to +\infty$ as $x \to 0^+$.

Step 4 (a), analytic case

By the Tougeron–Malgrange theorem on modules generated by analytic functions (Malgrange 1966, ch. VI, not read). For $K = q = 1$ the quotient $-L_0u/T_1u$ is meromorphic and formally holomorphic at each point, hence analytic.

Why it works. flatness hides the singularity of the law from every formal and pointwise test.

■

Uses: none.

Status

DICT-Q3(c) is reformulated, not reduced: a law exists iff the C^∞ division problem is solvable; for general C^∞ u there is no criterion. Open.

Dim. B144 **DICT-Q4-C**

Misfit certificates with non-local atoms [proof written, not audited] (the infinite-rank implication from Amitsur: [from cited source])

Let $D_0, \dots, D_N : C^\infty(\Omega) \to \mathcal{D}'(\Omega)$ be linearly independent and continuous. TFAE: (1) for every M, functionals $\ell_1..\ell_M$ and $u \in F_n$ there is $v \in U_n$ with $\ell_j(v) = \ell_j(u)$ (A(iii)); (2) for every closed finite-codimension subspace $Z = \bigcap \ker \ell_j$ there is $w \in Z$ with $D_0 w, \dots, D_N w$ independent. Moreover (2) holds if no non-trivial combination $\sum c_i D_i$ has finite rank, and (1) fails if one does.

Stated in the paper as no. 15, p. 77

Proof.

Step 1 (2)⇒(1)

The polynomial $s \mapsto \det(\ell_j'(D_i(u + sw)))$ has leading coefficient $\det(\ell_j'(D_i w)) \neq 0$; locality is used only to produce w.

Step 2 (1)⇒(2)

Take $u = 0 \in F_n$; a v with the same data lies in Z and in U_n.

Step 3 Infinite rank implies (2)

If every $w \in Z$ had dependent $D_i w$, the restrictions $D_i|_Z$ would be locally linearly dependent. By Amitsur's lemma (Trans. AMS 114, 1965; Brešar–Šemrl 1999, cited, not read) some $c \neq 0$ has $\operatorname{rank}(\sum c_i D_i|_Z) \leq N$, so $\operatorname{rank} \sum c_i D_i \leq N + \operatorname{codim} Z < \infty$.

Step 4 Finite rank implies failure

If $D = \sum c_i D_i \neq 0$ has finite rank r, $Z = \ker D$ is the intersection of r continuous functionals and the family is dependent on Z.

Step 5 Explicit example

On $\Omega = (0,1)$, $D_i f = \psi_i \int_0^1 f$ with $\psi_i = 1, x, x^2$: the single datum $\ell(u) = \int u = 0$ certifies the fit, since every v with $\ell(v) = 0$ has $D_i v = 0$.

Why it works. a certificate fails exactly when a combination of atoms collapses onto finitely many functionals.

■

Uses: S8-DICT §3.3.

Consequence

Non-zero differential operators have infinite rank, which recovers A(iii). The same holds for non-zero Fourier multipliers and Volterra convolutions $k*$ with $k \not\equiv 0$ (infinite rank by Titchmarsh), so for memory and symbols A(iii) holds conditionally on Amitsur's lemma. In the triage the infinite-rank implication was replaced by DICT-LOCDEP.

Dim. B145 DM-delta-minimax

The minimax rate $\varepsilon^{1/3}$

[proof written, not audited]

Let $\mathcal{U} = \mathcal{U}(M, \delta_0, K)$ be the $u \in C^3(\bar{K})$, $K = I \times J$, with $|u| \geq \delta_0$ and $\|\partial^\alpha u\|_{L^\infty} \leq M$ for $|\alpha| \leq 3$, $V_u = (iu_t + u_{xx})/u$, $\tilde{u}$ with $\|\tilde{u} - u\|_\infty \leq \varepsilon$. Assume $M > \delta_0$. The minimax risk satisfies $c\varepsilon^{1/3} \leq R(\varepsilon) \leq C\varepsilon^{1/3}$ on $K' \Subset K$ for $\varepsilon \leq \varepsilon_0$.

Stated in the paper as no. 15, p. 77

Proof.

Step 1 Upper bound

Let $\hat{V}(\tilde{u}) = V_{u^*}$ for any $u^* \in \mathcal{U}$ with $\|u^* - \tilde{u}\| \leq \varepsilon$ (u itself qualifies). Then $e = u - u^*$ has $\|e\| \leq 2\varepsilon$ and $\|e_{tt}\|, \|e_{xxx}\| \leq 2M$, and VAR-P3(c) gives $\|V_u - V_{u^*}\| \leq C\delta_0^{-1}(1 + \|V_{u^*}\|)\big((2\varepsilon)^{1/3}(2M)^{2/3} + (2\varepsilon)^{1/2}(2M)^{1/2} + 2\varepsilon\big)$ with $\|V_{u^*}\| \leq 2M/\delta_0$; for small ε the $\varepsilon^{1/3}$ term dominates.

Step 2 Lower bound, two points

Take $u_1 \equiv (M + \delta_0)/2$ (this is where $M > \delta_0$ enters) and x_0 interior to K'. Put $h = \varepsilon\varphi((x - x_0)/\varepsilon^{1/3})$ with $\varphi \in C_c^\infty$, $\varphi(0) = 0$, $\varphi''(0) \neq 0$, $\|\varphi\|_\infty \leq 1$. Then $\|h\|_\infty \leq \varepsilon$, $\|\partial_x^3 h\| = \|\varphi'''\| = O(1)$, and after rescaling φ, $u_2 = u_1 + h \in \mathcal{U}$.

Step 3 The gap

At (t, x_0), $h = 0$, so $V_{u_2} - V_{u_1} = h_{xx}/u_1 = \varepsilon^{1/3}\varphi''(0)/u_1$. The datum $\tilde{u} = u_1 + h/2$ is ε-close to both; every estimator errs by at least $\frac{1}{2}|V_{u_2} - V_{u_1}| \geq c\varepsilon^{1/3}$ on one of them.

Why it works. a bump of height ε and width $\varepsilon^{1/3}$ is invisible in the data yet changes the second derivative by $\varepsilon^{1/3}$.

■

Uses: VAR-P3(c), A39.

Scope

This covers the declared model only (complex V, no zeros, deterministic noise). It does not touch real V, the case with zeros, or stochastic noise.

Dim. B146 **S10B-C1**

Density fails around a hole: the rectangle functional [proof written, not audited]

Let $\Omega = (0,3)^2 \setminus [1,2]^2$ (bounded, connected, Lipschitz, non-convex), $P = \partial_x\partial_y$, $Q_1 = \partial_x$, $Q_2 = \partial_y$. For every $m \geq 0$, $\ker_{H^m(\Omega)} Q_1 + \ker_{H^m(\Omega)} Q_2$ is not dense in $\ker_{H^m(\Omega)} P$, not even in L^2. So the convexity in OPF5B-T1 cannot be dropped.

Stated in the paper as no. 16, p. 80

Proof.

Step 1 The datum

Take $\varphi \in C^\infty(\mathbb{R})$, $\varphi = 0$ on $(-\infty, 1]$, $\varphi = 1$ on $[2,\infty)$, and $u(x,y) = \varphi(x)\varphi(y)$. Then $u_{xy} = \varphi'(x)\varphi'(y)$ is supported in $[1,2]^2$, disjoint from Ω: $Pu = 0$ on Ω and $u \in H^m(\Omega)$ for all m.

Step 2 The functional

With $I_1 = (0.4, 0.6)$, $I_2 = (2.4, 2.6)$ and $F \in L^2(\Omega)$,

$$\Lambda(F) = 25^2\Big[\int_{I_2\times I_2} F - \int_{I_1\times I_2} F - \int_{I_2\times I_1} F + \int_{I_1\times I_1} F\Big].$$

The four squares lie in Ω and Λ is continuous on L^2.

Step 3 Λ kills both kernels

The strips $(0,3)\times I_j$ are connected and lie in Ω. If $\partial_x F = 0$ there, $F(x,y) = g_j(y)$ on each strip, so $\int_{I_2\times I_j} F = \int_{I_1\times I_j} F$ and $\Lambda(F) = 0$. Symmetrically for $\partial_y F = 0$.

Step 4 It does not kill u

$u = 1$ on $I_2 \times I_2$ and 0 on the other three squares, so $\Lambda(u) = 1$. Since Λ vanishes on the L^2-closure of $\ker Q_1 + \ker Q_2$, u is not in it.

Why it works. Λ is the monodromy of the form $u_x dx + u_y dy$ around the hole, and it is zero on every sum of a function of x and a function of y.

■

Uses: OPF5B-T1.

Sharper statement

The counterexample works as soon as there is a coordinate rectangle with its four edge-strips in Ω and a closed coordinate rectangle with non-empty interior in $\mathbb{R}^2 \setminus \Omega$ inside it. For a *point* hole $\Omega = (0,3)^2 \setminus \{p\}$ the claim holds only for $m = 0$ ($u = H(x-p_1)H(y-p_2)$, $u_{xy} = \delta_p$); for $m \geq 1$, $\delta_p \notin H^{-1}(\mathbb{R}^2)$ forces $u_{xy} = 0$ on the square and density holds. For which non-convex Ω density holds is open.

VAR-P11(e) Autonomous proof by Volterra and Gronwall. [proof written, not audited]

B121. VAR-L2-cor then [proof written, not audited] (it was from a cited source).

VAR-Q18-CE	Rank-2 counterexample, also inside $L^2(J)$ with Neumann H_0; "P11(e) with B merely continuous" is false.	[false]

B122. VAR-Q18 (multiplications) [open].

VAR-Q10-EX, VAR-T1(c)-Q10	A solution with $V = 0$, $\lvert Z\rvert > 0$ and $\operatorname{int} Z = \emptyset$ (Tychonoff plus a Cantor-type set).	[proof written, not audited]

B123; closes the caution of A38.

ENC-Q4 (reading DM3), ENC-Q10 (real residue)	Open; for ENC-Q10, the Galilean boost already gives $V_1 = 2/(x - 2kt + a)$.	[open]

F10-C.

DICT-Q4-C, DICT-U3d-v, DM-δ-minimax	See B144, B145 and Chapter 17.	[proof written, not audited]

F10-C.

P1-Q6 (Phase 10 form)	Proved with constants in A52 (S10A-T1, C1); the literal statement is false (reaction–diffusion $u_t = Du_{xx} + ru$ has E^{add} continuous from H^1).	[proof written, not audited]

F10-A; see also A52.

C3-Q1	Proved as a classification by data model: composition of P4-T5+P4-L7, OP-ALG-T3, OPR2-C2, OP-ALG-T1 mod (AC), CONV-T2; no new mathematics.	[proof written, not audited](classification)

F10-A.

18.6 Integration, examples and formalisation

EX10-a, b, c, d, e	Textbook examples checked against the theory: free Dirac $1+1$ (13/13), harmonic oscillator (19/19), Pöschl–Teller (18/18), fractional heat (13/13), Maxwell in vacuum (17/17). Total 80/80 PASS; three new arguments proved.	checked on examples

S10-INT; audit EX10 (ex10_b_nd 9/9, ex10_e_k 13/13, ex10_e_h 10/10).

F10-LEAN	Formalisation in Lean 4 + Mathlib: 37 entries, of which 27 faithful, 7 weaker and 3 different from the ledger statement, none vacuous; all 85 `#print axioms` use only the standard axioms.	verified

S10-LEAN; independent review S-CHECK-LEAN. Mathlib has no Weyl algebra and no C_0-semigroups.

F10-G9	Points delegated by Gate 9: ALG-TK (Trotter–Kato convergence only for relatively bounded V), and the renaming of the lower bound in VAR-P3 as κ (not δ).	[from cited source]; renaming

GATE9.

CHAPTER 19

Phases 6–8: variable coefficients, the remaining proofs

In this chapter	
Starting from	the notation of Chapter 6; $K = \mathbb{C}(x)$ with $\partial = d/dx$ and $\mathcal{D} = K[\partial]$ (a left and right Euclidean ring) for the ODE part; $A_2 = \mathbb{C}[t,x]\langle\partial_t,\partial_x\rangle$ (the Weyl algebra) for the last section.
Why	to give the proofs of the results of the variable-coefficient package that were only listed in Chapter 15: identification for linear ODEs, dynamical Lie algebras and the Pöschl–Teller propagator, non-autonomous splitting and Dyson–Phillips, Floquet, and identification in the Weyl algebra.
Role	Dim. B80–B87. They prove the corresponding statements of Chapter 12 of the paper.

19.1 Linear ODEs over $\mathbb{C}(x)$

Dim. B80 — **VAR-T2a**

Elementary identification for ODEs with rational coefficients [proved]

Let $u \neq 0$ lie in a differential extension of K with constant field $\mathbb{C}$, and $\mathrm{Ann}(u) = \{Q \in \mathcal{D} : Qu = 0\}$.

(i) $\mathrm{Ann}(u) \neq 0$ iff u is D-finite over K; then $\mathrm{Ann}(u) = \mathcal{D}L_u$ with L_u monic of minimal order (unique).

(ii) If $Lu = 0$, u identifies L ($\mathrm{Ann}(u) = \mathcal{D}L$) iff $\operatorname{ord} L_u = \operatorname{ord} L$ iff $u, u', \dots, u^{(n-1)}$ are linearly independent over K ($n = \operatorname{ord} L$).

(iii) If L is irreducible in $\mathcal{D}$, every solution $u \neq 0$ identifies L.

(iv) If u is hyperexponential ($u'/u \in K$), $L_u = \partial - u'/u$ has order 1: for $\operatorname{ord} L \geq 2$ no hyperexponential u identifies L.

(v) For $\operatorname{ord} L = 2$: u identifies L iff $u'/u \notin K$.

Proof.

Step 1 (i) A principal left ideal

$\mathcal{D}$ is right Euclidean ($Q = Q'L + R$ with $\operatorname{ord} R < \operatorname{ord} L$), so every left ideal is principal. $\operatorname{Ann}(u) \neq 0$ iff u satisfies an ODE with coefficients in K, i.e. u is D-finite.

Step 2 (ii)–(iii)

$L \in \operatorname{Ann}(u) = \mathcal{D}L_u$ gives $L = QL_u$, so $\operatorname{ord} L_u \leq \operatorname{ord} L$ with equality iff $Q \in K^\times$ iff $\mathcal{D}L = \mathcal{D}L_u$; and $\operatorname{ord} L_u$ is the least m with $u^{(m)} \in \operatorname{span}_K(u, \dots, u^{(m-1)})$. If $L = QL_u$ with $\operatorname{ord} L_u < \operatorname{ord} L$ (and $\operatorname{ord} L_u \geq 1$ since $u \neq 0$), L is reducible: this is (iii).

Step 3 (iv)–(v)

(iv) is direct. (v) By (ii) with $n = 2$: u identifies iff u, u' are independent over K, i.e. $u'/u \notin K$.

Why it works. the annihilator is the left ideal of the minimal ODE, and a solution identifies a law exactly when it satisfies no equation of lower order.

■

Uses: none.

> **Example 19.1** ∂^2 **is not identified**
>
> For $L = \partial^2$ the solutions are $a + bx$, and $(a + bx)\partial - b$ kills $a + bx$ for each such u: the minimal operator has order 1, so no solution identifies ∂^2 (case (iv)).

Dim. B81 **VAR-E1**

Airy and the oscillator: when every solution identifies [proved]

(E1) $L = \partial^2 - x$ is irreducible in $\mathbb{C}(x)[\partial]$, so every non-zero solution identifies it. (E2) For $L_E = -\partial^2 + x^2 - E$, $E \in \mathbb{C}$: L_E is reducible iff $E \in \pm(2\mathbb{N} + 1)$; for $E = 2n + 1$ a solution u identifies L_E iff $u \notin \mathbb{C}h_n$ (the Hermite function).

Proof.

Step 1 (E1) A Riccati equation without rational solution

A right factor of order 1 is $\partial - w$ with $w \in \mathbb{C}(x)$ and $w' + w^2 = x$ (Riccati). Since $w' + w^2 = x$ has no poles at finite distance, at a pole a of order $r \geq 1$ the dominant terms $-rc(x - a)^{-r-1}$ and $c^2(x - a)^{-2r}$ must cancel, which happens only for $r = 1$, $c = 1$; the exclusion of this case and of polynomial w is carried out in the package §2.3 (*condensed*). The first draft's argument "at the poles $w' + w^2$ has a pole, so w is a polynomial" was false ($w = 1/x$ gives $w' + w^2 = 0$) and was replaced. So L is irreducible; apply B80(iii).

Step 2 (E2) Rational solutions of the Riccati equation

A right factor $\partial - w$ needs $w' + w^2 = x^2 - E$. Since the right side has no poles, $w = \varepsilon x + \sum_j 1/(x - a_j)$ with $\varepsilon = \pm 1$ (simple poles of residue 1; at infinity $w \sim \varepsilon x$), so $u = p(x)e^{\varepsilon x^2/2}$ with p a polynomial. For $\varepsilon = -1$: $p'' - 2xp' + (E - 1)p = 0$, which has a non-zero polynomial solution iff $E - 1 = 2n$ (Hermite equation, $p = H_n$ unique up to a scalar); the case $\varepsilon = +1$ gives $E = -(2n + 1)$ in the same way.

Step 3 Which solutions identify

For $E \in \pm(2\mathbb{N}+1)$ the only right factor of order 1 is the one with solution h_n; a solution u identifies L_E iff it lies in no proper G-stable subspace, i.e. iff $u \notin \mathbb{C}h_n$ (B82 with V of dimension 2).

Why it works. identification fails exactly when u lies in the solution space of a proper right factor; for Airy there is none, and for the oscillator there is one for the Hermite energies.

■

Uses: B80.

19.2 Dynamical Lie algebras and the Pöschl–Teller propagator

Dim. B82 **VAR-P8(a,b)**

The Lie algebra generated by ∂^2 and V is finite-dimensional only for quadratic V [proof written, not audited]

Let $V \in C^\infty(J)$ and $\mathfrak{g}_V$ the complex Lie algebra generated by ∂^2 and V (multiplication) inside the differential operators with C^∞ coefficients. Then $\dim \mathfrak{g}_V < \infty$ iff V is a polynomial of degree ≤ 2. For quadratic V, $\mathfrak{g}_V \subset \operatorname{span}\{\partial^2, x^2, x\partial + \frac{1}{2}, \partial, x, 1\}$ (for $V = x^2$, $\mathfrak{g}_V = \operatorname{span}\{\partial^2, x^2, x\partial + \frac{1}{2}\} \cong \mathfrak{sl}_2$; for affine non-constant V the algebra is solvable, the Stark case). Consequently the Wei–Norman method applies to $i\partial_t = -\partial^2 + V$ iff $V \in \operatorname{span}\{1, x, x^2\}$.

Proof.

Step 1 (⇐) Direct closure

$[\partial^2, x^2] = 4x\partial + 2$, $[\partial^2, x] = 2\partial$, $[x\partial, \partial^2] = -2\partial^2$, $[x\partial, x^2] = 2x^2$, $[\partial, x] = 1$, $[\partial, x^2] = 2x$, $[x, \partial^2] = -2\partial$: the span of $\{\partial^2, x^2, x\partial + \frac{1}{2}, \partial, x, 1\}$ is closed.

Step 2 (⇒) Growth of the order

In finite dimension the order of the elements is bounded by some m. Put $c_0 = V$, $c_{j+1} = [\partial^2, c_j]$. For a function c, $\operatorname{ad}^j_{\partial^2}(c)$ has order j with leading coefficient $2^j c^{(j)}$ (induction on $[\partial^2, a\partial^k] = 2a'\partial^{k+1} + \cdots$). So the order $m+1$ would be reached unless $V^{(m+1)} \equiv 0$: V is a polynomial of degree $\le m$; then repeated brackets with V and ∂^2 raise the degree of the coefficients unless $\deg V \le 2$ (growth of degrees $4, 5, 6, 7$ checked for $V = x^3$; $V = e^x$ generates all e^{kx}).

Step 3 Wei–Norman

The method needs a finite-dimensional subalgebra containing ∂^2 and V, hence containing $\mathfrak{g}_V$; by Step 2 it exists iff V is quadratic. The converse is the classical Wei–Norman method for solvable algebras and $\mathfrak{sl}_2 \ltimes \mathfrak{heis}$ ([from cited source]; valid locally in t).

Why it works. every bracket with ∂^2 differentiates the coefficient and raises the order by one; a finite-dimensional algebra stops this only for a polynomial of degree at most two.

■

Uses: none.

Dim. B83 **VAR-P8(c,d)**

An explicit propagator for the modified Pöschl–Teller potential [proof written, not audited]

Let $H_\ell = -\partial^2 - \ell(\ell+1)\operatorname{sech}^2 x$ on $L^2(\mathbb{R})$, $U_\ell(t) = e^{-itH_\ell}$, $A_j = \partial + j\tanh x$, $A_j^\dagger = -\partial + j\tanh x$.

(i) $H_j = A_j^\dagger A_j - j^2$ and $A_j A_j^\dagger - j^2 = H_{j-1}$; $\ker A_j = \mathbb{C}\operatorname{sech}^j$ in L^2; $\ker A_j^\dagger \cap L^2 = 0$.

(ii) For $j \geq 1$ and $t \in \mathbb{R}$: $U_j(t) = A_j^\dagger U_{j-1}(t)(H_{j-1}+j^2)^{-1}A_j + e^{ij^2t}P_j$, with P_j the orthogonal projector on $\mathbb{C}\operatorname{sech}^j$. In particular $U_1(t) = A_1^\dagger e^{it\partial^2}(1-\partial^2)^{-1}A_1 + e^{it}\langle \operatorname{sech}, \cdot\rangle \operatorname{sech}/2$.

(iii) The Jost functions of H_1 are $A_1^\dagger e^{ikx} = (\tanh x - ik)e^{ikx}$ (eigenvalue k^2), and -1 is the only eigenvalue (bound state sech).

Moreover (VAR-P8(d)) with $A_1 = -i(-\partial^2)$, $A_2 = -iV$, $V = -2\operatorname{sech}^2$: $E^{\mathrm{add}}(t) = U_1(t) - e^{it\partial^2} - e^{2it\operatorname{sech}^2} + I$ in closed form with free ingredients, and for $f \in H^6$, $E^{\mathrm{add}}(t)f = \frac{t^2}{2}(A_1A_2 + A_2A_1)f + O(t^3)$ (checked numerically; not proved as an operator identity).

Proof.

Step 1 (i) A computation

$A_j^\dagger A_j = -\partial^2 - j(\tanh)' + j^2\tanh^2 = -\partial^2 - j\operatorname{sech}^2 + j^2(1-\operatorname{sech}^2) = H_j + j^2$, and $A_jA_j^\dagger = -\partial^2 + j\operatorname{sech}^2 + j^2 - j^2\operatorname{sech}^2 = H_{j-1} + j^2$. Also $A_j\operatorname{sech}^j = -j\operatorname{sech}^j\tanh + j\tanh\operatorname{sech}^j = 0$, while $\ker A_j^\dagger = \mathbb{C}\cosh^j \notin L^2$.

Step 2 (ii) Intertwining and the bound state

Since $H_{j-1} \geq -(j-1)^2$ as quadratic forms, $A_jA_j^\dagger = H_{j-1} + j^2 \geq 2j-1 > 0$: $(H_{j-1}+j^2)^{-1}$ is bounded with norm $\leq 1/(2j-1)$, $\operatorname{Ran} A_j^\dagger$ is closed, and $Q = A_j^\dagger(A_jA_j^\dagger)^{-1}A_j$ is the orthogonal projector onto it. From (i), $H_jA_j^\dagger = A_j^\dagger(A_jA_j^\dagger - j^2) = A_j^\dagger H_{j-1}$, so $A_j^\dagger$ intertwines H_{j-1} with H_j on $\operatorname{Ran} A_j^\dagger$, giving the first term (*condensed*: the complete argument with domains is in the package §3.3); on $\ker A_j = \mathbb{C}\operatorname{sech}^j$, $H_j = -j^2$, giving $e^{ij^2t}P_j$. Iterating, U_ℓ is expressed with the free propagator, free resolvents and multiplications only (Darboux chain; Crum 1955, [from cited source]).

Step 3 (iii)

Direct computation; by (ii), the spectrum of H_1 is that of H_0 (absolutely continuous $[0,\infty)$) together with $\{-1\}$. Numerically ($N = 2048$, periodic box of side 120) the formula for U_1 matches the dense diagonalisation of e^{-itH_1} to $< 10^{-8}$ at $t = 0.3, 1, 2.5$, and the iterated formula for $\ell = 2$ to $< 10^{-7}$.

Why it works. A_j and $A_j^\dagger$ intertwine H_j and H_{j-1} except for one bound state; each Darboux step removes a bound state and a sech^2 term, so the propagator reduces to the free one.

■

Uses: B82.

19.3 Non-autonomous evolution

Dim. B84 VAR-P9, VAR-P11

Non-autonomous interaction at second order, and Dyson–Phillips [proved]

Let $A_1, A_2 \in C([0,T];\mathcal{L}(X))$, U_A the solution of $U' = AU$, $U(0) = I$, $E^{\mathrm{mult}} = U_{A_1+A_2} - U_{A_1}U_{A_2}$, $a(t) = \int_0^t (\|A_1\| + \|A_2\|)$.

(a) $E^{\mathrm{mult}}(t) = \iint_{t>s_1>s_2>0} [A_2(s_1), A_1(s_2)]\,\mathrm{d}s_2\,\mathrm{d}s_1 + R(t)$, $\|R(t)\| \le 2\sum_{n\ge 3} a(t)^n/n!$.

(b) $E^{\mathrm{add}}(t) = \frac{t^2}{2}(A_1A_2 + A_2A_1)(0) + o(t^2)$.

(c) If $[A_1(s), A_2(r)] = 0$ for all s, r then $E^{\mathrm{mult}} \equiv 0$; commuting at equal times is *not* enough: for non-commuting X, Y and bumps f, g with disjoint supports in $[0, \frac12)$, $(\frac12, 1]$ of integral $\frac12$, $A_1 = fX$, $A_2 = gY$ have $[A_1(s), A_2(s)] = 0$ but $E^{\mathrm{mult}}(1) = e^{Y/2}e^{X/2} - e^{X/2}e^{Y/2} \neq 0$.

(Dyson–Phillips) Let A_0 generate a C_0-semigroup $T(t)$, $\|T(t)\| \le Me^{\omega t}$, and B piecewise strongly continuous with $b = \sup\|B\|$. (a′) $U = \sum U_n$, $U_0 = T(t-s)$, $U_{n+1}(t,s)x = \int_s^t T(t-r)B(r)U_n(r,s)x\,\mathrm{d}r$ converges in norm with $\|U_n\| \le Me^{\omega(t-s)}(Mb(t-s))^n/n!$ and solves the mild equation; (b′) the strongly continuous mild solution is unique; (c′) $U(t,r)U(r,s) = U(t,s)$.

Proof.

Step 1 (a)–(b) Compare the Dyson series

$U_A(t) = \sum_n \int_{\Delta_n} A(s_1)\cdots A(s_n)$, $\Delta_n = \{t > s_1 > \cdots > s_n > 0\}$, the n-th term bounded by $(\int\|A\|)^n/n!$. For $A = A_1 + A_2$ the mixed term of degree 2 is $\iint_{s_1>s_2}(A_1(s_1)A_2(s_2) + A_2(s_1)A_1(s_2))$; in $U_{A_1}U_{A_2}$ it is $(\int A_1)(\int A_2) = \iint_{s_1>s_2}(A_1(s_1)A_2(s_2) + A_1(s_2)A_2(s_1))$. The terms of degree $0, 1$ and the pure degree-2 terms agree, and the difference of the mixed terms is $\iint_{s_1>s_2}(A_2(s_1)A_1(s_2) - A_1(s_2)A_2(s_1))$, the stated double integral. The remainder is the sum of the terms of degree ≥ 3 of two series, each bounded by $a^n/n!$. (b) is the mixed degree-2 term of $U_{A_1+A_2} - U_{A_1} - U_{A_2} + I$.

Step 2 (c) The counterexample

With disjoint supports $A_1(s)A_2(s) = 0 = A_2(s)A_1(s)$, so $[A_1(s), A_2(s)] = 0$, yet $U_{A_1}(1) = e^{X/2}$, $U_{A_2}(1) = e^{Y/2}$ and $U_{A_1+A_2}(1) = e^{Y/2}e^{X/2}$ (first A_1 acts, then A_2), so $E^{\mathrm{mult}}(1) = e^{Y/2}e^{X/2} - e^{X/2}e^{Y/2} \neq 0$.

Step 3 Dyson–Phillips

(a′) By induction, $\|U_{n+1}(t,s)\| \le \int_s^t Me^{\omega(t-r)}b\,Me^{\omega(r-s)}(Mb(r-s))^n/n!\,\mathrm{d}r = Me^{\omega(t-s)}(Mb)^{n+1}(t-s)^{n+1}/(n+1)!$; strong continuity passes by induction, and summing the recursion gives the

mild equation. (b′) For two solutions $w(t) = (U - \tilde{U})(t,s)x$ satisfies $\|w(t)\| \le Mbe^{|\omega|T}\int_s^t \|w\|$, so $w = 0$ by Gronwall. (c′) is the splitting of the integral at r and uniqueness.

Why it works. the commutator of the two flows appears first at second order, with its time arguments in the *order* of the time integration; a simultaneous commutator cannot see it.

■

Uses: none.

Dim. B85 **VAR-T10, VAR-P12**

Explicit non-autonomous evolutions; Floquet logarithms [proof written, not audited]

(VAR-T10) (1) For the Stark potential $V = F(t)x$, $F \in C(\mathbb{R};\mathbb{R})$: with $p' = F$, $b' = -2p$, $\theta' = -p^2$, $p(0) = b(0) = \theta(0) = 0$, the solution is $\psi(t,x) = e^{i\theta(t)}e^{-ip(t)x}\varphi(t, x - b(t))$ with $\varphi = e^{it\partial^2}\psi_0$; hence $U(t) = e^{i\theta}e^{-ipx}T_b e^{it\partial^2}$ for every t (class A, global). (3) For $H = -\partial^2 + x^2$: $U(t) = c(t)e^{i\gamma x^2}e^{is\partial^2}e^{i\gamma x^2}$ with $\gamma = (\cos 2t - 1)/(2\sin 2t) = -\frac{1}{2}\tan t$, $s = \frac{1}{2}\sin 2t$; the chirp–free–chirp word exists iff $M_{12}(t) \neq 0$ or $M(t)$ is lower unipotent, i.e. iff $\cos t \neq 0$ (and at $t \in \pi\mathbb{Z}$, $M = I$); it breaks exactly at odd multiples of $\pi/2$ (where $U = -i\times$parity).

(VAR-P12) Let $H(t)$ be T-periodic with evolution U. (a) There is a bounded self-adjoint K with $\|K\| \le \pi/T$ and $U(T,0) = e^{-iTK}$, and $P(t) = U(t,0)e^{itK}$ is T-periodic. (b) A differential operator with C^∞ coefficients that is bounded on $L^2(J)$ has order 0. (c) For $V = v(t)$ on $J = \mathbb{R}$: $U(T,0) = e^{-iT\bar{v}}e^{iT\partial^2}$, the bounded K of (a) is the Fourier multiplier with symbol $\bar{v} + k^2$ reduced mod $2\pi/T$, which is not differential; the unbounded differential logarithm $K' = -\partial^2 + \bar{v}$ exists. (d) For generic $V(t,x)$ the existence of a differential logarithm is open (general case).

Proof.

Step 1 (T10.1) Direct substitution

Insert ψ in $i\psi_t + \psi_{xx} - Fx\psi = 0$ and use the free equation for φ; the terms in p, b, θ cancel by the three ODEs.

Step 2 (T10.3) Metaplectic factorisation

The Heisenberg equations $x' = 2p$, $p' = -2x$ give the symplectic matrix $M(t) = \begin{pmatrix} \cos 2t & \sin 2t \\ -\sin 2t & \cos 2t \end{pmatrix}$. A matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2$ factors as lower·upper·lower unipotent iff $b \neq 0$ or it is lower unipotent; here $b = \sin 2t$ vanishes iff $t \in \frac{\pi}{2}\mathbb{Z}$; at $t \in \pi\mathbb{Z}$, $M = I$ (the trivial word exists), while at odd multiples of $\frac{\pi}{2}$, $M = -I$ is not unipotent and the word breaks. Both sides of the stated identity are unitaries implementing the same symplectic transformation, so by Stone–von Neumann and Schur's lemma they differ by a scalar ([from cited source], Folland).

Step 3 (P12 a) Spectral theorem

$U(T,0) = \int_{(-\pi,\pi]} e^{-i\alpha}\,\mathrm{d}E(\alpha)$; put $K = T^{-1}\int \alpha\,\mathrm{d}E$, so $\|K\| \le \pi/T$ and $e^{-iTK} = U(T,0)$. By periodicity $U(t+T,0) = U(t,0)U(T,0)$, hence $P(t+T) = U(t,0)e^{-iTK}e^{i(t+T)K} = P(t)$.

Step 4 (P12 b, c)

(b) If $a_m \not\equiv 0$ with $m \ge 1$, on a compact interval where $|a_m| \ge c > 0$ take $f_\xi = \chi e^{i\xi x}$: $\|\sum a_k \partial^k f_\xi\| \ge c|\xi|^m\|\chi\| - O(|\xi|^{m-1})$, unbounded. (c) For $V = v(t)$ the evolution is $e^{i\partial^2 t}$ times a scalar phase, so $U(T,0) = e^{-iT\bar v}e^{iT\partial^2}$; in Fourier variables it is multiplication by $e^{-iT(\bar v + k^2)}$, and the principal branch of the logarithm reduces $\bar v + k^2$ modulo $2\pi/T$, a non-polynomial symbol; $K' = -\partial^2 + \bar v$ is a differential logarithm.

Why it works. a bounded logarithm always exists but is non-local; a differential logarithm is unbounded and needs the monodromy to be a function of the free Hamiltonian.

■

Uses: B82.

19.4 Identification in the Weyl algebra

Dim. B86 **VAR-C3Q2**

Identification when the coefficients are polynomials in t, x [proved]

Let $A_2 = \mathbb{C}[t,x]\langle\partial_t,\partial_x\rangle$, $B = \mathbb{C}[t,x]\langle\partial_x\rangle$, $L = \partial_t^m + \sum_{j<m} M_j\partial_t^j$ with $M_j \in B$, u real-analytic with $Lu = 0$.

(a) (Division) every $Q \in A_2$ is uniquely $Q = Q'L + \sum_{j<m} R_j\partial_t^j$ with $R_j \in B$.

(b) $\mathrm{Ann}(u) = A_2L$ iff $u, \partial_t u, \dots, \partial_t^{m-1}u$ are B-linearly independent.

(c) For $m = 1$: $\mathrm{Ann}(u) = A_2L$ iff u is not D-finite over $\mathbb{C}(t,x)$.

(d) If u is D-finite, $\mathrm{Ann}(u) \ne A_2L$ for every m. In particular the heat kernel $(1+4t)^{-1/2}e^{-x^2/(1+4t)}$ identifies $\partial_t - \partial_x^2$ in $\mathbb{C}[\partial_t,\partial_x]$ (it is not exponential-polynomial) but *not* in A_2: $(1+4t)\partial_x + 2x$ kills it; likewise $(1+4it)\partial_x + 2x$ for free Schrödinger.

(e) (Existence for free Schrödinger) with $\chi \in C_c^\infty((-1,1)) \ne 0$ and $\hat g(k) = \sum_{n\ge1} e^{-n^2}\chi(2^{n+2}(k-1/n))$, $u = \int \hat g(k)e^{ikx-itk^2}\,\mathrm{d}k$ is entire, solves $iu_t + u_{xx} = 0$, and $\mathrm{Ann}_{A_2}(u) = A_2(\partial_t - i\partial_x^2)$.

(f) (Non-constant leading coefficient) for $L = x\partial_t$, $\mathrm{Ann}(u) \supsetneq A_2L$ for every u: $xu_t = 0$ forces $u_t = 0$ by continuity, so $\partial_t \in \mathrm{Ann}(u)$, but $\partial_t \notin A_2x\partial_t$.

Proof.

Step 1 (a) Division by a monic operator

If $Q = \sum_{k\le K} q_k\partial_t^k$ with $K \ge m$, $q_K \ne 0$, subtract $q_K\partial_t^{K-m}L$: since $\partial_t^{K-m}M_j = \sum_i \binom{K-m}{i}(\partial_t^i M_j)\partial_t^{K-m-i}$ with $\partial_t^i M_j \in B$, it equals $q_K\partial_t^K +$ terms of ∂_t-degree $< K$; the degree drops and induction

concludes. Uniqueness: $Q'L = \sum R_j \partial_t^j$ with $Q' \neq 0$ of ∂_t-degree d would have degree $d+m \geq m$ on the left and $< m$ on the right.

Step 2 (b)–(d)

(b) $Q \in \mathrm{Ann}(u)$ iff its remainder $\sum R_j \partial_t^j$ kills u; so $\mathrm{Ann}(u) = A_2 L$ iff the remainder is zero whenever it kills u, i.e. iff B-independence. (c) For $m = 1$ the condition is $Ru = 0 \Rightarrow R = 0$ for $R \in B$, i.e. no $R \in \mathbb{C}[t,x]\langle\partial_x\rangle$ kills u: since $\partial_t u = -M_0 u$ and $M_0 \in B$, this says exactly that the $\mathbb{C}(t,x)$-span of all derivatives of u is infinite-dimensional, i.e. u is not D-finite. (d) is (c) for $m = 1$ (*condensed*: the package §5 gives the full argument), and for general m a D-finite u has a non-trivial B-relation among its ∂_t-derivatives. The heat kernel: $u_x = -2xu/(1+4t)$, so $(1+4t)u_x + 2xu = 0$.

Step 3 (e) and (f)

(e) The weight $\hat{g}$ is built to make u not D-finite (Fourier support in countably many tiny intervals around $1/n$ with super-exponentially small amplitudes); condensed, see the package §5. (f) as stated.

Why it works. in A_2 the leading coefficient must be a constant (monic form) and the data must not satisfy any ODE with polynomial coefficients: holonomic data, such as stationary states and Gaussian kernels, never identify in A_2.

■

Uses: B80.

Open

For $m \geq 2$ the criterion (b) is not reduced to an intrinsic property of u (relation with the holonomy of $A_2 u$); for non-constant leading coefficient only a saturated statement can be expected (VAR-C3Q2-open). Phase 10 closed a few cases (Chapter 18).

CHAPTER 20

Phases 7–8: encoder and convergence, the remaining proofs

In this chapter	
Starting from	the notation of Chapters 7 and 8.
Why	to give the proofs of results of the encoder and convergence packages that were only listed in Chapter 16: the structure of differential intertwiners, the interaction term with a potential, the spectral criterion for non-local laws, fractional and delay equations, the reduction to irreducible factors, the ambiguity in dimension $1+d$, and boundary conditions.
Role	Dim. B88–B93. They prove the corresponding statements of Chapters 14 and 13 of the paper.

20.1 The encoder

Dim. B88 — **ENC-K1p**

Differential intertwiners are Darboux chains [proved]

Let J be an open interval, $q \in C^\infty(J;\mathbb{C})$, $c \in \mathbb{C}$ and $W = \sum_{j=0}^{n} w_j \partial^j$ ($n \geq 1$, $w_n \not\equiv 0$) with $WL_q = (L_0 + c)W$. Then (i) w_n is constant; (ii) $N = \ker W$ has dimension n and is L_q-invariant; (iii) there is a closed countable set $Z \subset J$ such that on each component of $J \setminus Z$, $W = w_n A_n \cdots A_1$ with $A_k = \partial - \psi_k'/\psi_k$, where ψ_k is an eigenfunction of $L_{q_{k-1}}$ with eigenvalue λ_k, $q_0 = q$, $q_k = q_{k-1} - 2(\psi_k'/\psi_k)'$ and $q_n \equiv c$. Conversely every such Darboux chain gives a differential intertwiner (Crum). So, locally, a differential W with $WL_q = (L_0 + c)W$ exists iff q is obtained from c by a finite Darboux chain.

Idea. Factor W one first-order Darboux step at a time: an eigenvector of L_q on the finite-dimensional kernel of W gives the first factor, and the quotient is again an intertwiner, of order $n-1$ and for the transformed potential.

Proof.

Step 1 (i) and (ii)

The coefficient of ∂^{n+1} in $WL_q - (L_0 + c)W$ is $2w_n'$, so w_n is a non-zero constant, hence $\dim\ker W = n$ on J. If $Wf = 0$ then $WL_q f = (L_0 + c)Wf = 0$: N is L_q-invariant.

Step 2 (iii) One Darboux step

$\mathbb{C}$ is algebraically closed, so $L_q|_N$ has an eigenvector ψ_1 with eigenvalue λ_1. On $J_1 = J \setminus \psi_1^{-1}(0)$ (the zeros of a non-zero solution of a second-order linear ODE are isolated) put $g = \psi_1'/\psi_1$, $A_1 = \partial - g$, $A_1^\sharp = -\partial - g$. Since $g' + g^2 = \psi_1''/\psi_1 = q - \lambda_1$:

$$L_q - \lambda_1 = A_1^\sharp A_1, \qquad A_1 A_1^\sharp = L_{q_1} - \lambda_1 \ \ (q_1 = q - 2g'), \qquad \text{hence} \qquad A_1 L_q = L_{q_1} A_1.$$

$W\psi_1 = 0$, so right Euclidean division by the monic A_1 gives $W = W_1 A_1 + r$ with r a function and $r\psi_1 = 0$, so $r = 0$. From $W_1 A_1 L_q = (L_0 + c) W_1 A_1$ and $A_1 L_q = L_{q_1} A_1$: $(W_1 L_{q_1} - (L_0 + c) W_1) A_1 = 0$. A_1 is onto C^∞ of every subinterval (a first-order ODE), so $W_1 L_{q_1} = (L_0 + c) W_1$ with W_1 of order $n - 1$.

Step 3 Induction and the exceptional set

Iterate: at $n = 1$, $W_0 = w_0$ is a function with $w_0 L_{q_1} = (L_0 + c) w_0$; the coefficient of ∂ gives w_0 constant and that of ∂^0 gives $q_1 = c$. Let $Z = Z_1 \cup \cdots \cup Z_n$ with Z_k the zeros of ψ_k in $J \setminus (Z_1 \cup \cdots \cup Z_{k-1})$: each Z_k is discrete in its open set, so Z is countable, and closed because $Z_1 \cup \cdots \cup Z_k$ contains the closure of Z_k. (It is not claimed that Z is discrete.)

Why it works. a differential intertwiner has a finite-dimensional invariant kernel, which contains an eigenvector; dividing by the first-order operator built from it lowers the order by one.

■

Uses: B79.

Priority

The factorisation of an intertwiner into first-order Darboux steps is the subject of Andrianov–Sokolov (2003) and Sokolov (arXiv:0903.2835); the statement is almost certainly known. The global explicit classification for rational $V \to 0$ is Duistermaat–Grünbaum (1986) and holds only with decay at infinity (ENC-K1, from a cited source).

Dim. B89 **ENC-A9**

The interaction term for Schrödinger with a potential [proved]

Let $T_1 = -iH_0$ (kinetic), $T_2 = -iV$, $U(t) = e^{-it(H_0+V)}$, $E^{\mathrm{mult}}(t) = U(t) - U_0(t)e^{-itV}$ and $V \in W^{2,\infty}(\mathbb{R})$ real.

(a) For $f \in H^1$, $t \geq 0$: $\|E^{\mathrm{mult}}(t)f\| \leq \frac{t^2}{2}(\|V''\|_\infty \|f\| + 2\|V'\|_\infty \|f'\|) + \frac{2t^3}{3}\|V'\|_\infty^2 \|f\|$; for $f \in H^2$, $E^{\mathrm{mult}}(t)f = \frac{t^2}{2}[H_0, V]f + o(t^2)$ with $[H_0, V] = -V'' - 2V'\partial$; and $E^{\mathrm{mult}} \equiv 0$ near $t = 0$ iff V is constant.

(b) ($V \equiv c$) $E^{\mathrm{mult}} = 0$ and $E^{\mathrm{add}} = (e^{-ict} - 1)(U_0 - I)$.

(c) ($V = Fx$ on $L^2(\mathbb{R})$) $U(t) = e^{-iF^2t^3/12} e^{-itV/2} U_0(t) e^{-itV/2}$ (Strang, exact up to a phase), and $\|E^{\mathrm{mult}}(t)f\| \leq |F|t^2 \|f'\| + \frac{2}{3}F^2t^3\|f\|$, $\|E^{\mathrm{mult}}(t)\|_{L^2 \to L^2} = 2$.

The constant $2/3$ is optimal. (The first version wrote $F^2t^3/6$: false, because $e^{-itFx/2}$ translates p by $Ft/2$; for $F = t = 1$ and a Gaussian of width 25 the true value is 0.6549 against 0.1950.)

Proof.

Step 1 (a) Two Duhamel identities

For $f \in H^2$ put $\mathcal{S}(s) = U_0(s)e^{-isV}$. Then $s \mapsto U(t-s)\mathcal{S}(s)f$ is C^1 with derivative $iU(t-s)[V, U_0(s)]e^{-isV}f$ (the two terms with H_0 cancel because U_0 commutes with H_0), so

$$E^{\mathrm{mult}}(t)f = -i\int_0^t U(t-s)[V, U_0(s)]e^{-isV}f\,\mathrm{d}s, \qquad [V, U_0(s)]g = i\int_0^s U_0(r)[H_0, V]U_0(s-r)g\,\mathrm{d}r.$$

Step 2 The estimate and the leading term

$\|[H_0, V]h\| \le \|V''\|\|h\| + 2\|V'\|\|h'\|$, U_0 is unitary and commutes with ∂, and $\|(e^{-isV}f)'\| \le \|f'\| + s\|V'\|\|f\|$; hence

$$\|E^{\mathrm{mult}}(t)f\| \le \int_0^t s\big[\|V''\|\|f\| + 2\|V'\|(\|f'\| + s\|V'\|\|f\|)\big]\,\mathrm{d}s,$$

the stated bound; extend to H^1 by density. The double integral form shows $E^{\mathrm{mult}}(t)f/t^2 \to \frac{1}{2}[H_0, V]f$. If $E^{\mathrm{mult}} \equiv 0$ then $[H_0, V]f = 0$ for all $f \in H^2$; $f \equiv 1$ on an interval gives $V'' = 0$, then $f' \neq 0$ gives $V' = 0$. Conversely for constant V, e^{-itV} is a scalar.

Step 3 (c) The Stark case in Fourier variables

With $\hat{f}(p) = \int f e^{-ipx}$, $x = i\partial_p$ and the equation becomes $\partial_t\hat{u} - F\partial_p\hat{u} = -ip^2\hat{u}$; characteristics $p(s) = p + F(t-s)$ give

$$\hat{u}(t,p) = \hat{u}_0(p+Ft)\exp\big(-i(p^2t + pFt^2 + F^2t^3/3)\big).$$

The Strang product gives $e^{-i(p^2t+pFt^2+F^2t^3/4)}\hat{f}(p+Ft)$ (since $e^{-itFx/2}$ shifts $p \mapsto p + Ft/2$), so the phase $F^2t^3/12$ closes the identity. $\|E^{\mathrm{mult}}(t)f\|$ is the norm of $(1 - e^{i\theta(p)})$ applied after a translation, with θ affine and non-constant, whence the bound and the sup norm 2.

Why it works. the interaction of kinetic and potential terms is governed by the single commutator $[H_0, V]$, which is a first-order operator with coefficient V': it vanishes iff V is constant.

■

Uses: none.

A warning on E^{add}

E^{add} does *not* vanish for constant V: $E^{\mathrm{add}} = (e^{-ict} - 1)(U_0(t) - I)$, with expansion $-\frac{t^2}{2}\{H_0, V\} + O(t^3)$. The quantity that measures the genuine interaction between kinetic and potential terms is E^{mult}, and the right criterion is "$E^{\mathrm{mult}} \equiv 0 \iff V$ constant".

20.2 Locality, fractional laws and delays

Dim. B90 **CONV-T6**

A spectral criterion for the annihilator of a non-local datum [proved]

Let $u(t,x)=\int_{\mathbb{R}} e^{a(\kappa)t+i\kappa x}\hat f(\kappa)\,\mathrm{d}\kappa$ with $\hat f\in L^1$ of compact support, a continuous on $\operatorname{supp}\hat f$, $E=\{\hat f\neq 0\}$, Γ_E the graph $\{(a(\kappa),i\kappa):\kappa\in E\}$.

(1) $\operatorname{Ann}(u)=I(\Gamma_E)=\{Q: Q(a(\kappa),i\kappa)=0 \text{ a.e. on } E\}$.

(2) If $J\subseteq E$ is an open interval on which a is real-analytic, $\operatorname{Ann}(u)\subseteq I_J=\{Q: Q(a,i\kappa)=0 \text{ on } J\}$, a prime ideal; $I_J=0$ if $a|_J$ is transcendental over $\mathbb{C}(\kappa)$, otherwise $I_J=(m_J)$ with m_J irreducible. If E is a finite union of such intervals, $\operatorname{Ann}(u)=\bigcap I_{J_i}$: zero if some branch is transcendental, otherwise principal, generated by the product of the distinct m_{J_i}.

(3) Any multiplier $b=a$ a.e. on E gives the same u: the non-local law is determined by u only on Γ_E; if $a|_E$ is algebraic, the local law $m(\partial)u=0$ and the non-local law are indistinguishable.

(4) If $u=\iint e^{i(\omega t+\kappa x)}\hat g$ with $\hat g\in L^1(\mathbb{R}^2)$ of compact support and $\hat g\neq 0$ a.e. on a non-empty open set, then no real-analytic $m\not\equiv 0$ satisfies $m(D)u=0$ on Ω; in particular $\operatorname{Ann}(u)=0$.

Proof.

Step 1 (1) Differentiate under the integral

$Q(\partial)u=\int Q(a(\kappa),i\kappa)e^{a(\kappa)t+i\kappa x}\hat f\,\mathrm{d}\kappa$. For fixed t this is, in x, the inverse Fourier transform of a compactly supported L^1 function, hence entire; if it vanishes on Ω it vanishes on a segment, hence on $\mathbb{R}$, and by injectivity of the Fourier transform $Q(a,i\kappa)e^{at}\hat f=0$ a.e. The converse is immediate.

Step 2 (2) Primality and principality

For $Q\in\operatorname{Ann}(u)$, $\kappa\mapsto Q(a(\kappa),i\kappa)$ is real-analytic on J and zero a.e., hence identically zero. I_J is the kernel of the homomorphism $R\to\mathcal{O}(J)$, $Q\mapsto Q(a,i\kappa)$ into an integral domain, so it is prime. If $a|_J$ is transcendental, $I_J=0$. Otherwise $I_J\neq 0$ and not maximal (a maximal ideal contains $\xi-\xi_0$, impossible since $i\kappa$ is not constant on J), and a non-zero non-maximal prime of $\mathbb{C}[\tau,\xi]$ has height one, hence is generated by an irreducible polynomial (unique factorisation). For a finite union, $\operatorname{Ann}(u)=\bigcap I_{J_i}$ is the intersection of principal primes: generated by the product of the distinct generators.

Step 3 (3) and (4)

(3) is immediate from (1). (4) $m(D)u=\iint m\hat g e^{i(\omega t+\kappa x)}$ is entire; zero on Ω implies zero everywhere, so $m\hat g=0$ a.e., so $m=0$ a.e. on U, so $m\equiv 0$ on U and, by analyticity on $\mathbb{R}^2$, everywhere.

Why it works. a datum remembers only the Zariski closure of the graph of its symbol; a law is local exactly when that graph is algebraic.

■

Uses: A2, A41.

Example 20.1 Poisson, Klein–Gordon, telegraph, fractional

Poisson $e^{-t|D|}$ ($a = -|\kappa|$, two intervals): $m = \tau - i\xi$ on $\kappa > 0$ and $\tau + i\xi$ on $\kappa < 0$; the product $\tau^2 + \xi^2$: u is harmonic. $e^{-t\sqrt{1-\partial^2}}$ gives $\tau^2 + \xi^2 - 1$ (elliptic Klein–Gordon). Exponential memory $u_t = \int_{-\infty}^{t} e^{-a(t-s)} u_{xx}(s)\,ds$ gives $\tau(\tau + a) = \xi^2$ (telegraph). A fractional $a = -\kappa^{p/q}$ on an interval of $(0,\infty)$ with $\gcd(p,q) = 1$ has minimal polynomial dividing $(-\tau)^q - (-i\xi)^p$.

Dim. B91 **CONV-P6, CONV-P7**

Irrational fractional power and delay: Ann $= 0$; locality is a property of the pair $(u, \mathcal{K})$ [proved]

Let $J \subset (0,\infty)$ be an open interval and $\hat{f} \neq 0$ a.e. on J.

(a) If $a(\kappa) = -\kappa^\alpha$ with $\alpha > 0$ irrational, $\mathrm{Ann}(u) = 0$.

(b) If $\sigma > 0$, $J \subset (0, (\sigma e)^{-1/2})$ and $a(\kappa) = W_0(-\sigma\kappa^2)/\sigma$ (principal Lambert branch), u solves $u_t(t,x) = u_{xx}(t-\sigma,x)$ and $\mathrm{Ann}(u) = 0$.

(c) (the delay is local after augmentation) $w(t,\theta,x) = u(t-\theta,x)$ solves $w_t + w_\theta = 0$ with boundary coupling $\partial_t w(t,0,x) = \partial_x^2 w(t,\sigma,x)$; conversely $w = w(t-\theta,0,x)$ and $u = w(\cdot,0,\cdot)$ solves the delay equation.

(d) (P7) Without the non-triviality condition every u lies in $\mathcal{L}^+_{\mathcal{K}}$ (take $Y = \mathbb{R}$, $S = \{\partial_y v = 0\}$, $\gamma(v) = v(\cdot,\cdot,0)$). With it: the delay is in $\mathcal{L}^+_{\text{transport}} \setminus \mathcal{L}$; exponential memory is in $\mathcal{L} \cap \mathcal{L}^+_{\text{ODE}}$; a datum with a two-dimensional spectrum is in $\mathcal{N}_{\mathcal{K}}$ for $\mathcal{K}$ real-analytic translation-invariant multipliers.

Proof.

Step 1 (a) Independence of exponentials

If $Q = \sum c_{ij}\tau^i\xi^j \neq 0$ vanishes on the graph, $H(\kappa) = \sum c_{ij}(-1)^i \kappa^{i\alpha+j}$ is analytic on $(0,\infty)$ and zero on J, so zero on $(0,\infty)$. The exponents $i\alpha + j$ are distinct for distinct (i,j) (α irrational); with $\kappa = e^s$ the functions $e^{(i\alpha+j)s}$ are exponentials with distinct frequencies, hence independent (A2): all $c_{ij} = 0$. Apply B90(2).

Step 2 (b) Lambert

For $y \in [-1/e, 0)$, $W_0(y)e^{W_0(y)} = y$ is real, real-analytic and non-constant on $(-1/e, 0)$. With $a = W_0(-\sigma\kappa^2)/\sigma$: $a = -\kappa^2 e^{-\sigma a}$, so $u_t = \int a e^{at + i\kappa x}\hat{f}$ equals $u_{xx}(t-\sigma) = \int (-\kappa^2) e^{a(t-\sigma)+i\kappa x}\hat{f}$. For transcendence: if $Q(a, i\kappa) = 0$ then a is algebraic over $\mathbb{C}(\kappa)$, $a \neq 0$ on J, so $e^{\sigma a} = -\kappa^2/a$ is algebraic over $\mathbb{C}(\kappa)$; as a is non-constant, κ is algebraic over $\mathbb{C}(a)$, so $e^{\sigma a}$ would be algebraic over $\mathbb{C}(a)$, impossible (a transcendental over $\mathbb{C}$ and $e^{\sigma a}$ transcendental over $\mathbb{C}(a)$). So $I_J = 0$ and $\mathrm{Ann}(u) = 0$ by B90(2).

Step 3 (c) and (P7)

(c) $w_t + w_\theta = u_t(t-\theta) - u_t(t-\theta) = 0$; $w(t,0,x) = u$, $w(t,\sigma,x) = u(t-\sigma,x)$ gives the coupling; conversely w is constant along $(t+s,\theta+s)$. (P7)(a) is the given trivial system; (b): the traces of solutions of the delay system are only the solutions of the delay equation (for example $e^t \sin x$ is not one: $1 \neq -e^{-\sigma}$), and $\mathrm{Ann}(u) = 0$ by (b); memory: $v = \int_{-\infty}^{t} e^{-a(t-s)} u_{xx}(s)\,ds$ solves $v_t = -av + u_{xx}$ and $u_t = v$; the two-dimensional spectrum case is B90(4).

Why it works. "local or non-local" is not a property of u but of the pair (u, a declared class of laws): without a non-triviality condition the class of augmented laws is empty of content.

■

Uses: A2, B90.

20.3 Dimension $1+d$

Dim. B92 — **CONV-P8, CONV-D1**

Reduction to irreducible factors, and the ambiguity in $1+d$ [proved]

(P8) Let $P = \prod_i P_i \in R_n$ be reduced (P_i pairwise non-associate irreducibles), F any R_n-module, $u \in F$ with $Pu = 0$, $w_i = (P/P_i)u$. Then $\mathrm{Ann}(u) = (P) \iff \mathrm{Ann}(w_i) = (P_i)$ for every i.

(D1) Let $\Omega \subset \mathbb{R}^n$ be any open set and $\mathcal{A} = C^\infty(\Omega)$ or $\mathcal{D}'(\Omega)$. (a) For $Q_1, Q_2 \in R_n$, each $\zeta \in V(Q_1,Q_2)$ gives $e_\zeta = e^{\langle \zeta, z\rangle} \in \ker Q_1 \cap \ker Q_2$, with distinct ζ independent; for $n \geq 3$, $\dim R_n/(Q_1,Q_2) < \infty \iff (Q_1,Q_2) = R_n \iff V(Q_1,Q_2) = \emptyset$, otherwise $\ker Q_1 \cap \ker Q_2$ is infinite-dimensional. (b) The ambiguity space $\mathcal{K} = \{(v_j) \in \bigoplus \ker Q_j : \sum v_j = 0\}$ is 0 iff the Q_j are pairwise co-maximal; for $n \geq 3$ and $\mathcal{K} \neq 0$, $\dim \mathcal{K} = \infty$. (c) (a non-EP exceptional class) in $\mathbb{R}^3$ with $P_i = \tau - \xi_1$, $w = f(t + x_1)$, f not exponential-polynomial, has $\mathrm{Ann}(w) = (\tau - \xi_1, \xi_2) \supsetneq (P_i)$ and $\dim R_3 w = \infty$.

So in $1+d$, $d \geq 2$, $\delta(P) \in \{0, \infty\}$, with $\delta = 0$ iff the factors are pairwise co-maximal, on every open set without convexity.

Proof.

Step 1 (P8)

($\Rightarrow$) $P_i w_i = Pu = 0$, so $(P_i) \subseteq \mathrm{Ann}(w_i)$. If $Qw_i = 0$ then $Q(P/P_i) \in \mathrm{Ann}(u) = (P)$, so $P_i \mid Q(P/P_i)$; P_i is prime and does not divide P/P_i, so $P_i \mid Q$. ($\Leftarrow$) $(P) \subseteq \mathrm{Ann}(u)$; if $Qu = 0$ then $Qw_i = (P/P_i)Qu = 0$, so $P_i \mid Q$ for all i, and distinct primes give $P \mid Q$.

Step 2 (D1)(a)

$Q(\partial)e_\zeta = Q(\zeta)e_\zeta$; independence is A2 restricted to a generic line on which the linear forms $\langle \zeta, \cdot \rangle$ stay distinct. For $n \geq 3$ and $V(Q_1,Q_2) \neq \emptyset$ every irreducible component has dimension $\geq n-2 \geq 1$ (Krull's principal ideal theorem, twice), so V is infinite, $R_n/(Q_1,Q_2)$ is infinite-dimensional (Nullstellensatz) and there are infinitely many independent exponentials in the intersection. If $V = \emptyset$, $(Q_1,Q_2) = R_n$.

Step 3 (D1)(b) and (c)

(b) If the Q_j are pairwise co-maximal then Q_i and $\prod_{j\neq i} Q_j$ are co-maximal: $1 = AQ_i + B\prod_{j\neq i} Q_j$. From $\sum v_j = 0$, applying $\prod_{j\neq i} Q_j$ gives $\prod_{j\neq i} Q_j v_i = 0$, and with $Q_i v_i = 0$: $v_i = AQ_i v_i + B\prod_{j\neq i} Q_j v_i = 0$. Conversely if $(Q_i, Q_j) \neq R_n$ pick $\zeta \in V(Q_i, Q_j)$: $(v_i, v_j) = (e_\zeta, -e_\zeta)$ is a non-zero element of $\mathcal{K}$, and for $n \geq 3$ infinitely many independent ones exist. (c) $(\tau - \xi_1)w = f' - f' = 0$, $\xi_2 w = 0$; for $Q \in \mathrm{Ann}(w)$ reduce modulo $(\tau - \xi_1, \xi_2)$ to $q(\xi_1)$, and $q(\partial)f = 0$ forces $q = 0$ because f is not exponential-polynomial; $R_3 w \cong \mathbb{C}[\xi_1]$.

Why it works. in more than two variables two coprime polynomials still meet along a variety of positive dimension, so the overlap of two factors is either empty or infinite.

■

Uses: A2, A13.

20.4 Boundary conditions

Dim. B93 — **CONV-B1, CONV-B2**

Boundary conditions are invisible to $\mathrm{Ann}(u)$; stability with a dictionary of traces [proved]

(B1) For u real-analytic on connected Ω, $\mathrm{Ann}(u)$ is determined by the germ of u at any interior point (A41(a)). On $(0, L)$ with $\mu = \pi/L$, $\kappa_0 = D\mu^2$, $u_D = e^{-\kappa_0 t}\sin \mu x$ (Dirichlet) and $u_N = e^{-\kappa_0 t}\cos \mu x$ (Neumann) have the same Ann $= (\tau + \kappa_0,\ \xi^2 + \mu^2)$.

(B2) Let Ω be bounded Lipschitz, $\Gamma \subseteq \partial\Omega$ relatively open, $\gamma_\Gamma : H^1(\Omega) \to L^2(\Gamma)$ the trace (constant C_{tr}), $\Phi_\Gamma(u)\lambda = \gamma_\Gamma(\sum_{|\alpha|\leq n_0} \lambda_\alpha \partial^\alpha u)$, with $\Phi_\Gamma(u)\lambda^* = 0$, $|\lambda^*| = 1$, $\sigma_2(\Phi_\Gamma(u)) \geq \rho_\Gamma > 0$. If $\|\tilde{u} - u\|_{H^{n_0+1}} \leq \delta$ and $\tilde{\lambda}$ minimises $\|\Phi_\Gamma(\tilde{u})\lambda\|$ on $|\lambda| = 1$, then $\sin\angle(\tilde{\lambda}, \lambda^*) \leq 2C_{\mathrm{tr}}\sqrt{D+1}\,\delta/\rho_\Gamma$; the misfit certificate and discrepancy rule of A43 carry over with δ replaced by $C_{\mathrm{tr}}\sqrt{D+1}\,\delta$.

Proof.

Step 1 (B1)

Both $\tau + \kappa_0$ and $\xi^2 + \mu^2$ kill both functions; for each dim $R \cdot u = 2$ (u and u_x are independent: their Wronskian in x is non-zero), and dim $R/(\tau + \kappa_0, \xi^2 + \mu^2) = 2$. Since $(\tau + \kappa_0, \xi^2 + \mu^2) \subseteq \mathrm{Ann}(u)$ and $R \cdot u \cong R/\mathrm{Ann}(u)$, the inclusion is an equality.

Step 2 (B2)

$\|(\Phi_\Gamma(\tilde{u}) - \Phi_\Gamma(u))\lambda\| \leq \sum |\lambda_\alpha| C_{\mathrm{tr}} \|\partial^\alpha e\|_{H^1} \leq C_{\mathrm{tr}}|\lambda|(\sum_\alpha \|\partial^\alpha e\|_{H^1}^2)^{1/2}$; each ∂^γ with $|\gamma| \leq n_0 + 1$ appears at most $D + 1$ times (as ∂^α itself or as $\partial_j \partial^\alpha$), so the sum is $\leq (D+1)\|e\|_{H^{n_0+1}}^2$. Then the proof of A43 (T4a) word for word.

Why it works. the annihilator is a germ-level object, and a boundary condition selects among the solutions without changing the equation they satisfy.

■

Uses: A41, A43.

Dim. B155 **CONV-B3**

Tangential obstruction: no boundary margin above the minimal level [proof written, not audited]

Let $\Gamma = \{x_1 = c\} \cap \bar{\Omega}$ be a flat boundary piece transversal to t, $u \in C^{n_0+2}$ up to Γ, and $\Phi_\Gamma(u)\lambda = \gamma_\Gamma(\sum_{|\alpha|\le n_0} \lambda_\alpha \partial^\alpha u)$. If $\lambda^* \in \ker \Phi_\Gamma(u)$ corresponds to Q^* with $\deg Q^* \le n_0 - 1$, then τQ^* is also in the kernel, $\dim \ker \Phi_\Gamma(u) \ge 2$ and $\sigma_2(\Phi_\Gamma(u)) = 0$: no margin. With Dirichlet data and $n_0 \ge 1$ the kernel contains 1 and τ.

Stated in the paper as no. 13, p. 71

Proof.

Step 1 **The trace of Q^*u vanishes**

$w = Q^*(\partial)u$ is C^1 up to Γ and $w|_\Gamma = 0$.

Step 2 **A tangential derivative preserves it**

∂_t is tangent to Γ, so $(\partial_t w)|_\Gamma = \partial_t(w|_\Gamma) = 0$: $(\tau Q^*)(\partial)u$ has zero trace.

Step 3 **Independence**

Q^* and τQ^* have different degrees, hence are linearly independent in $R_{\le n_0}$. So $\dim \ker \ge 2$ and $\sigma_2 = 0$.

Why it works. a boundary condition is automatically accompanied by all its tangential consequences.

■

Uses: none.

Reading

Identifying a boundary condition with a margin requires the minimal level (the rule of T5), or a dictionary of normal derivatives only, or the quotient by tangential consequences. Example: $u = e^{-t}\sin x$ on $x = 0$ has traces $(0, 0, e^{-t})$ of $(1, \partial_t, \partial_x)$.

CHAPTER 21

Phase 8: splitting with a boundary, periodic resonance, Mori–Zwanzig and the contract

In this chapter	
Starting from	the notation of Chapter 7. For B94: $I \subset \mathbb{R}$ open interval, $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, $X = C^\infty(I \times \mathbb{T})$. For B97: $X = L^2(0, L)$, $A_0 = D\partial^2$ with Dirichlet conditions, $V = -c\partial_x$.
Why	to give the proofs of the remaining convergence results: how periodic boundaries break the factor decomposition, the exact Mori–Zwanzig form for constant-coefficient systems, the error of a chain of steps, and the order of operator splitting with a Dirichlet boundary.
Role	Dim. B94–B97. They prove the corresponding statements of Chapter 13 and of the extended algorithm (Chapter 17).

Dim. B94 — **CONV-R1**

Periodic boundary: a finite-dimensional failure of the factor decomposition [proved]

Let $f, g \in R = \mathbb{C}[\tau, \xi]$ be coprime, $f_k(\tau) = f(\tau, ik)$, $g_k(\tau) = g(\tau, ik)$. Call $k \in \mathbb{Z}$ *resonant* if $f_k \not\equiv 0$, $g_k \not\equiv 0$ and $\gcd(f_k, g_k) \neq 1$. Then:

(a) $\ker_X(fg) = \ker_X f + \ker_X g$ iff there is no resonant k;

(b) the set of resonant k is finite and $\dim \ker_X(fg)/(\ker_X f + \ker_X g) = \sum_{k \text{ resonant}} \deg \gcd(f_k, g_k)$;

(c) on $I \times J$ with J an interval (convex Ω) the equality $\ker(fg) = \ker f + \ker g$ always holds (R2-T3, C12-T1′): the failure on $\mathbb{T}$ is a boundary phenomenon.

Proof.

Step 1 Bezout and finitely many resonances

f, g coprime in $\mathbb{C}[\xi][\tau]$ are coprime in $\mathbb{C}(\xi)[\tau]$ (Gauss), so $Af + Bg = r(\xi)$ with $0 \neq r \in \mathbb{C}[\xi]$. Let $Z = \{k : r(ik) = 0\}$, finite. For $k \notin Z$ a common root of f_k, g_k would be a root of $r(ik)$: k is not resonant. So the resonant k lie in Z.

Step 2 Reduction to Fourier modes

For $v \in X$ put $\hat v_k(t) = (2\pi)^{-1}\int v\, e^{-ikx}\,\mathrm{d}x$; then $(Q(\partial)v)^\wedge_k = Q(\partial_t, ik)\hat v_k$ and, by integration by parts in x, $\sup_K |\partial_t^j \hat v_k| \le C_{K,j,M}(1+|k|)^{-M}$; conversely a sequence with these bounds defines $\sum v_k e^{ikx} \in X$. For non-zero $p, q \in \mathbb{C}[\tau]$ on $C^\infty(I)$: $\dim\ker p(\partial_t) = \deg p$ and $\ker p \cap \ker q = \ker \gcd(p,q)$, so $\operatorname{codim}(\ker p + \ker q)$ in $\ker(pq)$ is $\deg\gcd(p,q)$.

Step 3 Sufficiency: an explicit splitting

Let $u \in \ker_X(fg)$, so $\hat u_k \in \ker(f_k g_k)$. For $k \notin Z$ put $v_k = B_k g_k \hat u_k / r(ik)$, $w_k = A_k f_k \hat u_k / r(ik)$: then $v_k + w_k = \hat u_k$, $f_k v_k = 0$, $g_k w_k = 0$. For the finitely many non-resonant $k \in Z$ split by the ODE facts. Regularity: $|r(ik)| \ge c > 0$ for large $|k|$ (a non-zero polynomial), and A_k, B_k, f_k, g_k are differential operators of fixed order, so the decay in k is preserved and $\sum v_k e^{ikx}, \sum w_k e^{ikx} \in X$.

Step 4 Necessity and the dimension

If k is resonant take $y \in \ker(f_k g_k) \setminus (\ker f_k + \ker g_k)$ and $u = y(t)e^{ikx}$. A splitting $u = v + w$ would give $y = \hat v_k + \hat w_k$ with $f_k \hat v_k = 0$, $g_k \hat w_k = 0$: contradiction. More precisely, $u \mapsto (\hat u_k)_{k\ \text{resonant}}$ induces an isomorphism $\ker(fg)/(\ker f + \ker g) \cong \bigoplus_{k\ \text{res.}} \ker(f_k g_k)/(\ker f_k + \ker g_k)$, onto by finite sums, with kernel $\ker f + \ker g$ by Step 3 on non-resonant modes.

Why it works. on a circle every mode is an ODE in t, and a common root of the two factors at a frequency ik is a genuine overlap that Bezout in one variable cannot split.

■

Uses: A13, A32 (Bezout).

Example 21.1 Resonance at $k = 0$

$f = \tau - \xi^2$, $g = \tau + \xi$: $f - g = -\xi(\xi + 1)$, the only resonant mode is $k = 0$ ($f_0 = g_0 = \tau$), the codimension is 1, and the witness is $u = t$ (on $\mathbb{R}$, $t = x - (x - t)$). For $f = \tau - \xi^2$, $g = \tau + \xi + 1$ the common roots satisfy $\xi^2 + \xi + 1 = 0$, outside $i\mathbb{Z}$: the decomposition always exists. For $f = \xi$, $g = \tau$ the line $\xi = 0$ lies in $V(f)$, so $(0,0)$ is not resonant and $\ker(\tau\xi) = \{A(t) + B(x)\} = \ker\xi + \ker\tau$.

Counterexample 21.2 The criterion without the clause on contained lines

[false] The criterion "no resonance $V(f) \cap V(g) \cap (\mathbb{C} \times i\mathbb{Z})$" of the route RT-R2a is correct only up to lines contained in $V(f)$ or $V(g)$; without that clause the example $f = \xi$, $g = \tau$ refutes it (CONV-R1-F).

Dim. B95 **CONV-MZ**

Mori–Zwanzig with constant coefficients [proved]

(MZ1) If $D \in \mathrm{Mat}_k(R_n)$, $\det D \neq 0$, $\mathcal{A}$ an R_n-module and $DU = 0$, then $(\det D)U_i = 0$ for every i: each component of a square non-degenerate system has a *local* law of order at most $\deg \det D$.

(MZ2) For $(\hat{u}, \hat{v})' = A(i\kappa)(\hat{u}, \hat{v})$, $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$, $a_{ij} \in \mathbb{C}[\xi]$:

$$\hat{u}'(t) = a_{11}\hat{u}(t) + a_{12}\int_0^t e^{(t-s)a_{22}} a_{21}\hat{u}(s)\,\mathrm{d}s + a_{12}e^{ta_{22}}\hat{v}(0)$$

(Markov term, memory, "noise" from unresolved data), and $(\partial_t - a_{22})(\partial_t - a_{11})\hat{u} = a_{12}a_{21}\hat{u}$, i.e. $\det(\tau - A(\xi))(\partial)u = 0$.

(MZ3) The memory kernel $K(t,\xi) = a_{12}e^{ta_{22}}a_{21}$ vanishes iff $a_{12}a_{21} = 0$; $[\mathcal{P}, A] = 0$ with $\mathcal{P} = \mathrm{diag}(1,0)$ iff $a_{12} = a_{21} = 0$. The criterion "$E_{\mathrm{int}} = 0 \iff [\mathcal{P}, L] = 0$", read as "memory zero $\iff [\mathcal{P}, A] = 0$", is **false**, and so is the reading "memory and noise zero $\iff [\mathcal{P}, A] = 0$".

Proof.

Step 1 (MZ1)

$\mathrm{adj}(D)D = \det(D)I$, so $\det(D)U = \mathrm{adj}(D)(DU) = 0$.

Step 2 (MZ2) Duhamel

$\hat{v}(t) = e^{ta_{22}}\hat{v}(0) + \int_0^t e^{(t-s)a_{22}}a_{21}\hat{u}(s)\,\mathrm{d}s$; substitute in $\hat{u}' = a_{11}\hat{u} + a_{12}\hat{v}$. Applying $\partial_t - a_{22}$ to $\hat{u}' - a_{11}\hat{u} = a_{12}\hat{v}$ gives $a_{12}(\hat{v}' - a_{22}\hat{v}) = a_{12}a_{21}\hat{u}$.

Step 3 (MZ3) and the counterexamples

$\mathbb{C}[\xi]$ is a domain and $e^{ta_{22}} \neq 0$, so $K \equiv 0 \iff a_{12}a_{21} = 0$; $\mathcal{P}A - A\mathcal{P} = \begin{pmatrix} 0 & a_{12} \\ -a_{21} & 0 \end{pmatrix}$. With $a_{21} = 0$, $a_{12} = 1$ the memory is zero but $[\mathcal{P}, A] \neq 0$. With $a_{12} = 0$, $a_{21} = 1$ the component $\hat{u}$ is exactly closed ($\hat{u}' = a_{11}\hat{u}$, memory and noise zero) but $[\mathcal{P}, A] = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix} \neq 0$.

Why it works. with constant coefficients, eliminating a variable gives back a local law (the determinant), and "memory" is the same information written as an integral.

■

Uses: none.

Reading

For under-determined systems the eliminated law may be missing ($u_t = v_x$: every u is a component of a solution on convex Ω, $v = \int u_t \, dx$), consistent with the class $\mathcal{N}_{\mathcal{K}}$ of B91. With variable coefficients the converse "closed local law $\iff$ finite elimination" is open (CONV-Q11).

Dim. B96 **CONV-T7**

The error of a chain of steps [proved]

Let $X_0, \dots, X_k$ be metric spaces, $F_i : X_{i-1} \to X_i$ the exact maps and $\tilde{F}_i$ the computed ones, $\tilde{x}_0 = x_0$, $\tilde{x}_i = \tilde{F}_i(\tilde{x}_{i-1})$, $x_i = F_i(x_{i-1})$. If $d(\tilde{F}_i(\tilde{x}_{i-1}), F_i(\tilde{x}_{i-1})) \le \varepsilon_i$ and each F_j is L_j-Lipschitz on a set containing the points that occur below, then

$$d(\tilde{x}_k, x_k) \le \sum_{i=1}^{k} \Big(\prod_{j>i} L_j \Big) \varepsilon_i .$$

Proof.

Step 1 Telescoping

Put $y_i = F_k \circ \cdots \circ F_{i+1}(\tilde{x}_i)$, so $y_k = \tilde{x}_k$ and $y_0 = x_k$. Then $d(\tilde{x}_k, x_k) \le \sum_{i=1}^{k} d(y_i, y_{i-1})$.

Step 2 Each term

$y_{i-1} = F_k \circ \cdots \circ F_{i+1}(F_i(\tilde{x}_{i-1}))$ and $y_i = F_k \circ \cdots \circ F_{i+1}(\tilde{F}_i(\tilde{x}_{i-1}))$, so $d(y_i, y_{i-1}) \le (\prod_{j>i} L_j)\varepsilon_i$.

Why it works. an error made at step i is amplified only by the Lipschitz constants of the steps that follow.

■

Uses: none.

Use in the contract

The guaranteed chain contains only steps of class (A) (finite or exact: elimination, Wei–Norman, E_{int} in the class $\mathfrak{B}$, commuting multipliers, MZ1) and (B) (explicit ε_i: Dyson–Phillips with a bounded perturbation, Magnus only conditionally). A step of class (C) has no ε_i and is outside the guarantee; a step of class (Lim) is exact from K^* on, but K^* is not certifiable, so the chain guarantee holds "under the event $K \ge K^*$", which must be declared to the user. As a theorem the proposition is trivial; its value is as a specification of Step 7.

Dim. B97 **CONV-T8**

Splitting with a Dirichlet boundary: orders in L^2 [proved]

Let $e_n = \sqrt{2/L}\sin(\mu_n x)$, $\mu_n = n\pi/L$, $\lambda_n = -\kappa n^2$, $\kappa = D\pi^2/L^2$, $V_{mn} = \langle e_m, Ve_n\rangle = -4cmn/(L(m^2-n^2))$ for $m+n$ odd, 0 otherwise; $S^{(1)}(t) = \int_0^t S_0(t-s)VS_0(s)\,\mathrm{d}s = \partial_\varepsilon S_\varepsilon|_{\varepsilon=0}$; $h>0$ the step, $n_h = T/h$. Let G_h^{Lie}, G_h^{Str} be the first-order-in-ε global errors of the Lie and Strang splittings for modal data $u_0 = \sum_{n\le N} c_n e_n$.

(a) (closed form) with $x_{mn} = \kappa(m^2-n^2)h$, $\varphi_L(x) = 1/x - 1/(e^x-1)$, $\varphi_S(x) = 1/x - \frac{1}{2}\coth(x/2)$:

$$\langle e_m, G_h^{\mathrm{Lie}}\rangle = \sum_n c_n V_{mn}(e^{\lambda_n T} - e^{\lambda_m T})\, h\,\varphi_L(x_{mn}), \qquad \langle e_m, G_h^{\mathrm{Str}}\rangle = \sum_n c_n V_{mn}(e^{\lambda_n T} - e^{\lambda_m T})\, h\,\varphi_S(x_{mn}).$$

(b) (Lie: global order 1) $\|G_h^{\mathrm{Lie}}\| \le h\sqrt{N}(\sum |c_n|^2 c^2 \mu_n^2)^{1/2}$ and, for $u_0 = e_1$, $\|G_h^{\mathrm{Lie}}\|/h \to C_L > 0$.

(c) (Strang: order reduced from 2 to 5/4 in L^2) $\|G_h^{\mathrm{Str}}\| \le Ch^{5/4}$ for $0 < h \le 1$, and for $u_0 = e_1$ and $h \le 1/(36\kappa)$ a lower bound $\ge c_* h^{5/4}$.

Proof.

Step 1 (a) Explicit sums

By linearity take $u_0 = e_n$; $\langle e_m, S_0(r)VS_0(s)e_n\rangle = e^{\lambda_m r}V_{mn}e^{\lambda_n s}$ and $\langle e_m, S^{(1)}(T)e_n\rangle = V_{mn}(e^{\lambda_n T} - e^{\lambda_m T})/(\lambda_n - \lambda_m)$. With $a = e^{\lambda_n h}$, $b = e^{\lambda_m h}$ the Lie sum is $V_{mn}hb(a^{n_h} - b^{n_h})/(a-b)$, and $b/(a-b) = 1/(e^x-1)$, $1/\Delta = h/x$ with $x = \Delta h$, giving φ_L; the Strang sum gives φ_S in the same way.

Step 2 Properties of φ_L, φ_S

$\varphi_L \in (0,1)$ for $x \ne 0$ and $\varphi_L(x) \to \frac{1}{2}$ as $x \to 0$. φ_S is odd, and for $y > 0$, $\varphi_S(y) = -\frac{1}{2}\mathcal{L}(y/2)$ with the Langevin function $\mathcal{L}(z) = \coth z - 1/z$, increasing with $0 < \mathcal{L}(z) < \min(z/3, 1)$; so $|\varphi_S(x)| \le \min(|x|/12, \frac{1}{2})$ and $|\varphi_S|$ increases in $|x|$.

Step 3 (b) Lie

$|e^{\lambda_n T} - e^{\lambda_m T}| \le 1$ and $|\varphi_L| \le 1$ give $|\langle e_m, G^{\mathrm{Lie}}\rangle| \le h\sum_n |c_n||V_{mn}|$; Cauchy–Schwarz and Parseval ($\|Ve_n\| = |c|\mu_n$) give the bound. For $u_0 = e_1$ and fixed m, $x_{m1} \to 0$ so $\varphi_L \to \frac{1}{2}$; dominated convergence gives $\|G^{\mathrm{Lie}}\|^2/h^2 \to C_L^2 = \frac{1}{4}\sum_{m\ \mathrm{even}} V_{m1}^2(e^{\lambda_1 T} - e^{\lambda_m T})^2 > 0$ since $V_{21} \ne 0$.

Step 4 (c) Strang: the upper bound

For $m \le 2N$, $|\varphi_S(x_{mn})| \le \kappa 4N^2 h/12$ and $|V_{mn}| \le 8|c|N^2/L$, so the contribution is $O(h^4)$ in the squared norm. For $m > 2N$, $m^2 - n^2 \ge \frac{3}{4}m^2$ gives $|\langle e_m, G^{\mathrm{Str}}\rangle| \le \frac{16|c|A}{3L}m^{-1}h\min(\kappa m^2 h/12, \frac{1}{2})$; splitting at $M = \lfloor h^{-1/2}\rfloor$, $\sum_{m\le M} \lesssim \kappa^2 h^{1/2}$ and $\sum_{m>M} \lesssim h^{1/2}$, so $\|G^{\mathrm{Str}}\|^2 \le Ch^{5/2}$. (The lower bound uses the window of even m with $1 \le x_{m1} \le 4$ and is *condensed*: package §6.)

Why it works. the transport step does not preserve the Dirichlet condition ($Ve_1 \notin H_0^1$), producing a boundary layer on the modes $m \sim h^{-1/2}$; the parabolic smoothing absorbs it for Lie, but for Strang it lowers the order from 2 to 5/4.

■

Uses: none.

Scope and priority

First order in ε only, modal data, transport step truncated at first order. In L^∞ near $x = 0$ the observed Strang slope is $1.004 \to 1.001$ (order 1, checked on an example, not proved). The mechanism is that of Einkemmer–Ostermann (2015) for diffusion–reaction splitting with non-compatible boundary conditions; the norm in which Dang–Einkemmer–Ostermann (2025) state their order is not verified (CONV-Q6′).

APPENDIX A

Index of results

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